SOCIETY OS

Enterprise Documentation

v4.5 | Consolidated Master Edition

EDITION CONTROL

Published	04 November 2026
Status	Engineering product specification; not a statement of deployed software.
Retention	v4.4 consolidated master retained in full; no existing baseline sections removed.
New content	44-page v4.5 extension: Sections 37-64.
Coverage	86 document / record candidates, 33 triggering events, 78 additional QA cases.
Production constraint	Zero paid generative-AI API dependency for core document workflows.

HOW TO READ THIS EDITION

1. Legacy consolidated blueprint and v4.4 trust addendum: PDF pages 2-160.
2. New v4.5 document and record operation specification: PDF pages 161-204.
3. Use the PDF bookmarks to jump directly to the 28 new numbered sections.
4. Where rules conflict, require formal approval; do not silently change deployed policy.

Critical legal and verification boundary

Society-issued documents, cryptographic authenticity, uploaded IDs, and government or regulated signatures carry different assurances. Statutory applicability must be reviewed per location and legal entity.

SOCIETY OS

ENTERPRISE PRODUCT DOCUMENTATION

v4.4 revision package | Document Trust & Automation

Release information

Release date	04 November 2026
Document class	Consolidated product blueprint with v4.4 engineering addendum
Source baseline	133-page Society OS consolidated enterprise blueprint
Additional specification	25-page Document Trust & Automation engineering addendum
Engineering status	Requirements specified; no implementation is claimed
Paid LLM/AI endpoints	Not required for core functionality

How to read this consolidated release

Pages 2-134: original consolidated product blueprint retained unchanged.

Pages 135-159: v4.4 detailed requirements and implementation controls.

Original pages display legacy v4.0 headers and internal v4.3 consolidation notes.

The v4.4 supplement clarifies document workflows but does not remove prior scope.

Design/role matrices and actual implementation progress are maintained separately.

Scope & assurance boundary

A scanned ID, OCR match or document format check is not official identity verification.

Issued PDFs and NOCs require governed approvals and current-status verification.

Offline Aadhaar verification and legal electronic signatures require separate controls.

Society OS can operate without OpenAI or other paid LLM APIs.

SOCIETY OS

Enterprise Product Blueprint & Functional Specification

For Indian Residential Communities, Standalone Buildings, Apartments, Villas, Complexes and Townships

VERSION 4.0 | Consolidated Enterprise Product Blueprint & Functional Specification

Document Control	Details
Document type	Product blueprint, functional specification, implementation guide and roadmap
Product	Society OS
Target market	Indian residential societies, RWAs, CHS, standalone buildings, gated communities, villas, mixed-use complexes and townships
Commercial direction	Free launch for selected societies; subscription-ready architecture retained privately
Document version	v4.0 — Consolidated Enterprise Production Blueprint
Status	Approved for Engineering Execution, AI Code Generation (Codex/Antigravity), Design & QA

CONFIDENTIAL — PRODUCT & IMPLEMENTATION DOCUMENT

VERSION 4.3 CONSOLIDATION: Existing 19-section structure and v3 content are retained; v4 requirements are integrated as production-level expansions. The blueprint deliberately avoids feature-to-role/application mapping and screen-by-screen UI specification.

VERSION 4.3 SCENARIO-FIRST OBJECTIVE - This master blueprint is designed for Product, Engineering, QA, Management and pitch/demo use. Every major capability is expected to be explained through real-world scenarios, explicit normal/alternate/exception paths, authoritative data behaviour, measurable acceptance criteria, and aligned use-case/data-flow diagrams.

VERSION 4.3 EXPANSION NOTE — Existing v4 content is retained. This revision adds foundational day-to-day workflows, real-world scenarios, exception handling, reusable platform engines, operational test dimensions, migration/recovery detail, and pitch-ready narrative depth. The document may intentionally exceed 109 pages.

Table of Contents

1. Executive Summary

2. Product Vision and Principles

3. User Personas and Applications

4.3 Feature Definition Standard

Each feature in the catalogue should be understood as a capability contract with five mandatory dimensions: trigger, business rules, state/lifecycle, exceptions, and measurable outcome. A feature is not considered fully specified merely because its name appears in the catalogue.

Capability area	Normal scenario	Important exception	Business outcome
Property & resident foundation	New owner registers and is linked to Unit A-1203	Ownership transfer occurs while a tenant is active	Accurate occupancy and historical traceability
Access & gate	Pre-approved visitor arrives and pass is validated	Resident is unreachable and visitor is not pre-approved	Secure admission with auditable exception handling
Finance	Monthly invoice is generated and paid by UPI	Gateway sends duplicate callback after timeout	Exactly-once financial posting
Operations	Lift AMC work order becomes due	Vendor misses SLA and uploads incomplete service evidence	Preventive maintenance and escalation visibility

4. Complete Feature Catalogue

5. Architecture and Data Flow Diagrams

6. Role-based Permission Model

7. Deep Functional Specification by Module

8. Owner, Tenant and Occupancy History Management

9. Biometric Attendance Integration

10. Document Vault, NOC and Secure Access

11. Workflows and User Journeys

12. Data Model and API Design

13. Feature Flags and Configuration

14. Security, Privacy and Audit Logging

15. Reports and Society Health Score

16. Smart Automation and Future Integrations

17. Roadmap and Delivery Priorities

18. Acceptance Criteria and Implementation Backlog

19. Appendix

Document discipline This specification defines product behaviour and engineering intent. It deliberately avoids screen-by-screen UI specifications and avoids assigning individual features to particular apps or roles; those details belong in dedicated UX and authorization specifications.

This blueprint defines behaviour, state machines, contracts, data rules, architecture, acceptance criteria and engineering intent. It intentionally does not prescribe screen-by-screen layouts or a feature-to-role/application visibility matrix; those belong in dedicated UX and authorization specifications.

1.6 Document Discipline

Convenience first; configurable society profiles; India-first payment and operational workflows; DPDP-oriented purpose/consent and retention boundaries; immutable financial history; hardware/offline resilience; and enterprise-grade traceability.

1.5 Core Guarantees

Target runtimes are cross-platform native mobile for iOS and Android, Android Tablet/Android Go for guard operations, and a React/Next.js administrative web console. The distributed backend is API-first and must keep authoritative domain, financial, access, workflow and audit state deterministic.

1.4 Production Runtime Contract

Guard workflows should minimize typing, use large touch targets and high contrast, provide camera/OCR assistance, support local-language/audio prompts and remain usable through network outages.

2.7 Low-Cognitive-Load Guard Operations

Accounting standards, fund treatment, interest rules, statutory exports, regional language, verification and society governance rules must be represented as configurable rule packs. Maharashtra, Karnataka and other state-specific assumptions must not be hard-coded as universal rules.

2.6 India-First Rule Packs

Profiles are configurable presets rather than separate products: Standalone/Small CHS (approximately 8–50 units), Mid-tier Gated Society (approximately 50–500 units), and Integrated Multi-Phase Township (approximately 500–10,000+ units). Individual society settings can enable or disable capabilities subject to dependency and authorization checks.

2.5 Dynamic Society Sizing

The application ecosystem is described for runtime and operational context only. This blueprint deliberately avoids mapping individual features to individual roles or applications.

3.4 Boundary Rule

Mobile clients are battery/network aware; tablets support landscape operational workflows; low-memory Android Go devices suppress nonessential animation and cache pressure; desktop web supports information-dense administrative workflows. All surfaces use the same domain model, configuration and authorization contract.

3.3 Device Classes and Runtime Boundaries

Standalone/Small CHS: core billing, notices, complaints, basic visitor and documents. Mid-tier Gated: pre-approval, delivery handling, facilities, parking, vendor/AMC and dispute workflows. Township: multi-gate ANPR/RFID, boom barriers, smart sub-metering, biometric shift calculations, multi-committee governance, e-voting and zone-based security. Profiles can be overridden by validated society settings.

4.2 Configuration Profiles

Visitor state machine, resident revocation race handling, dynamic parcel cubby PINs, overstay rules, offline reconciliation, double-entry financial reversals, explainable rental/tenant assessment, owner/tenant historical privacy, biometric log-only integration, cryptographic NOC verification, Smart Poll Engine, Vendor Sinking Fund/Milestone Escrow, acoustic/seepage dispute evidence, Smart Lost & Found, Community Carpool, Festival/Event Automation, Society Micro-Commerce, Domestic Worker Replacement Network, Pet Management, Guest Stay Management, Housekeeping Quality, intercom replacement, payroll readiness, water/electricity/solar/EV operations, utility connectors and enterprise intelligence are included as configurable capabilities.

4.1 v4 Enterprise Additions

PostgreSQL is authoritative for transactions. Redis supports cache/pub-sub/idempotency/session use cases but is not the financial or access source of truth. Object storage is private. Long-running jobs use queues/outbox workers. Integrations are adapter-based and circuit-breaker protected. Audit events are emitted by sensitive domain mutations.

5.9 Architecture Principles

Critical field events are recorded locally with client UUIDs, timestamps and synchronization metadata. On reconnect, the server validates tenant context, deduplicates, applies deterministic ordering/conflict rules and returns per-record outcomes. Third-party hardware failures must degrade to safe manual or queued modes rather than disabling unrelated functions.

5.8 Offline / Hardware Reliability

X-Society-Id is an input, not proof of authorization. The server validates the authenticated session, society context and entity relationship before accessing data. Repository/ORM operations carry societyId, and PostgreSQL RLS reinforces isolation.

Figure 12. Tenant resolution, authorization and PostgreSQL isolation boundary.

5.7 Multi-Tenant Security Boundary

The resident, guard and administrative clients communicate through a secure gateway. Local persistence uses encrypted SQLite/SQLCipher or WatermelonDB-backed SQLite. Secure storage holds tokens and encryption keys. Real-time synchronization may use WebSockets/MQTT with backoff. High-priority approval calling may use platform-supported VoIP/foreground mechanisms where permitted.

Figure 11. Cross-platform runtime, gateway and service architecture.

5.6 Cross-Platform Runtime Architecture

Figure 13. Original blueprint feature-flag and permission control flow retained for continuity.

Current unit association does not grant access to previous occupants' KYC, agreements, phone numbers or sensitive cases. Historical access is separately authorized and audited.

6.6 Historical Data Boundary

Frontend restrictions improve usability only. Backend authorization remains authoritative. A feature being visible does not make it executable, and a hidden feature must still be blocked server-side.

6.5 Authorization Independence

Every protected request evaluates active session, trusted society context, role/permission claims, feature state, entity relationship and sensitivity, valid workflow transition, configured approval requirements and audit requirements.

6.4 Eight Authorization Filters

A later occupant does not inherit prior occupant KYC, rent agreements, phone numbers or sensitive tickets merely because the unit is the same.

8.9 Historical Privacy

The rental/tenant assessment is an explainable operational readiness indicator based on authorized criteria such as verification completeness, agreement validity and configured society requirements. It must not use protected or unrelated personal characteristics and must show the contributing factors.

8.8 Rental / Tenant Assessment

Clearance evaluates authoritative dues, physical access assets, parking, damage, configured complaints/disputes and required documents. A blocked clearance creates an explicit exception; a permitted

override requires a signed reason and audit record. NOC issuance creates a signed PDF and verification digest/code. Access revocation invalidates sessions and configured gate mappings before archival.

Figure 15. Detailed move-out, clearance, NOC, access revocation and archival flow.

8.7 Move-Out Lifecycle

Request → identity/profile capture → KYC/document collection → owner approval where required → society validation → rule acknowledgement → lift/truck logistics where configured → access activation → ACTIVE occupancy.

8.6 Move-In Lifecycle

Corrections preserve old/new values, reason, actor, approval and audit. Daily/monthly summaries support shift adherence and outsourced security/housekeeping invoice verification.

9.6 Attendance Correction and Billing Support

Biometric capture/matching stays in the device/vendor ecosystem. Society OS receives attendance logs such as employee code, device identifier, punch timestamp, punch type, source and synchronization status. Duplicate keys, tenant scope and staff mapping are validated before persistence.

Figure 16. Biometric attendance synchronization without raw biometric storage.

9.5 Production Sync Boundary

No-dues, move-out, tenant, parking, renovation and residence certificates are configurable. Each defines clearance checks, approval chain, document template, verification method and retention policy.

10.6 NOC Catalogue

Sensitive documents store category, owning entity, sensitivity, version, checksum, storage key, expiry and retention. Object storage remains private. Downloads require a fresh authorization decision and a short-lived signed URL/controlled stream. NOCs/certificates can carry SHA-256 verification data and QR codes.

Figure 17. Secure document lifecycle and controlled retrieval.

10.5 Cryptographic Integrity and Retrieval

Trigger → classify → dispatch → acknowledge → respond → escalate → close → post-incident review. Delivery failure uses configured fallback paths.

11.7 Emergency Flow

Request → purpose → recipient decision → persistent channel on acceptance; report routes to audited moderation. Phone numbers remain masked.

11.6 Resident Connect First Contact

Pass revocation versus offline admission: server-authoritative state wins when the revocation is determined to precede the admission; the admission becomes UNAUTHORIZED_ENTRY_FLAGGED. Duplicate punches are silently deduplicated by deterministic key. Concurrent edits use server-monotonic ordering and workflow state; stale clients receive explicit conflict responses.

11.5 Conflict Resolution Rules

Offline records carry client UUID, device identifier, local timestamp and vector/ordering metadata. Reconnect processing authenticates, validates tenant scope, deduplicates and resolves each record independently. One conflict must not fail an entire batch.

Figure 18. Offline queue, idempotency and deterministic conflict resolution.

11.4.1 Import, Export and Data Migration Workflow

Operational deployments often begin with spreadsheets. Import must therefore be a product capability: Upload → Format Validation → Column Mapping → Duplicate Check → Business Validation → Preview → Confirm → Async Processing → Success/Failure Report → Audit. A failed row must be downloadable with an actionable error reason.

Real-world scenario — 5,000-resident migration: a society uploads owners, tenants, vehicles, parking and opening balances from legacy spreadsheets. 4,912 records pass, 88 fail because of duplicate unit associations or invalid payment references. The system imports only the confirmed valid set, exposes the 88 errors, and preserves import history so the same file is not accidentally applied twice.

11.4.2 Backup, Restore and Disaster Recovery Workflow

Production operations require database backups, object/document backups, encryption, retention policies, point-in-time recovery where supported, restore testing, disaster recovery procedures and preservation of audit integrity. Backup success must be monitored; “backup job ran” is not equivalent to “restore is possible”.

Real-world scenario — accidental data corruption: a bad deployment corrupts a configuration table. The recovery process restores the affected state to a known point, validates tenant isolation, checks financial consistency and confirms that audit records remain intact before traffic is returned to the recovered environment.

11.4 Offline Gate Sync

Use versioned endpoints, society-scoped context, idempotency for retryable mutations, signature verification for callbacks, stable identifiers, deterministic error contracts, secure upload/retrieval separation and trace IDs.

11.4.4 Account, Session and Device Lifecycle

Authentication must include login/OTP/password flows as configured, device registration, session issuance, expiry, token revocation, force logout, lost-device handling and suspicious-login controls. Device trust is separate from user identity.

Real-world scenario — lost guard tablet: the tablet is reported lost. Existing sessions/tokens are revoked, the device is marked untrusted, and a replacement device must complete registration before offline operational data can be synchronized.

- Sessions have explicit expiry and revocation state.
- Device identifiers must not be treated as permanent identities for a human.
- Offline capability must use bounded credentials and local data protection.
- Suspicious login signals should be observable and auditable.

11.4.3 File and Media Management

Files should be treated as managed objects rather than arbitrary URLs. The lifecycle includes upload initiation, chunk/resume where supported, validation, malware scanning, checksum, metadata normalization, storage, versioning, access control, download audit and retention.

Real-world scenario — gate photo upload: a guard captures a damaged vehicle photo while offline. The attachment enters a local queue, is retried when connectivity returns, and is attached to the violation record only after server validation confirms integrity.

- Allowed MIME types, size limits and retention are configurable.
- EXIF metadata should be stripped where privacy policy requires it.
- Failed scans quarantine the object and prevent normal retrieval.

- Access URLs should be short-lived/signed where appropriate and must not become authorization bypasses.

12.8 API Rules

POST /api/v1/gate/sync-batch — offline batch reconciliation; per-record COMMITTED/CONFLICT results.
POST /api/v1/gate/visitor-checkin — visitor creation/check-in with secure pass token and server monotonic time.
POST /api/v1/finance/adjustments — compensating/reversal journal linked to original transaction.
POST /api/v1/visitors/preapprovals; POST /api/v1/gate/entries; POST /api/v1/bills/generate; POST /api/v1/payments/callback; POST /api/v1/complaints; POST /api/v1/bookings; POST /api/v1/integrations/biometric/punches/sync; GET /api/v1/attendance/monthly; PATCH /api/v1/attendance/{id}/correction.

12.7 Contract APIs

```
CREATE EXTENSION IF NOT EXISTS "uuid-ossp";
CREATE EXTENSION IF NOT EXISTS "citext";

-- Core tables: societies, units, users, occupancies, visitors, financial_accounts,
-- ledger_transactions, ledger_lines, document_records, staff_profiles,
-- biometric_attendance_punches and disputes.
-- Production rule: every tenant-scoped table carries society_id; ledger_lines
-- also carries society_id so RLS can be enforced directly.
-- Posted financial records are append-only; corrections are reversal journals.

ALTER TABLE units ENABLE ROW LEVEL SECURITY;
ALTER TABLE visitors ENABLE ROW LEVEL SECURITY;
ALTER TABLE ledger_transactions ENABLE ROW LEVEL SECURITY;
ALTER TABLE ledger_lines ENABLE ROW LEVEL SECURITY;
ALTER TABLE document_records ENABLE ROW LEVEL SECURITY;
ALTER TABLE occupancies ENABLE ROW LEVEL SECURITY;

CREATE POLICY society_isolation_units ON units FOR ALL USING
(society_id = NULLIF(current_setting('app.current_society_id', true), '')::UUID);
CREATE POLICY society_isolation_visitors ON visitors FOR ALL USING
(society_id = NULLIF(current_setting('app.current_society_id', true), '')::UUID);
CREATE POLICY society_isolation_ledger_transactions ON ledger_transactions FOR ALL USING
(society_id = NULLIF(current_setting('app.current_society_id', true), '')::UUID);
CREATE POLICY society_isolation_ledger_lines ON ledger_lines FOR ALL USING
(society_id = NULLIF(current_setting('app.current_society_id', true), '')::UUID);
CREATE POLICY society_isolation_documents ON document_records FOR ALL USING
(society_id = NULLIF(current_setting('app.current_society_id', true), '')::UUID);
CREATE POLICY society_isolation_occupancies ON occupancies FOR ALL USING
(society_id = NULLIF(current_setting('app.current_society_id', true), '')::UUID);
```

12.6 Consolidated Core DDL

The v4 production schema must scope uniqueness and idempotency by society where business identifiers are society-local; carry society_id onto ledger_lines for direct isolation; apply RLS to every sensitive tenant-scoped table; and enforce balanced journal posting in a transaction. Original posted ledger rows are not updated or deleted.

12.5 Production DDL Hardening Notes

POST /api/v1/finance/adjustments — Request contains originalTransactionId, reversalReason, supportingDocumentId and effectiveDate. Response 201 returns reversalTransactionId, referenceNumber, posted debit/credit lines and balanceStatus=BALANCED.

POST /api/v1/gate/visitor-checkin — Request contains unitId, visitorName, phoneNumber, visitorType, optional deliveryBrand, vehicleRegistration, gateId and offlineCreatedTimestamp. Response 201 returns visitorId, status, requiresCallBypass, secure passValidationToken, residentActionTimeoutSeconds, traceId and serverMonotonicTime.

POST /api/v1/gate/sync-batch — Headers: Authorization Bearer <JWT>, X-Society-Id <UUID>, Idempotency-Key <UUID>. Request contains deviceIdentifier, batchTimestamp and records with idempotencyKey, unitId, visitorName, phoneNumber, visitorType, vehicleNumber, action, offlineEventTimestamp and vectorClock. Response 200 returns PROCESSED, processedCount, per-record COMMITTED/CONFLICT resolution, visitorId/currentStatus and serverSyncTimestamp.

12.10 Exact API Contract Examples

```
CREATE EXTENSION IF NOT EXISTS "uuid-ossp";
CREATE EXTENSION IF NOT EXISTS "citext";

CREATE TABLE societies (
  id UUID PRIMARY KEY DEFAULT uuid_generate_v4(), slug CITEXT UNIQUE NOT NULL,
  name VARCHAR(255) NOT NULL, registration_number VARCHAR(100), state_code VARCHAR(10) NOT NULL,
  currency_code VARCHAR(3) NOT NULL DEFAULT 'INR', accounting_standard VARCHAR(100),
  settings JSONB NOT NULL DEFAULT '{}'::jsonb, is_active BOOLEAN NOT NULL DEFAULT TRUE,
  created_at TIMESTAMPTZ NOT NULL DEFAULT NOW(), updated_at TIMESTAMPTZ NOT NULL DEFAULT NOW()
);
CREATE TABLE units (
  id UUID PRIMARY KEY DEFAULT uuid_generate_v4(), society_id UUID NOT NULL REFERENCES societies(id) ON DELETE RESTRICT,
  phase VARCHAR(100), tower VARCHAR(100) NOT NULL, block VARCHAR(100), wing VARCHAR(100), floor_number INT NOT NULL,
  unit_number VARCHAR(50) NOT NULL, unit_type VARCHAR(50) NOT NULL DEFAULT 'FLAT',
  super_builtup_sqft NUMERIC(10,2), carpet_sqft NUMERIC(10,2), occupancy_status VARCHAR(50) NOT NULL DEFAULT 'VACANT',
  created_at TIMESTAMPTZ NOT NULL DEFAULT NOW(), updated_at TIMESTAMPTZ NOT NULL DEFAULT NOW(),
  CONSTRAINT uk_society_unit UNIQUE (society_id, tower, unit_number)
);
CREATE TABLE users (
  id UUID PRIMARY KEY DEFAULT uuid_generate_v4(), society_id UUID REFERENCES societies(id) ON DELETE RESTRICT,
  phone_number VARCHAR(20) NOT NULL, email CITEXT, full_name VARCHAR(255) NOT NULL, password_hash VARCHAR(255),
  dpdp_consent_record JSONB NOT NULL DEFAULT '{}'::jsonb, is_suspended BOOLEAN NOT NULL DEFAULT FALSE,
  created_at TIMESTAMPTZ NOT NULL DEFAULT NOW(), updated_at TIMESTAMPTZ NOT NULL DEFAULT NOW(),
  CONSTRAINT uk_society_user_phone UNIQUE (society_id, phone_number)
);
CREATE TABLE occupancies (
  id UUID PRIMARY KEY DEFAULT uuid_generate_v4(), society_id UUID NOT NULL REFERENCES
```

```
societies(id) ON DELETE RESTRICT,
  unit_id UUID NOT NULL REFERENCES units(id) ON DELETE RESTRICT, user_id UUID NOT NULL
REFERENCES users(id) ON DELETE RESTRICT,
  occupant_role VARCHAR(50) NOT NULL, start_date DATE NOT NULL, end_date DATE,
  status VARCHAR(50) NOT NULL DEFAULT 'PENDING_APPROVAL', created_at TIMESTAMPTZ NOT NULL
DEFAULT NOW(), updated_at TIMESTAMPTZ NOT NULL DEFAULT NOW()
);
CREATE TABLE visitors (
  id UUID PRIMARY KEY DEFAULT uuid_generate_v4(), society_id UUID NOT NULL REFERENCES
societies(id) ON DELETE RESTRICT,
  unit_id UUID NOT NULL REFERENCES units(id) ON DELETE RESTRICT, visitor_name VARCHAR(255) NOT
NULL, phone_number VARCHAR(20),
  visitor_type VARCHAR(50) NOT NULL, vehicle_registration VARCHAR(50), pass_code_hash VARCHAR(255),
  entry_gate_id VARCHAR(50), exit_gate_id VARCHAR(50), check_in_at TIMESTAMPTZ, check_out_at
TIMESTAMPTZ,
  current_status VARCHAR(50) NOT NULL DEFAULT 'EXPECTED', idempotency_key UUID NOT NULL,
sync_vector_clock BIGINT NOT NULL DEFAULT 0,
  created_at TIMESTAMPTZ NOT NULL DEFAULT NOW(), CONSTRAINT uk_visitor_idempotency UNIQUE
(society_id, idempotency_key)
);
CREATE TABLE financial_accounts (
  id UUID PRIMARY KEY DEFAULT uuid_generate_v4(), society_id UUID NOT NULL REFERENCES
societies(id) ON DELETE RESTRICT,
  account_code VARCHAR(50) NOT NULL, account_name VARCHAR(255) NOT NULL, account_type
VARCHAR(50) NOT NULL,
  is_statutory_fund BOOLEAN NOT NULL DEFAULT FALSE, CONSTRAINT uk_society_account UNIQUE
(society_id, account_code)
);
CREATE TABLE ledger_transactions (
  id UUID PRIMARY KEY DEFAULT uuid_generate_v4(), society_id UUID NOT NULL REFERENCES
societies(id) ON DELETE RESTRICT,
  unit_id UUID REFERENCES units(id) ON DELETE RESTRICT, transaction_date TIMESTAMPTZ NOT NULL
DEFAULT NOW(),
  reference_number VARCHAR(100) NOT NULL, description TEXT NOT NULL, is_reversal BOOLEAN NOT
NULL DEFAULT FALSE,
  reversed_transaction_id UUID REFERENCES ledger_transactions(id), created_by UUID NOT NULL
REFERENCES users(id), created_at TIMESTAMPTZ NOT NULL DEFAULT NOW(),
  CONSTRAINT uk_society_reference UNIQUE (society_id, reference_number)
);
CREATE TABLE ledger_lines (
  id UUID PRIMARY KEY DEFAULT uuid_generate_v4(), society_id UUID NOT NULL REFERENCES
societies(id) ON DELETE RESTRICT,
  transaction_id UUID NOT NULL REFERENCES ledger_transactions(id) ON DELETE RESTRICT, account_id
UUID NOT NULL REFERENCES financial_accounts(id) ON DELETE RESTRICT,
  amount NUMERIC(14,2) NOT NULL CHECK (amount > 0), entry_type VARCHAR(10) NOT NULL CHECK
(entry_type IN ('DEBIT','CREDIT'))
);
CREATE TABLE document_records (
  id UUID PRIMARY KEY DEFAULT uuid_generate_v4(), society_id UUID NOT NULL REFERENCES
societies(id) ON DELETE RESTRICT,
  unit_id UUID REFERENCES units(id) ON DELETE RESTRICT, category VARCHAR(50) NOT NULL,
file_storage_key VARCHAR(512) NOT NULL,
  file_sha256_checksum CHAR(64) NOT NULL, mime_type VARCHAR(100) NOT NULL, is_encrypted
```

```
BOOLEAN NOT NULL DEFAULT TRUE,
  access_level VARCHAR(50) NOT NULL DEFAULT 'RESTRICTED', retention_valid_until DATE, created_at
TIMESTAMPTZ NOT NULL DEFAULT NOW()
);
CREATE TABLE staff_profiles (
  id UUID PRIMARY KEY DEFAULT uuid_generate_v4(), society_id UUID NOT NULL REFERENCES
societies(id) ON DELETE RESTRICT,
  user_id UUID NOT NULL REFERENCES users(id) ON DELETE RESTRICT, staff_type VARCHAR(50) NOT
NULL, employee_code VARCHAR(100) NOT NULL,
  is_verified BOOLEAN NOT NULL DEFAULT FALSE, created_at TIMESTAMPTZ NOT NULL DEFAULT NOW(),
CONSTRAINT uk_society_employee UNIQUE (society_id, employee_code)
);
CREATE TABLE biometric_attendance_punches (
  id UUID PRIMARY KEY DEFAULT uuid_generate_v4(), society_id UUID NOT NULL REFERENCES
societies(id) ON DELETE RESTRICT,
  staff_id UUID NOT NULL REFERENCES staff_profiles(id) ON DELETE RESTRICT, device_identifier
VARCHAR(100) NOT NULL,
  punch_timestamp TIMESTAMPTZ NOT NULL, punch_type VARCHAR(20) NOT NULL CHECK (punch_type
IN ('IN','OUT')),
  punch_hash CHAR(64) NOT NULL, created_at TIMESTAMPTZ NOT NULL DEFAULT NOW(), CONSTRAINT
uk_society_punch_hash UNIQUE (society_id, punch_hash)
);
CREATE TABLE disputes (
  id UUID PRIMARY KEY DEFAULT uuid_generate_v4(), society_id UUID NOT NULL REFERENCES
societies(id) ON DELETE RESTRICT,
  initiator_unit_id UUID NOT NULL REFERENCES units(id) ON DELETE RESTRICT, target_unit_id UUID NOT
NULL REFERENCES units(id) ON DELETE RESTRICT,
  category VARCHAR(50) NOT NULL, status VARCHAR(50) NOT NULL DEFAULT 'OPEN', evidence_payload
JSONB NOT NULL DEFAULT '{}'::jsonb,
  created_at TIMESTAMPTZ NOT NULL DEFAULT NOW(), updated_at TIMESTAMPTZ NOT NULL DEFAULT
NOW()
);
ALTER TABLE units ENABLE ROW LEVEL SECURITY; ALTER TABLE visitors ENABLE ROW LEVEL SECURITY;
ALTER TABLE ledger_transactions ENABLE ROW LEVEL SECURITY; ALTER TABLE ledger_lines ENABLE
ROW LEVEL SECURITY;
ALTER TABLE document_records ENABLE ROW LEVEL SECURITY; ALTER TABLE occupancies ENABLE ROW
LEVEL SECURITY;
CREATE POLICY society_isolation_units ON units FOR ALL USING (society_id =
NULLIF(current_setting('app.current_society_id', true), '')::UUID);
CREATE POLICY society_isolation_visitors ON visitors FOR ALL USING (society_id =
NULLIF(current_setting('app.current_society_id', true), '')::UUID);
CREATE POLICY society_isolation_ledger_transactions ON ledger_transactions FOR ALL USING (society_id =
NULLIF(current_setting('app.current_society_id', true), '')::UUID);
CREATE POLICY society_isolation_ledger_lines ON ledger_lines FOR ALL USING (society_id =
NULLIF(current_setting('app.current_society_id', true), '')::UUID);
CREATE POLICY society_isolation_documents ON document_records FOR ALL USING (society_id =
NULLIF(current_setting('app.current_society_id', true), '')::UUID);
CREATE POLICY society_isolation_occupancies ON occupancies FOR ALL USING (society_id =
NULLIF(current_setting('app.current_society_id', true), '')::UUID);
```

12.9 Full v4 DDL Reference

{"tier":"GATED_APARTMENT_COMPLEX","modules":{"visitorManagement":
{"enabled":true,"enforceVehicleOCR":true,"requireResidentApprovalOverstayMinutes":120,"allowGuardMan

ualOverride":false},"financialLedger":
{"enabled":true,"lateFeeAnnualInterestPercent":21.0,"lateFeeType":"SIMPLE_INTEREST","reversalRequiresDualSignoff":true},"hardwareGateway":
{"biometricSyncEnabled":false,"anprBarrierEnabled":false},"interFlatChat":
{"enabled":true,"maskContactNumbers":true,"requireFirstContactApproval":true}}}

13.7 Example JSON Configuration

Figure 19. Feature flag evaluation followed by independent authorization and workflow validation.

13.6 Runtime Control Flow

Enablement requires dependency validation. Disabling a module blocks new business actions but does not delete historical records. Configuration changes are audited with old value, new value, actor and timestamp. Feature flags never replace authorization.

13.5 Dependency and Runtime Rules

Core launch flags remain visitorManagement, maintenanceBilling, complaints, documentVault, ownerTenantHistory, moveOutNocMandatory and residentMessenger enabled; public directory, biometric and hardware integrations, subscription UI and smart automation remain disabled by default. Water, energy, EV, governance voting, marketplace, carpool, lost & found, festival automation, guest stays, housekeeping, payroll, utility integrations and advanced intelligence are configurable.

13.4 v4 Configuration Contract

Sensitive gate/security events can be chained in a Merkle-style append-only audit structure. Document issuance can store SHA-256 of serialized bytes. Audit records include tenant, actor, action, target, timestamp, result and trace context without unnecessary duplication of personal content.

14.6 Cryptographic Audit

Policy baselines such as 180-day retention for selected transient visitor/security PII and approximately 8-year retention for selected financial/audit records may be configured, but production values must be validated against the applicable legal, contractual and society policy requirements.

14.7 Backup, Disaster Recovery and Business Continuity

Backup and disaster recovery are production capabilities. The platform should protect relational data, document/object storage and critical configuration with encrypted backups and defined retention. Recovery procedures must be rehearsed rather than assumed.

Real-world scenario — regional outage: the primary hosting region becomes unavailable. The recovery plan restores services according to defined RPO/RTO targets, validates tenant isolation and financial consistency, and records the recovery event. The target is controlled continuity, not an untested promise of zero downtime.

- Define RPO/RTO per environment and criticality tier.
- Perform periodic restore verification.
- Preserve audit integrity during restoration.
- Have explicit procedures for corruption, accidental deletion, credential compromise and infrastructure loss.

14.5 Retention Baselines

The system records purpose/consent context where required and evaluates retention before deletion/anonymisation. An erasure request does not automatically remove records that must legitimately be retained; instead the retention/legal-hold decision is recorded and personal fields are minimized or scrubbed where eligible.

Figure 20. Purpose, retention, erasure/anonymisation and retained-record separation.

14.4 DPDP-Oriented Retention Architecture

Every score movement must be traceable to source metrics, time window, formula/version and weighting. AI may summarize but cannot invent the underlying metric.

15.5 Explainability

Collection efficiency, complaint SLA, security discipline, compliance health, operational efficiency, resident satisfaction, workforce health and utility/sustainability health may contribute to a transparent society-level score.

15.5.1 Health Score Calculation and Guardrails

The Society Health Score should combine measurable dimensions such as collections, complaint SLA, preventive maintenance, vendor performance, facility utilization, incident history, document/compliance status and operational responsiveness. It should explain which factors drove the score.

Real-world scenario — score falls from 82 to 68: the explanation identifies increased overdue dues, two missed critical maintenance tasks and a complaint SLA breach. Management can drill into source records instead of receiving an opaque AI judgement.

- Weights are configurable and versioned.
- Low data quality reduces confidence rather than being silently treated as a zero score.
- AI-generated narratives must link back to source metrics and should not invent causes.

15.4 Expanded Health Dimensions

Payment, messaging, biometric and utility integrations use circuit breakers, queues and retry state. A baseline of three consecutive failures may open a circuit; synchronous unavailable services return controlled 503/Retry-After behaviour where applicable.

16.9 Integration Failure Isolation

AI never becomes the source of truth. Permissions are evaluated before retrieval. Uploaded documents/messages are untrusted. Low-confidence results require uncertainty/manual review. Financial, access, approval and compliance mutations continue through deterministic workflows. Core operations work if the AI provider is unavailable.

16.9.2 API and Schema Versioning

API contracts must support backward compatibility and explicit deprecation. Database migrations must be forward-safe where possible, and mobile/offline clients must tolerate server evolution.

Real-world scenario — offline app after release: a guard tablet reconnects two weeks after an application update. The server receives an older sync schema. Compatibility logic validates and migrates the payload or rejects it with a recoverable error; it must not partially apply unknown fields.

- Use explicit API versions for breaking changes.
- Contract tests protect mobile/server compatibility.
- Database migrations are ordered, observable and reversible where technically possible.
- Feature-flag/configuration migrations must not leave orphaned states.

16.9.1 Observability and Operational Control Plane

Production observability must cover API health, database health, queues, scheduled jobs, notification delivery, payment callbacks, sync conflicts, biometric failures, storage, integration latency, suspicious activity and feature-flag/configuration changes.

Real-world scenario — notification queue backlog: payment reminders are delayed because a provider is throttling requests. Operations should see queue depth, retry rate, dead-letter volume and affected notification types without confusing this infrastructure problem with payment failure.

- Every cross-service operation carries a trace/correlation identifier.
- Metrics must be tied to business outcomes where possible, not only CPU/memory.
- Dead-letter items require retry/replay controls with authorization and audit.
- Alert thresholds should distinguish transient spikes from sustained incidents.

16.8 AI Guardrails

Smart Lost & Found, Community Carpool, Festival/Event Automation, Society Micro-Commerce, Domestic Worker Replacement Network, Pet Management, Classifieds, Guest Stay Management, Housekeeping Quality, Emergency Local Network, intercom replacement, water, electricity, solar, EV charging and utility connectors are configurable extensions.

16.7 Community Intelligence and Utility Extensions

AI Society Manager; Society Intelligence Engine; Society Digital Twin; Society Memory/Knowledge Graph; Predictive Maintenance; AI Financial Advisor; Vendor Intelligence; Smart Complaint/Incident Intelligence; Community Health/Anomaly Detection; No-Code Automation; Smart Security; Emergency Command Center; and resident-facing timeline/personal-assistant capabilities are planned as downstream intelligence services.

16.6 Enterprise Intelligence Portfolio

A feature is production-ready only after data model, authorization, migrations, observability, failure handling, feature-flag paths, acceptance tests, security review and operational documentation are complete.

17.4 Production Activation Gate

MVP: setup, units, onboarding, visitor/gate, billing, complaints, notices, staff, documents, notifications, feature flags and offline foundations. Phase 1A: Resident Connect, history, move lifecycle, archive, parcel handover, inter-flat issues, lost & found. Phase 2: NOC, parking, facilities, vendors/AMC, assets, inventory, renovation, governance, compliance, housekeeping and advanced reporting. Phase 3: biometric, workforce automation, senior care, emergency volunteers, marketplace, commerce, carpool, events, payroll readiness and Health Score. Phase 4: RFID, ANPR, boom barriers, CCTV, smart meters, solar/energy, EV, smart automation, e-voting and advanced connected devices.

17.3 Consolidated v4 Roadmap

Test concurrency, idempotency, offline replay, RLS isolation, backup/restore, queue recovery, circuit-breaker opening/closing, document checksum verification, audit-chain continuity, low-memory devices, regional formatting and deterministic report reconciliation.

18.3 Additional Non-Functional Acceptance

TC-SEC-01 resident denial locks admission; TC-FIN-01 reversal leaves original ledger unchanged and balances; TC-NOC-01 outstanding dues return deterministic 422 and block NOC; TC-SYNC-01 200 offline visitors reconcile with zero loss/duplicates; TC-DPDP-01 eligible erasure scrubs personal data while preserving legitimately retained records; TC-HARD-01 low-memory device avoids OOM; TC-SEC-02 revocation race flags unauthorized entry; TC-CHAT-01 first-contact approval masks phone; TC-PRIV-01 later occupant cannot retrieve prior occupant KYC/private history; TC-FLAG-01 disabled feature rejects backend call; TC-AI-01 AI recommendation is permission-filtered and cannot mutate authoritative state.

18.2 Expanded End-to-End Tests

Society OS should make a society easier to operate, easier to govern and easier to live in without sacrificing privacy, financial integrity, physical security, transparency or traceability. The moat is one trusted

operational system in which people, units, money, documents, security, services, governance, assets, utilities and intelligence share a consistent foundation.

19.7 Final Product Principle

Provision PostgreSQL with RLS and migrations; scaffold Flutter/React Native with encrypted local storage; implement NestJS or Go API/service boundaries; publish OpenAPI contracts; implement queues/outbox and connector circuit breakers; create state-machine tests; run cross-society isolation tests; test low-end tablets; validate privacy/retention policies; complete end-to-end acceptance matrix.

19.6 Implementation Checklist

Never execute an unscoped tenant mutation. Reject null societyId. Never UPDATE/DELETE posted ledger_transactions or ledger_lines. Use client UUID idempotency for gate/attendance events. Put external integrations behind circuit breakers. Keep offline local writes transactional. Hash document bytes where cryptographic proof is required. Never expose historical tenant KYC through current occupancy. AI cannot bypass permissions, workflow state, approvals or financial controls.

19.5 Master AI / Developer Guardrails

1. Executive Summary

Society OS is a configurable operating platform for Indian residential communities. It is designed to support the full lifecycle of a society—from initial setup and resident onboarding through daily security, billing, complaints, facilities, staff operations, governance, compliance, asset management, communication and community services.

The product is intentionally broader than a basic visitor-and-maintenance application. It is designed to become the operational backbone of a society while remaining modular enough for a small standalone building and powerful enough for a large township. The same core architecture should support different society structures, policies, workflows, charge models, approval chains and integrations without requiring a separate product codebase.

The platform combines structured operational records with configurable workflows, secure documents, notifications, auditability, analytics and an intelligent automation layer. The intelligence layer is intended to understand patterns, explain information, assist with drafting and recommend actions; authoritative financial, access, approval and audit state remains controlled by deterministic application logic.

1.1.1 What “Complete” Means in This Blueprint

Completeness is defined operationally, not cosmetically. Every major capability is described through its business purpose, normal path, exception path, data that changes, audit implications, user-visible outcome, and testable acceptance behaviour. This makes the document usable by engineering, product, QA, implementation teams and management without relying on unstated assumptions.

Real-world scenario — Monday morning society operations: at 8:45 AM, a resident reports a water leak, a tenant has a move-in scheduled, a delivery rider arrives at the gate, an AMC vendor misses a scheduled visit, and a monthly maintenance payment fails. Society OS must treat these as separate domain events while keeping the shared unit, person, access, finance, notification and audit records consistent.

1.1.2 Pitch Narrative: Why Society OS Is More Than a Resident App

Society OS is positioned as an operating system for residential communities: a common operational record for property structure, people, access, money, work, documents, communication, compliance and community services. The pitch is therefore not “another society app”; it is a configurable operational platform that reduces manual coordination and fragmented records.

1.1 Product Scope

- Society and property master data, units, occupancy, ownership and resident lifecycle.
- Visitor, gate, delivery, vendor, staff and material movement management.
- Maintenance billing, collections, receipts, accounting records, budgets and reconciliation readiness.
- Complaints, helpdesk, SLA management, escalation, evidence and resolution workflows.
- Facilities, amenities, reservations, deposits, inspections and usage management.
- Parking, vehicles, violations, visitor parking and integration readiness.
- Resident communication, controlled messaging, announcements, community groups and privacy controls.
- Inter-flat issues, mediation, evidence and resolution tracking.
- Governance, meetings, notices, resolutions, polls, elections and e-voting where configured.
- Safety, emergency response, senior citizen support, incident management and local emergency coordination.
- Staff, domestic help, attendance, shift management, verification, payroll readiness and workforce operations.
- Vendors, AMC, assets, preventive maintenance, inventory and procurement workflows.
- Document vault, compliance records, certificates, NOCs and secure access.
- Marketplace, community classifieds, local services, carpool and community commerce.
- Water, electricity, energy, solar, EV charging and utility-management readiness.
- Smart automation, analytics, intelligence, knowledge retrieval and future connected-device integrations.

1.2 Supported Society Models

Society model	Product approach
Standalone building	Simple configuration with the same core data model; unnecessary modules remain disabled.
Cooperative housing society / CHS	Governance, billing, records, notices, compliance and committee workflows are supported.
Gated apartment community	Multiple towers, gates, parking areas, amenities, vendors and operational teams are supported.
Villa / bungalow community	Unit hierarchy, visitor management, utilities, security and community services remain configurable.
Mixed-use complex	Residential and commercial units can coexist with separate rules and charge structures.
Large township	Multiple phases, towers, zones, gates, amenities, staff teams and operational dependencies can be represented.

1.3 Core Product Promise

Promise	Meaning
Simple	Frequent society tasks should be quick, understandable and consistent.
Configurable	The product adapts to society-specific rules instead of forcing one operating model.
India-first	UPI, cash/cheque, regional language readiness, local verification workflows and Indian society processes are native considerations.
Privacy-first	Sensitive resident, tenant, financial, communication and historical data is protected by default.
Traceable	Important transactions and decisions are auditable from initiation through closure.
Operationally complete	The product should reduce dependency on paper registers, fragmented spreadsheets and unmanaged messaging groups.
Future-ready	Integrations and smart capabilities are added through controlled interfaces rather than hard-coded dependencies.

2. Product Vision and Principles

2.1 Mission

To simplify, digitize and continuously improve the daily operations of Indian residential communities through a configurable, trustworthy and easy-to-use platform.

2.2 Vision

To become the operating system for Indian housing communities—covering people, property, money, security, services, governance, compliance, assets and community life—while creating a strong foundation for automation, connected devices and intelligent decision support.

2.3 Product Philosophy

Principle	Product interpretation
Convenience first	High-frequency actions should require minimal effort and provide clear confirmation.
Configurable, not forced	Societies should enable only what they need and define their own rules.
Transparency by default	Bills, complaints, approvals, attendance, NOCs, governance decisions and operational changes should leave a traceable record.
Privacy by design	Access is granted by explicit permission and data sensitivity rather than by convenience.
Deterministic core	Money, permissions, access, workflow state and audit records are controlled by application logic.
Automation with control	Automation can reduce repetitive work but must respect approvals, exceptions and audit requirements.
Low-connectivity readiness	Critical operational workflows should degrade gracefully and synchronise when connectivity returns.
India-first operations	Payment methods, society practices, verification, local vendors and regional language considerations are treated as first-class requirements.
Modular engineering	Domains remain independently configurable with clear interfaces and bounded responsibilities.
Data as a product asset	Clean historical records enable reporting, automation, forecasting and future intelligence.

2.4 Product Differentiation

- Complete operational coverage rather than a single-purpose visitor or billing tool.
- Strong owner/tenant lifecycle history and controlled historical document archive.
- Mandatory move-out clearance and NOC framework with configurable exceptions.
- Privacy-first resident communication that does not require exposing personal phone numbers.
- Workforce and biometric attendance readiness for outsourced security and housekeeping operations.
- Society-wide asset, AMC, inventory and preventive-maintenance management.
- Community utility layer covering marketplace, classifieds, carpool, events and local services.
- Utility and sustainability layer covering water, electricity, solar, EV and energy intelligence.
- Intelligence layer designed around the society's own structured records, rules and history.
- Subscription-ready architecture without forcing monetisation into the initial product experience.

2.4.1 Localization and Accessibility Principles

Society OS should be usable by residents, senior citizens, guards and staff with different language, literacy, device and accessibility needs. Localization must cover English, Hindi, Marathi, Kannada, Tamil, Telugu, Bengali, Gujarati, Malayalam, Punjabi and future languages through externalized content.

Real-world scenario — senior citizen receives an emergency notice: the notification should be understandable without relying on tiny text or color-only indicators. Where supported, audio/local-language assistance can improve comprehension.

- Use scalable typography and large touch targets.
- Provide high contrast and non-color-only status indicators.
- Support screen-reader semantics and keyboard navigation on web.
- Externalize notification/message templates and terminology for localization.
- Use reduced motion preferences where supported.

3. User Personas and Applications

Society OS is a multi-party operational platform. Different participants have different responsibilities, but the product is designed around a single shared society data model and consistent governance, security and audit principles.

3.1 Personas

Persona	Primary context
Resident owner	Lives in or owns a unit and participates in day-to-day society life, payments and governance.
Tenant	Occupies a unit under an authorised tenancy and interacts with society processes during occupancy.
Family member	Participates in household activities under controlled association with the unit.
Security guard / supervisor	Performs gate, visitor, incident and operational security activities.
Facility manager / supervisor	Coordinates complaints, staff, vendors, assets, inspections and day-to-day operations.
Committee / office bearer	Oversees governance, approvals, notices, compliance and society administration.
Treasurer / accountant	Maintains financial operations, billing, collections, expenses, budgets and records.
Vendor / service workforce	Performs authorised society services, contracts, maintenance and operational work.
Super administrator	Maintains platform-level configuration, onboarding, support and subscription-ready controls.

3.2 Application Ecosystem

Application surface	Purpose
Resident mobile experience	Primary digital interaction layer for residents and household members.
Security / gate experience	Fast operational interface for physical entry, visitor and incident workflows.
Administrative web experience	Central operational workspace for configuration, records, approvals, reporting and governance.
Facility operations experience	Focused operational workspace for maintenance, assets, workforce, vendors and incidents.
Platform administration experience	Cross-society configuration, onboarding, support and platform controls.

Scope boundary This section intentionally defines the product ecosystem without prescribing which individual feature belongs to which application or role, and without defining screen-by-screen layouts.

4. Complete Feature Catalogue

The catalogue below consolidates the product scope into a delivery-oriented feature inventory. Phases indicate intended maturity rather than a promise that every feature must be enabled for every society.

Feature area	Functional scope	Phase
Rental / Tenant Assessment	Explainable rental/tenant readiness assessment using configurable verification, document completeness, agreement and operational criteria; no opaque profiling.	1A
Society & Unit Setup	Society hierarchy, phases, towers, blocks, wings, floors, units, unit types, areas, occupancy, billing attributes and configuration.	MVP
Owner & Tenant Management	Current ownership, tenancy, family members, KYC, agreements, vehicles, pets, approvals and lifecycle status.	MVP
Owner/Tenant History	Historical ownership and occupancy timeline, previous resident records, archive and controlled historical references.	1A
Document Vault	Secure files, metadata, versions, expiry, verification, controlled access and audit history.	MVP
Move-in / Move-out	Onboarding, document collection, approvals, access activation/revocation, move logistics and clearance.	1A
Digital NOC & Certificates	No-dues, move-out, tenant, parking, renovation, residence and other configurable certificates.	2
Visitor & Gate Management	Guest, delivery, cab, vendor, staff, material passes, QR/OTP, gate logs and offline operation.	MVP
Staff & Domestic Help	Guards, housekeeping, maids, drivers, cooks, technicians and other workforce records, verification and shifts.	MVP
Biometric Attendance	Device integration, punch sync, shift calculation, corrections, reports and vendor invoice verification.	3
Maintenance Billing & Accounting	Charge heads, recurring bills, penalties, UPI, cash/cheque, receipts, ledger, reconciliation, budgets and exports.	MVP/2
Complaints & Helpdesk	Tickets, categories, priorities, evidence, SLA, assignment, escalation, vendor work and feedback.	MVP
Facility & Amenity Booking	Clubhouse, gym, pool, sports courts, halls, guest rooms, deposits, QR check-in, inspection and refunds.	2
Parking & Vehicles	Vehicle master, allocation, visitor parking, violations, blocking workflow and RFID/ANPR readiness.	2
Resident Connect	Privacy-first messaging, first-contact approval, controlled group communication, announcements and moderation.	1A
Privacy Directory	Configurable visibility, opt-out and privacy-aware resident discovery.	1A
Inter-flat Issues	Leakage, noise, renovation disturbance, parking, pets, damage claims, evidence and mediation.	1A
Governance	AGM/SGM, agenda, RSVP, proxies, minutes, resolutions, polls, elections and optional e-voting.	2/4
Safety & Emergency	SOS, medical, fire, lift, water/electrical	2/3

	incidents, volunteers, senior support and incident timeline.	
Vendor, AMC & Assets	Vendor records, contracts, AMC, warranties, assets, service history, inspections and scorecards.	2
Inventory & Procurement	Consumables, stock levels, issue/return, purchase requests, approvals, reorder levels and vendor linkage.	2
Marketplace & Community Commerce	Approved service directory, local services, micro-commerce, community deals and future verified partners.	3
Community Services	Lost & found, carpool, classifieds, domestic-worker replacement network, pet services and local utility services.	3
Events & Festival Automation	Events, RSVPs, contributions, volunteer coordination, reminders, notices and recurring festival workflows.	3
Water Management	Meter/readings readiness, tanker tracking, consumption, leakage incidents, maintenance and conservation reporting.	3/4
Energy & Sustainability	Electricity tracking, solar generation, common-area consumption, EV charging and sustainability metrics.	4
Legal & Compliance	Compliance register, document expiry, statutory records, notices, action tracking and renewal reminders.	2/3
Security Workforce Management	Shift rosters, attendance, post/beat assignment, deployment records, incident logs and vendor verification.	2/3
Payroll Readiness	Staff/vendor workforce attendance inputs, pay-period summaries, deductions/adjustments readiness and exports.	3
Intercom Replacement	In-app emergency and service communication workflows replacing fragmented intercom dependency.	3
Guest Stay Management	Guest room/stay requests, approvals, deposits, occupancy records and checkout.	2
Housekeeping Quality	Checklists, inspection scores, recurring tasks, issue logging and vendor/service quality trends.	2/3
Regional Localization	Language-ready content, configurable labels, date/time/number formats and region-specific communication.	MVP/1A
Offline Support	Offline-safe critical operations with queued actions, sync status and conflict handling.	MVP/2
Smart Automation	Notice drafting, document search, bill explanation, complaint routing, meeting summaries and operational recommendations.	4
Intelligent Society Insights	Pattern detection, anomaly identification, risk alerts, forecasts and explainable recommendations.	4
Hardware Integration Layer	Biometric devices, RFID, ANPR, CCTV, smart meters, EV chargers, boom barriers and IoT-ready connectors.	3/4

Commercial direction The launch experience remains free for selected societies. Subscription, pricing, trial and upgrade mechanics remain hidden until monetisation is intentionally activated.

5. Architecture and Data Flow Diagrams

The architecture is intentionally layered. Society OS should separate presentation, domain services, persistence, integrations, notifications, document storage, analytics and future intelligence. This keeps operational workflows reliable while allowing advanced capabilities to evolve independently.

5.1 High-level Architecture

Figure 1. High-level Society OS architecture.

5.2 Context-level Data Flow

Figure 2. Context-level data flow and external integration boundaries.

5.3 Level-1 Data Flow

Figure 3. Level-1 data flow showing core processing and data stores.

5.4 Architectural Principles

- API-first domain services with clear ownership of business rules.
- Multi-tenant isolation enforced server-side on every business request.
- Relational transactional storage for authoritative operational records.
- Object storage for documents with controlled access and signed/authorised retrieval.
- Event/outbox pattern for reliable notifications and external integrations.
- Background workers for large imports, document indexing, attendance synchronisation, reports and scheduled automation.
- Integration adapters for payment gateways, messaging providers, biometric devices, RFID/ANPR, utility data and future IoT.
- Feature flags and configuration evaluated consistently across the product and backend.
- Audit events generated for sensitive mutations and configuration changes.
- Intelligence services remain downstream of authoritative data and cannot bypass domain authorization.

5.5 Reliability Model

Concern	Design requirement
Transaction integrity	Critical changes are committed atomically and remain traceable.
Idempotency	Callbacks, imports and retried commands use deterministic idempotency keys.
Async processing	Long-running work is queued and monitored rather than blocking transactional APIs.
Offline operation	Critical field workflows can queue events locally and reconcile later.
Failure isolation	An unavailable third-party provider must not make unrelated core functions unavailable.
Observability	Every integration and scheduled job exposes status, failure reason, retry state and audit context.

6. Role-based Permission Model

Role-based access control is a platform security foundation. The purpose of the model is not to document screen visibility; it is to establish a consistent authorization boundary around society data, actions and sensitive records.

6.1 Authorization Layers

1. Authenticate the requesting identity and validate the active session.
2. Resolve the society and business context from trusted server-side data.
3. Evaluate role and granular permission claims.
4. Evaluate feature flag and society configuration.
5. Evaluate entity relationship and data sensitivity.
6. Validate workflow state and whether the requested transition is legal.
7. Apply approval requirements where configured.
8. Record an audit event for sensitive operations.

6.2 Permission Dimensions

Dimension	Examples
Tenant / society scope	Society, phase, tower, unit and operational entity boundaries.
Action permission	Create, view, update, approve, reject, publish, export, correct, close or administer.
Data sensitivity	Public, resident-visible, unit-restricted, staff-restricted, financial, security, personal-sensitive and historical.
Workflow state	A request may be editable in draft but immutable after approval or accounting lock.
Relationship	Ownership, occupancy, assignment, operational responsibility or explicit administrative authority.
Feature state	Disabled capability must be blocked regardless of UI visibility.
Approval authority	Actions may require one or more configured approvals.
Audit requirement	Sensitive actions must produce immutable audit records.

6.3 Sensitive Data Classes

- Identity and contact information.
- Owner, tenant and historical occupancy records.
- Agreements, KYC, police verification and sensitive documents.
- Financial transactions, bills, receipts and ledgers.
- Private communication metadata and reported content.
- Security incidents, watchlists and controlled camera evidence.
- Attendance and workforce records.
- Emergency and senior-support information.
- Legal and compliance records.

Security rule Frontend restrictions are for usability only. The backend remains the authoritative enforcement point for society scope, role, permission, feature state, data sensitivity and workflow state.

7. Deep Functional Specification by Module

Each module below defines business purpose, functional scope, business rules and implementation guidance. The intent is to provide a stable product contract without coupling the specification to a particular screen design.

7.1 Society, Unit and Resident Foundation

This is the master-data backbone. Every operational record must ultimately resolve to a society and, where applicable, a unit or property entity.

Key capabilities

- Society hierarchy: society, phase, tower, block, floor, wing, flat, villa, shop or office.
- Unit attributes: area, unit type, occupancy, billing category, parking linkage and status.

- Resident profile with ownership/tenancy relationship, family association and lifecycle state.
- Vehicle, pet, emergency contact, KYC and rule-acknowledgement records.
- Bulk import with validation, duplicate detection, preview, error reporting and rollback.
- Configurable unit numbering and hierarchy rules.

Implementation guidance

Unit is the central business entity. All applicable transactions should carry a strong society scope and unit relationship so reporting, security and lifecycle history remain consistent.

7.1.4 Family and Household Management

Household membership is a dated relationship model. Support owner, co-owner, tenant, spouse, child, parent, dependent and other authorized member relationships, along with emergency contact, vehicle, pet, visitor and facility associations where configured.

Real-world scenario — elderly parent moves in: the parent is added as an authorized household member with an effective date. Emergency and communication preferences can be configured without changing the legal ownership of the unit.

- Every relationship has start/end semantics.
- Household membership must not automatically imply access to financial or sensitive records.
- Changes to household composition should trigger only the notifications and access recalculations defined by policy.

7.1.3 Resident Registration, Verification and Account Recovery

Resident onboarding must support invitation, self-registration where enabled, OTP/mobile or email verification, profile completion, unit association, owner/tenant/family relationship, document submission, verification, approval, activation, suspension and recovery. Registration state must be separate from occupancy state.

Real-world scenario — duplicate registration: a tenant receives an invitation and also registers manually using the same phone number. The system should identify a possible duplicate, require confirmation/verification, and avoid creating two active identities for the same person without an explicit business reason.

- Verification attempts and failures are rate-limited and auditable.
- Account recovery must revoke compromised sessions/tokens where appropriate.
- Suspension disables configured access without deleting historical relationships.
- Identity matching must be conservative; false-positive merges are more damaging than a duplicate review case.

7.1.2 Property Master: Full Hierarchy and Lifecycle

The property master is the backbone for every other domain. It must model Society → Phase → Tower/Wing → Floor → Unit and independently track parking, storage, common areas, facilities, utility areas, meters and shared infrastructure. A unit may be a flat, villa, plot, shop or another configured property type.

Real-world scenario — redevelopment or renumbering: a society changes Tower B unit numbering during redevelopment. Historical invoices, complaints and visitor records must continue to resolve to the original identity while the current display name reflects the new numbering. The system must use stable identifiers rather than relying only on human-readable unit numbers.

- Never use a mutable unit label as the primary identity.
- Property hierarchy changes require effective dates and audit history.
- Common assets and facilities may belong to the society or to a defined property node.
- Property deactivation must not cascade-delete financial, legal or historical records.

7.1.1 Society Onboarding and Initial Configuration

Onboarding is the first production workflow and must be treated as a controlled configuration process rather than a simple registration form. A society record becomes operational only after its legal identity, geographic settings, property structure, billing rules, operating policies and initial data have passed validation.

Real-world scenario — a 6-tower apartment society: the implementation team creates the society, selects Maharashtra and IST, configures the financial year, adds six towers with 24 floors each, defines flat sizes, imports 1,152 units, maps parking slots, configures maintenance charge heads and late-fee rules, loads emergency contacts, and validates imported owners before activation.

Failure scenario — incomplete property master: if 1,152 units are imported but 37 units have duplicate numbers within the same tower, activation must stop or require an explicit correction workflow. The platform must never silently overwrite one unit with another.

- Capture society name, legal/registration identifiers, address hierarchy, pincode, timezone, currency, financial year and society type.
- Define property hierarchy: Society → Phase → Tower/Wing → Floor → Unit, with optional villas, plots, shops and shared/common infrastructure.
- Configure parking inventory, common areas, facilities, meters, charge heads, billing frequency, late fees, payment methods, notification defaults and operational policies.
- Support initial import with validation, preview, duplicate detection, business-rule validation, error report and controlled activation.
- Preserve configuration history so changes to numbering, charges or structure never destroy historical references.

7.2 Visitor and Gate Management

Digitises physical entry operations while keeping the process fast and auditable.

Key capabilities

- Guest pre-approval with QR/OTP.
- Walk-in visitor registration and resident verification.
- Delivery, cab, vendor, staff and material movement records.
- Gate-wise logs, entry/exit timestamps and purpose.
- Watchlist/blacklist with controlled escalation.
- Offline entry and later synchronisation.
- Visitor parking and temporary vehicle association.

Implementation guidance

The entry workflow must remain resilient on low-end devices and unstable networks. Duplicate events and sync conflicts must be safely handled.

7.2.1 Everyday Gate Scenarios and Exception Handling

Gate management must work for the messy reality of residential security: pre-approved guests, unknown visitors, delivery agents, domestic workers, service technicians, cabs, school transport, emergency responders, recurring visitors and residents returning after a long absence.

Real-world scenario — pre-approved guest: a resident creates a visitor invitation for Saturday 7:00–10:00 PM. The visitor arrives at 7:35 PM, the guard verifies the pass and identity, records the vehicle, and admission is logged. If the visitor leaves at 9:10 PM, the visit is closed and the system can show who is currently inside the premises.

Real-world scenario — resident unreachable: a plumber arrives for Unit B-804 without pre-approval. The resident does not answer. The system must not convert “resident did not answer” into automatic admission. The configured exception path may allow a temporary request/approval process, but final admission remains auditable.

Real-world scenario — emergency: an ambulance arrives without normal visitor credentials. Emergency policy may bypass ordinary approval checks, but the bypass reason, gate, time, operator/device and resulting access event must be captured.

- Visitor types should be configurable and not hard-coded.
- Every visit must have explicit entry and exit semantics; current occupancy must be derivable from valid events.
- Offline gate events require device timestamps, idempotency keys and deterministic reconciliation when connectivity returns.
- Vehicle details, recurring visitor patterns, blacklists and exceptions must be handled as policy-controlled data rather than ad hoc notes.
- Gate workflows should support camera/OCR assistance without making OCR correctness a prerequisite for safe operation.

7.3 Maintenance Billing, Payments and Accounting

Provides an auditable financial foundation for recurring society operations.

Key capabilities

- Recurring monthly bills and ad-hoc charges.
- Fixed, area-based, meter-based, parking, facility and penalty charge heads.
- UPI/payment gateway support and manual cash/cheque recording.
- Receipts, unit ledger, outstanding balance and defaulter reporting.
- Bank reconciliation readiness and accounting exports.
- Budget versus actual and fund-wise reporting.
- Correction/reversal instead of destructive deletion.

Implementation guidance

Resident maintenance billing must remain logically separate from SaaS subscription billing. Financial records should be append-corrected and never silently deleted.

7.3.2 Financial Exception Catalogue

Financial workflows must explicitly define exceptions instead of treating them as generic errors. Important cases include partial payments, advance payments, credit balances, waivers, discounts, late fees, interest, bounced cheques, failed UPI payments, duplicate callbacks, refunds, reversals, charge corrections and unmatched bank transactions.

Real-world scenario — advance payment: a resident pays ₹20,000 when only ₹15,000 is due. The ₹5,000 surplus becomes a controlled credit/advance according to configuration; it is not silently discarded or incorrectly posted as revenue.

Real-world scenario — bounced cheque: a cheque initially marked received later bounces. The system reverses the settlement effect through an auditable financial transaction and restores the outstanding balance plus configured charges.

- Never edit a posted financial journal to correct history.
- All adjustments require reason, actor, timestamp and supporting evidence where policy requires.
- Reconciliation exceptions must be visible until resolved.
- Receipt numbering and reversal relationships remain traceable.

7.3.1 Complete Billing-to-Ledger Business Lifecycle

Finance is the system-of-record domain and must be described at transaction level. The product must distinguish an invoice, a payment attempt, a settled payment, a receipt, an allocation to an invoice, a ledger posting, an adjustment and a refund. These are related but not interchangeable concepts.

Real-world scenario — normal monthly billing: on the first day of the billing cycle, the engine calculates maintenance, parking, water and sinking-fund charges for each unit. Validation completes, invoices are

generated and published, residents are notified, payments are received, receipts are issued and balanced ledger entries are posted.

Real-world scenario — partial payment: a unit owes ₹12,500 and pays ₹7,500. The system records the actual payment, allocates it according to configured allocation rules, leaves the remaining ₹5,000 outstanding, and recalculates ageing without pretending the invoice was fully settled.

Real-world scenario — duplicate gateway callback: the payment provider retries the same callback after a network timeout. Signature verification and idempotency must cause the second callback to become a no-op from the ledger perspective while retaining an audit record of the duplicate event.

Real-world scenario — refund after reconciliation: a ₹10,000 payment is later refunded by ₹2,000. The original posted transaction remains immutable; a compensating/refund transaction records the reversal and the unit balance changes deterministically.

- Support fixed, area-based, usage-based and configurable charge heads, including maintenance, parking, water, electricity/common-area recovery, funds, penalties, interest, amenity charges, move-in/out and miscellaneous charges.
- Support partial payments, advances, credits, waivers, discounts, penalties, failed payments, bounced cheques, refunds, reversals and manual adjustments with approvals where configured.
- Use transaction serialization and balanced journal posting; never resolve financial conflicts with last-write-wins.
- Bank reconciliation must distinguish unmatched, partially matched, matched, reversed and exception transactions.
- Invoice numbering, tax configuration, rounding and currency precision must be configurable and jurisdiction-aware rather than hard-coded.

7.3.7 Financial Reporting and Audit Traceability

Every financial KPI, statement and report must trace to source transactions, period, account and unit where applicable. Users must be able to explain why an amount changed without relying on manually edited history.

7.3.6 Financial Integrity Invariants

Posted financial journals are balanced and immutable; corrections are represented by reversal/adjustment transactions; callbacks are idempotent; refunds reference their source payment; partial payments retain residual outstanding; credits retain source provenance; effective-dated configuration determines calculations; reconciliation cannot duplicate settlement; month-end controls prevent uncontrolled back-posting.

7.3.5 Financial Exception Catalogue

The product must explicitly distinguish failed payment, unpaid invoice, duplicate callback, partial payment, advance/credit, refund, reversal, bounced cheque, late fee, interest, waiver, discount, credit/debit adjustment, unmatched bank movement, disputed transaction, failed refund and period-close exception.

7.3.4 Financial End-to-End Lifecycle

Billing configuration -> billing run -> charge calculation -> validation -> invoice generation -> publication -> notification -> payment initiation -> provider confirmation -> signature/idempotency validation -> allocation -> receipt -> ledger posting -> reconciliation -> ageing and collections -> adjustments/refunds/reversals -> period close.

7.3.3 Financial Product as the System of Record

Finance is a first-class operational subsystem. Charges, invoices, payments, receipts, credits, waivers, adjustments, refunds, reversals, payables, reconciliation and close controls must be explainable from source records and ledger effects.

7.4 Complaints and Helpdesk

Converts informal complaints into structured, owned and measurable work.

Key capabilities

- Complaint categories, priorities, evidence and affected location.
- Assignment to internal teams or approved vendors.
- SLA start, pause, escalation, breach and resolution.
- Resident confirmation, feedback and reopen.
- Private/sensitive complaint mode with restricted visibility.
- Recurring issue correlation and parent incident capability.

Implementation guidance

Every complaint should have status, owner, timestamps, SLA, audit trail and notification rules. Sensitive complaints require stricter data visibility.

7.4.2 Complaint Intake Channels and Duplicate Handling

Complaints may originate from resident apps, staff, imports, integrations or operational teams. Regardless of channel, the resulting record follows one canonical lifecycle and source attribution.

Real-world scenario — duplicate complaint: a resident reports a broken streetlight at 8:00 PM and another resident reports the same pole at 8:05 PM. The second ticket can be linked to the parent incident while retaining its own reporter, acknowledgement and communication history.

- Preserve original source and submitted timestamp.
- Duplicate linking must never delete the reporter's original evidence.
- Parent incidents may broadcast common updates while individual records retain unit-specific communication.

7.4.1 Complaint Lifecycle, SLA and Escalation

A complaint is a service workflow, not merely a message. It must carry enough context to identify what failed, where it failed, how urgent it is, who owns it, what evidence exists, what SLA applies, what actions were taken and why it was finally closed.

Real-world scenario — lift complaint: a resident reports that the lift stops at the 7th floor and uploads a video. The ticket is classified as Lift → Operational Failure, marked high priority, assigned to the lift AMC workflow, acknowledged, worked, placed on hold while a spare part is sourced, resumed after the part arrives, resolved with service evidence and finally closed after the configured closure rule.

Real-world scenario — recurring complaint: ten residents report “water pressure low” in the same tower within 20 minutes. The system should detect possible correlation and allow a parent incident/outage record, preventing ten independent technician visits while preserving each resident complaint for communication and history.

Real-world scenario — reopen: a ticket is closed but the resident reports that the issue persists the next morning. Reopening must create a new state transition and preserve the original closure evidence rather than editing history.

- Lifecycle: CREATED → CLASSIFIED → ASSIGNED → ACKNOWLEDGED → IN_PROGRESS → WAITING/HOLD → RESOLVED → REOPENED → CLOSED.
- Capture category, subcategory, location, evidence, priority, SLA, assignment, escalation, comments, internal notes, root cause and closure reason.
- Support reassignment, duplicate detection, parent incidents, recurring complaints and SLA breach escalation.
- Keep resident-facing communication separate from internal operational notes while preserving auditability.
- Closure should be evidence-based and measurable; “technician marked done” is not automatically equivalent to “service outcome verified”.

7.5 Facility and Amenity Booking

Manages shared resources with capacity, schedule and financial controls.

Key capabilities

- Configurable facilities, operating hours and blackout periods.
- Slot availability and capacity rules.
- Free/paid bookings, deposits and refunds.
- Approval workflows where required.
- QR check-in/check-out.
- Damage deposit and post-use inspection.

Implementation guidance

Database-level constraints must prevent double booking. Booking state must drive payment, deposit and refund behaviour.

7.5.1 Booking Engine and Conflict Rules

Facility booking must be treated as a reservation engine with concurrency controls, not a calendar of informal requests.

Real-world scenario — clubhouse booking: a resident requests the clubhouse from 6:00–8:00 PM. The facility has a 30-minute cleanup buffer, a maximum capacity of 50 and a cancellation cutoff of 24 hours. A second request for 7:30–9:00 PM must be rejected or placed on a waitlist if overlap is prohibited by policy.

Real-world scenario — blackout period: the pool is unavailable for maintenance every Monday 6:00–10:00 AM. Booking requests during the blackout are rejected with a policy reason, not treated as an arbitrary application error.

Real-world scenario — no-show: a resident reserves a court but does not check in. The system records the no-show and can apply a configurable penalty or booking restriction without altering unrelated bookings.

- Facility master must define operating hours, holidays, capacity, slot duration, buffers, booking window, cancellation window, deposits, charges, penalties, approval requirements and blackout periods.
- Confirmed reservations must use atomic conflict prevention where overlap is prohibited.
- Support recurring bookings, waitlists, cancellation, refund, no-show, check-in, usage and damage inspection where applicable.
- Booking changes must be versioned so concurrent updates cannot silently overwrite each other.

7.6 Parking and Vehicle Management

Maintains vehicle ownership and allocation while supporting future access automation.

Key capabilities

- Vehicle registration and verification.
- Parking slot allocation and reassignment.
- Visitor and temporary parking passes.
- Wrong parking, blocking and violation workflows.
- Repeat-offence escalation.
- RFID/ANPR readiness.

Implementation guidance

Parking slots are treated as managed assets linked to units and vehicles. Violation records remain controlled operational data.

7.6.2 Parking Master, Allocation and Enforcement Rules

Parking allocation must distinguish ownership/entitlement, physical slot, temporary use, vehicle association and actual occupancy. A slot can have a historical allocation chain while only one valid active allocation applies at a given time.

Real-world scenario — EV conversion: a resident requests an EV charging-enabled parking arrangement. The system records the vehicle's EV status, the slot capability and any required approval/charging configuration without changing the legal identity of the parking slot.

- Detect conflicting active allocations.
- Track temporary and visitor allocations with explicit expiry.
- Keep violation evidence separate from the permanent parking master.
- Support future RFID/ANPR integration without making hardware availability a prerequisite for manual operation.

7.6.1 Parking, Vehicle and Violation Operations

Parking is both a property master and an operational enforcement domain. The platform must distinguish a parking slot from the vehicle currently using it and from the historical allocation of that slot.

Real-world scenario — vehicle replacement: a resident sells MH-12-AB-1234 and buys MH-14-CD-5678. The new vehicle is linked to the same unit and parking slot while the old vehicle remains historically recorded as inactive; historical gate and parking events must continue to resolve correctly.

Real-world scenario — visitor parking abuse: a visitor vehicle occupies a reserved resident slot. Security records the incident with slot, vehicle, evidence and time. The system can trigger a violation workflow and configured penalty/towing readiness without changing the permanent parking allocation.

Real-world scenario — temporary allocation: a resident has a second vehicle for two weeks. The society grants temporary parking with a start/end date. The temporary allocation must automatically expire and must not overwrite the permanent slot ownership.

- Support covered/open, two/four-wheeler, reserved, visitor, EV, accessibility and temporary slots.
- Vehicle master should include registration, type, make/model, unit/person association, parking, RFID/ANPR identifiers, EV flag and active/inactive state.
- Prevent duplicate active vehicle registration where policy requires uniqueness; allow legitimate historical duplicates across time.
- Capture violations with evidence, status, resolution and optional penalty/towing workflow.

7.7 Resident Connect and Controlled Communication

Provides community communication without forcing residents to expose personal phone numbers.

Key capabilities

- First-contact request with subject and reason.
- Accept, reject, block and report controls.
- Persistent chat after acceptance.
- Controlled group communication.
- Announcements and notice acknowledgement.
- Moderation workflow for reported content.

Implementation guidance

Private communication should remain private by default. Moderation access must be exceptional, policy-driven and audited.

7.7.1 Communication as a Controlled Operational System

Communication must cover both broad announcements and privacy-sensitive person-to-person contact. The platform should make the safe path the easy path: users can communicate without exposing private phone numbers by default.

Real-world scenario — first-contact request: a committee member wants to contact a resident about a parking dispute. Instead of exposing the resident’s phone number, the platform creates a contact request. The resident approves or declines; only then does the permitted communication channel open.

Real-world scenario — emergency broadcast: a major water pipeline failure affects Tower C. The society publishes a high-priority notice, pushes it to affected residents, records delivery/acknowledgement where supported and keeps the notice available for the duration of the incident.

Real-world scenario — moderation: a community message contains abusive content. Moderation actions must preserve evidence, identify the action taken and follow configurable retention and escalation rules.

- Support announcements, direct communication, groups, emergency broadcasts, delivery/read/acknowledgement states and channel fallback.
- Provide templates, localization, scheduling, expiry, withdrawal and revision history.
- Apply privacy boundaries to discovery, contact initiation, exports and notification payloads.
- Prepare integration boundaries for push, SMS, email and WhatsApp-like channels without coupling the domain model to a single provider.

7.8 Inter-flat Issues and Dispute Mediation

Creates a calm, evidence-based process for disputes that involve more than one property or resident.

Key capabilities

- Leakage, noise, renovation, parking, pets and damage cases.
- Evidence upload and affected-property tagging.
- Inspection and mediation assignment.
- Resolution proposal, acceptance and closure proof.
- Escalation and documented exceptions.

Implementation guidance

Dispute cases remain separate from normal helpdesk tickets because visibility, evidence and multi-party handling are more sensitive.

7.8.1 Dispute Case Lifecycle

Disputes require neutral case management rather than becoming unstructured chat threads. A case should capture subject, parties, property/context, evidence, chronology, communications, actions, mediation status, resolution and closure reason.

Real-world scenario — water seepage between flats: Unit A-701 reports seepage believed to originate from A-801. The case records evidence and inspection findings, creates tasks for maintenance/vendor inspection, allows controlled communication, and closes only after a documented resolution or formal outcome.

- Support evidence, comments, internal notes and controlled participant communication.
- Preserve an immutable case timeline.
- Allow escalation to configured governance/legal processes without exposing unrelated resident information.
- Dispute closure should record whether the outcome was resolved, mediated, rejected, withdrawn or unresolved.

7.9 Governance, AGM and Elections

Supports formal society governance and auditable decisions.

Key capabilities

- AGM/SGM notices, agendas, attendance and RSVP.
- Proxy authorisation and question submission.
- Minutes and resolution tracker.
- Polls and voting where enabled.
- Election nomination, candidate information and result publication.
- Decision archive and acknowledgement.

Implementation guidance

E-voting remains optional. Eligibility, quorum, voting rules and compliance assumptions must be configured before activation.

7.9.1 Governance and Decision Lifecycle

Governance features must support agenda preparation, meeting scheduling, notices, quorum/attendance capture, resolutions, voting, minutes, action items and archival of approved records.

Real-world scenario — AGM resolution: an agenda item proposes a major repair fund. The system publishes the notice, records eligible participation, captures the configured voting process, records the resolution outcome and links resulting actions to finance/procurement workflows.

- Governance records must be versioned and tamper-evident.
- Voting eligibility and quorum rules are configuration-driven and jurisdiction-aware.
- Approved minutes are not silently edited; corrections use versioned amendments.

7.10 Safety, Emergency and Senior Citizen Care

Provides a unified operational framework for urgent events.

Key capabilities

- SOS, medical, fire and lift-stuck alerts.
- Water/electrical emergency categories.
- Emergency volunteer coordination.
- Senior citizen support and check-in workflows.
- Incident timeline, acknowledgement and escalation.
- Post-incident review and evidence.

Implementation guidance

Critical alerts must not depend on ordinary notification preferences. Emergency workflows require reliable acknowledgement and escalation.

7.10.2 Incident, Evacuation and Post-Incident Review

Safety operations should support incident declaration, severity, affected zones, responders, tasks, evacuation/check-in information where applicable, communication, status updates and post-incident corrective actions.

Real-world scenario — basement flooding: water ingress is detected in the basement. The incident is declared, affected parking is marked unavailable, residents receive a notice, maintenance creates work orders for pumps and electrical inspection, and the final incident record contains response duration and corrective actions.

- Incident status must be independent from individual complaints that may be linked to it.
- Emergency instructions are versioned so later edits do not erase what was communicated during the incident.
- Critical actions require audit evidence.

7.10.1 Emergency Operations and Incident Command

Emergency handling must prioritize speed while preserving an auditable chain of events. Incidents should support declaration, severity, affected area, responders, communication, tasks, status updates, resolution and post-incident review.

Real-world scenario — fire alarm: an alarm is triggered in Tower D. The system records the incident, broadcasts an emergency notice, surfaces emergency contacts and operational tasks, tracks updates, and later records the post-incident report. Normal notification quiet hours must not suppress critical alerts when policy permits override.

- Emergency events must have higher delivery priority than routine notifications.

- Critical access bypasses must be explicitly reason-coded.
- Post-incident review should capture root cause, response time and corrective actions.
- Senior-citizen/emergency-care data must be purpose-limited and protected as sensitive information.

7.11 Vendor, AMC, Asset and Inventory Management

Manages the physical and contractual resources that keep the society running.

Key capabilities

- Vendor master and contract records.
- AMC dates, renewals, warranties and service history.
- Asset register with location, category, condition and lifecycle.
- Preventive-maintenance schedules and inspection checklists.
- Inventory stock, issue/return and purchase requests.
- Vendor performance scorecards.

Implementation guidance

Asset records should preserve lifecycle history. Vendor scoring should use objective operational data rather than opaque ratings.

7.11.3 Procurement Lifecycle and Three-Way Verification

Procurement should connect demand, approval, purchase order, receipt, service/inspection and invoice verification. Where applicable, finance should compare ordered quantity/value, received quantity/value and invoiced quantity/value before payment.

Real-world scenario — plumbing spares: a PO orders 100 valves. Only 90 arrive. The goods receipt records 90, inventory increases by 90, and an invoice for 100 cannot be treated as fully received without an exception or authorized variance.

- Support purchase requests, approvals, PO revisions, partial receipts, returns and cancellations.
- Link procurement lines to inventory items, assets/work orders and financial commitments.
- Keep supplier and contract history even after vendor deactivation.

7.11.2 Preventive Maintenance Engine

Preventive maintenance converts asset registers into planned work. Each asset may have one or more maintenance plans with frequency, due-date calculation, inspection checklist, SLA, assigned vendor/team, parts and cost capture.

Real-world scenario — diesel generator: the generator requires weekly visual inspection, monthly test run and annual servicing. The scheduler creates the correct work orders, tracks missed tasks, records readings and service evidence, and calculates the next due date from the completed event.

- Assets should support criticality, location, installation date, warranty, AMC, meter/readings and lifecycle state.
- Maintenance frequency can be calendar-based, usage-based or rule-driven.
- Missed maintenance must trigger escalation according to configured SLA.
- Completion must record evidence and next-due calculation.

7.11.1 Vendor-to-Work-Order-to-Payment Lifecycle

Vendor management must connect commercial records with operational service delivery. A vendor is not “active” merely because a profile exists; contracts, insurance, KYC, AMC dates, work orders, service evidence, invoices and performance history form one lifecycle.

Real-world scenario — lift AMC: an AMC expires on 31 March. Before expiry, the system creates a renewal reminder. After renewal, a scheduled preventive maintenance work order is created. The technician/vendor records inspection findings, replaces a component, uploads the service report, and submits an invoice. Finance verifies the work completion before payment.

Real-world scenario — stock-out during repair: a pump repair requires a spare that is below reorder level. The work order can create a material request, procurement can raise a PO, goods receipt increases stock, the spare is issued to the work order, and consumption is reflected in inventory and cost records.

Real-world scenario — poor vendor performance: a vendor repeatedly misses SLA. Performance scoring aggregates breach history, unresolved defects and response time, enabling renewal/penalty/termination decisions based on evidence.

- Vendor lifecycle: onboarding → KYC/docs → contract → AMC/PO → service → work order → inspection → invoice → verification → payment → performance → renew/terminate.
- Inventory lifecycle: item master → opening stock → purchase request → approval → PO → receipt → stock-in → issue → consumption → return/adjustment → reorder.
- Support batch, serial, expiry, damaged stock, stock locations, min/reorder levels and stock audit.
- Link asset, preventive maintenance plan, work order, spare consumption, vendor service and financial cost for traceability.

7.12 Marketplace and External Services

Creates a controlled community utility layer around approved local services.

Key capabilities

- Society-approved service directory.
- Cleaning, car wash, pest control, laundry, appliance services, pet care, tutors and elder support.
- Packers/movers, interiors and renovation services.
- Community classifieds and local service discovery.
- Micro-commerce / society marketplace for permitted products and services.
- Verified partner and payment readiness.

Implementation guidance

The marketplace should feel like a useful community utility rather than an intrusive advertising surface. Partner onboarding and commercial controls must remain configurable.

Society is the tenant root; unit identifiers are unique within the society hierarchy; occupancy periods are time-bounded; active overlap is rejected unless explicitly supported; unit status derives from authoritative occupancy. Bulk import follows validate → normalize → deduplicate → preview → transactional commit → audit/rollback.

Low-memory guard mode: release image buffers, suppress blur/shadows/animations, throttle camera capture and preserve text/OCR operations.

Tailgating/expiry: a cleared gate pass has a short configurable opening window (baseline example 90 seconds); if passage sensors fail, the pass does not remain indefinitely valid.

Delivery cubby protocol: “Drop at Gate” allocates a physical bin and a single-use pickup PIN delivered only to the intended unit recipient.

Resident revocation race: an offline guard admission is provisional; on reconciliation, authoritative server state and event ordering determine the outcome. If revocation precedes admission, the event becomes UNAUTHORIZED_ENTRY_FLAGGED and triggers configured security/resident alerts.

Figure 14. Deterministic visitor entry state machine.

7.2.1 Visitor State Machine

Every posted journal must balance debits and credits. Corrections use new compensating/reversal entries linked to the original. Charge rules store calculation inputs for reproducibility and bill explanation. Payment callbacks require signature verification and idempotency; cash/cheque entries remain traceable. Statutory fund/interest/export rules are configurable.

Complaint lifecycle supports created/classified/assigned/in-progress/resolved/confirmed/closed plus reopen and escalation exceptions. Duplicate/parent incident correlation can be intelligence-assisted but final state remains deterministic.

Database/service constraints prevent overlapping bookings. Deposits, payments, cancellations, refunds, inspection and damage claims follow booking state. Guest rooms are included alongside clubhouse, gym, pool, sports courts and halls.

Vehicle and slot records support temporary passes, wrong-parking/blocking incidents, repeat-offence escalation and RFID/ANPR adapters. Hardware absence must not stop manual parking operations.

First-contact requests capture subject/reason. Accept/reject/block/report controls are explicit. Phone numbers remain masked. Reported content can enter audited moderation; ordinary private communication is not exposed through administrative browsing.

Disputes are separate from ordinary complaints. Seepage, noise/vibration, renovation, parking, pets and damage can carry photos, documents, inspection notes and configured measurement evidence. Mediation proposals, acknowledgements and legal escalation remain time-stamped.

Governance covers AGM/SGM, agenda, RSVP, proxies, minutes, resolutions, polls, elections and decision archive. Smart Poll Engine may assist with poll drafting and response summarization, but eligibility, quorum, counting and final results are deterministic.

SOS, medical, fire, lift, water and electrical incidents use urgent acknowledgement/escalation. Emergency alerts bypass ordinary notification preferences. Senior support can include simple mode and configurable check-ins while preserving data minimization.

Vendor lifecycle includes verification, contract/AMC, service, inspection, invoice, scorecard and renewal. Assets preserve lifecycle history. Vendor Sinking Fund/Milestone Escrow links capital-project milestones, supporting documents, approvals and financial movements. Inventory supports reorder levels, purchase requests and issue/return.

Community services include cleaning, car wash, pest control, laundry, appliance service, pet care, tutors, elder support, packers/movers and renovation. Society Micro-Commerce supports approved products/services and payment readiness without turning the product into an intrusive ad surface.

7.12.3 Service Requests, Quotes, Orders and Refunds

External service transactions should use an explicit state model such as REQUESTED → QUOTED → ACCEPTED → SCHEDULED → IN_PROGRESS → COMPLETED → CANCELLED/DISPUTED. Commercial settlement should remain distinct from service completion.

Real-world scenario — appliance repair: a resident requests a service, receives a quote, accepts it, schedules a visit, the provider completes the work, the resident confirms completion and a payment is settled. If the provider does not arrive, the order enters an exception state rather than being marked completed.

- Cancellation windows, provider no-show, resident no-show and refund rules are configurable.
- Ratings are linked to completed orders and cannot be manipulated by creating fictitious service completions.
- Provider failures must not corrupt society-level finance or resident records.

7.12.2 Notice, Announcement and Emergency Communication Lifecycle

Announcements are governed content objects. They require authoring, audience targeting, priority, effective time, expiry, attachments, localization, scheduled publication, withdrawal, revision history and delivery/acknowledgement tracking where supported.

Real-world scenario — planned water shutdown: the society schedules a notice for Tuesday 9:00 AM explaining that Tower B water supply will stop from 11:00 AM to 2:00 PM. At 10:55 AM the notice becomes active, affected residents receive it, and after restoration a follow-up notice closes the incident. The original notice remains historically available.

Real-world scenario — emergency override: a critical fire-related notice must be delivered immediately even when a resident has enabled quiet hours. The emergency policy explicitly permits the override and the audit record captures why it occurred.

- Separate draft, scheduled, published, expired, withdrawn and superseded states.
- Target by configured audience criteria without leaking the audience definition to recipients who should not see it.
- Attachments inherit document security and retention rules.
- Localized templates must preserve meaning rather than simply translating labels independently.

7.12.1 Marketplace and Service Order Lifecycle

Marketplace functionality should separate discovery from contracting and fulfilment. A resident may request a service, receive a quote, place an order, track fulfilment, provide feedback and raise a dispute. Society-sponsored vendors may have different commercial rules from external providers.

Real-world scenario — electrician booking: a resident requests an electrician for a Saturday slot. The service provider confirms availability, the job is completed, the resident approves the outcome, payment is captured and the service history is retained without exposing unrelated resident data.

- Provider onboarding, availability, service area, pricing/quote, order state, cancellation and refund rules should be configurable.
- External service data must remain logically separated from society financial records unless a formal settlement relationship exists.
- Provider ratings should be traceable to actual completed orders.

8. Owner, Tenant and Occupancy History Management

Owner and tenant lifecycle history is a core differentiator. The system should preserve a reliable historical timeline for each unit rather than overwriting current resident information.

8.1 Business Purpose

For every unit, the platform should be able to establish the current ownership/occupancy state, previous ownership and tenancy periods, relevant documents, move-in and move-out events, NOC history, clearance status and access lifecycle.

8.2.1 Occupancy and Household Model

Occupancy is time-bound. The same unit may have an owner, co-owner, tenant, family members, authorized occupants and historical occupants at different times. The model must therefore use dated relationships rather than a single mutable “current resident” field.

Real-world scenario — owner becomes landlord: Owner A purchases Unit A-1203, later rents it to Tenant B, then terminates the tenancy and moves back in. Historical records must show all three periods. Visitor records, invoices, documents and NOC decisions must resolve to the correct period without exposing unrelated historical data.

Real-world scenario — family member exits: a child previously associated with the household moves out. The relationship ends on a date; historical facility or visitor events remain intact, while future household permissions no longer inherit from the previous association.

8.2.2 Resident Registration and Verification Lifecycle

Registration should support invitation, mobile/email verification, identity/profile completion, unit association, owner/tenant/family relationship, document submission, verification, approval where configured, activation, suspension, recovery and eventual exit. Duplicate-account detection must be based on configured identity signals and must never merge records blindly.

8.2 Functional Scope

Capability	Expected behaviour
Current ownership	Maintain active owner/co-owner relationship with effective dates.
Current tenancy	Maintain tenant, agreement metadata, move-in date, verification status and owner approval context.
Previous ownership	Preserve historical owners, transfer dates, transfer reason and linked documents.
Previous tenancy	Preserve previous tenants, occupancy period, exit status, NOC and clearance context.
Document archive	Retain historical agreements, KYC, verification and handover documents without exposing them by default to later occupants.
Unit timeline	Provide chronological lifecycle events including ownership, move-in, move-out, NOC and access activation/revocation.
Mandatory move-out clearance	Do not close the lifecycle until configured clearances are complete or a permitted override is recorded with reason.

8.3 Lifecycle Rules

9. Create or update the occupancy relationship with effective dates.
10. Collect required onboarding records and approvals.
11. Activate the occupancy only after required validation.
12. Record operational events against the current occupancy period.
13. Initiate move-out with a defined exit date and clearance checklist.
14. Resolve dues, access items, parking, damage, complaints and required documents.
15. Generate the applicable NOC/certificate.
16. Close the occupancy period and revoke applicable access.

8.4 Rental / Tenant Assessment

The platform may provide a configurable, explainable assessment of tenancy readiness for society operations. It should be based only on authorised, relevant operational criteria such as verification status, required document completeness, agreement validity and configured society requirements.

The assessment must not use protected or unrelated personal characteristics, must show the factors contributing to the result, and must not be presented as an infallible prediction of a person's character or future behaviour.

17. 8.5 Data Entities

8.4 Data Entities

Entity	Purpose	Important fields
UnitOccupancyHistory	Historical occupancy period.	unitId, ownerId, tenantId, startDate, endDate, status, moveInRequestId, moveOutRequestId
OwnerHistory	Ownership timeline.	unitId, ownerProfileId, ownershipStart, ownershipEnd, transferReason, documentIds
TenantHistory	Tenant occupancy and exit context.	unitId, tenantProfileId, agreementStart, agreementEnd, verificationStatus, nocStatus
MoveInRequest	Onboarding workflow.	unitId, requestedBy, approvals, liftSlot, truckPass, accessStatus
MoveOutRequest	Exit and clearance workflow.	unitId, requestedBy, checklist, approvalStatus, exitDate
NocRecord	Issued certificate metadata.	requestId, nocType, issuedBy, issuedAt, documentId, verificationCode
DocumentAccessLog	Sensitive document access trail.	documentId, actorUserId, action, timestamp, ipAddress, deviceId

Historical privacy rule A new occupant of a unit must not automatically inherit access to documents or sensitive records belonging to previous occupants.

Figure 6. Move-out and mandatory NOC lifecycle flow.

9. Biometric Attendance Integration

The platform should support biometric/fingerprint attendance devices primarily for operational attendance, workforce verification and vendor invoice reconciliation.

9.1 Privacy Boundary

Society OS should not store raw fingerprint images or biometric templates. Registration and biometric matching should remain inside the device/vendor ecosystem. Society OS should receive attendance events and associated operational metadata.

9.2 Functional Scope

Capability	Detailed behaviour
Device registration	Register device code, location, society, operational context and synchronisation mode.
Staff mapping	Map device employee code to a workforce profile, vendor and shift context.
Punch synchronisation	Import punch events through API, file import, sync agent or vendor connector.
Duplicate prevention	Use a deterministic duplicate key such as employeeCode + deviceId + punchTime.
Shift calculation	Compare attendance with roster to derive present, late, early checkout, half day or absent.
Manual correction	Allow controlled correction of missed or invalid punches with reason, approval where configured and audit.
Reporting	Generate daily, monthly, staff-wise, vendor-wise and location-wise attendance summaries.
Vendor billing support	Provide attendance summaries that can support outsourced workforce invoice verification.

9.3 Data Entities

Entity	Purpose	Important fields
BiometricDevice	Device metadata.	id, societyId, locationId, deviceCode, vendorName, syncType, status
BiometricStaffMapping	Employee-code mapping.	deviceId, employeeCode, staffProfileId, effectiveFrom, effectiveTo
AttendancePunch	Imported attendance event.	staffId, deviceId, punchTime, punchType, source, syncJobId, duplicateKey
AttendanceDailySummary	Computed attendance.	staffId, date, shiftId, status, firstIn, lastOut, lateMinutes, earlyMinutes
AttendanceCorrectionRequest	Correction workflow.	attendanceId, requestedBy, reason, oldValue, newValue, approvalStatus

9.4 Sync Flow

Figure 7. Biometric attendance synchronisation flow.

10. Document Vault, NOC and Secure Access

Documents are among the most sensitive records in Society OS. The document architecture must treat storage, metadata, access, versioning, expiry and audit as one controlled lifecycle.

10.1 Document Categories

Category	Examples	Control requirements
Society documents	Registration, bylaws, AGM minutes, fire NOC, lift certificate.	Controlled publication, versioning and expiry.
Owner documents	Sale deed, share certificate, identity/address proof.	Sensitive and explicitly authorised access.
Tenant documents	Rent agreement, KYC, police verification, owner approval.	Sensitive; isolated from unrelated residents and later occupants.
Move-in documents	Move-in form, vehicle details, rule acknowledgement, truck pass.	Linked to move-in lifecycle.
Move-out documents	No-dues, move-out NOC, handover, damage inspection.	Required for lifecycle closure.
Staff/vendor documents	Verification, contracts, licences, insurance, AMC.	Controlled operational access and expiry reminders.
Compliance documents	Fire, lift, insurance, legal notices and certificates.	Expiry tracking and compliance register.

10.2 NOC and Certificate Catalogue

NOC / certificate	Purpose	Typical clearance
No-dues certificate	Financial clearance.	Maintenance, penalties, facility charges and deposits.
Move-out NOC	Formal exit clearance.	Dues, access cards, parking, complaints, damage and documents.
Tenant NOC	Tenancy onboarding/exit validation.	Owner approval, KYC, agreement and verification.
Parking NOC	Parking/vehicle clearance.	Parking dues, sticker and violations.
Renovation NOC	Controlled renovation completion.	Damage inspection, debris clearance and contractor closure.
Residence certificate	Society address/occupancy proof.	Active occupancy and document verification.

10.3 Secure Document Access Flow

Figure 8. Secure document access flow.

10.4.1 Document Upload, Versioning and Retrieval

Document handling is a security boundary. Uploads should support configured file types and limits, checksum calculation, malware scanning, optional image compression/thumbnails, metadata minimization, encryption, access control, versioning, retention and download audit.

Real-world scenario — rental agreement: a tenant uploads a new agreement. The previous agreement remains immutable as version 1; the new document becomes version 2 with effective date and uploader recorded. A user without permission cannot retrieve either document merely by guessing its URL.

Failure scenario — interrupted upload: a mobile user loses connectivity at 80%. The client may resume or retry safely. The backend must not create a half-valid document record that appears complete.

10.4.2 NOC as a Business Workflow

NOC issuance must be treated as a controlled clearance process. A move-out request can require financial clearance, document/asset return, parking status, access revocation, meter readings, pending complaint/vendor obligations and configurable approvals before an NOC is issued.

Real-world scenario — move-out with outstanding dues: the resident requests NOC while ₹18,000 is outstanding. The system calculates the current ledger position, flags the clearance condition, allows settlement, posts the payment, verifies all required checks, and only then issues the NOC. The original debt and clearance evidence remain auditable.

10.4 Document Rules

- Never expose raw object-storage paths publicly.
- Use signed URLs or controlled download APIs with short-lived access.
- Maintain document type, owner entity, visibility, version and expiry metadata.
- Record sensitive view/download events.
- Retain historical versions where required by the document lifecycle.
- Support controlled verification of issued NOCs and certificates.
- Apply retention and deletion policies without breaking legal/audit requirements.

11. Workflows and User Journeys

Workflows should be implemented through configurable state machines or workflow services wherever practical. The objective is to support different society policies without duplicating business logic.

11.1.1 Reusable Workflow Engines

The platform should avoid implementing approval, SLA, notification and scheduling logic separately inside every module. Reusable engines reduce inconsistent behaviour and make society-specific policy configuration possible.

11.1.2 Reusable Approval Engine

Generic flow: Request → Rule Evaluation → Approver Resolution → Pending → Approve/Reject/Return → Multi-level Approval where required → Final State → Audit. Examples include tenant onboarding, move-out/NOC, procurement, vendor onboarding, facility exceptions, financial adjustments and governance decisions.

11.1.3 Reusable SLA and Escalation Engine

Generic flow: Domain Event → SLA Policy → Due Time → Reminder → Escalation 1 → Escalation 2 → Breach → Audit. The same engine can serve complaints, vendor AMC, preventive maintenance, compliance expiry, document expiry and operational exceptions.

11.1.4 Reusable Rules Engine

Generic flow: Input → Ruleset → Conditions → Action → Exception → Audit. Rules should drive late fees, booking constraints, tenant requirements, parking rules, move-out clearance, notification policies and approval routing without hard-coding society-specific logic into application services.

11.1.5 Calendar and Scheduling Engine

A shared scheduling capability should handle facility bookings, inspections, maintenance, vendor visits, meetings, events, move-in/out appointments, drills and shifts. Recurrence, blackout periods, buffers and conflict detection should be first-class concepts.

11.1 Workflow Principles

- Every workflow has a clear start state, valid transitions, terminal states and exception handling.
- Approval requirements are configuration-driven.
- Every state change is traceable.
- Notifications are derived from state changes rather than hard-coded UI actions.

- Long-running work is asynchronous and observable.
- Exceptions require an explicit reason and, where configured, approval.
- Closed financial and compliance records are corrected through controlled adjustments rather than deletion.

Workflow	Main steps	Important edge cases
Monthly maintenance billing	Prepare charge heads → generate draft → validate → approve → publish → collect → receipt → reconcile.	Duplicate payment, partial payment, late penalty, manual cheque/cash, accounting lock.
Complaint lifecycle	Create → classify → assign → SLA → work → resolve → confirm/reopen → close.	Repeated complaint, private complaint, escalation, vendor delay, wrong category.
Move-in	Submit → validate → upload documents → approvals → rule acknowledgement → activate occupancy/access.	Incomplete KYC, missing verification, slot conflict, missing document.
Move-out	Request → calculate clearances → resolve dues/assets/access → approvals → issue NOC → revoke access → close occupancy.	Pending dues, damage claim, lost access card, override, missing NOC.
Biometric attendance	Device punch → sync → validate → deduplicate → map → calculate shift → summarise.	Duplicate punch, missing checkout, offline device, wrong employee code, correction.
Wrong parking	Report → identify vehicle → notify → investigate → resolve → record violation.	Unregistered vehicle, emergency blocking, repeat violation, false report.
Visitor entry	Pre-approval or walk-in → validation → entry → optional exit → close record.	Network failure, expired pass, duplicate entry, invalid QR/OTP.
NOC issuance	Request → checklist → approvals → document generation → verification code → issue.	Incomplete clearance, approval rejection, document error.
Vendor AMC	Contract register → reminder → service → inspection → invoice → performance update → renewal/closure.	Expired contract, repeated failure, disputed invoice, emergency service.
Emergency incident	Trigger → acknowledge → classify → notify → respond → escalate if needed → close → review.	No acknowledgement, simultaneous incidents, communication failure.

11.1.7 Calendar, Recurrence and Conflict Management

The scheduling engine should represent events independently from calendars and views. A recurring maintenance schedule, facility booking, vendor visit and move-in appointment are all time-bound domain commitments with different conflict rules.

Real-world scenario — move-in versus maintenance: a move-in appointment requires loading access on Saturday, but a lift maintenance window is also scheduled. The scheduling engine identifies the conflict and allows policy-driven rescheduling rather than leaving operations to discover the collision at the gate.

- Use explicit timezone and local-date semantics.
- Recurrence rules must have end conditions and exception dates.
- Buffers and blackout windows participate in conflict evaluation.
- Changing an event must preserve its history and source of change.

11.1.6 Notification Engine

Notifications are a reusable event-driven capability. Domain event → notification rule → audience resolution → channel selection → delivery → retry/fallback → status/audit. The engine should support user preferences, quiet hours, critical overrides, templates and localization.

Real-world scenario — invoice reminder: an invoice becomes overdue. The rules engine sends a push notification first, then a configured fallback channel if delivery fails. The system records the attempt and final delivery state without creating a second invoice reminder event for the same rule occurrence.

- Notifications must be idempotent at the business-event/rule level.
- Critical notifications may override quiet hours according to explicit policy.
- Delivery failure must not be confused with business failure; the invoice remains overdue regardless of notification status.

- Templates must be versioned.

11.2 Visitor Pre-approval Flow

Figure 9. Visitor pre-approval workflow.

11.3 Resident Communication Approval Flow

Figure 10. Privacy-first first-contact communication flow.

12. Data Model and API Design

The data model must be multi-tenant from day one. Every business record must be directly scoped by societyId or strongly linked through a parent entity whose society boundary is validated server-side.

12.1.1 Master Data vs Transactional Data

Master/reference data changes relatively slowly and defines allowed vocabulary: complaint categories, visitor types, facility types, charge heads, payment methods, document categories, NOC types, vehicle types, vendor categories, asset categories, inventory categories, templates, SLA policies and escalation rules. Transactional data records what actually happened: visits, invoices, payments, complaints, bookings, work orders, messages and attendance events.

12.1.2 Entity Timeline and Historical Boundary

A unit timeline may include creation, ownership, parking, tenant move-in, visitors, complaints, payments, documents, move-out and NOC events. This timeline is an operational reconstruction aid, not unrestricted access to every historical record. Privacy and authorization boundaries remain applicable at every lookup.

12.1.3 Time, Timezone, Financial Year and Rounding

Store authoritative timestamps in UTC where appropriate and render them in the society/user context. Business rules must explicitly handle IST/local midnight, month-end, leap years, financial-year boundaries, billing cut-offs, INR precision and configurable rounding. Date-only business events must not be accidentally shifted by timezone conversion.

12.1 Core Data Entities

Entity	Purpose
Society	Tenant/account root containing hierarchy, configuration, policies and subscription-ready metadata.
Unit	Flat/villa/shop/office entity linked to residents, bills, visitors, documents, parking and complaints.
ResidentProfile	Owner, tenant or family profile linked to a user and unit relationship.
UnitOccupancyHistory	Historical ownership and tenancy periods.
Document	Secure file metadata, storage reference, visibility, verification and expiry.
NocRequest / NocRecord	Certificate request workflow and issued-document metadata.
StaffProfile	Guard, housekeeping, facility, domestic-help or vendor workforce record.
AttendancePunch / AttendanceDailySummary	Raw imported event and calculated attendance state.
Bill / Payment / LedgerEntry	Maintenance billing and financial transaction model.
Complaint	Structured issue with SLA, assignment, evidence and status.
Facility / Booking	Amenity definition, schedule and reservation lifecycle.
Vehicle / ParkingSlot	Vehicle and allocation model.
Visitor / GateEntry	Visitor identity, pass and physical entry record.

Vendor / Contract / AMC	Supplier and contractual service lifecycle.
Asset / MaintenanceRecord	Asset lifecycle, warranty, AMC and service history.
InventoryItem / StockTransaction	Consumable inventory and movement.
GovernanceMeeting / Resolution / Poll / Election	Formal governance records.
Incident / EmergencyEvent	Operational and emergency incident lifecycle.
AuditLog	Immutable sensitive-action trail.

12.1.4 Master and Reference Data Governance

Master data requires ownership, effective dates, validation and dependency awareness. Examples include complaint categories, visitor types, facility types, charge heads, payment methods, document categories, NOC types, vehicle types, staff types, vendor categories, asset categories, inventory categories, templates and escalation/SLA definitions.

Real-world scenario — complaint category rename: “Lift Problem” becomes “Vertical Transport”. Historical tickets retain their original classification semantics while new tickets use the new category. Reports must be able to group across versions when configured.

- Do not hard-delete referenced master values.
- Use inactive/effective-dated states for changed vocabulary.
- Configuration changes require audit and, where necessary, approval.
- Dependencies must be validated before a master record is retired.

12.2 API Design Principles

- Version APIs explicitly.
- Use consistent resource naming and HTTP semantics.
- Scope requests to the authenticated society context server-side.
- Use idempotency keys for retryable mutations and callbacks.
- Return stable identifiers for long-running operations.
- Keep payment, billing, NOC and workflow transitions behind domain services.
- Separate file upload initiation from secure file retrieval.
- Expose audit and operational status where appropriate without exposing sensitive internal metadata.

12.2.1 Error Contract and Validation Semantics

APIs must distinguish validation failure, authentication/authorization failure, business-rule rejection, concurrency conflict, idempotent replay, integration failure and unexpected server failure. Clients and testers need deterministic categories to handle these states correctly.

Real-world scenario — booking conflict: two users submit the same facility slot. One succeeds; the other receives a deterministic business conflict response with a reference to the failed reservation attempt. The second request must not partially create a booking.

- Use stable error codes in addition to human-readable messages.
- Validation errors identify the affected field or business rule where possible.
- Concurrency failures expose retry-safe semantics.
- Do not leak sensitive information through error messages.

12.3 Representative API Catalogue

Endpoint	Purpose
POST /api/v1/societies	Create society.
PATCH /api/v1/societies/{id}/configuration	Update society configuration.
POST /api/v1/units/import	Bulk import units/residents.
GET /api/v1/units/{id}/occupancy-history	Retrieve unit lifecycle history.
POST /api/v1/move-in-requests	Create move-in workflow.
POST /api/v1/move-out-requests	Create move-out workflow.
POST /api/v1/noc-requests	Request NOC/certificate.
POST /api/v1/documents	Upload document metadata/file reference.
GET /api/v1/documents/{id}/download	Controlled document retrieval.

POST /api/v1/visitors/preapprovals	Create visitor pass.
POST /api/v1/gate/entries	Record gate entry.
POST /api/v1/bills/generate	Generate maintenance bills.
POST /api/v1/payments/callback	Receive payment callback.
POST /api/v1/complaints	Create complaint.
POST /api/v1/bookings	Create facility booking.
POST /api/v1/integrations/biometric/devices	Register biometric device.
POST /api/v1/integrations/biometric/punches/sync	Import attendance logs.
GET /api/v1/attendance/monthly	Attendance reporting.
PATCH /api/v1/attendance/{id}/correction	Attendance correction workflow.

12.4.1 Domain-Specific Concurrency Strategy

Conflict resolution must be selected by business semantics rather than applying one universal CRDT or last-write-wins rule. Financial ledgers require transaction serialization; payment callbacks require idempotency and signature verification; occupancy requires validated state transitions; facility booking requires atomic reservation; attendance requires idempotent event ingestion; documents require version conflict detection; configuration uses optimistic concurrency; chat is append-oriented; audit is append-only.

12.4.2 Idempotency and Exactly-Once Business Effects

API retries are normal in mobile and distributed systems. Every externally retried command that can create a business effect should have an idempotency strategy. A repeated request may produce the same business result, but the system must not create duplicate invoices, payments, visitor admissions, stock issues or attendance events.

12.4 Data Integrity Rules

- Do not hard-delete financial transactions.
- Do not overwrite historical occupancy periods.
- Do not expose previous resident documents through current occupancy relationships.
- Do not permit duplicate booking or duplicate attendance events.
- Do not accept payment callbacks without signature/verification and idempotency handling.
- Do not allow feature-disabled APIs to execute business actions.

12.4.3 Financial, Reservation and Occupancy Invariants

Critical domains require explicit invariants. Financial journals must balance; active occupancy periods must not overlap when prohibited; confirmed reservations must not conflict; active parking allocations must respect slot constraints; visitor state transitions must be valid; attendance events must be deduplicated; audit events must not be overwritten.

These invariants should be enforced at the strongest practical layer—database constraints for structural rules, transactional services for cross-record rules, and workflow/rules engines for configurable policy.

- Every invariant has at least one automated regression test.
- A failed invariant leaves no partial business effect.
- Constraint violations are observable and diagnosable.

13. Feature Flags and Configuration

Feature flags allow one codebase to support societies with different operational complexity. Flags should control availability without creating inconsistent backend behaviour.

13.1 Core Feature Flags

Feature flag	Purpose	Initial direction
visitorManagement	Visitor passes, gate entries and gate logs.	true

maintenanceBilling	Maintenance billing and payment workflows.	true
complaints	Complaint/helpdesk workflows.	true
documentVault	Secure document management.	true
ownerTenantHistory	Historical occupancy and document archive.	true
moveOutNocMandatory	Require NOC/clearance before move-out closure.	true
residentMessenger	Privacy-first messaging.	true
publicResidentDirectory	Optional public directory.	false
biometricAttendanceIntegration	Biometric synchronisation.	false at launch
hardwareIntegration	Common integration framework.	false at launch
subscriptionBillingUI	Pricing/subscription UI.	false during free launch
smartAutomation	Smart automation layer.	false until maturity/readiness criteria are met
waterManagement	Water operational module.	configurable
energyManagement	Energy/solar/utility module.	configurable
evCharging	EV charging management.	false until enabled
governanceVoting	Configured voting workflows.	configurable
marketplace	Community services and commerce.	configurable
carpool	Community carpool.	configurable
lostAndFound	Community lost & found.	configurable
festivalAutomation	Event and festival workflows.	configurable

13.2 Configuration Principles

- Disabled modules must not appear as active capabilities in the product experience.
- Backend APIs must reject access to disabled capabilities.
- Configuration changes must be audited with old value, new value, actor and timestamp.
- Configuration must be tenant/society scoped.
- Dependencies must be validated before enabling a feature.
- Feature flags must not be used as a substitute for authorization.
- Commercial flags must remain hidden until monetisation is activated.

13.2.2 Configuration Safety and Dependency Checks

A feature flag or policy cannot be considered safe merely because it is a boolean. Dependencies must be validated before enabling a capability. For example, enabling automated payment reminders requires notification templates and channel configuration; enabling preventive maintenance requires asset and schedule structures.

Real-world scenario — partial rollout: a society enables biometric attendance but one device is not configured. The feature remains available for configured devices while the missing device is surfaced as an operational readiness exception.

- Feature enablement must validate prerequisites.
- Configuration changes must be reversible or have a documented migration path.
- Runtime evaluation is separate from authorization.
- Flag state changes are observable and auditable.

13.2.1 Configuration Governance and Effective Dating

Configuration is production data. Changes to charge heads, SLA values, booking windows, notification templates, approval rules, retention periods and feature flags should support versioning/effective dates where historical interpretation matters.

Real-world scenario — late-fee change: the society changes late fees from 1% to 1.5% starting 1 October. September invoices retain the old rule; October calculations use the new rule. Historical invoices must not be silently recalculated.

- Do not apply future configuration retroactively unless an explicit migration is executed.

- Configuration changes are audited.
- Dependent configurations are validated before activation.
- A configuration rollback must have defined semantics.

13.3 Example Configuration Structure

features: { visitorManagement: true, maintenanceBilling: true, complaints: true, documentVault: true, ownerTenantHistory: true, moveOutNocMandatory: true, residentMessenger: true, publicResidentDirectory: false, biometricAttendanceIntegration: false, hardwareIntegration: false, subscriptionBillingUI: false, smartAutomation: false, waterManagement: true, energyManagement: false, marketplace: false, carpool: false, lostAndFound: true, festivalAutomation: false }

14. Security, Privacy and Audit Logging

Society OS contains identity, financial, resident, tenant, staff, security and communication data. Security and privacy therefore form part of the product definition rather than an implementation afterthought.

14.1 Security Controls

Area	Required control
Tenant isolation	Every request is scoped to societyId and validated server-side.
RBAC	Role and permission checks apply to sensitive operations.
Document security	Use controlled storage and signed/authorised retrieval; never expose public document paths.
Audit logging	Record sensitive access, approvals, corrections, configuration changes and financial controls.
Privacy settings	Directory visibility, contact exposure and communication preferences are server-enforced.
Chat privacy	Private communication is not readable by administrators by default; moderation requires an explicit workflow.
Biometric privacy	Do not store raw fingerprint templates or images.
Data retention	Retention periods are configurable by data class and operational/legal requirement.
Session security	Secure tokens, expiry, device/session management and forced logout support.
Financial integrity	Use correction/reversal entries; do not delete posted financial history.
Secrets	Keep credentials and integration secrets outside source code and user-facing configuration.
Operational monitoring	Security events, failed authentication, unusual access and integration failures are observable.

14.2 Privacy Principles

- Collect only information necessary for a defined business purpose.
- Separate sensitive data from general profile information where practical.
- Make privacy defaults restrictive.
- Prevent historical data from leaking through shared unit relationships.
- Treat uploaded documents and user-generated content as untrusted input.
- Maintain access logs for sensitive documents and administrative operations.
- Provide configurable retention and controlled deletion where legally and operationally appropriate.

14.2.1 Privacy-by-Design Scenarios

Privacy controls should be tested against realistic misuse, not only documented permissions. Search, exports, notifications, communication, documents, historical timelines and offline caches can all become data-leak channels.

Real-world scenario — shared device: a guard logs out after using a shared tablet. The next guard must not see the previous operator's resident searches, visitor history or cached personal information.

Real-world scenario — notification preview: a lock-screen notification should not reveal sensitive document or financial details if the configured privacy level forbids it.

- Minimize sensitive data in push/SMS/email payloads.
- Encrypt sensitive local/offline data.
- Log access to high-sensitivity documents and records.
- Deletion/anonymization must respect legitimate retention requirements.

14.3.1 Operational Audit vs Business History

Audit answers “who did what, when, from where, through which channel, and with what result?”. Business history answers “what was the valid state over time?”. These should be correlated but not conflated. A correction to a financial record is a new compensating transaction, not an edit that erases the original event.

14.3.2 Data Lifecycle and Retention

A generic lifecycle is CREATED → ACTIVE → INACTIVE → ARCHIVED → RETENTION → ANONYMIZED/PURGED, subject to legal, contractual and operational requirements. The retention engine should be configurable by record category and purpose.

Real-world scenario — former tenant: a tenant leaves in June. Their operational association becomes inactive, but required financial records and legally necessary audit evidence may remain retained. Personal information that no longer has a valid purpose may be anonymized or purged according to policy; the platform must not treat all records as either “delete everything” or “keep forever”.

14.3 Audit Model

Event category	Examples
Identity	Login, logout, forced logout, password/credential changes.
Authorization	Permission changes, role changes, access denial.
Financial	Bill lock, correction, reversal, payment reconciliation, refund.
Document	View, download, upload, version, expiry, permission change.
Lifecycle	Move-in approval, move-out closure, NOC issue, access revocation.
Attendance	Correction, manual entry, mapping change.
Governance	Resolution approval, poll configuration, election result publication.
Security	Watchlist change, incident escalation, emergency action.
Configuration	Feature flag or society policy change.

15. Reports and Society Health Score

Reporting should support decisions, not merely display counts. The reporting layer should combine current status, trends, exceptions and historical comparisons.

15.1.1 Operational Dashboard Model

Dashboards should be derived from authoritative source records and should answer operational questions, not merely display attractive charts. Typical domains include dues and collections, outstanding ageing, complaints and SLA, current visitors, incidents, facility utilization, parking, attendance, assets, preventive maintenance, inventory, vendors, compliance and emergencies.

Real-world scenario — committee meeting: the committee wants to know why expenses increased. The dashboard should allow the aggregate number to be traced to invoices, vendor work orders, inventory consumption and ledger postings rather than producing an unexplained AI-generated narrative.

15.1.2 Search and Global Retrieval

Global search should cover units, residents, vehicles, visitors, complaints, invoices, payments, vendors, assets, documents, parking and bookings with filters, date/status/category/unit/reference, sorting, pagination and export. Search results must respect privacy and authorization boundaries; discoverability is not permission.

15.1 Report Catalogue

Category	Example reports
Financial	Collection, defaulters, expenses, budget vs actual, fund balances, reconciliation.
Operations	Complaint SLA, vendor performance, housekeeping quality, asset maintenance, recurring issues.
Security	Visitor trends, gate load, incident counts, watchlist events, workforce attendance.
Staff	Daily/monthly attendance, shift adherence, absence, vendor-wise summaries.
Owner/Tenant lifecycle	Move-in status, pending move-out clearance, agreement expiry, verification pending.
Compliance	Expired documents, fire/lift renewals, legal actions, governance records, rule acknowledgement.
Community	Notice acknowledgement, event participation, poll response, complaint feedback.
Utilities	Water consumption, tanker usage, electricity consumption, solar generation, EV usage where enabled.

15.2 Society Health Score

The Society Health Score is a composite operational indicator. It should be transparent, trend-based and explainable. It must not become an opaque personal score for residents.

Component	Example metrics
Collection efficiency	Percentage collected within due date, ageing and trend.
Complaint resolution	SLA compliance, average resolution time, reopen rate and feedback.
Security discipline	Gate process adherence, incidents, workforce attendance and unresolved security tasks.
Compliance health	Validity of mandatory documents and renewal timeliness.
Operational efficiency	Vendor performance, asset uptime, repeated issue reduction and checklist completion.
Resident satisfaction	Feedback, response time, participation and complaint sentiment where appropriate.
Workforce health	Attendance adherence, shift coverage and repeated absence patterns.
Utility health	Water/energy variance, leakage events, consumption trends and sustainability indicators where enabled.

15.3 Intelligence-Ready Reporting

- Explain material changes rather than only showing the new value.
- Highlight exceptions requiring attention.
- Allow historical comparison against the society's own baseline.
- Keep financial calculations deterministic.
- Show the source records behind derived indicators.
- Retain report generation metadata for audit and reproducibility.

15.3.1 KPI Definitions and Source-of-Truth Traceability

Every executive KPI should have a written definition, calculation period, source entities, exclusions, refresh behaviour and drill-down path. This is essential for trust in committee, management and investor-facing conversations.

Real-world scenario — collection efficiency: the KPI must distinguish billed amount, collectible amount, settled amount, reversed/refunded amount and outstanding amount. A dashboard showing one “collection %” without these definitions can create misleading conclusions.

- KPI definitions are versioned when business formulas change.
- Aggregations must reconcile to source records within documented tolerances.
- Exports and dashboards use the same authoritative metric definitions.

16. Smart Automation and Future Integrations

Smart automation is a controlled extension of the operational platform. It should be introduced after the core data model is clean, permissions are stable and historical records are reliable.

16.1 Smart Capabilities

Capability	Functional purpose	Dependency
Smart Complaint Routing	Suggest category, priority and operational routing using complaint context and rules.	Complaint history and category rules.
Automated Notice Drafting	Draft clear, multilingual society notices from structured context and approved intent.	Communication module.
Smart Document Search	Answer questions over permitted society documents and structured records.	Document vault and permission model.
Bill Explanation Assistant	Explain changes in a maintenance bill using authoritative charge data.	Structured billing model.
Meeting Summary	Summarise meeting notes/transcripts into actions and decisions.	Governance records.
Maintenance Risk Alerts	Identify assets with rising operational risk.	Asset and maintenance history.
Duplicate Incident Detection	Correlate similar complaints into a common operational incident.	Complaint history and location/time data.
Vendor Intelligence	Summarise SLA, cost, complaint and service performance.	Vendor and work-order history.
Collection Intelligence	Explain collection trends and highlight unusual variance.	Billing and payment history.
Community Operations Insights	Surface recurring issues, bottlenecks and recommended actions.	Cross-module operational data.

16.1.1 AI/Automation Decision Boundaries

AI may assist with classification, summarization, anomaly detection, forecasting and recommendations, but authoritative business state remains deterministic. AI output must never silently create a payment, approve a resident, issue an NOC, admit a visitor or alter an audit record.

Real-world scenario — complaint classification: AI predicts “water leakage” with 91% confidence. The prediction pre-fills classification, but the operational workflow still validates the category and permits correction. The final ticket state remains governed by deterministic rules.

- Store model/version and confidence metadata where AI materially affects a workflow.
- Provide human-review paths for consequential recommendations.
- Never fabricate missing source data in summaries.
- AI-generated explanations must link to authoritative records.

16.2 Intelligence Guardrails

- Authoritative values come from structured application data, not model invention.

- Permissions are applied before sensitive information is retrieved.
- Sensitive state changes require the existing deterministic workflow and appropriate approval.
- Uploaded documents and messages are treated as untrusted content.
- Every important recommendation includes evidence, confidence or source context where applicable.
- Low-confidence output should be presented as uncertain or routed for manual review.
- Core society operations must continue if an external intelligence provider is unavailable.

16.3.1 Integration Reliability Contract

Every external integration should define authentication, request/response contract, timeout, retry policy, idempotency, rate limits, failure states, reconciliation path, observability and versioning. Integrations must fail closed where security requires it and fail gracefully where availability is more important.

Real-world scenario — payment provider outage: residents can initiate payments but the provider is temporarily unavailable. Society OS must avoid marking invoices paid based on client-side success alone. Pending payment attempts remain pending until verified evidence arrives, while reconciliation exposes exceptions for operations/finance.

Real-world scenario — biometric device offline: a biometric terminal stores punches locally while connectivity is unavailable. When the connection returns, events are uploaded with device identifiers and event timestamps. Duplicate sync attempts do not create duplicate attendance records.

16.3.2 Localization and Accessibility as Platform Capabilities

Localization should cover English, Hindi, Marathi, Kannada, Tamil, Telugu, Bengali, Gujarati, Malayalam, Punjabi and future languages through externalized labels/templates rather than duplicated business logic. Accessibility should include scalable text, high contrast, screen-reader semantics, keyboard navigation on web, adequate touch targets, voice/audio assistance where appropriate, reduced motion and non-color-only status communication.

16.3 Integration Catalogue

Integration	Purpose	Design direction
Biometric devices	Attendance synchronisation.	Attendance logs only; no raw biometric templates.
RFID	Vehicle/staff access automation.	Map tags to authorised records.
ANPR	Automatic vehicle recognition.	Optional integration with explicit privacy controls.
CCTV	Restricted incident evidence.	Highly controlled and audited access.
Smart meters	Water/electricity/gas data.	Sensor-ready schema with manual fallback.
EV charging	Slot, usage and billing.	Integrate with facility and financial workflows.
Payment gateways	Digital collections.	Verified callbacks and idempotency.
Messaging providers	SMS/WhatsApp-style fallback and notifications.	Provider abstraction and delivery tracking.
Utility providers	Electricity/water/energy data where APIs are available.	Connector-based integration with graceful fallback.
Property-tax services	Property-related payment/status integration where available.	External-service adapter; do not make core operations dependent on it.

16.4 Community Intelligence Extensions

- Smart Lost & Found: structured item registration, matching, verification and closure.
- Community Carpool: ride requests, recurring routes, matching and controlled contact exchange.
- Festival & Event Automation: planning templates, contributions, volunteers, reminders and closure.
- Society Micro-Commerce: controlled local commerce with order/payment readiness.
- Domestic Worker Replacement Network: availability and verified-service discovery without exposing unnecessary personal information.
- Pet Management: pet registry, service discovery and incident workflows.
- Classifieds: controlled community listings with reporting and moderation.

- Guest Stay Management: stay records, approvals, deposits and checkout.
- Housekeeping Quality: recurring checklists, inspection scores and corrective actions.
- Emergency Local Network: configurable local contacts, volunteers and response coordination.

16.5 Water, Energy and Sustainability

Area	Scope
Water	Meter/readings, tanker logs, consumption, leakage, maintenance, shortage alerts and conservation reporting.
Electricity	Common-area consumption, meter data readiness, billing inputs and anomaly reporting.
Solar	Generation, usage, savings and maintenance records.
EV	Charging points, slot booking, usage, billing and load management readiness.
Energy analytics	Trend, variance, efficiency and sustainability reporting.

16.5.1 Utility and Meter Operations

Utility management should support meter masters, readings, reading source, validation, anomalies, billing/recovery and historical trends. Water, electricity, solar generation, diesel and common-area consumption can be modelled as related but distinct utility domains.

Real-world scenario — abnormal water usage: Tower A consumption doubles compared with its baseline. The anomaly engine flags the change, maintenance receives an investigation task, and the final outcome records whether it was a leak, meter fault, genuine usage spike or data error.

- Meter readings require timestamp/source semantics.
- Outliers should be reviewable before becoming financial charges where policy requires.
- Utility corrections should be versioned and auditable.
- Energy data can feed sustainability and cost reporting.

17. Roadmap and Delivery Priorities

The roadmap is designed to establish a dependable operational core before introducing advanced intelligence and hardware dependencies.

Phase	Feature set	Outcome
MVP	Society setup, unit master, resident onboarding, visitor/gate, billing, complaints, notices, staff management, document vault, feature flags, core notifications and low-connectivity foundations.	Launch a dependable society platform with the essential daily workflows.
Phase 1A	Resident Connect, privacy directory, owner/tenant history, smart move-in/move-out, document archive, parcel handover, inter-flat issues, lost & found and basic community services.	Differentiate the product through privacy, lifecycle history and daily resident utility.
Phase 2	Digital NOC, parking, facility booking, vendor/AMC, assets, inventory, renovation, governance, compliance, housekeeping quality and advanced reporting.	Strengthen operational maturity and administrative control.
Phase 3	Biometric attendance, workforce automation, senior care, emergency volunteers, marketplace, community commerce, carpool, events, staff/payroll readiness and Society Health Score.	Add measurable operational intelligence and broader community utility.
Phase 4	RFID, ANPR, boom barrier, CCTV integrations, smart meters, solar/energy, EV charging, smart automation, e-voting and advanced connected-device integrations.	Make the platform enterprise- and premium-community ready.

17.1 Delivery Priorities

18. Establish clean society, unit, resident and financial master data.
19. Make security, tenant isolation and auditability production-grade before expanding integrations.
20. Complete core visitor, billing, complaint, document and staff workflows.
21. Introduce lifecycle history and NOC capabilities without compromising privacy.
22. Expand operational modules such as parking, facilities, vendors, assets and governance.
23. Add workforce, attendance, compliance and emergency capabilities.
24. Introduce community services and commerce after operational foundations are stable.
25. Introduce smart automation only after sufficient structured data and evaluation criteria exist.
26. Add hardware and utility integrations through stable connector interfaces.
27. Continuously measure adoption, reliability, operational savings and data quality.

17.2.1 Delivery Slice Definition

A delivery slice is complete only when the normal scenario, important exceptions, persistence rules, audit behaviour, notifications, failure handling, observability and acceptance tests are covered. “API implemented” or “screen completed” is not a production-complete slice.

Example slice — visitor admission: the slice includes pre-approval, gate validation, resident action, offline capture, duplicate prevention, exit recording, notification, audit, emergency exception and reconciliation—not just a visitor create endpoint.

17.2.2 Versioning and Migration Discipline

Database migrations, API versions, backward compatibility, feature-flag migrations, mobile/server compatibility and offline schema compatibility must be planned before release. A mobile client may remain offline across a deployment boundary, so sync payloads require compatibility and migration strategy.

17.2 MVP Discipline

Build principle Do not overload the first release with every advanced capability. Build the operational backbone well, keep advanced modules configurable, and add complexity only when the data, workflow and support maturity justify it.

18. Acceptance Criteria and Implementation Backlog

Acceptance criteria must cover positive cases, negative cases, permissions, audit behaviour, edge cases, data integrity and recovery from failure.

Epic	User story / requirement	Acceptance criteria
Resident onboarding	Import residents and units in bulk.	Duplicates detected; validation preview shown; valid records imported; errors reported; import job logged; rollback available.
Visitor management	Create and validate a visitor pass.	QR/OTP or walk-in validation works; entry is logged; invalid/expired passes are rejected; duplicate entry is handled.
Billing	Generate monthly maintenance bills.	Correct charge heads applied; draft validation available; publication is controlled; payments update ledger; receipts are generated.
Complaints	Create and resolve a complaint.	Ticket identifier created; SLA starts; assignment is traceable; evidence retained; resolution and reopen are auditable.
Owner/tenant history	Preserve historical occupancy.	Occupancy periods do not overlap incorrectly; previous records remain available under controlled access; current occupant cannot inherit sensitive historical documents.

Move-out NOC	Prevent premature lifecycle closure.	Clearance checklist evaluated; NOC not issued when mandatory conditions fail; authorised override requires reason and audit.
Document vault	Secure a sensitive document.	Metadata required; storage is controlled; access is authorised; download is logged; expiry/version are retained.
Biometric sync	Import attendance punches.	Duplicate events rejected; employee mapping validated; daily summary calculated; sync failures visible; job is traceable.
Attendance correction	Correct a missed punch.	Reason required; approval rules applied; old/new values retained; audit event created.
Resident communication	Initiate controlled contact.	First-contact request captures purpose; acceptance/rejection/block/report works; phone exposure is not required.
Private communication	Protect private content.	Unreported private conversations are not exposed through ordinary administrative access; moderation follows an audited workflow.
Parking	Record a violation.	Vehicle/property association validated; evidence can be attached; status and history are retained; repeat behaviour can be identified.
Facility booking	Prevent double booking.	Database constraint prevents overlap; payment/deposit behaviour follows booking state; cancellation/refund rules are deterministic.
Emergency	Trigger an urgent incident.	Alert is generated quickly; acknowledgement is tracked; escalation works when unacknowledged; incident history is retained.
Feature flags	Disable a module.	Capability is unavailable in the product experience; backend rejects calls; configuration change is audited.
Financial integrity	Correct an existing transaction.	Original record remains; correction/reversal is linked; audit trail identifies the reason and actor.
Security	Prevent cross-society access.	Requests using another society's identifiers fail even if the resource ID is known.
Offline operation	Continue critical field workflow without network.	Event is queued safely; user sees sync state; reconciliation prevents duplicates after reconnect.
Smart automation	Provide a recommendation or draft.	Source context is valid; permissions are enforced before retrieval; low-confidence cases are handled safely; sensitive state changes still require normal authorization.
Reporting	Produce repeatable operational reports.	Filters are deterministic; totals reconcile with source records; report generation metadata is retained.

18.1.1 Test Scenario Taxonomy

QA should test at five levels: happy path, validation/negative path, concurrency/idempotency, failure/recovery, and audit/privacy. For major workflows, tests should also cover offline behaviour, localization, boundary dates and configuration variants.

Test dimension	Example	Expected result	Evidence
Happy path	Resident pays full invoice	Invoice settled and receipt issued	Ledger + receipt + audit
Negative	Visitor arrives with invalid/expired pass	Admission blocked with reason	Gate event + audit

Concurrency	Two users reserve same slot simultaneously	At most one confirmed reservation	Reservation constraint/log
Idempotency	Same payment callback sent twice	One financial effect	Idempotency record
Failure recovery	Network fails during gate sync	Event retained and reconciled later	Sync log
Privacy	Unauthorized user searches resident	Sensitive data not exposed	Authorization/audit evidence

18.1 Definition of Done

- Business rules documented and implemented.
- Positive and negative cases tested.
- Permission and tenant-isolation tests completed.
- Audit events implemented for sensitive mutations.
- Feature-flag behaviour tested in both enabled and disabled states.
- Failure/retry behaviour documented for integrations.
- Data migration and rollback strategy documented where applicable.
- Operational metrics and logging available.
- User-facing terminology consistent with the product glossary.
- Security review completed for sensitive modules.

19. Appendix

19.1 Example Feature Flag Structure

```
features: {
    visitorManagement: true,
    maintenanceBilling: true,
    complaints: true,
    documentVault: true,
    ownerTenantHistory: true,
    occupancyHistory: true,
    moveOutNocMandatory: true,
    residentDocumentArchive: true,
    residentMessenger: true,
    residentDirectory: true,
    publicResidentDirectory: false,
    controlledGroupChat: true,
    digitalNocRequests: true,
    societyRulesAcknowledgement: true,
    biometricAttendanceIntegration: false,
    attendanceDeviceSync: false,
    staffShiftRoster: true,
    attendanceCorrectionWorkflow: true,
    cctvAccess: false,
    anprRfidIntegration: false,
    smartMeters: false,
    waterManagement: true,
    energyManagement: false,
    evCharging: false,
    marketplace: false,
    carpool: false,
    lostAndFound: true,
    festivalAutomation: false,
    subscriptionBillingUI: false,
    pricingPage: false,
    upgradePrompts: false
```

```
}
```

19.2 Developer Rules

- Never hardcode visible pricing in the user-facing product during the free-launch period.
- All modules must enforce feature flags on both frontend and backend.
- Every sensitive administrative action must create an audit record.
- Do not delete posted financial records; use reversal or correction entries.
- Documents must have access control, expiry metadata and access logs.
- Critical gate workflows must support low connectivity and synchronisation.
- Keep SaaS subscription billing separate from resident maintenance payments.
- Do not store raw fingerprint data in Society OS; synchronise attendance events only.
- Previous owner/tenant documents must never be visible to later occupants by default.
- Move-out closure requires the configured NOC/clearance unless an authorised override with reason is permitted.
- All external integrations must have retry, failure and observability handling.
- AI/smart capabilities must not bypass deterministic permissions, workflow rules or financial controls.

19.3 Product Glossary

Term	Definition
Society	The top-level tenant/account representing a residential community.
Unit	A flat, villa, shop, office or other occupiable property within a society.
Occupancy	A time-bound relationship between a unit and its resident/tenant context.
NOC	No-objection/no-dues or other configured certificate generated through an approval workflow.
AMC	Annual maintenance contract or similar recurring service agreement.
Feature flag	Configuration switch controlling whether a capability is active.
Audit log	Immutable record of a sensitive action, decision or configuration change.
Smart automation	Rule- or intelligence-assisted functionality that reduces repetitive work or provides recommendations.
Society Health Score	Explainable composite indicator of operational health.
Integration layer	Connector architecture used to communicate with external systems and devices.

19.5 Real-World Scenario Library

The following scenarios should remain part of product discovery, demos, QA regression and implementation workshops because they demonstrate the difference between a feature list and a real operating platform.

Scenario A — New Resident Move-In

28. Society creates or imports the unit and parking master.
29. Owner/tenant receives an invitation and completes verification.
30. Occupancy record becomes active for a defined period.
31. Required documents are uploaded and versioned.
32. Move-in appointment is scheduled and gate instructions are generated.
33. Vehicle and parking are linked.
34. Notifications and audit events are recorded.
35. Any configured deposit/charge is posted to the financial workflow.

Scenario B — Tenant Move-Out with NOC

36. Tenant submits move-out request with target date.
37. Rules engine evaluates notice period and required documents.
38. Finance calculates outstanding balance and pending adjustments.
39. Parking, assets, access and document clearances are checked.

40. Required approvals complete.
41. Final dues/refund are posted through immutable financial transactions.
42. NOC is generated as a controlled document version.
43. Access is revoked at the effective time and occupancy is closed.

Scenario C — Lift Breakdown During Peak Hours

44. Resident reports failure with evidence.
45. Ticket is classified and SLA starts.
46. Correlation engine identifies whether related complaints exist.
47. AMC vendor/work order is assigned.
48. Residents receive an incident update.
49. Vendor diagnoses, requests spare parts and records service evidence.
50. SLA/escalation engine tracks delay.
51. Resolution is verified and cost/inventory records are linked.
52. Ticket closes while the asset maintenance history is updated.

Scenario D — Monthly Billing and Payment Failure

53. Billing period closes and charge calculation runs.
54. Validation detects missing configuration before invoices are published.
55. Invoices are generated and notified.
56. Resident initiates UPI payment.
57. Gateway callback is verified and posted idempotently.
58. A second callback is ignored as a duplicate business effect.
59. A failed/partial payment remains accurately represented.
60. Reconciliation and ageing dashboards update from source records.

19.6 Product Pitch Message

Society OS brings the society's fragmented operational world into one configurable system of record. It connects the property hierarchy with people, access, money, work, assets, documents, communication, governance and intelligence—while preserving auditability, privacy, offline resilience and India-first operational realities. The platform is designed so a small standalone building and a multi-tower township can use the same core model with different configuration, rules and scale.

19.9 How This Blueprint Is Used by Each Discipline

Engineering uses the specification as the domain and implementation contract: states, entities, invariants, API/event semantics, idempotency, concurrency, failure handling and migration rules. Product owners use it to validate business scope, policy options, user outcomes and future roadmap decisions. QA uses it to derive happy-path, negative, concurrency, recovery, privacy, offline, boundary-date and configuration-variant tests. Managers use it to understand delivery scope, operational dependencies, risk, readiness and measurable outcomes. Pitch/demo teams use the real-world scenarios to explain value through complete stories rather than isolated feature names.

A capability should not be considered "specified" until the relevant scenario, exception path, data behaviour, security/privacy boundary and acceptance criteria can be traced back to this blueprint or to an explicitly referenced detailed specification.

19.8 Foundational Capability Coverage Index

This index is a completeness control for future product reviews. It does not assign features to roles or applications; it identifies where a foundational capability is specified and what lifecycle it must cover.

Capability	Primary specification area	Minimum lifecycle / scenario
Society onboarding	1, 7.1	Create → configure → validate → import → activate
Property/unit master	7.1, 8, 12	Create → change → historical

		state → archive
Resident onboarding	7.1, 8	Invite → verify → associate → approve → active → exit
Household/family	7.1, 8	Add → effective period → update → end
Visitor/gate	7.2, 11	Pre-approve → admit → exit → exception → audit
Billing/payments	7.3, 12	Calculate → invoice → pay → reconcile → adjust/refund
Complaints	7.4	Create → assign → work → resolve → reopen/close
Facility booking	7.5	Request → conflict check → confirm → use → cancel/refund
Parking/vehicles	7.6	Register → allocate → operate → violate → reassign/history
Communication	7.7, 11	Compose → target → deliver → acknowledge/moderate
Disputes	7.8	Open → investigate → mediate → resolve → close
Governance	7.9	Agenda → meeting → vote → resolution → minutes/action
Emergency	7.10	Detect → declare → respond → communicate → recover/review
Vendor/AMC/assets	7.11	Onboard → contract → work → evidence → invoice → performance
Inventory/procurement	7.11	Request → approve → PO → receive → issue → reconcile
Documents/NOC	10	Upload → scan → version → access → retain → NOC
Attendance	9	Ingest → validate → dedupe → correct → report
Imports/exports	11, 12	Upload → validate → preview → process → report
Notifications	11, 16	Event → rule → channel → retry/fallback → audit
Search	15	Query → filter → authorize → retrieve → export
Backup/DR	14	Backup → verify → restore → validate → resume
Sessions/devices	11, 14	Authenticate → trust → expire/revoke → recover
SLA/escalation	11, 16	Event → due → remind → escalate → breach
Rules/approval	11, 13	Request → evaluate →

		approve/reject → audit
Calendar/scheduling	11	Schedule → detect conflict → execute → reschedule
Localization/accessibility	2, 16	Configure language/accessibility → render → validate
Data lifecycle	14	Active → inactive → archive → retain → anonymize/purge
Observability	16	Emit → collect → alert → diagnose → recover
Versioning/migration	16, 17	Change → migrate → compatibility → deprecate

19.7 Master Documentation Rule

Whenever a new capability is added to Society OS, its specification should include: business problem, real-world scenario, normal workflow, alternate workflow, exception cases, state model, business rules, data entities, API/event contract, permissions boundary, notifications, audit requirements, offline/failure behaviour where relevant, reporting impact, acceptance criteria, security/privacy impact and operational observability. This rule prevents future versions from becoming feature-name catalogues without implementation depth.

19.4 Final Product Recommendation

Society OS should launch as a clean, free and configurable society management platform while keeping its architecture enterprise-ready. The product should solve standard society needs exceptionally well—visitors, billing, complaints, documents and staff operations—while progressively adding deeper capabilities such as owner/tenant lifecycle history, mandatory move-out NOC, privacy-first communication, workforce attendance, governance, safety, compliance, assets, utilities, community services and smart automation.

The strongest product strategy is not to compete by accumulating isolated features. It is to create one trusted operational system in which people, units, money, documents, security, services, assets, governance and history are connected. Once that foundation is reliable, automation and intelligence can deliver meaningful value because they operate on structured, permission-safe society data.

Final principle Society OS should make the society easier to operate, easier to govern and easier to live in—without sacrificing privacy, financial integrity, security or transparency.

19.10 Scenario-First Functional & Complete Diagram Compendium

Purpose: This section is the authoritative scenario and diagram companion for Product, Engineering, QA, Management and pitch/demo teams. Every scenario contains both a complete use-case diagram and a complete data flow diagram. Diagrams explicitly include authenticated entry points for protected human actions, approval/signature steps where required, external participants, processing services, authoritative storage, notifications and exception paths.

Diagram completeness standard: A diagram is considered complete only when every participant that initiates, authorizes, approves, signs, supplies external evidence, or receives an outcome is represented; every protected administrative action has an explicit sign-in/authentication path; every authoritative state change has a processing component and destination store; every external callback/retry has a verification/idempotency boundary; and exception/retry/conflict paths are shown where the written scenario requires them.

Source-of-truth rule: The written scenario, state progression, use-case diagram and DFD must describe the same business outcome. If a diagram omits a participant or business step, it is considered incomplete even if the narrative mentions it.

Human / External Actor	Application / Process	Authoritative Data Store	Exception / Risk Path

F01 - Monthly Maintenance Billing

Domain: Finance / Billing

Primary participants: Billing Administrator, Resident, Notification Service

Business objective: Generate a reproducible billing cycle from effective-dated rules and post receivables once.

Real-world scenario: A 180-unit society closes September billing. Maintenance, parking, water and sinking-fund charges are calculated; anomalous bills are held for review before invoices are published.

Preconditions: Unit/property master is active; billing rules and effective dates are configured; the financial period is open.

Normal flow: Admin signs in → selects billing period → engine snapshots applicable rules → calculates charges → validates exceptions → admin reviews exceptions → invoices are generated → receivables are posted → residents are notified → run is closed.

Exception / alternate flow: Missing rate, duplicate charge, negative/zero result, mid-cycle configuration change or partial run failure must produce a recoverable exception without duplicate posting.

State progression: DRAFT → CALCULATING → VALIDATING → REVIEW → PUBLISHED → POSTED → CLOSED

Key invariant: A billing run is idempotent for the same society/period/rule-version; posted invoices are immutable except through controlled adjustments.

QA acceptance intent

- Verify the protected human actor signs in before performing privileged operations such as approval, verification, financial adjustment or NOC issuance.
- Verify the happy path creates exactly one authoritative business outcome and downstream notifications are triggered from confirmed state.
- Verify retries, duplicates and provider callbacks cannot create duplicate business effects.
- Verify exception states are visible, recoverable and auditable, without corrupting historical records.

Figure F01-A. Complete use-case diagram - participants, authenticated entry points and business interactions.

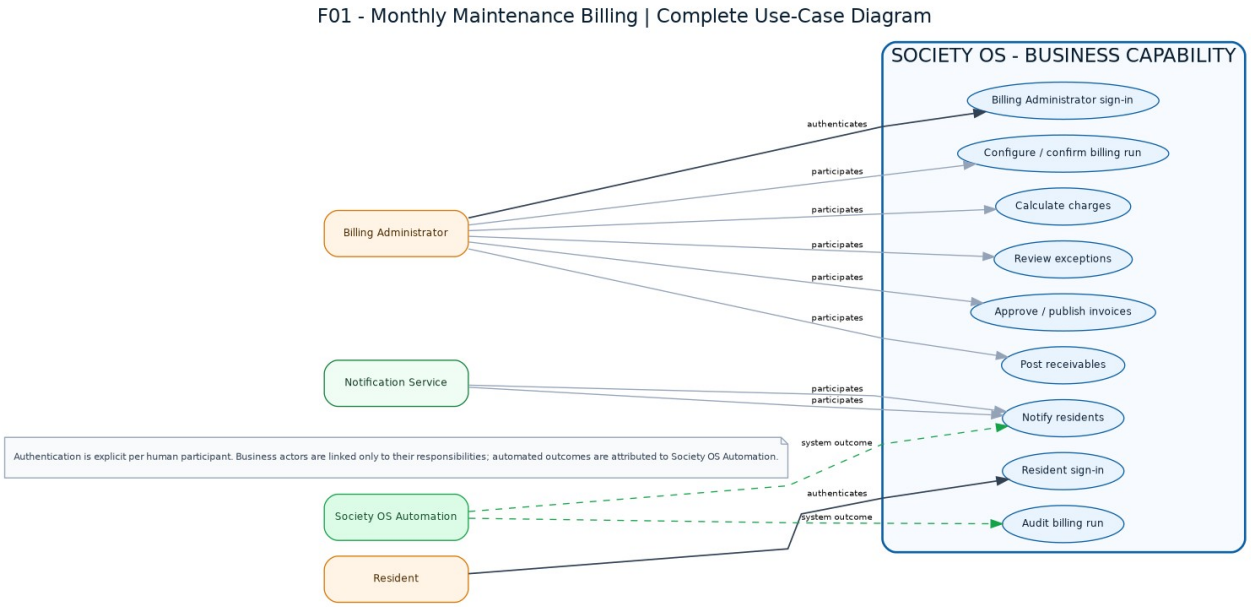
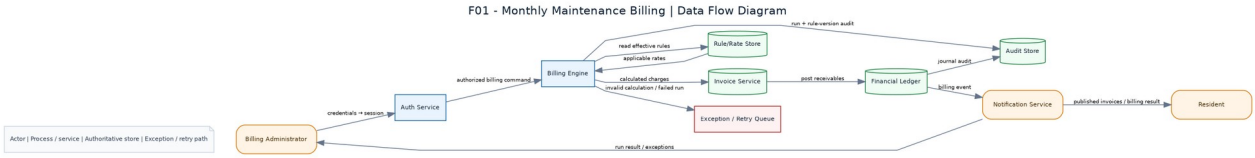


Figure F01-B. Complete data flow diagram - actors, services, authoritative stores, external boundaries and exception path.


F01 - Monthly Maintenance Billing | Data Flow Diagram
Actor | Process / service | Authoritative store | Exception / retry path
Billing Administrator
Auth Service
Billing Engine
Rule/Rate Store
Invoice Service
Exception / Retry Queue
Financial Ledger
Audit Store
Notification Service
Resident
credentials → session
authorized billing command
read effective rules
applicable rates
calculated charges
invalid calculation / failed run
run + rule/version audit
post receivables
journal audit
billing event
published invoices / billing result
run result / exceptions

F02 - Normal UPI Payment

Domain: Finance / Payments

Primary participants: Resident, Payment Gateway, Finance Administrator

Business objective: Turn a verified UPI payment into exactly one allocation, ledger posting and receipt.

Real-world scenario: A resident pays a ₹12,500 maintenance invoice using UPI. The gateway callback is signature-checked and the payment is matched before the ledger is updated.

Preconditions: An invoice is payable; payment gateway and callback credentials are configured; resident session is valid.

Normal flow: Resident signs in → views invoice → initiates UPI → gateway processes payment → callback reaches API → signature and amount/reference are validated → payment is allocated → ledger posts → receipt is generated → resident is notified.

Exception / alternate flow: Gateway timeout, unmatched reference, amount mismatch or provider outage creates a pending/exception record; no payment is posted on client-side success alone. If an exception remains unresolved, the Finance Administrator signs in to review the exception and retry/reconcile it.

State progression: REQUESTED → GATEWAY_PENDING → VERIFIED → ALLOCATED → POSTED → RECEIPTED / EXCEPTION

Key invariant: Callback verification and idempotency are required before financial posting.

QA acceptance intent

- Verify the protected human actor signs in before performing privileged operations such as approval, verification, financial adjustment or NOC issuance.
- Verify the happy path creates exactly one authoritative business outcome and downstream notifications are triggered from confirmed state.
- Verify retries, duplicates and provider callbacks cannot create duplicate business effects.
- Verify exception states are visible, recoverable and auditable, without corrupting historical records.

Figure F02-A. Complete use-case diagram - participants, authenticated entry points and business interactions.

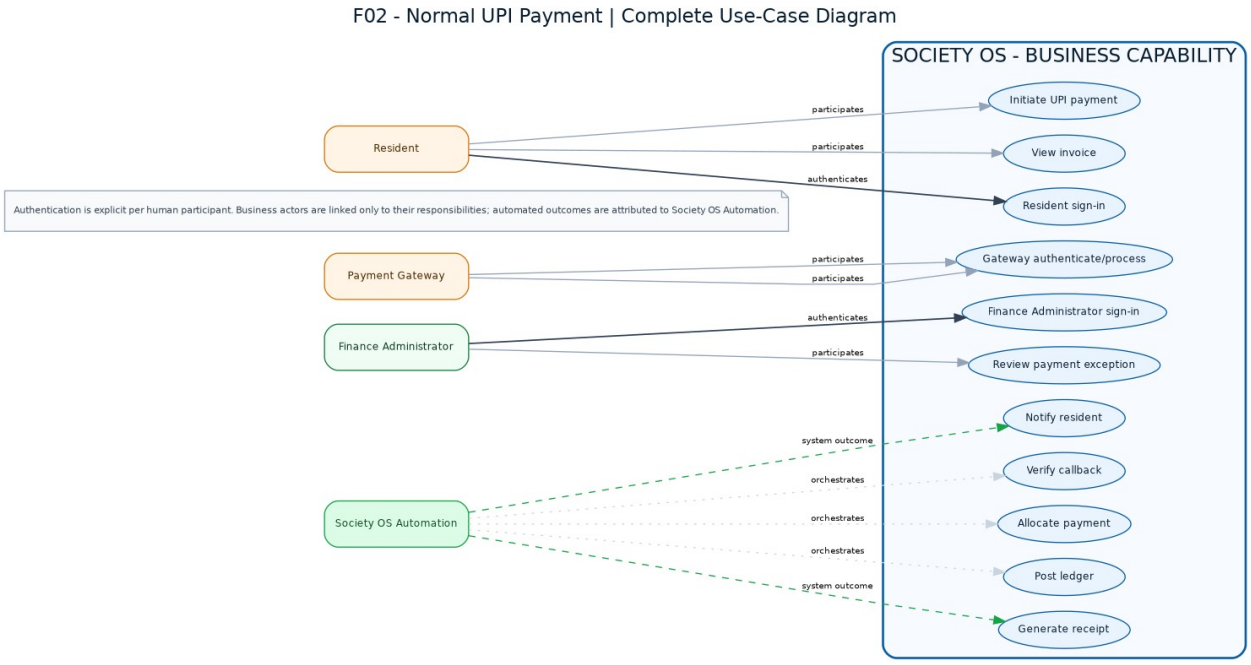
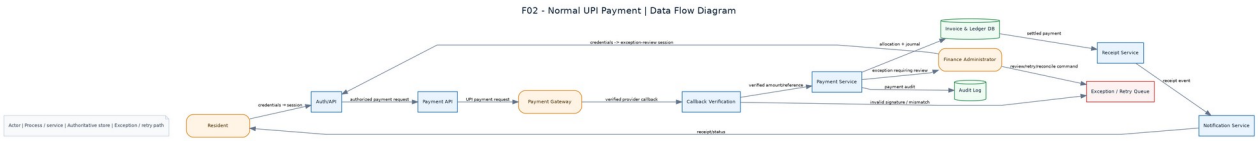


Figure F02-B. Complete data flow diagram - actors, services, authoritative stores, external boundaries and exception path.


F02 - Normal UPI Payment | Data Flow Diagram
Actor | Process / service | Authoritative store | Exception / retry path
Resident
Auth/API
Payment API
Payment Gateway
Callback Verification
Payment Service
Invoice & Ledger DB
Finance Administrator
Audit Log
Receipt Service
Exception / Retry Queue
Notification Service
credentials -> session
credentials -> exception-review session
Authorized payment request
UPI payment request
verified provider callback
verified amount/reference
allocation + journal
exception requiring review
payment audit
invalid signature / mismatch
settled payment
review/retry/reconcile command
receipt/status
receipt event

F03 - Partial Payment

Domain: Finance / Payments

Primary participants: Resident, Finance Administrator

Business objective: Accept a payment lower than the outstanding amount while preserving the remaining balance.

Real-world scenario: A unit owes ₹12,500 and pays ₹7,500. The system allocates the payment using configured priority rules and leaves ₹5,000 outstanding.

Preconditions: Open invoice and allocation policy exist.

Normal flow: Resident signs in → selects amount → payment succeeds → system allocates ₹7,500 → ledger posts settled portion → balance remains ₹5,000 → receipt reflects partial settlement → ageing continues on remaining amount.

Exception / alternate flow: Amount exceeds balance, allocation rule unavailable, duplicate callback or payment reversal must not distort outstanding.

State progression: OPEN → PARTIALLY_PAID → PARTIALLY_RECONCILED → FULLY_PAID or OVERDUE

Key invariant: Allocated amount + remaining outstanding must equal the authoritative invoice balance after all approved adjustments.

QA acceptance intent

- Verify the protected human actor signs in before performing privileged operations such as approval, verification, financial adjustment or NOC issuance.
- Verify the happy path creates exactly one authoritative business outcome and downstream notifications are triggered from confirmed state.
- Verify retries, duplicates and provider callbacks cannot create duplicate business effects.
- Verify exception states are visible, recoverable and auditable, without corrupting historical records.

Figure F03-A. Complete use-case diagram - participants, authenticated entry points and business interactions.

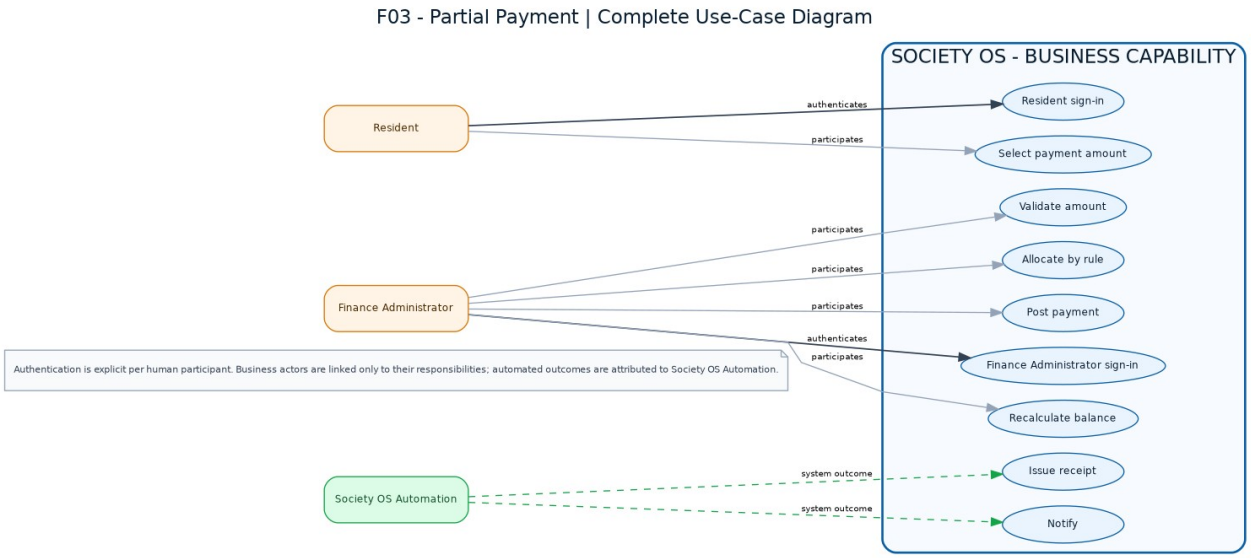
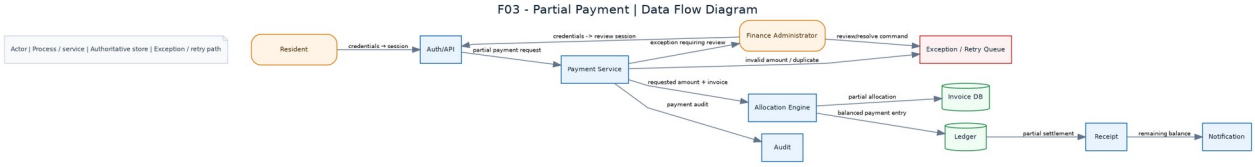


Figure F03-B. Complete data flow diagram - actors, services, authoritative stores, external boundaries and exception path.



F04 - Advance / Credit Payment

Domain: Finance / Payments

Primary participants: Resident, Finance Administrator

Business objective: Store excess payment as controlled resident credit rather than unclassified cash.

Real-world scenario: A resident pays ₹20,000 against ₹15,000 due. ₹15,000 settles current dues and ₹5,000 becomes a traceable credit balance for future allocation or refund.

Preconditions: Payment exceeds current collectible balance; credit policy is configured.

Normal flow: Resident signs in → payment verified → current dues allocated → excess identified → credit account created/updated → ledger records source transaction → resident receives receipt showing settlement and credit.

Exception / alternate flow: Credit expiry, restricted refund, duplicate credit or invalid source must be handled by explicit policy.

State progression: REQUESTED → VERIFIED → DUES_ALLOCATED → CREDIT_CREATED → RECONCILED

Key invariant: Every credit must point to the source payment and may not be created by overwriting the ledger.

QA acceptance intent

- Verify the protected human actor signs in before performing privileged operations such as approval, verification, financial adjustment or NOC issuance.
- Verify the happy path creates exactly one authoritative business outcome and downstream notifications are triggered from confirmed state.
- Verify retries, duplicates and provider callbacks cannot create duplicate business effects.
- Verify exception states are visible, recoverable and auditable, without corrupting historical records.

Figure F04-A. Complete use-case diagram - participants, authenticated entry points and business interactions.

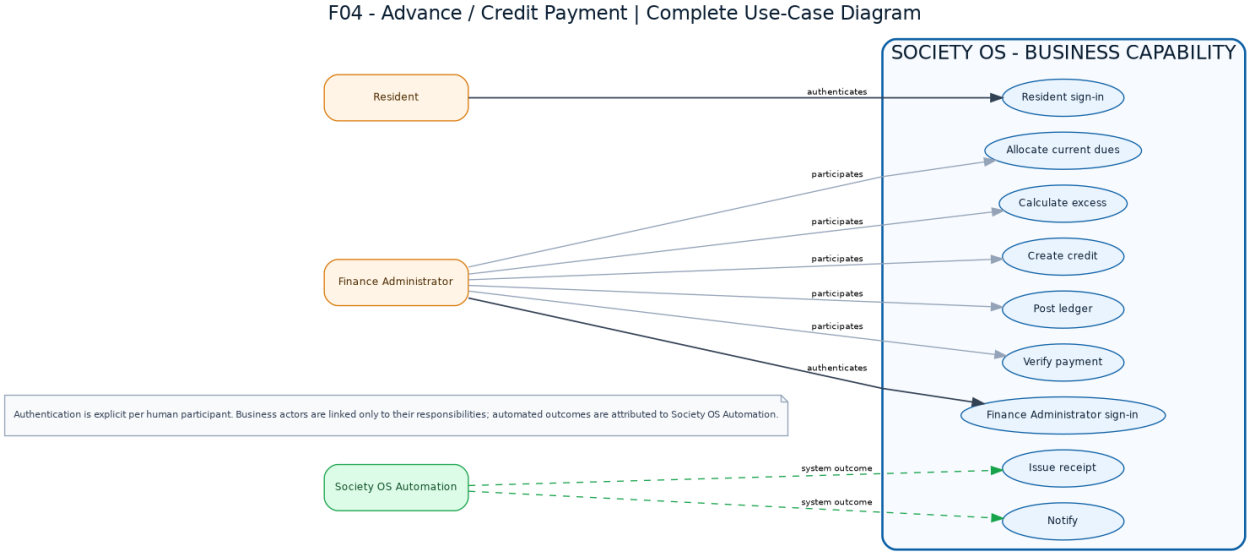
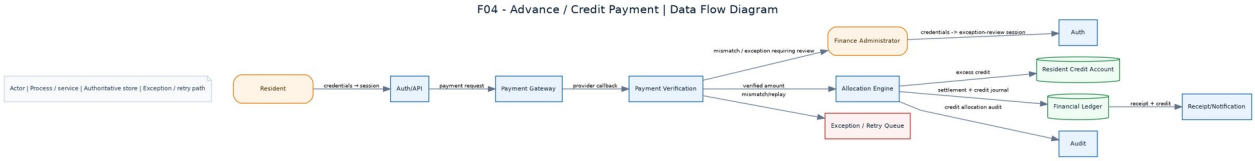


Figure F04-B. Complete data flow diagram - actors, services, authoritative stores, external boundaries and exception path.



F05 - Failed Payment and Retry

Domain: Finance / Payments

Primary participants: Resident, Payment Gateway, Finance Administrator

Business objective: Make payment failure recoverable without creating a false paid status.

Real-world scenario: A UPI attempt fails because the bank declines it. The invoice remains outstanding and the resident can retry or use another method.

Preconditions: Invoice is open; provider is reachable enough to accept the request.

Normal flow: Resident signs in → attempts payment → provider returns failure/timeout → payment attempt is marked failed/pending as appropriate → no ledger posting occurs → retry remains available → successful retry creates a new verified transaction.

Exception / alternate flow: Provider callback arrives after client timeout; an earlier attempt later succeeds; repeated retries must not double-settle.

State progression: INITIATED → PROCESSING → FAILED/PENDING → RETRY or CANCELLED → SUCCESS

Key invariant: Failed client attempts never become ledger postings unless an authoritative verified success is received.

QA acceptance intent

- Verify the protected human actor signs in before performing privileged operations such as approval, verification, financial adjustment or NOC issuance.
- Verify the happy path creates exactly one authoritative business outcome and downstream notifications are triggered from confirmed state.
- Verify retries, duplicates and provider callbacks cannot create duplicate business effects.
- Verify exception states are visible, recoverable and auditable, without corrupting historical records.

Figure F05-A. Complete use-case diagram - participants, authenticated entry points and business interactions.

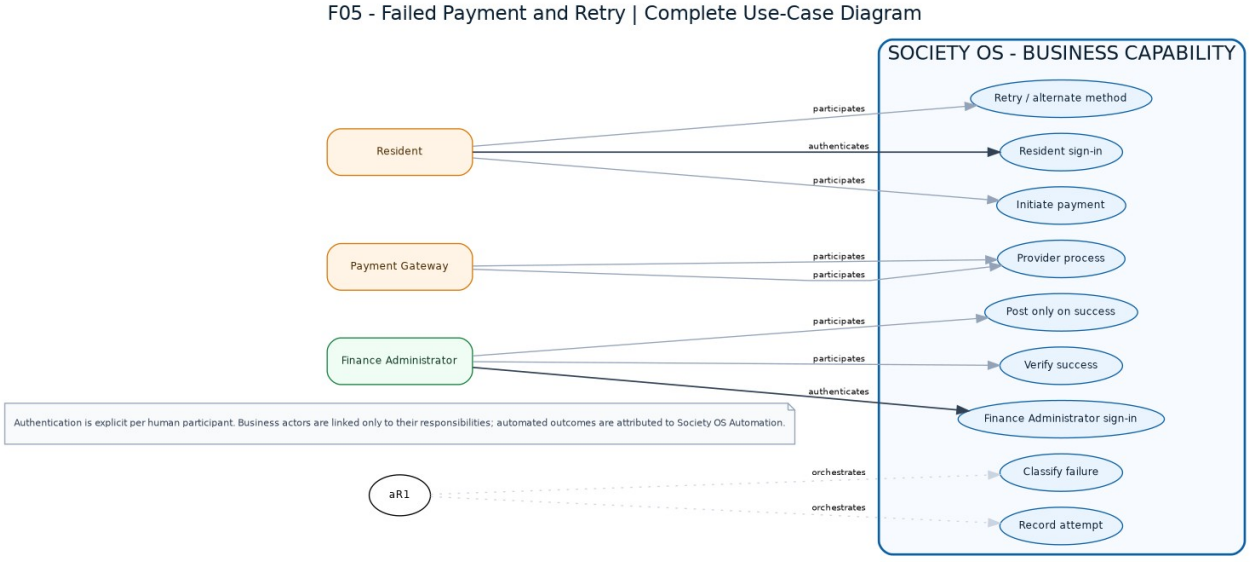
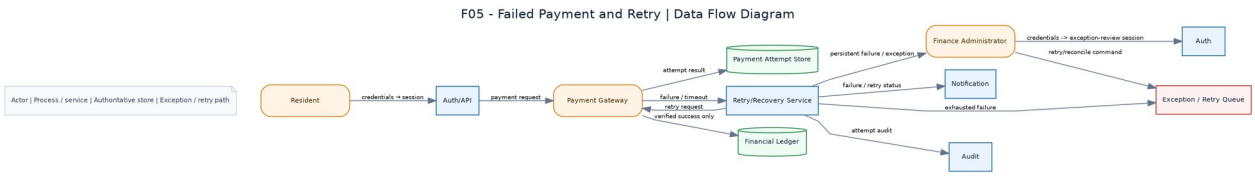


Figure F05-B. Complete data flow diagram - actors, services, authoritative stores, external boundaries and exception path.



F06 - Duplicate Payment Callback

Domain: Finance / Payments

Primary participants: Payment Gateway, Finance Administrator

Business objective: Make provider retries harmless to financial state.

Real-world scenario: The gateway retries the same callback after a network timeout. The second callback returns a safe replay result and creates no second receipt or journal.

Preconditions: An earlier callback with the same provider reference/idempotency key was processed or is in-flight.

Normal flow: Gateway sends callback → authentication/signature validation → idempotency lookup → existing transaction found → return prior outcome → no duplicate allocation/journal/receipt → audit replay.

Exception / alternate flow: Same reference with different amount/signature is treated as suspicious and rejected/escalated rather than merged.

State progression: RECEIVED → VERIFYING → FIRST_POST / DUPLICATE_REPLAY or SUSPICIOUS_REJECT

Key invariant: Provider reference and idempotency key uniqueness is scoped to the society/provider integration.

QA acceptance intent

- Verify the protected human actor signs in before performing privileged operations such as approval, verification, financial adjustment or NOC issuance.
- Verify the happy path creates exactly one authoritative business outcome and downstream notifications are triggered from confirmed state.
- Verify retries, duplicates and provider callbacks cannot create duplicate business effects.
- Verify exception states are visible, recoverable and auditable, without corrupting historical records.

Figure F06-A. Complete use-case diagram - participants, authenticated entry points and business interactions.

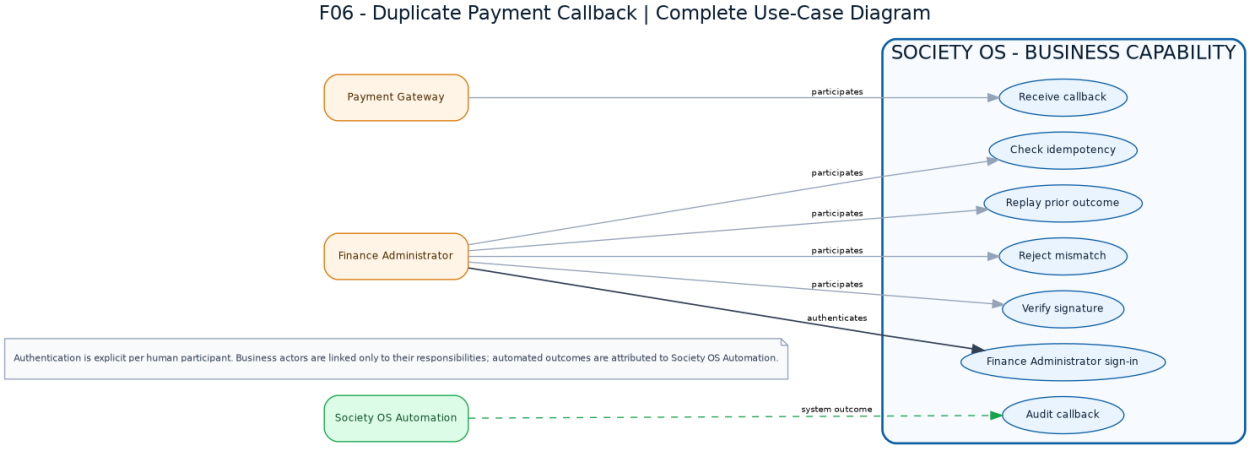
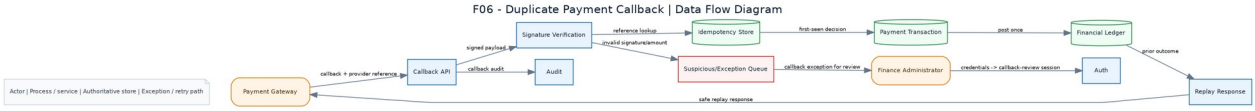


Figure F06-B. Complete data flow diagram - actors, services, authoritative stores, external boundaries and exception path.



F07 - Refund

Domain: Finance / Payments

Primary participants: Resident, Finance Administrator, Payment Gateway

Business objective: Return money without mutating the original posted payment history.

Real-world scenario: A ₹10,000 payment is refunded by ₹2,000 after an approved service cancellation. The original transaction remains intact and the refund is a linked compensating transaction.

Preconditions: Original payment exists; refund policy and available refundable amount are validated.

Normal flow: Resident/admin raises refund → admin signs in → eligibility checked → amount approved → gateway refund initiated → confirmation received → refund transaction posted → linked balance recalculated → receipt/notification issued.

Exception / alternate flow: Provider timeout, partial refund, repeated refund request or refund rejection must remain traceable and retry-safe.

State progression: REQUESTED → APPROVED → PROCESSING → REFUNDED/PARTIALLY_REFUNDED → RECONCILED or EXCEPTION

Key invariant: Refund amount cannot exceed refundable balance and must reference the originating payment.

QA acceptance intent

- Verify the protected human actor signs in before performing privileged operations such as approval, verification, financial adjustment or NOC issuance.
- Verify the happy path creates exactly one authoritative business outcome and downstream notifications are triggered from confirmed state.
- Verify retries, duplicates and provider callbacks cannot create duplicate business effects.
- Verify exception states are visible, recoverable and auditable, without corrupting historical records.

Figure F07-A. Complete use-case diagram - participants, authenticated entry points and business interactions.

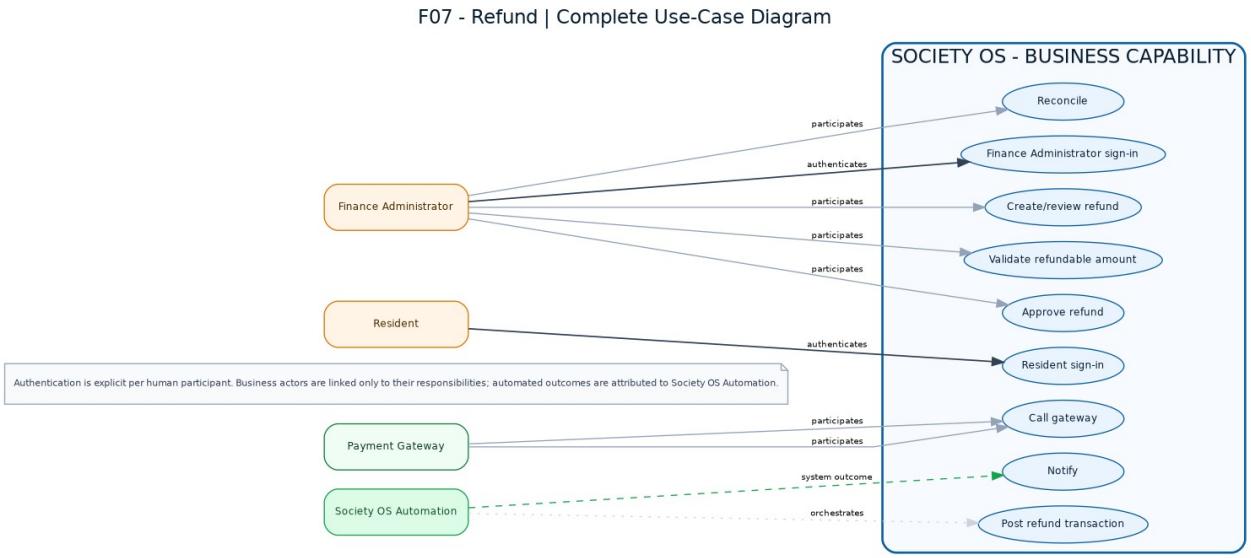
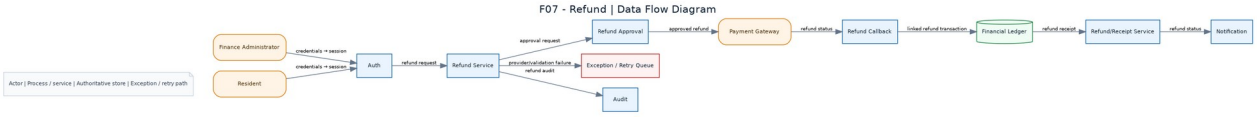


Figure F07-B. Complete data flow diagram - actors, services, authoritative stores, external boundaries and exception path.



F08 - Payment Reversal

Domain: Finance / Accounting

Primary participants: Finance Administrator, Resident

Business objective: Reverse an incorrect posted payment through a controlled compensating transaction.

Real-world scenario: A payment was posted to the wrong unit. Finance discovers the issue and posts a reversal, then the correct payment is allocated separately.

Preconditions: Original transaction is posted and identifiable; reversal reason is mandatory.

Normal flow: Admin signs in → selects transaction → validates reversible status → enters reason/support → approval if configured → balanced reversal journal posted → original remains unchanged → affected balances recalculate → audit captures linkage.

Exception / alternate flow: Already reversed, period closed, partial downstream allocation or missing support document requires an exception workflow.

State progression: POSTED → REVERSAL_REQUESTED → APPROVED → REVERSED

Key invariant: Original journal rows are never edited or deleted; reversal must balance and link to original transaction.

QA acceptance intent

- Verify the protected human actor signs in before performing privileged operations such as approval, verification, financial adjustment or NOC issuance.
- Verify the happy path creates exactly one authoritative business outcome and downstream notifications are triggered from confirmed state.
- Verify retries, duplicates and provider callbacks cannot create duplicate business effects.
- Verify exception states are visible, recoverable and auditable, without corrupting historical records.

Figure F08-A. Complete use-case diagram - participants, authenticated entry points and business interactions.

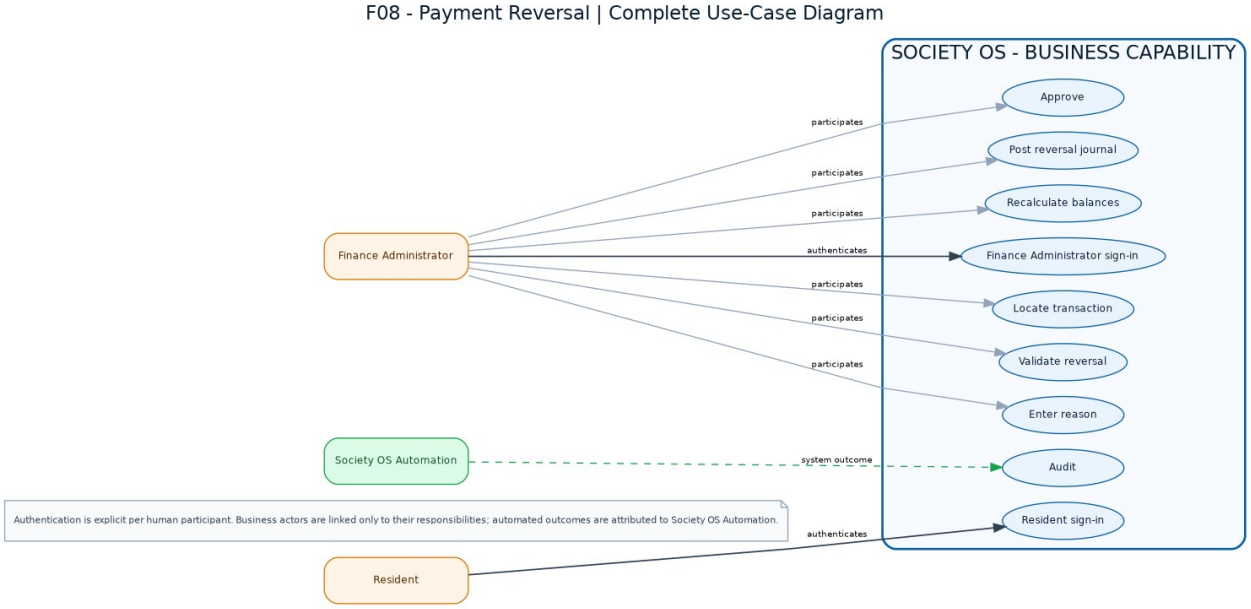
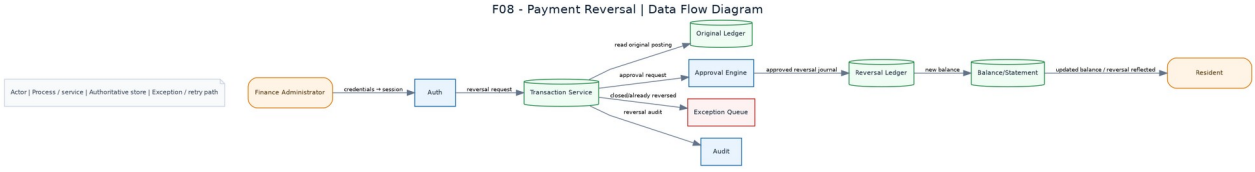


Figure F08-B. Complete data flow diagram - actors, services, authoritative stores, external boundaries and exception path.



F09 - Cheque Bounce

Domain: Finance / Payments

Primary participants: Finance Administrator, Resident, Bank

Business objective: Handle a cheque that was initially recorded as received but later rejected by the bank.

Real-world scenario: A ₹25,000 cheque is recorded and later bounces. The settlement effect is reversed, the outstanding amount returns, and configured bounce charges may be added.

Preconditions: Cheque payment exists in pending/received state with bank reference.

Normal flow: Bank/finance records bounce → admin signs in → system matches cheque → payment status becomes bounced → compensating entry reverses settlement → charges applied if configured → outstanding recalculated → resident notified.

Exception / alternate flow: Unknown cheque, duplicate bounce notice, already-refunded cheque or closed period require controlled exception.

State progression: RECEIVED → PRESENTED → CLEARED or BOUNCED → REVERSED/CHARGED → OUTSTANDING

Key invariant: A bounce must not silently delete the original receipt or create an unbalanced journal.

QA acceptance intent

- Verify the protected human actor signs in before performing privileged operations such as approval, verification, financial adjustment or NOC issuance.
- Verify the happy path creates exactly one authoritative business outcome and downstream notifications are triggered from confirmed state.
- Verify retries, duplicates and provider callbacks cannot create duplicate business effects.
- Verify exception states are visible, recoverable and auditable, without corrupting historical records.

Figure F09-A. Complete use-case diagram - participants, authenticated entry points and business interactions.

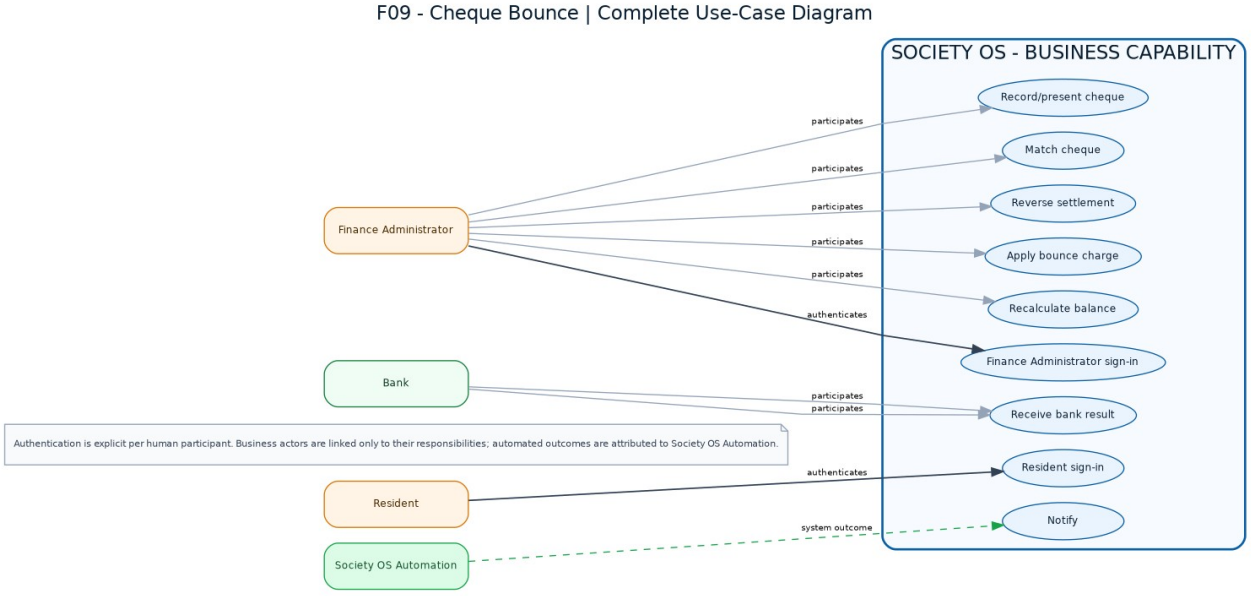
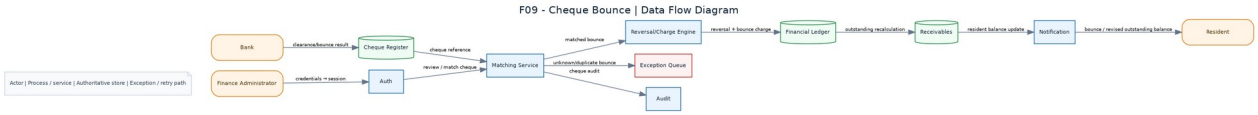


Figure F09-B. Complete data flow diagram - actors, services, authoritative stores, external boundaries and exception path.



F10 - Late Fee / Interest

Domain: Finance / Billing

Primary participants: Billing Administrator, Resident

Business objective: Apply overdue charges using configured effective-dated rules and transparent calculation evidence.

Real-world scenario: An invoice due on 5 October remains unpaid after the configured grace period. Late fee or interest is calculated according to the active policy.

Preconditions: Invoice is overdue; grace period and calculation basis are configured.

Normal flow: Scheduler identifies overdue invoices → rule engine evaluates grace period → calculates late fee/interest → validates cap/rounding → posts charge → notifies resident → ageing updates.

Exception / alternate flow: Holiday/cut-off boundary, partial payment, waiver, rate change or calculation correction must use the applicable effective-date rules.

State progression: DUE → GRACE → OVERDUE → CHARGE_CALCULATED → CHARGE_POSTED

Key invariant: Historical invoices must use the rule version effective for the relevant period; charges must be separately traceable.

QA acceptance intent

- Verify the protected human actor signs in before performing privileged operations such as approval, verification, financial adjustment or NOC issuance.
- Verify the happy path creates exactly one authoritative business outcome and downstream notifications are triggered from confirmed state.
- Verify retries, duplicates and provider callbacks cannot create duplicate business effects.
- Verify exception states are visible, recoverable and auditable, without corrupting historical records.

Figure F10-A. Complete use-case diagram - participants, authenticated entry points and business interactions.

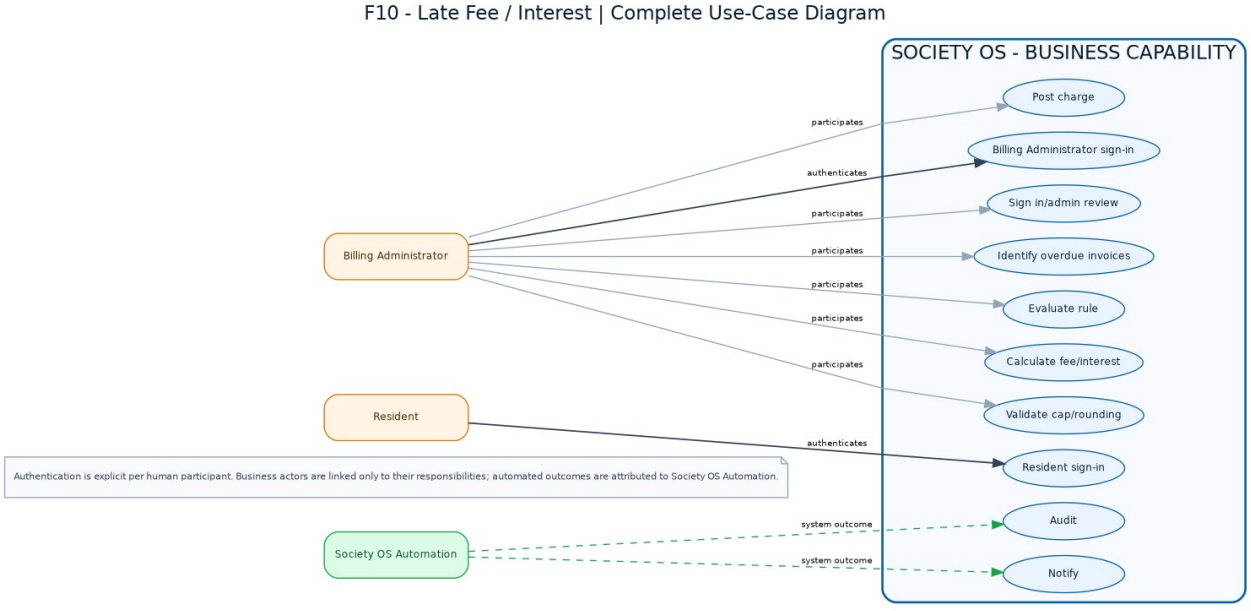
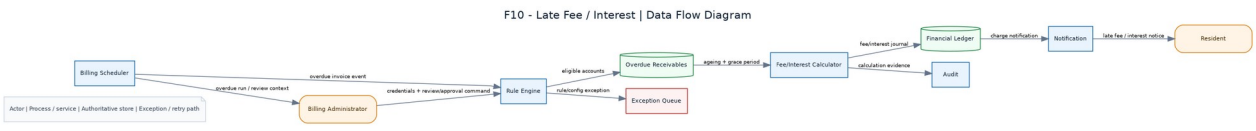


Figure F10-B. Complete data flow diagram - actors, services, authoritative stores, external boundaries and exception path.



F11 - Waiver / Discount

Domain: Finance / Adjustments

Primary participants: Resident, Finance Administrator, Committee Approver

Business objective: Reduce an eligible charge through an explicit, auditable waiver or discount process.

Real-world scenario: A resident requests waiver of a late fee due to a documented exceptional event. The approver reviews the request and the system posts the approved reduction.

Preconditions: Charge exists; waiver/discount policy and approval thresholds are configured.

Normal flow: Resident/admin requests → admin signs in → eligibility evaluated → approval chain runs → approved amount posted as adjustment → original charge retained → new balance/statement generated → notification.

Exception / alternate flow: Exceeds threshold, duplicate request, unsupported reason or revoked approval must not alter the ledger.

State progression: REQUESTED → UNDER_REVIEW → APPROVED/REJECTED → POSTED → RECONCILED

Key invariant: Waiver is not a silent edit; it is a separate auditable financial entry linked to the original charge.

QA acceptance intent

- Verify the protected human actor signs in before performing privileged operations such as approval, verification, financial adjustment or NOC issuance.
- Verify the happy path creates exactly one authoritative business outcome and downstream notifications are triggered from confirmed state.
- Verify retries, duplicates and provider callbacks cannot create duplicate business effects.
- Verify exception states are visible, recoverable and auditable, without corrupting historical records.

Figure F11-A. Complete use-case diagram - participants, authenticated entry points and business interactions.

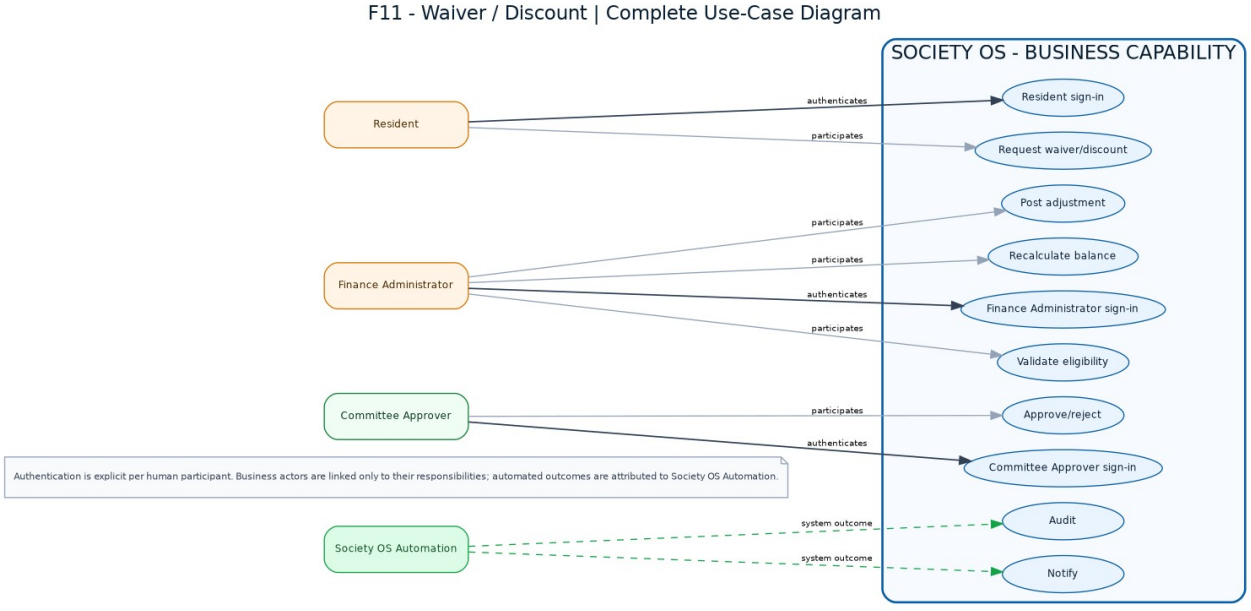
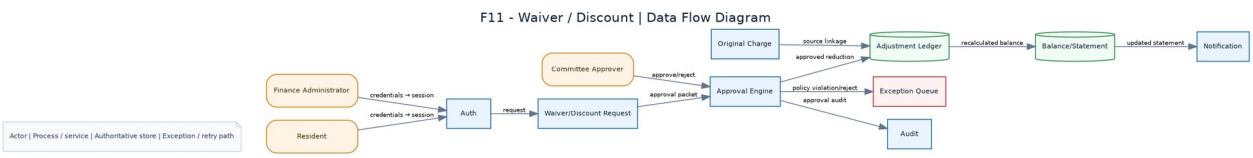


Figure F11-B. Complete data flow diagram - actors, services, authoritative stores, external boundaries and exception path.



F12 - Bank Reconciliation

Domain: Finance / Accounting

Primary participants: Finance Administrator, Bank

Business objective: Match bank statement entries against internal transactions and expose unresolved exceptions.

Real-world scenario: The society imports its bank statement. 97% of entries auto-match by amount/reference/date; unmatched entries enter a review queue.

Preconditions: Statement format configured; bank account and ledger mappings exist.

Normal flow: Admin signs in → imports statement → parser validates rows → matcher compares references/amounts → exact matches reconcile → unmatched/ambiguous rows queue for review → admin resolves → reconciliation closes.

Exception / alternate flow: Duplicate statement row, amount mismatch, missing bank reference, reversed payment or timing difference requires exception handling.

State progression: IMPORTED → PARSED → MATCHED/PENDING_REVIEW → RECONCILED or EXCEPTION

Key invariant: Reconciliation marks a relationship between bank evidence and internal transactions; it must not overwrite the underlying ledger.

QA acceptance intent

- Verify the protected human actor signs in before performing privileged operations such as approval, verification, financial adjustment or NOC issuance.
- Verify the happy path creates exactly one authoritative business outcome and downstream notifications are triggered from confirmed state.
- Verify retries, duplicates and provider callbacks cannot create duplicate business effects.
- Verify exception states are visible, recoverable and auditable, without corrupting historical records.

Figure F12-A. Complete use-case diagram - participants, authenticated entry points and business interactions.

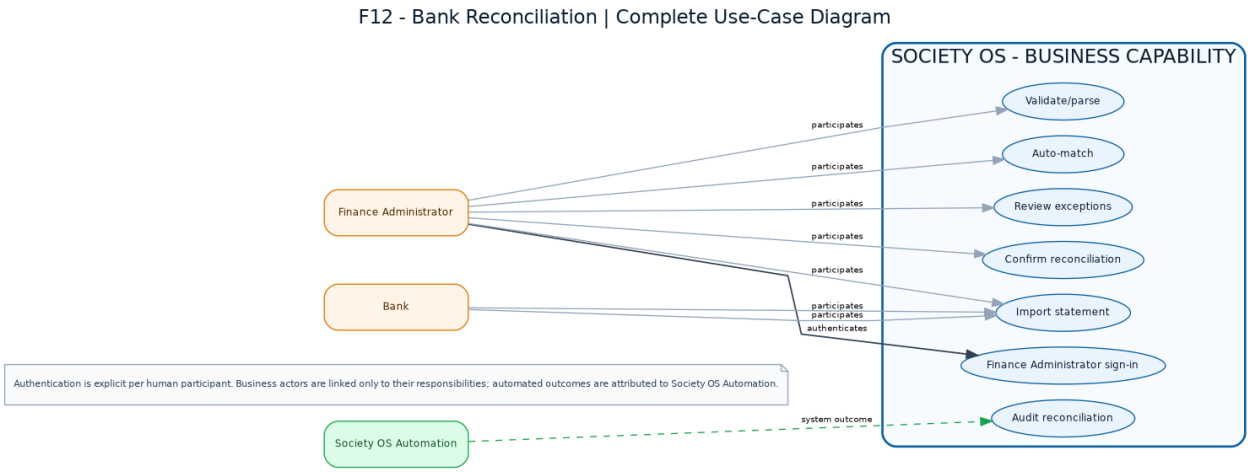
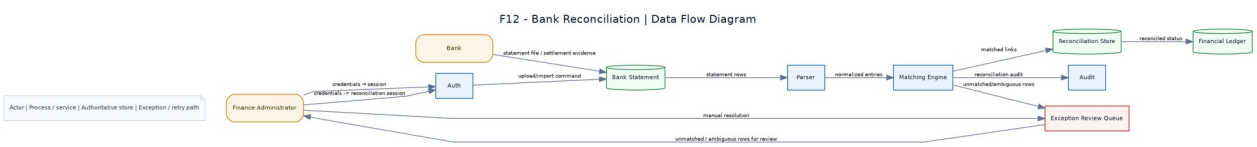


Figure F12-B. Complete data flow diagram - actors, services, authoritative stores, external boundaries and exception path.



F13 - Move-Out Final Settlement

Domain: Finance / Move-Out

Primary participants: Resident, Finance Administrator, Society Administrator

Business objective: Compute the final amount payable/receivable at move-out using all authoritative financial obligations.

Real-world scenario: A resident requests move-out while ₹18,000 is outstanding. Final maintenance, parking, utilities, approved damages and credits are consolidated before clearance.

Preconditions: Move-out request exists; occupancy dates and financial records are available.

Normal flow: Resident requests move-out → admin signs in → system freezes relevant period → calculates outstanding + pending charges - credits/deposits → finance reviews → payment/refund/settlement occurs → financial clearance recorded → downstream NOC can proceed.

Exception / alternate flow: Pending invoice, disputed damage, missing meter reading, unapplied credit or refund due creates a clearance exception.

State progression: REQUESTED → CALCULATING → REVIEW → SETTLEMENT_PENDING → CLEARED / EXCEPTION

Key invariant: Final settlement must be reproducible from source transactions and effective-dated rules.

QA acceptance intent

- Verify the protected human actor signs in before performing privileged operations such as approval, verification, financial adjustment or NOC issuance.
- Verify the happy path creates exactly one authoritative business outcome and downstream notifications are triggered from confirmed state.
- Verify retries, duplicates and provider callbacks cannot create duplicate business effects.
- Verify exception states are visible, recoverable and auditable, without corrupting historical records.

Figure F13-A. Complete use-case diagram - participants, authenticated entry points and business interactions.

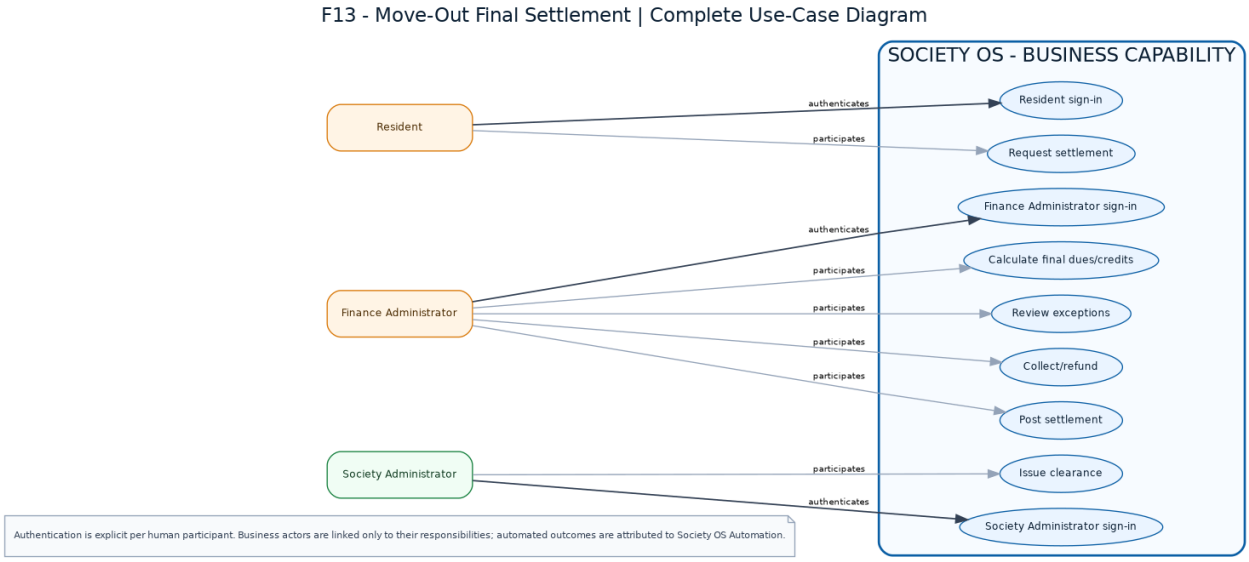
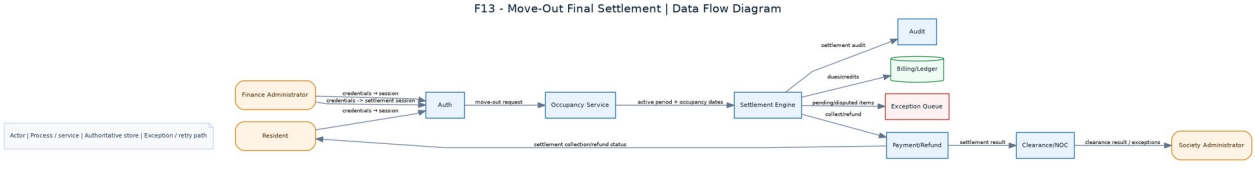


Figure F13-B. Complete data flow diagram - actors, services, authoritative stores, external boundaries and exception path.



F14 - Vendor Invoice / Three-Way Match

Domain: Finance / Procurement

Primary participants: Finance Administrator, Procurement Administrator, Vendor

Business objective: Pay vendors only when invoice, purchase order and goods/service receipt agree under configured tolerance.

Real-world scenario: A plumbing vendor invoices 100 valves against a PO for 100, but only 90 were received. The system blocks full payment and exposes the mismatch.

Preconditions: PO and goods receipt exist; vendor is active; invoice uploaded.

Normal flow: Admin signs in → invoice captured → OCR/manual validation → PO matched → receipt matched → tolerance evaluated → approve/reject/hold → approved liability posted → payment workflow continues.

Exception / alternate flow: Quantity mismatch, price variance, duplicate invoice, inactive vendor or missing receipt leads to hold and exception resolution.

State progression: RECEIVED → VALIDATING → MATCHED / EXCEPTION → APPROVED → PAYABLE → PAID

Key invariant: A payable cannot be approved merely because an invoice exists; required matching evidence must satisfy policy.

QA acceptance intent

- Verify the protected human actor signs in before performing privileged operations such as approval, verification, financial adjustment or NOC issuance.
- Verify the happy path creates exactly one authoritative business outcome and downstream notifications are triggered from confirmed state.
- Verify retries, duplicates and provider callbacks cannot create duplicate business effects.
- Verify exception states are visible, recoverable and auditable, without corrupting historical records.

Figure F14-A. Complete use-case diagram - participants, authenticated entry points and business interactions.

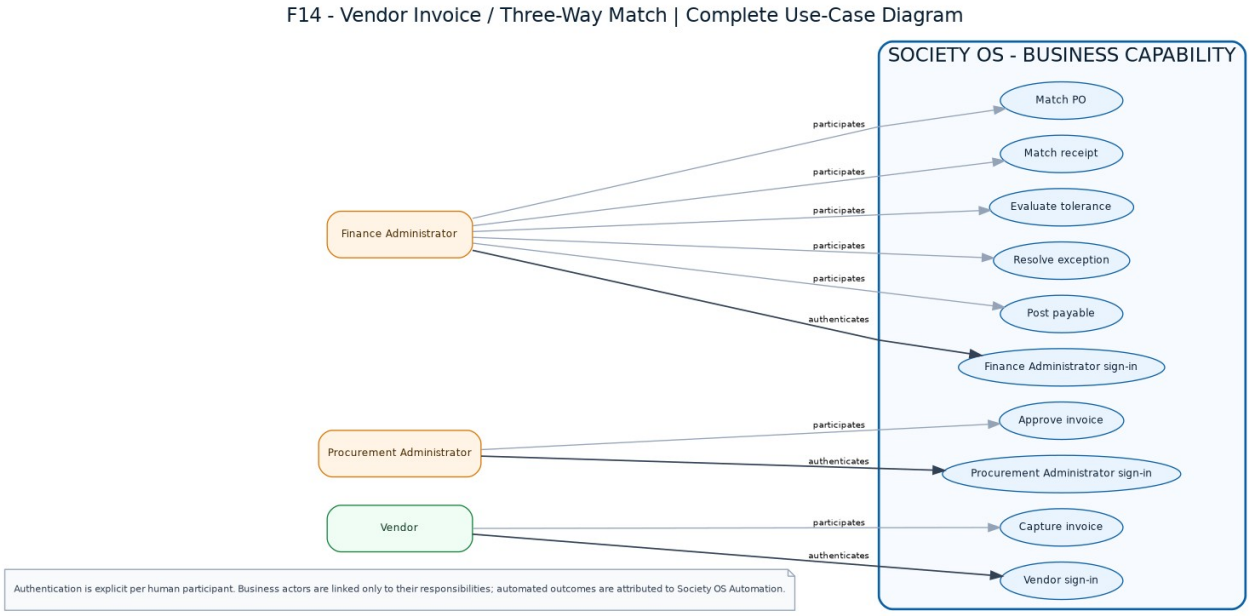
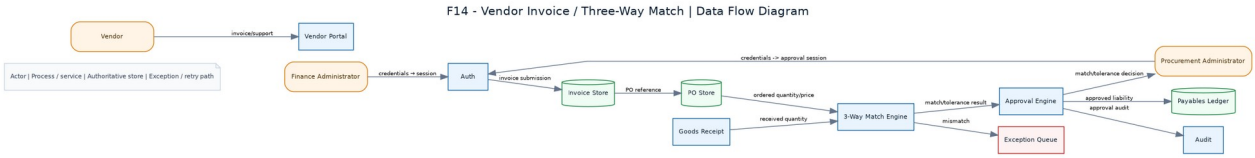


Figure F14-B. Complete data flow diagram - actors, services, authoritative stores, external boundaries and exception path.



F15 - Budget vs Actual

Domain: Finance / Planning

Primary participants: Finance Administrator, Committee / Management

Business objective: Compare approved budget against actual committed and paid amounts with drill-down traceability.

Real-world scenario: Management sees that lift AMC spending is 18% above budget. The report traces variance to specific vendor invoices and maintenance work orders.

Preconditions: Budget period and accounts are defined; actuals are posted.

Normal flow: Admin signs in → selects period/account → system aggregates approved budget, commitments and actuals → variance calculated → user drills from summary to source transactions → notes/actions recorded if needed.

Exception / alternate flow: Late postings, reversed invoices, budget revision or closed period must remain transparent.

State progression: BUDGET_DRAFT → APPROVED → TRACKING → VARIANCE_REVIEW → CLOSED

Key invariant: Every reported actual must be traceable to source financial transactions; budget revisions must be versioned.

QA acceptance intent

- Verify the protected human actor signs in before performing privileged operations such as approval, verification, financial adjustment or NOC issuance.
- Verify the happy path creates exactly one authoritative business outcome and downstream notifications are triggered from confirmed state.
- Verify retries, duplicates and provider callbacks cannot create duplicate business effects.
- Verify exception states are visible, recoverable and auditable, without corrupting historical records.

Figure F15-A. Complete use-case diagram - participants, authenticated entry points and business interactions.

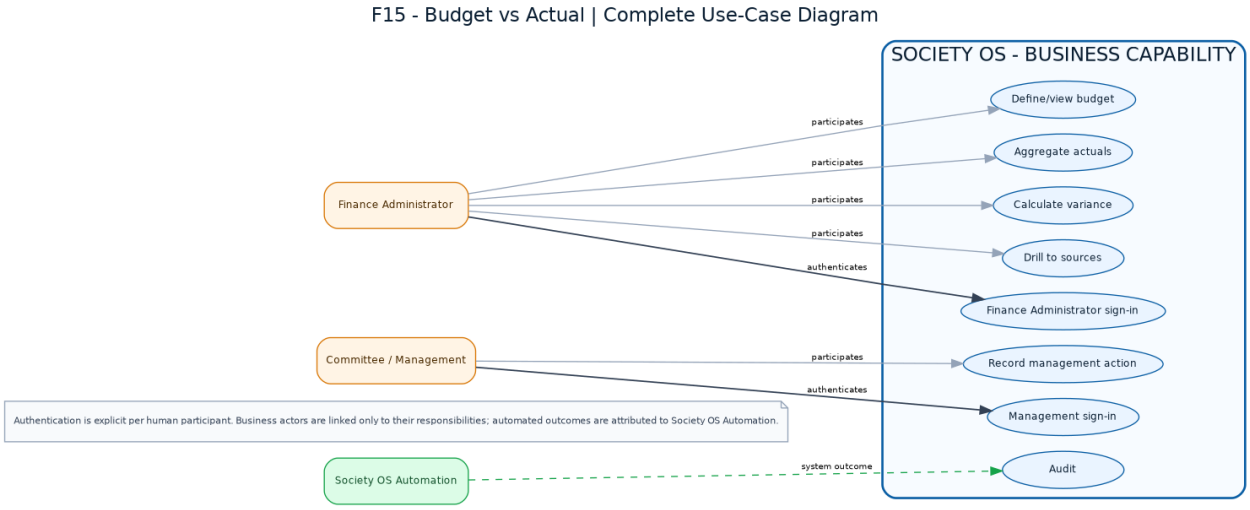
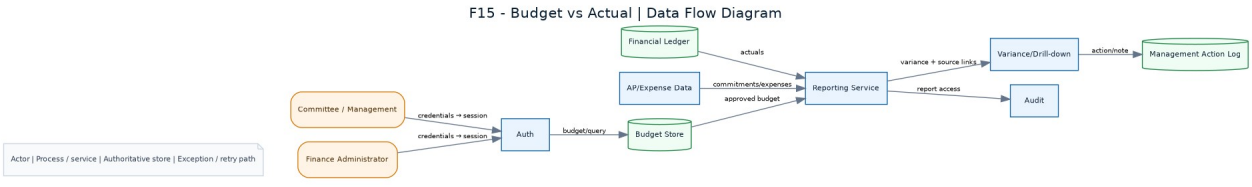


Figure F15-B. Complete data flow diagram - actors, services, authoritative stores, external boundaries and exception path.



F16 - Expense Reimbursement

Domain: Finance / Expenses

Primary participants: Employee / Staff, Finance Administrator, Approver

Business objective: Reimburse legitimate employee expenses through an approval and accounting workflow.

Real-world scenario: A site engineer uploads a ₹3,200 emergency plumbing purchase receipt. Finance validates the expense, approver approves it, and the reimbursement becomes payable.

Preconditions: Employee account active; expense policy configured.

Normal flow: Staff signs in → creates claim → attaches evidence → policy checks → admin/approver signs in → approves/rejects → payable posted → payment initiated → reimbursement reconciled.

Exception / alternate flow: Missing receipt, policy limit exceeded, duplicate claim or rejected payment must stop posting/payment at the correct stage.

State progression: DRAFT → SUBMITTED → REVIEW → APPROVED/REJECTED → PAYABLE → PAID/RECONCILED

Key invariant: The expense evidence and approval decision remain linked to the resulting payable.

QA acceptance intent

- Verify the protected human actor signs in before performing privileged operations such as approval, verification, financial adjustment or NOC issuance.
- Verify the happy path creates exactly one authoritative business outcome and downstream notifications are triggered from confirmed state.
- Verify retries, duplicates and provider callbacks cannot create duplicate business effects.
- Verify exception states are visible, recoverable and auditable, without corrupting historical records.

Figure F16-A. Complete use-case diagram - participants, authenticated entry points and business interactions.

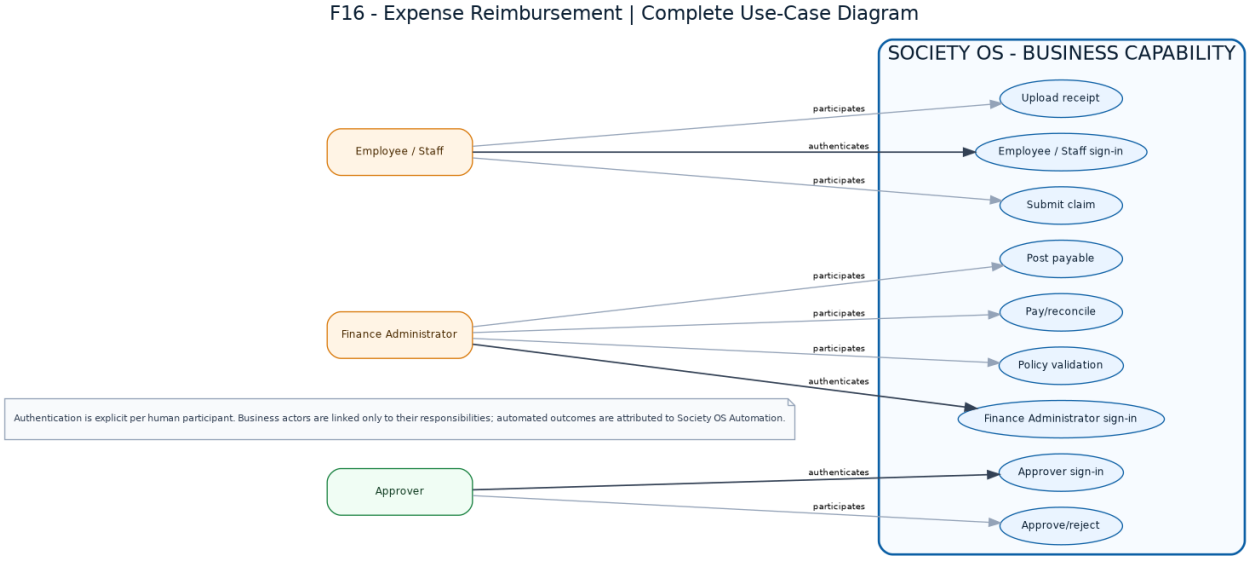
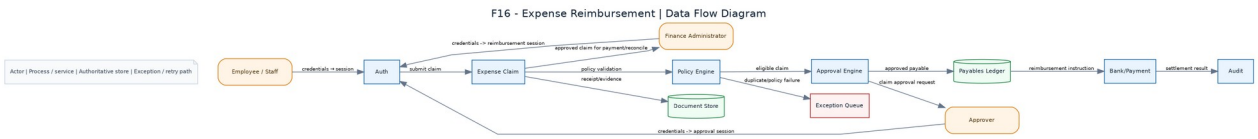


Figure F16-B. Complete data flow diagram - actors, services, authoritative stores, external boundaries and exception path.



F17 - Outstanding / Ageing / Collections

Domain: Finance / Collections

Primary participants: Finance Administrator, Resident, Notification Service

Business objective: Turn overdue receivables into a controlled collection workflow without losing financial traceability.

Real-world scenario: The society reviews 0-30, 31-60 and 60+ day ageing buckets, sends reminders, records commitments and escalates long-overdue units.

Preconditions: Invoices and payments are current; ageing rules configured.

Normal flow: Admin signs in → ageing snapshot generated → residents grouped by bucket → notices sent → commitments/disputes recorded → payment updates ageing → unresolved accounts escalate according to policy.

Exception / alternate flow: Disputed charge, legal hold, incorrect allocation or payment pending settlement must be excluded or flagged rather than incorrectly escalated.

State progression: CURRENT → DUE → OVERDUE → REMINDED → ESCALATED → RESOLVED / DISPUTED

Key invariant: Ageing is derived from authoritative receivables, not manually maintained balances.

QA acceptance intent

- Verify the protected human actor signs in before performing privileged operations such as approval, verification, financial adjustment or NOC issuance.
- Verify the happy path creates exactly one authoritative business outcome and downstream notifications are triggered from confirmed state.
- Verify retries, duplicates and provider callbacks cannot create duplicate business effects.
- Verify exception states are visible, recoverable and auditable, without corrupting historical records.

Figure F17-A. Complete use-case diagram - participants, authenticated entry points and business interactions.

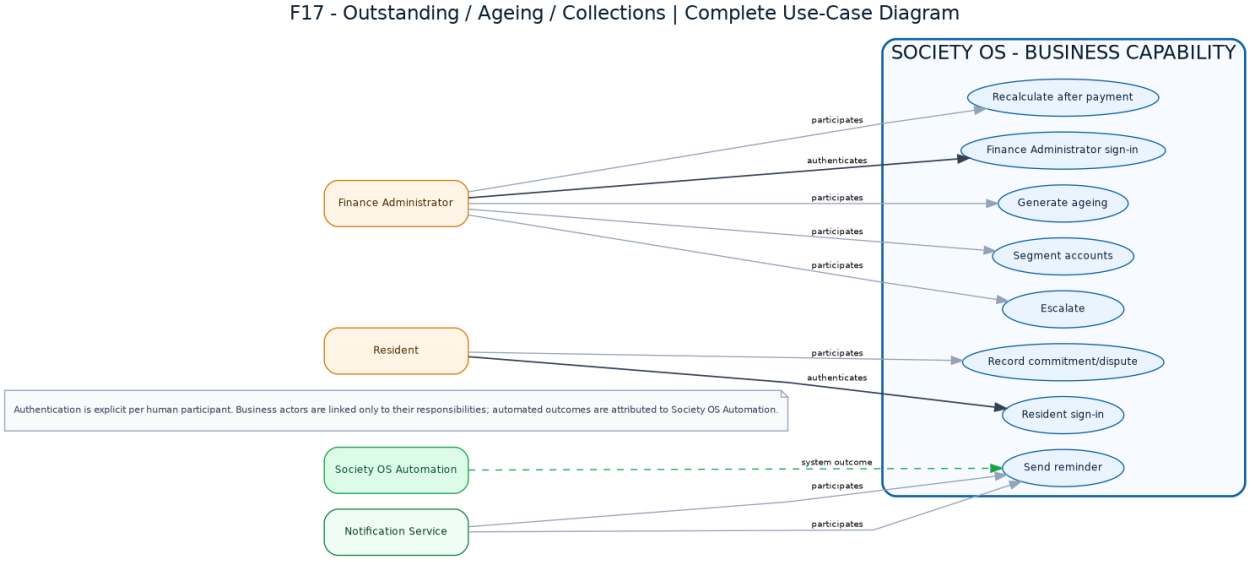
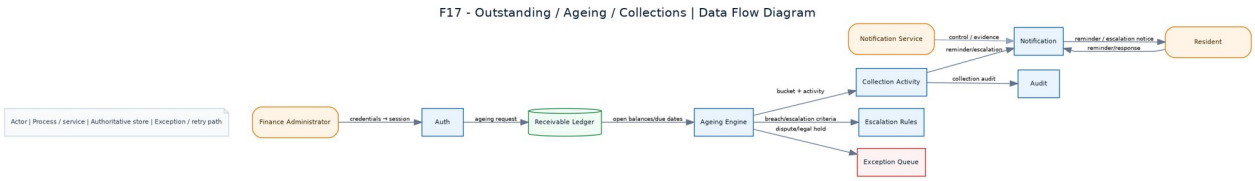


Figure F17-B. Complete data flow diagram - actors, services, authoritative stores, external boundaries and exception path.



F18 - Credit Note / Debit Adjustment

Domain: Finance / Adjustments

Primary participants: Finance Administrator, Approver

Business objective: Correct a bill or account through a linked adjustment transaction rather than rewriting history.

Real-world scenario: A resident was charged ₹1,500 for a service that was cancelled. Finance issues a ₹1,500 credit note against the original charge.

Preconditions: Original invoice/charge exists and adjustment policy is active.

Normal flow: Admin signs in → selects source charge → chooses credit/debit reason → approval runs if required → adjustment posted → balance recalculated → revised statement references original charge and adjustment.

Exception / alternate flow: Source not found, amount exceeds permitted correction, duplicate adjustment or closed period creates exception.

State progression: REQUESTED → REVIEW → APPROVED → POSTED → RECONCILED

Key invariant: Every adjustment carries source linkage, reason and approval evidence as required.

QA acceptance intent

- Verify the protected human actor signs in before performing privileged operations such as approval, verification, financial adjustment or NOC issuance.
- Verify the happy path creates exactly one authoritative business outcome and downstream notifications are triggered from confirmed state.
- Verify retries, duplicates and provider callbacks cannot create duplicate business effects.
- Verify exception states are visible, recoverable and auditable, without corrupting historical records.

Figure F18-A. Complete use-case diagram - participants, authenticated entry points and business interactions.

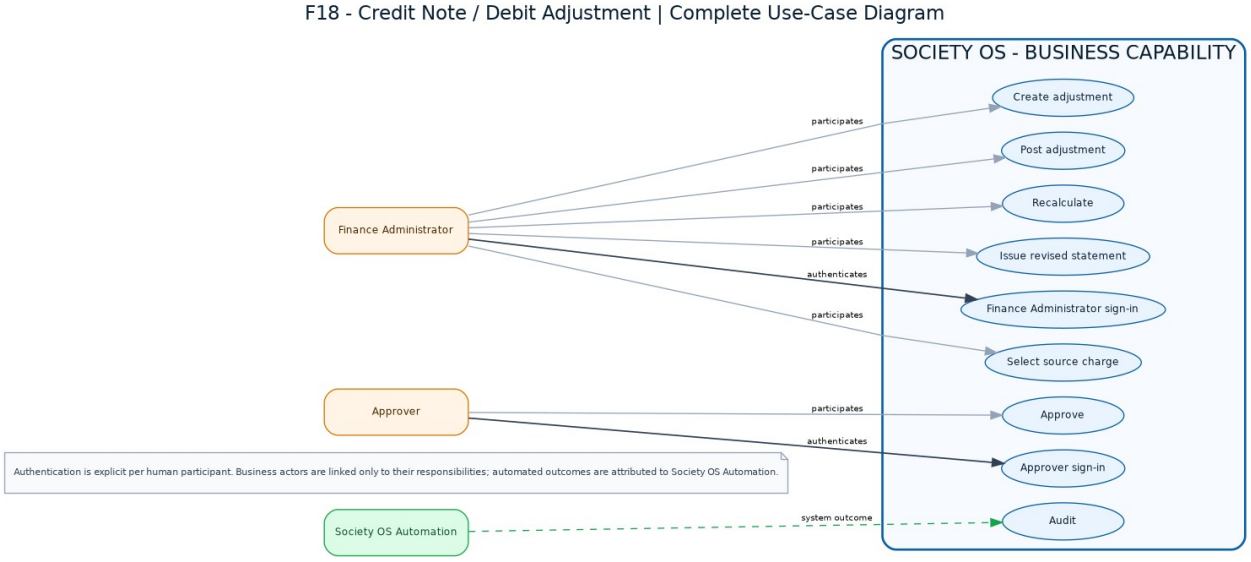
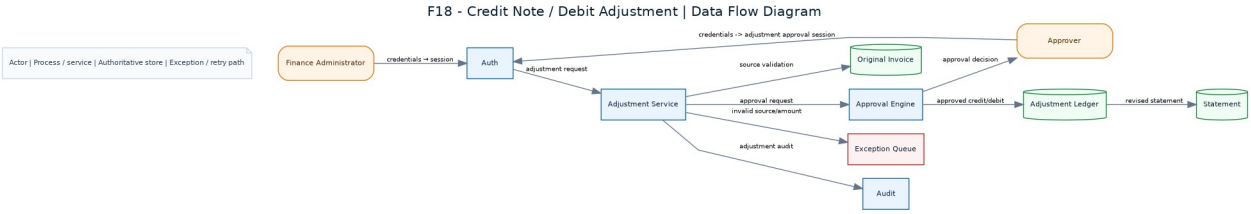


Figure F18-B. Complete data flow diagram - actors, services, authoritative stores, external boundaries and exception path.



F19 - Month-End Close

Domain: Finance / Closing

Primary participants: Finance Administrator, Society Management

Business objective: Close a financial period only after defined validation and reconciliation checks pass.

Real-world scenario: At month end, finance verifies bank reconciliation, open payment callbacks, unposted expenses, suspense balances and draft adjustments before closing September.

Preconditions: Period exists and close checklist is configured.

Normal flow: Admin signs in → close checklist generated → exceptions reviewed → reconciliation checks run → pending jobs verified → period locked → closing snapshot recorded → reports generated.

Exception / alternate flow: Open settlement, unresolved bank item, failed batch job or attempted post-close transaction must block or create controlled approval.

State progression: OPEN → CHECKING → EXCEPTIONS_REVIEW → READY_TO_CLOSE → CLOSED

Key invariant: Post-close transactions follow explicit reopening/adjustment policy and never silently rewrite closed data.

QA acceptance intent

- Verify the protected human actor signs in before performing privileged operations such as approval, verification, financial adjustment or NOC issuance.
- Verify the happy path creates exactly one authoritative business outcome and downstream notifications are triggered from confirmed state.
- Verify retries, duplicates and provider callbacks cannot create duplicate business effects.
- Verify exception states are visible, recoverable and auditable, without corrupting historical records.

Figure F19-A. Complete use-case diagram - participants, authenticated entry points and business interactions.

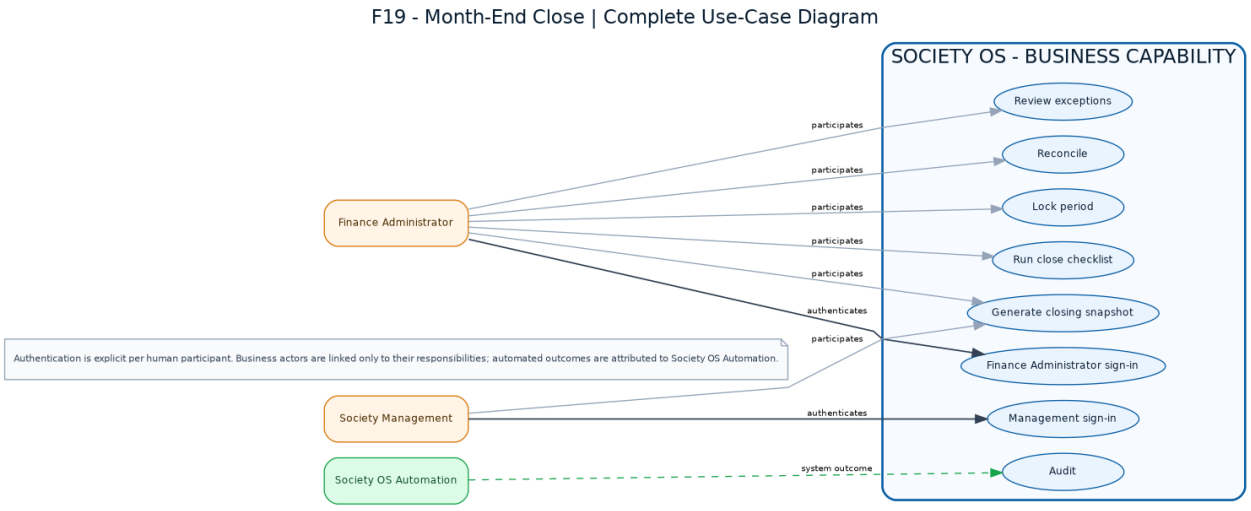
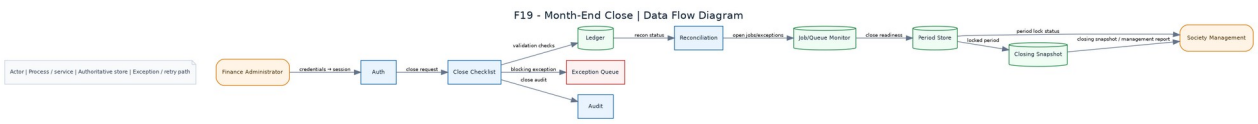


Figure F19-B. Complete data flow diagram - actors, services, authoritative stores, external boundaries and exception path.



F20 - Financial Data Import / Opening Balances

Domain: Finance / Migration

Primary participants: Migration Administrator, Finance Administrator

Business objective: Load legacy financial balances and validate them before becoming opening balances.

Real-world scenario: A new society imports 5,000 resident records and opening balances. 4,912 pass validation and 88 rows require correction before commit.

Preconditions: Import template and mapping are configured; source file available.

Normal flow: Admin signs in → upload → schema validation → duplicate check → account/unit mapping → debit/credit balance validation → preview → confirm → transactional import → exception report → opening balances posted.

Exception / alternate flow: Unmapped unit, duplicate account, unbalanced opening trial, invalid amount or partial import failure requires rollback/compensation.

State progression: UPLOADED → VALIDATED → PREVIEWED → COMMITTING → COMMITTED / FAILED_WITH_ERRORS

Key invariant: Opening balances must balance and remain traceable to the import batch and source row.

QA acceptance intent

- Verify the protected human actor signs in before performing privileged operations such as approval, verification, financial adjustment or NOC issuance.
- Verify the happy path creates exactly one authoritative business outcome and downstream notifications are triggered from confirmed state.
- Verify retries, duplicates and provider callbacks cannot create duplicate business effects.
- Verify exception states are visible, recoverable and auditable, without corrupting historical records.

Figure F20-A. Complete use-case diagram - participants, authenticated entry points and business interactions.

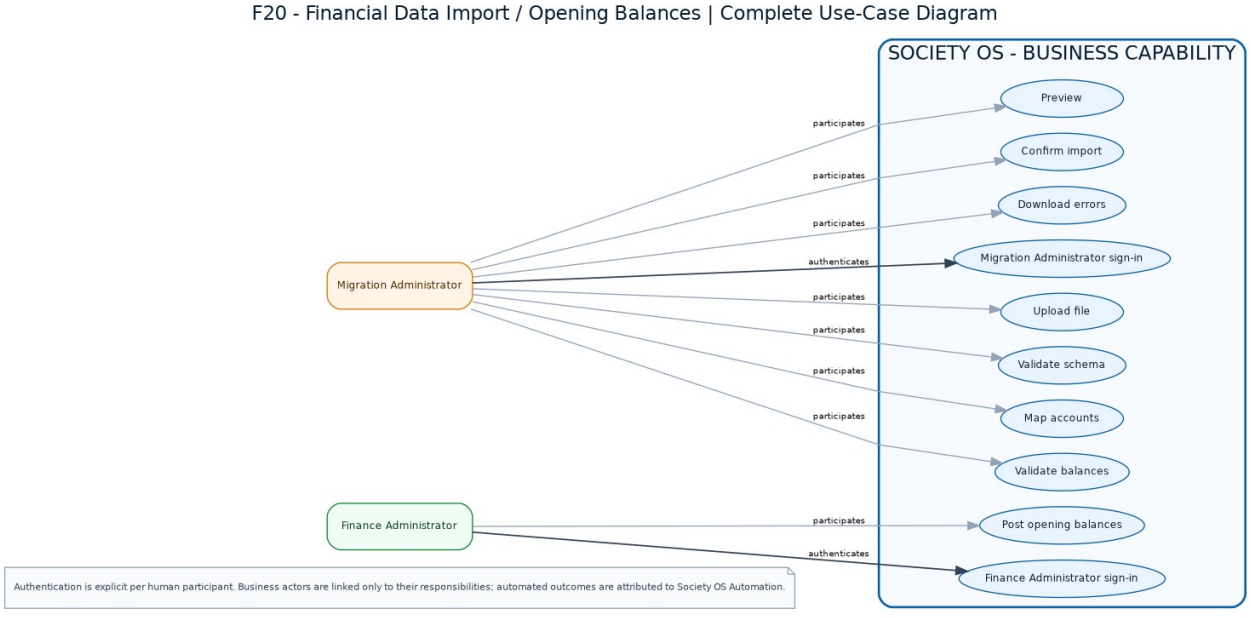
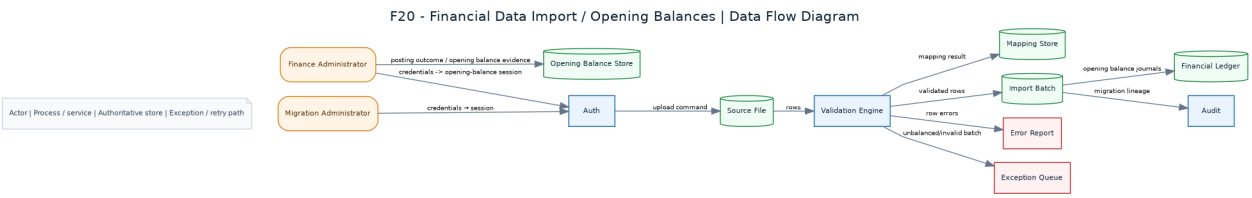


Figure F20-B. Complete data flow diagram - actors, services, authoritative stores, external boundaries and exception path.



O01 - Society Onboarding

Domain: Platform / Onboarding

Primary participants: Society Administrator, Platform Administrator

Business objective: Create a fully configured society tenant from legal identity through activation.

Real-world scenario: A six-tower Maharashtra society is created with registration details, IST, FY rules, units, parking, amenities, charge heads and notification defaults before activation.

Preconditions: Authorized onboarding user exists.

Normal flow: Admin signs in → creates society → captures registration/address → selects rule pack/FY/timezone → builds property hierarchy → imports units/parking → configures charge heads and policies → validates readiness → activates tenant.

Exception / alternate flow: Duplicate slug, incomplete property tree, conflicting unit numbers or missing mandatory configuration blocks activation.

State progression: DRAFT → CONFIGURING → VALIDATING → READY → ACTIVE

Key invariant: Tenant cannot be active until mandatory structural and configuration validations pass.

QA acceptance intent

- Verify the protected human actor signs in before performing privileged operations such as approval, verification, financial adjustment or NOC issuance.
- Verify the happy path creates exactly one authoritative business outcome and downstream notifications are triggered from confirmed state.
- Verify retries, duplicates and provider callbacks cannot create duplicate business effects.
- Verify exception states are visible, recoverable and auditable, without corrupting historical records.

Figure O01-A. Complete use-case diagram - participants, authenticated entry points and business interactions.

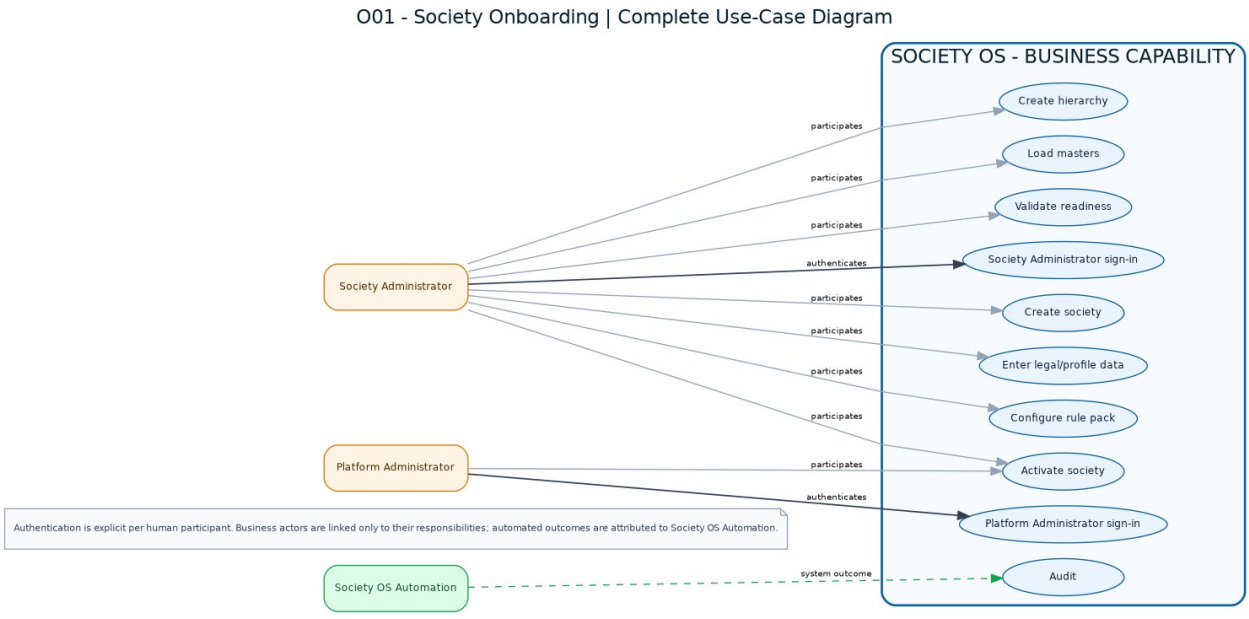
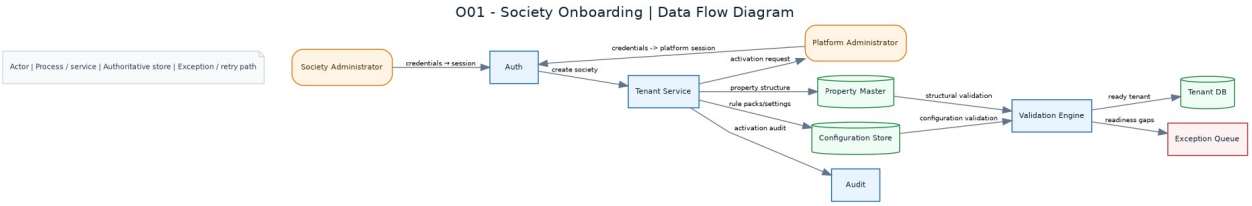


Figure O01-B. Complete data flow diagram - actors, services, authoritative stores, external boundaries and exception path.



O02 - Resident Registration and Verification

Domain: Identity / Resident

Primary participants: Resident, Society Administrator, Identity/Email Provider

Business objective: Register a resident, verify identity and obtain society approval where configured.

Real-world scenario: A tenant receives an invitation, verifies mobile by OTP, uploads KYC/agreement, and waits for the admin to approve. Admin must sign in before verification/approval.

Preconditions: Invitation or open-registration policy is enabled; unit exists.

Normal flow: Resident signs in/registers → OTP/email verified → profile completed → unit/relationship selected → documents submitted → admin signs in → admin reviews identity/documents → approve/reject/resubmit → optional admin digital approval/signature → account activates.

Exception / alternate flow: Duplicate phone, wrong unit, missing document, rejected KYC or expired invitation requires resubmission without creating duplicate occupancy.

State progression: INVITED → REGISTERED → IDENTITY_VERIFIED → DOCUMENTS_SUBMITTED → ADMIN_REVIEW → APPROVED/RESUBMIT/REJECTED → ACTIVE

Key invariant: Registration identity state is separate from occupancy; admin approval must be authenticated and auditable.

QA acceptance intent

- Verify the protected human actor signs in before performing privileged operations such as approval, verification, financial adjustment or NOC issuance.
- Verify the happy path creates exactly one authoritative business outcome and downstream notifications are triggered from confirmed state.
- Verify retries, duplicates and provider callbacks cannot create duplicate business effects.
- Verify exception states are visible, recoverable and auditable, without corrupting historical records.

Figure O02-A. Complete use-case diagram - participants, authenticated entry points and business interactions.

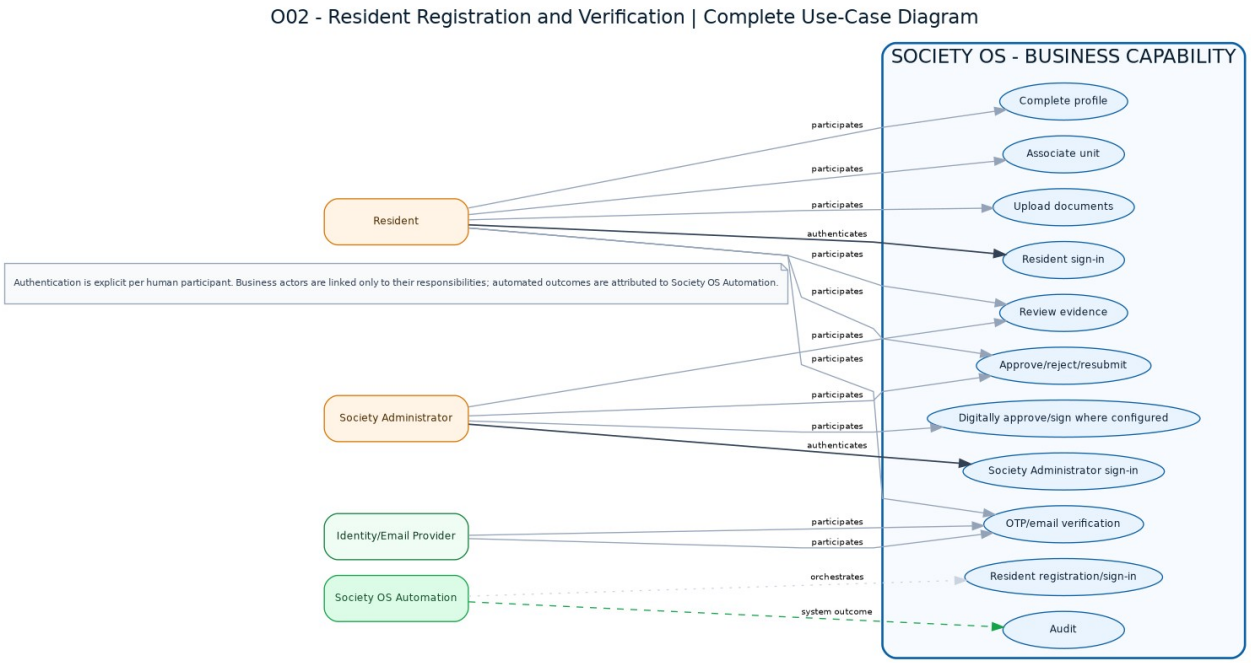
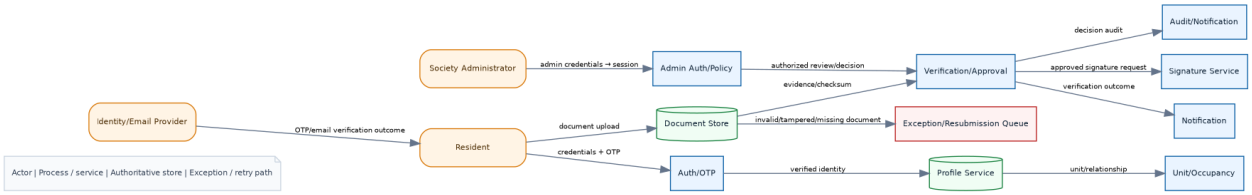


Figure O02-B. Complete data flow diagram - actors, services, authoritative stores, external boundaries and exception path.

O02 - Resident Registration and Verification | Data Flow Diagram



O03 - Owner/Tenant/Household Lifecycle

Domain: Identity / Occupancy

Primary participants: Resident, Society Administrator

Business objective: Maintain accurate owner, tenant, co-owner and household relationships over time.

Real-world scenario: An owner rents Unit A-1203 to Tenant B, later terminates the tenancy and moves back in. Historical occupancy and household relationships remain intact.

Preconditions: Unit master exists; identity records exist or are being created.

Normal flow: Resident/admin signs in → create/modify relationship → validate overlap dates → approve if configured → update active occupancy/household → close prior relationship at effective date → audit timeline.

Exception / alternate flow: Overlapping occupancies, deleting historical members, orphaned household members or privacy leakage must be prevented.

State progression: REQUESTED → VALIDATED → ACTIVE → ENDED / SUPERSEDED

Key invariant: Historical relationships are immutable facts with effective dates; current active relationships are derived from valid periods.

QA acceptance intent

- Verify the protected human actor signs in before performing privileged operations such as approval, verification, financial adjustment or NOC issuance.
- Verify the happy path creates exactly one authoritative business outcome and downstream notifications are triggered from confirmed state.
- Verify retries, duplicates and provider callbacks cannot create duplicate business effects.
- Verify exception states are visible, recoverable and auditable, without corrupting historical records.

Figure O03-A. Complete use-case diagram - participants, authenticated entry points and business interactions.

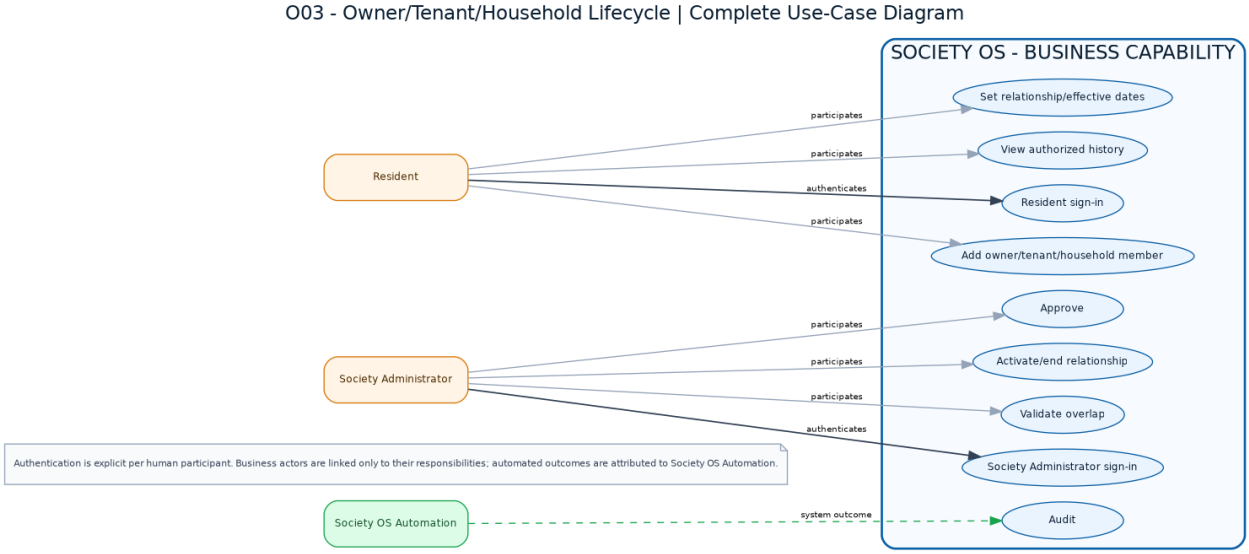
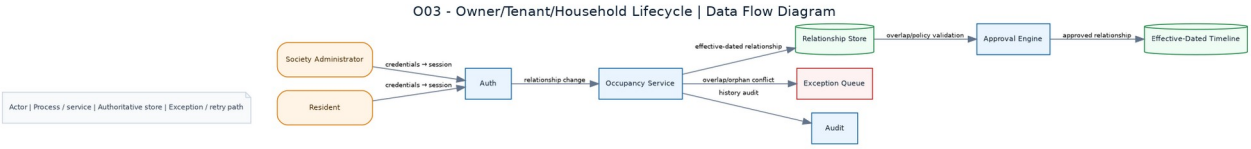


Figure O03-B. Complete data flow diagram - actors, services, authoritative stores, external boundaries and exception path.



O04 - Visitor Pre-Approval and Gate Check-In

Domain: Gate / Visitor

Primary participants: Resident, Guard, Gate Device

Business objective: Allow an approved visitor to enter with rapid, traceable gate validation.

Real-world scenario: A resident pre-approves a guest for 7:00-10:00 PM. At 7:35 PM the guard scans the pass, validates identity and records entry.

Preconditions: Visitor pass exists and is within validity window.

Normal flow: Resident signs in → creates invitation → system issues pass → visitor arrives → guard signs in/device authenticates → pass scanned → gate validates unit, window, status and restrictions → entry logged → pass state updates → resident notified.

Exception / alternate flow: Expired/revoked pass, duplicate entry, resident request to deny entry or offline gate operation must be handled safely.

State progression: CREATED → APPROVED → PRESENTED → CHECKED_IN → CHECKED_OUT / EXPIRED / REVOKED

Key invariant: Gate admission is security-sensitive and server-authoritative when current connectivity is available.

QA acceptance intent

- Verify the protected human actor signs in before performing privileged operations such as approval, verification, financial adjustment or NOC issuance.
- Verify the happy path creates exactly one authoritative business outcome and downstream notifications are triggered from confirmed state.
- Verify retries, duplicates and provider callbacks cannot create duplicate business effects.
- Verify exception states are visible, recoverable and auditable, without corrupting historical records.

Figure O04-A. Complete use-case diagram - participants, authenticated entry points and business interactions.

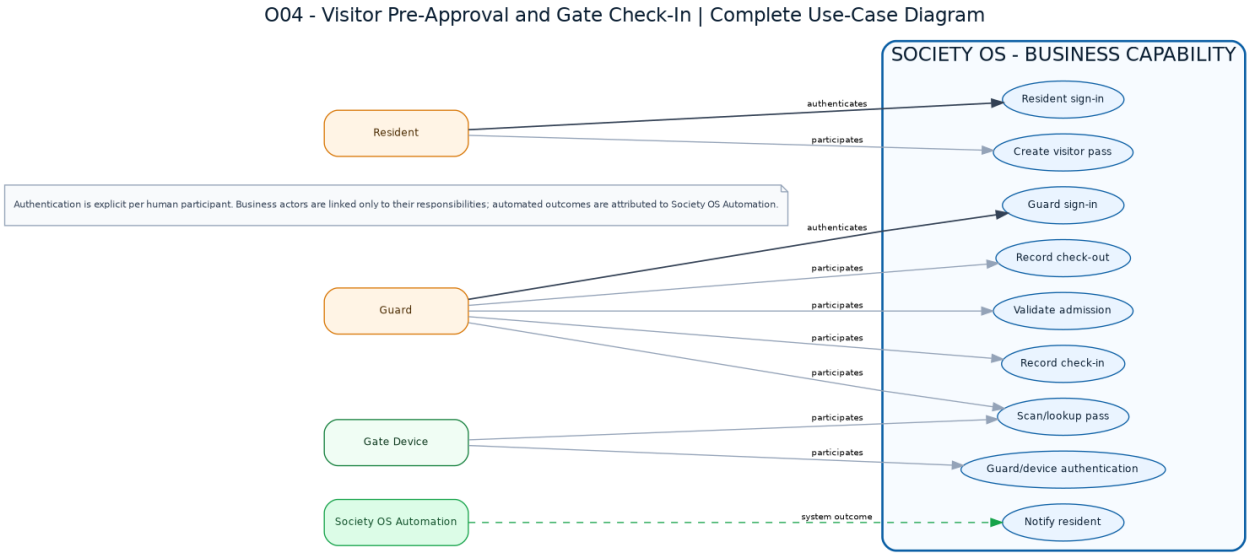
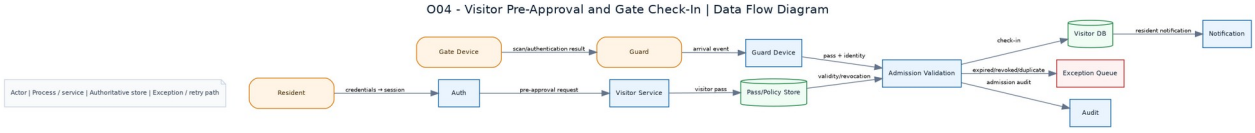


Figure O04-B. Complete data flow diagram - actors, services, authoritative stores, external boundaries and exception path.



O05 - Unknown Visitor / Resident Unreachable

Domain: Gate / Security

Primary participants: Guard, Resident, Security Administrator

Business objective: Resolve an unapproved visitor request without converting non-response into automatic approval.

Real-world scenario: A plumber arrives for Unit B-804 without pre-approval; the resident does not answer. The guard follows the configured deny/wait/escalate policy.

Preconditions: Visitor is not pre-approved; emergency override policy may or may not apply.

Normal flow: Guard signs in → searches unit → system shows approval state → calls/request sent → timer starts → resident responds approve/deny or remains unreachable → configured policy decides wait/deny/escalate → decision and reason logged.

Exception / alternate flow: Emergency arrival, false identity, suspicious behaviour or resident response after denial require explicit exception handling.

State progression: PRESENTED → AWAITING_RESIDENT → APPROVED / DENIED / ESCALATED / EMERGENCY_BYPASS

Key invariant: No response is not equivalent to approval.

QA acceptance intent

- Verify the protected human actor signs in before performing privileged operations such as approval, verification, financial adjustment or NOC issuance.
- Verify the happy path creates exactly one authoritative business outcome and downstream notifications are triggered from confirmed state.
- Verify retries, duplicates and provider callbacks cannot create duplicate business effects.
- Verify exception states are visible, recoverable and auditable, without corrupting historical records.

Figure O05-A. Complete use-case diagram - participants, authenticated entry points and business interactions.

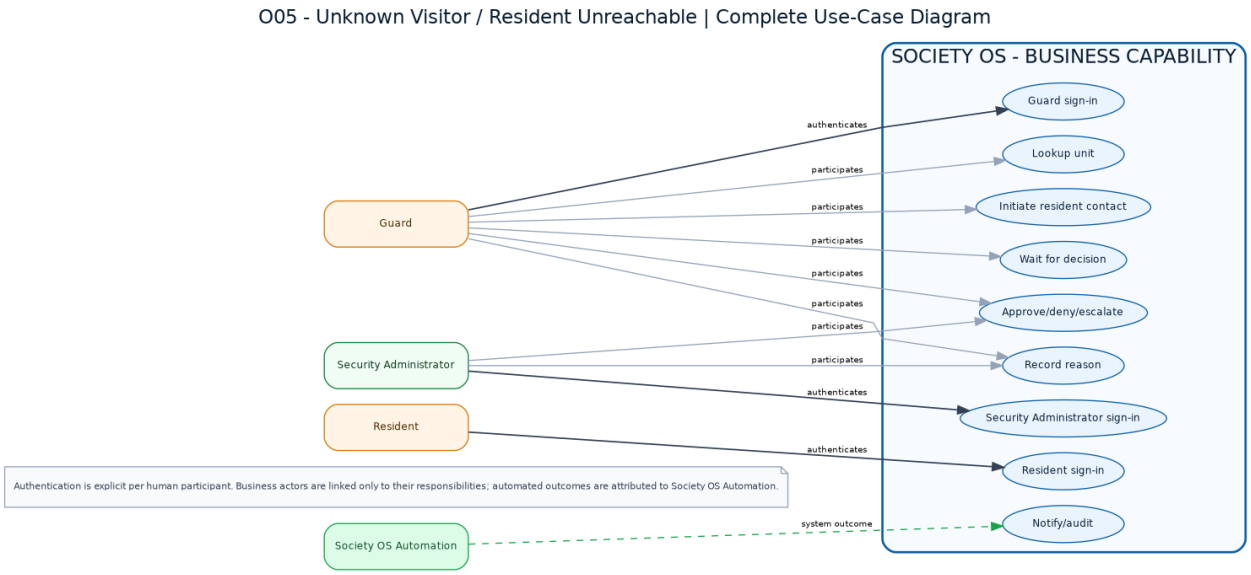
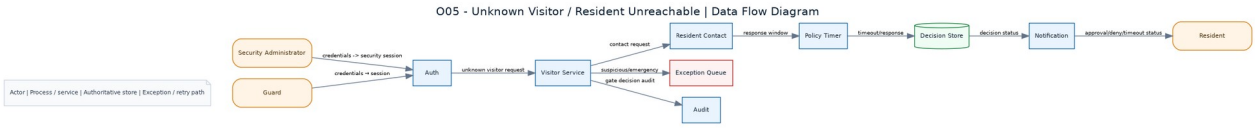


Figure O05-B. Complete data flow diagram - actors, services, authoritative stores, external boundaries and exception path.



O06 - Complaint Lifecycle and SLA

Domain: Helpdesk

Primary participants: Resident, Helpdesk Administrator, Technician

Business objective: Manage a complaint from creation to verified resolution with measurable SLA.

Real-world scenario: A resident reports a lift stopping at the 7th floor, adds a video, the ticket is assigned to a technician, placed on hold for a spare, resolved and later confirmed by the resident.

Preconditions: Complaint categories and SLA policies exist.

Normal flow: Resident signs in → creates complaint → classification/SLA calculated → admin/engine assigns → technician acknowledges → work starts → wait/hold as needed → resolution evidence added → resident notified → resident confirms or ticket closes → SLA metrics recorded.

Exception / alternate flow: Duplicate, parent incident, SLA breach, reassignment, reopen or missing resolution evidence must follow explicit rules.

State progression: CREATED → CLASSIFIED → ASSIGNED → ACKNOWLEDGED → IN_PROGRESS → WAITING/HOLD → RESOLVED → REOPENED/CLOSED

Key invariant: SLA clock treatment must be deterministic for hold, pause and reopen semantics.

QA acceptance intent

- Verify the protected human actor signs in before performing privileged operations such as approval, verification, financial adjustment or NOC issuance.
- Verify the happy path creates exactly one authoritative business outcome and downstream notifications are triggered from confirmed state.
- Verify retries, duplicates and provider callbacks cannot create duplicate business effects.
- Verify exception states are visible, recoverable and auditable, without corrupting historical records.

Figure O06-A. Complete use-case diagram - participants, authenticated entry points and business interactions.

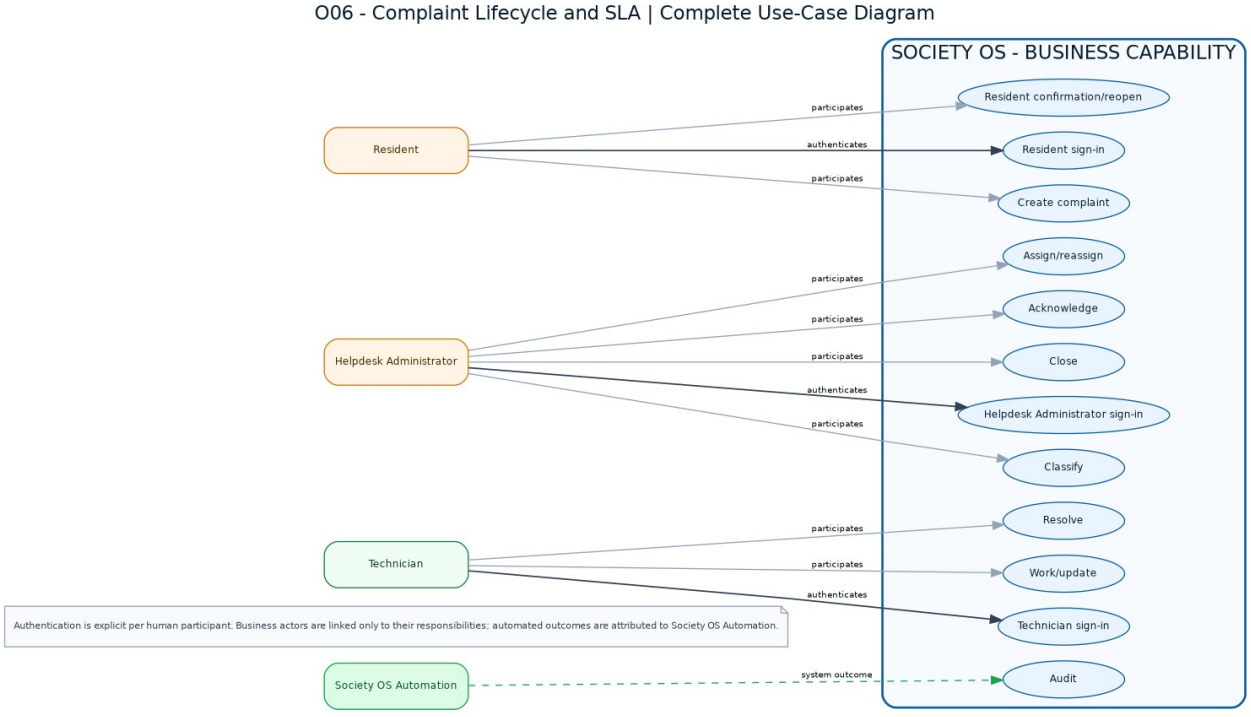
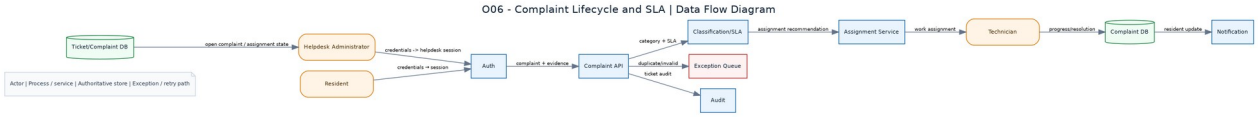


Figure O06-B. Complete data flow diagram - actors, services, authoritative stores, external boundaries and exception path.


O06 - Complaint Lifecycle and SLA | Data Flow Diagram
Ticket/Complaint DB
open complaint / assignment state
Helpdesk Administrator
Resident
credentials -> helpdesk session
credentials -> session
Auth
complaint + evidence
Complaint API
category + SLA
duplicate/invalid
ticket audit
Classification/SLA
Exception Queue
Audit
assignment recommendation
Assignment Service
work assignment
Technician
progress/resolution
Complaint DB
resident update
Notification
Actor | Process / service | Authoritative store | Exception / retry path

O07 - Parent Incident / Duplicate Complaints

Domain: Helpdesk / Incidents

Primary participants: Resident, Helpdesk Administrator, Technician

Business objective: Correlate multiple complaints into a parent incident while preserving individual reporters.

Real-world scenario: Ten residents report low water pressure in Tower C within 20 minutes. The platform proposes a parent incident and broadcasts common updates.

Preconditions: Multiple related complaints exist; correlation rules enabled.

Normal flow: Complaints arrive → correlation engine evaluates location/category/time → parent incident suggested/created → admin signs in/reviews → linked complaints retain reporter evidence → common updates distributed → parent resolution closes/reconciles children.

Exception / alternate flow: False correlation, complaint-specific resolution or incident split must be supported.

State progression: CHILD_CREATED → CORRELATION_CANDIDATE → LINKED_PARENT → IN_PROGRESS → RESOLVED → CLOSED

Key invariant: Linking never deletes individual complaint history or reporter identity.

QA acceptance intent

- Verify the protected human actor signs in before performing privileged operations such as approval, verification, financial adjustment or NOC issuance.
- Verify the happy path creates exactly one authoritative business outcome and downstream notifications are triggered from confirmed state.
- Verify retries, duplicates and provider callbacks cannot create duplicate business effects.
- Verify exception states are visible, recoverable and auditable, without corrupting historical records.

Figure O07-A. Complete use-case diagram - participants, authenticated entry points and business interactions.

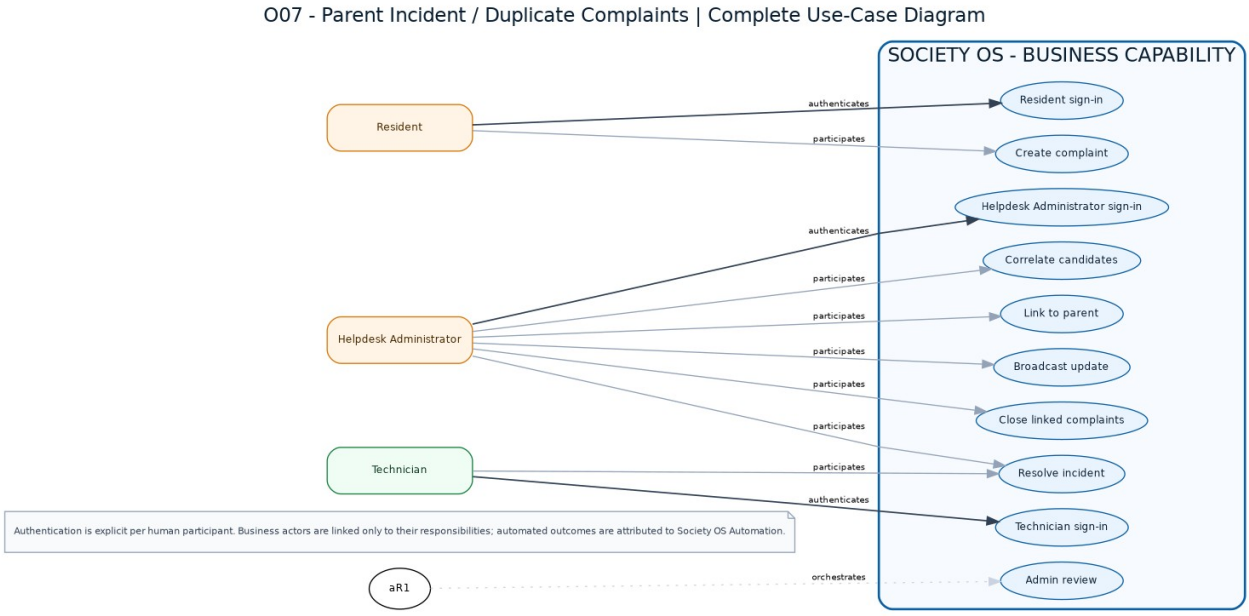
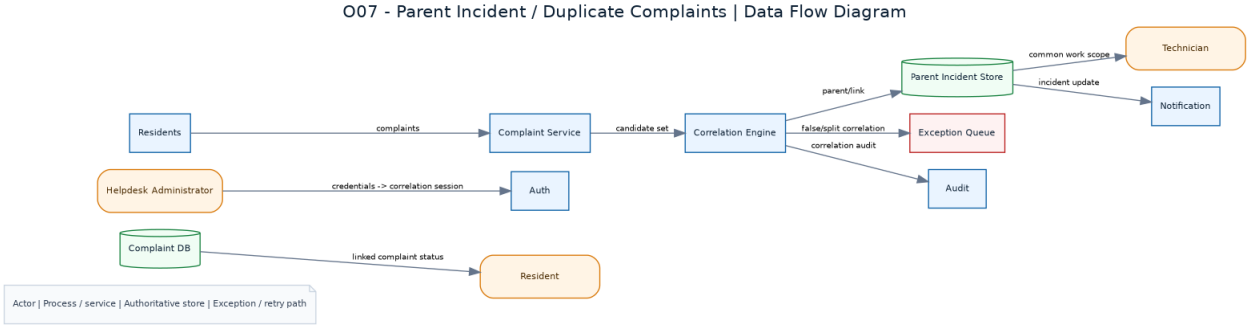


Figure O07-B. Complete data flow diagram - actors, services, authoritative stores, external boundaries and exception path.

O07 - Parent Incident / Duplicate Complaints | Data Flow Diagram



O08 - Preventive Maintenance

Domain: Maintenance

Primary participants: Maintenance Manager, Technician, Vendor

Business objective: Convert asset maintenance plans into scheduled, auditable work orders.

Real-world scenario: A generator has weekly inspections, monthly test runs and annual servicing. The scheduler creates work orders and advances the next-due dates.

Preconditions: Assets and maintenance plans exist; technician/vendor availability is known.

Normal flow: Manager signs in → defines/edits plan → scheduler generates due work order → assigns technician/vendor → inspection/service performed → parts/cost recorded → completion evidence captured → next due calculated → asset history updated.

Exception / alternate flow: Missed schedule, vendor no-show, failed inspection or parts shortage must create follow-up tasks rather than silently closing work.

State progression: PLAN_ACTIVE → WORK_ORDER_CREATED → ASSIGNED → IN_PROGRESS → COMPLETED / FAILED → NEXT_DUE_SCHEDULED

Key invariant: Every completed maintenance event must remain linked to asset, plan, worker/vendor and evidence.

QA acceptance intent

- Verify the protected human actor signs in before performing privileged operations such as approval, verification, financial adjustment or NOC issuance.
- Verify the happy path creates exactly one authoritative business outcome and downstream notifications are triggered from confirmed state.
- Verify retries, duplicates and provider callbacks cannot create duplicate business effects.
- Verify exception states are visible, recoverable and auditable, without corrupting historical records.

Figure O08-A. Complete use-case diagram - participants, authenticated entry points and business interactions.

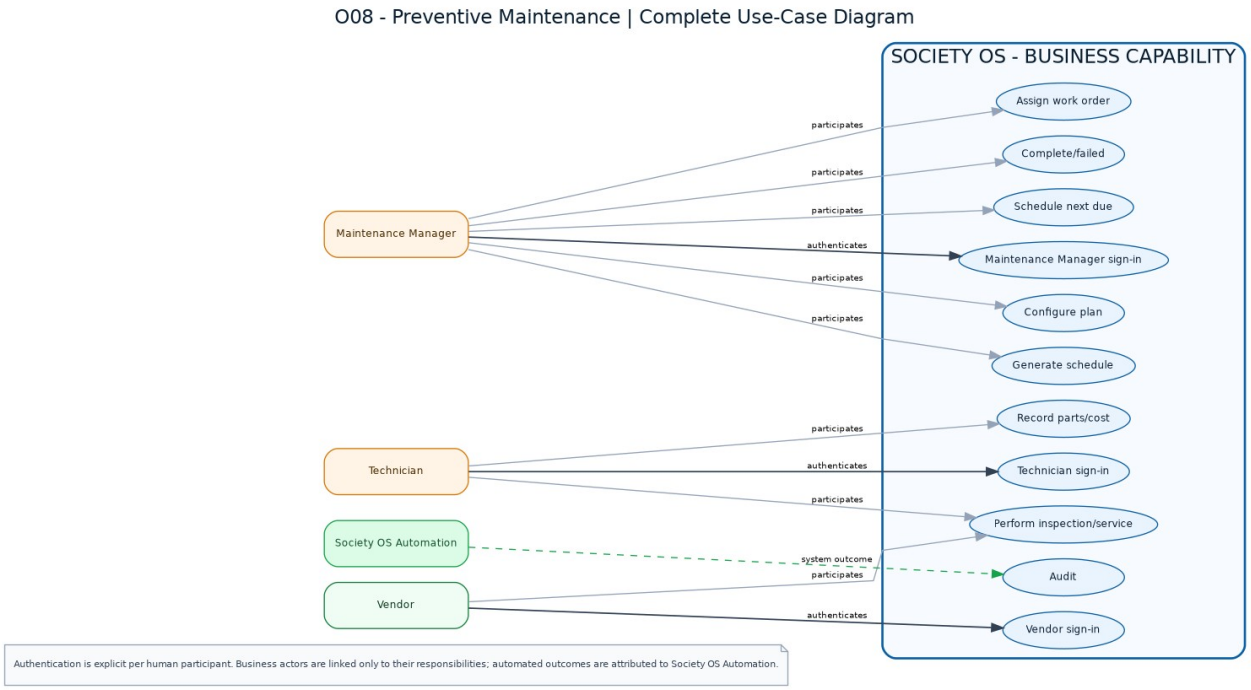
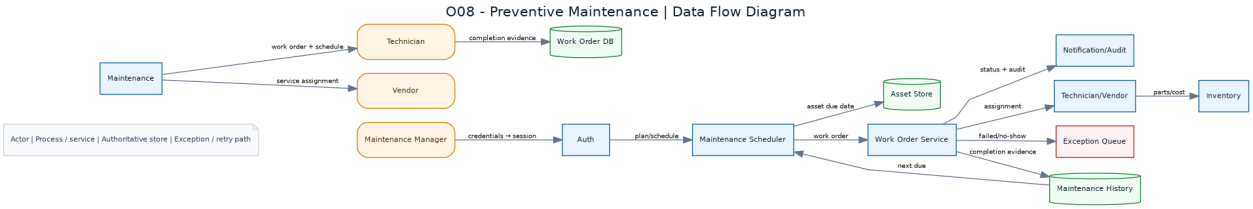


Figure O08-B. Complete data flow diagram - actors, services, authoritative stores, external boundaries and exception path.


O08 - Preventive Maintenance | Data Flow Diagram
Maintenance
work order + schedule
service assignment
Technician
Vendor
completion evidence
Work Order DB
Maintenance Manager
credentials → session
Auth
plan/schedule
Maintenance Scheduler
asset due date
Asset Store
work order
Work Order Service
status + audit
assignment
completion evidence
next due
Notification/Audit
Technician/Vendor
parts/cost
Inventory
failed/no-show
Exception Queue
Maintenance History
Actor | Process / service | Authoritative store | Exception / retry path

O09 - Inventory Reorder and Stock Issue

Domain: Inventory / Procurement

Primary participants: Storekeeper, Maintenance Technician, Procurement Administrator

Business objective: Maintain accurate stock and replenish materials based on reorder thresholds.

Real-world scenario: A pump repair consumes the last spare seal below reorder level. The storekeeper issues stock, and procurement raises a purchase request.

Preconditions: Item master, stock locations and reorder levels exist.

Normal flow: Storekeeper signs in → view stock → issue material to work order → stock decremented → reorder rule triggered → purchase request created → approval/PO → receipt → stock incremented.

Exception / alternate flow: Insufficient stock, duplicate receipt, damaged material or serial mismatch must be recorded explicitly.

State progression: AVAILABLE → LOW_STOCK → REQUESTED → ORDERED → RECEIVED → AVAILABLE

Key invariant: Stock balance is derived from controlled movements, not manual overwrites.

QA acceptance intent

- Verify the protected human actor signs in before performing privileged operations such as approval, verification, financial adjustment or NOC issuance.
- Verify the happy path creates exactly one authoritative business outcome and downstream notifications are triggered from confirmed state.
- Verify retries, duplicates and provider callbacks cannot create duplicate business effects.
- Verify exception states are visible, recoverable and auditable, without corrupting historical records.

Figure O09-A. Complete use-case diagram - participants, authenticated entry points and business interactions.

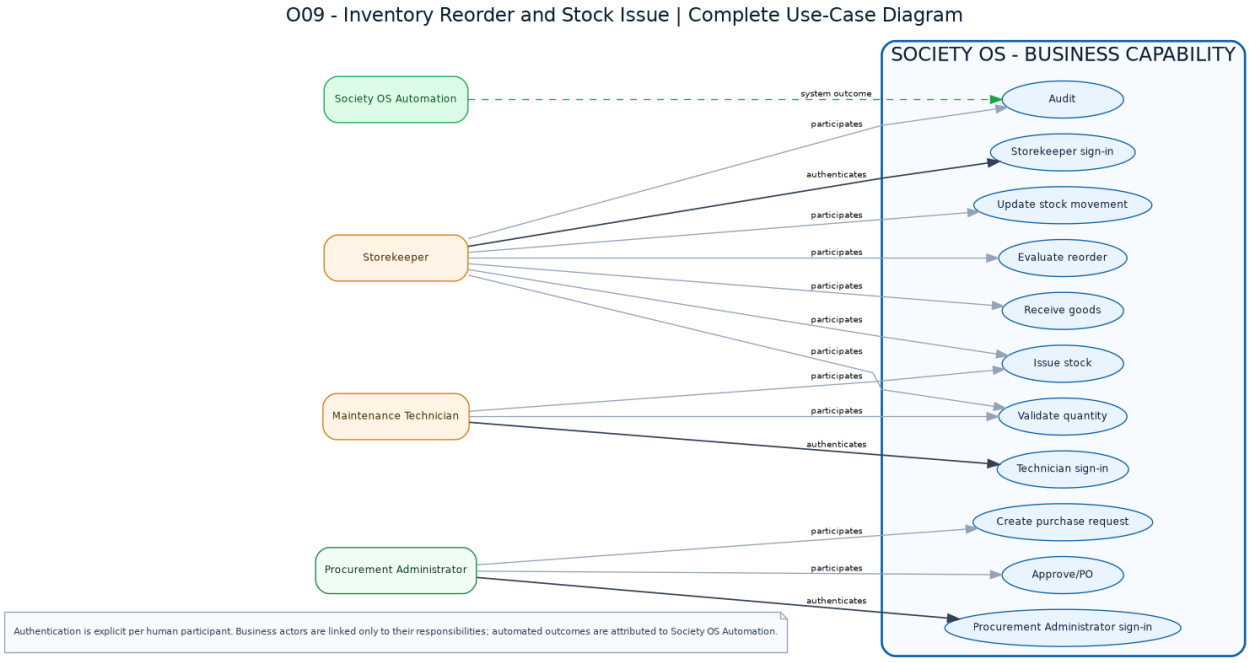
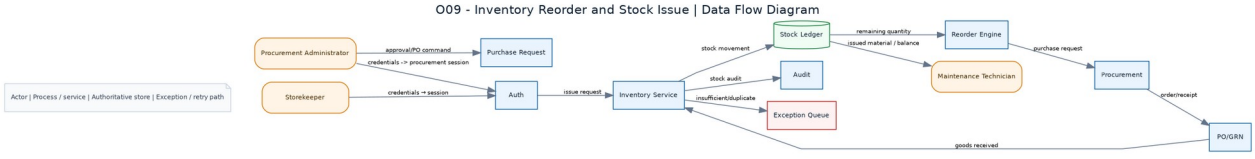


Figure O09-B. Complete data flow diagram - actors, services, authoritative stores, external boundaries and exception path.



O10 - Vendor Onboarding and AMC Renewal

Domain: Vendor / Contracts

Primary participants: Society Administrator, Vendor, Finance Administrator

Business objective: Onboard vendors, validate documents and manage AMC lifecycle to renewal/termination.

Real-world scenario: A lift AMC expires on 31 March. The system reminds the society, captures renewal terms and preserves the previous contract version.

Preconditions: Vendor category and required document rules exist.

Normal flow: Admin signs in → creates vendor → collects KYC/GST/PAN/bank/docs → validates → contract/AMC created → service history tracked → renewal reminder → new version/contract approved → expiry/renewal state updated.

Exception / alternate flow: Expired insurance, failed verification, price change or missed renewal must escalate.

State progression: DRAFT → VERIFICATION → ACTIVE → EXPIRING → RENEWED / EXPIRED / TERMINATED

Key invariant: Vendor master changes and contract versions are historical; active contract validity must be unambiguous.

QA acceptance intent

- Verify the protected human actor signs in before performing privileged operations such as approval, verification, financial adjustment or NOC issuance.
- Verify the happy path creates exactly one authoritative business outcome and downstream notifications are triggered from confirmed state.
- Verify retries, duplicates and provider callbacks cannot create duplicate business effects.
- Verify exception states are visible, recoverable and auditable, without corrupting historical records.

Figure O10-A. Complete use-case diagram - participants, authenticated entry points and business interactions.

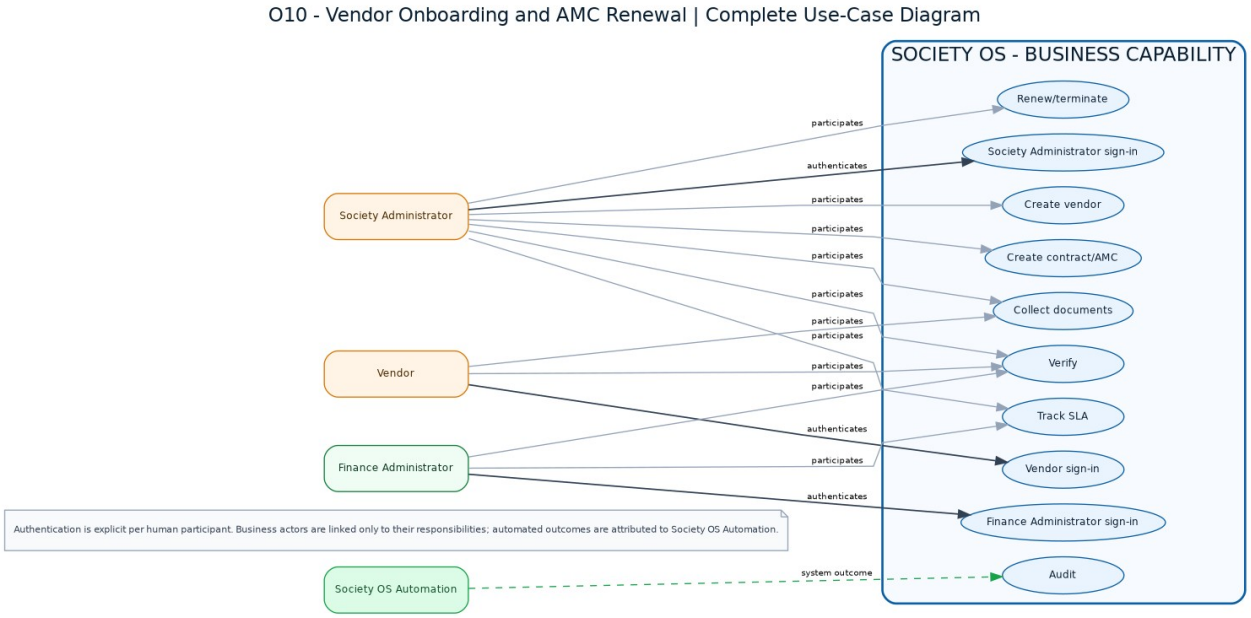
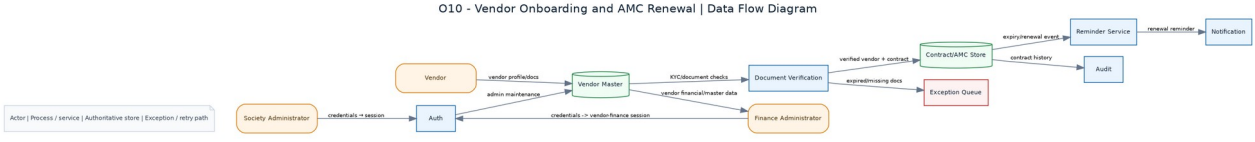


Figure O10-B. Complete data flow diagram - actors, services, authoritative stores, external boundaries and exception path.



O11 - Facility Booking Conflict

Domain: Facilities

Primary participants: Resident, Facility Administrator

Business objective: Reserve shared facilities atomically while enforcing time, capacity and blackout rules.

Real-world scenario: A resident requests the clubhouse 6:00-8:00 PM. The system applies a 30-minute cleanup buffer and rejects overlapping confirmed bookings.

Preconditions: Facility hours, booking windows, capacity, blackout and cancellation policies exist.

Normal flow: Resident signs in → selects facility/time → system checks blackout/capacity/conflicts → payment/deposit if configured → reservation atomically created → confirmation sent → check-in/usage tracked → completion or cancellation handled.

Exception / alternate flow: Concurrent booking, blackout, no-show, over-capacity or cancellation-window violation must return deterministic business outcomes.

State progression: REQUESTED → VALIDATING → CONFIRMED / REJECTED → CHECKED_IN → COMPLETED/CANCELLED/NO_SHOW

Key invariant: Confirmed reservations require atomic conflict protection; cleanup buffers are part of conflict evaluation.

QA acceptance intent

- Verify the protected human actor signs in before performing privileged operations such as approval, verification, financial adjustment or NOC issuance.
- Verify the happy path creates exactly one authoritative business outcome and downstream notifications are triggered from confirmed state.
- Verify retries, duplicates and provider callbacks cannot create duplicate business effects.
- Verify exception states are visible, recoverable and auditable, without corrupting historical records.

Figure O11-A. Complete use-case diagram - participants, authenticated entry points and business interactions.

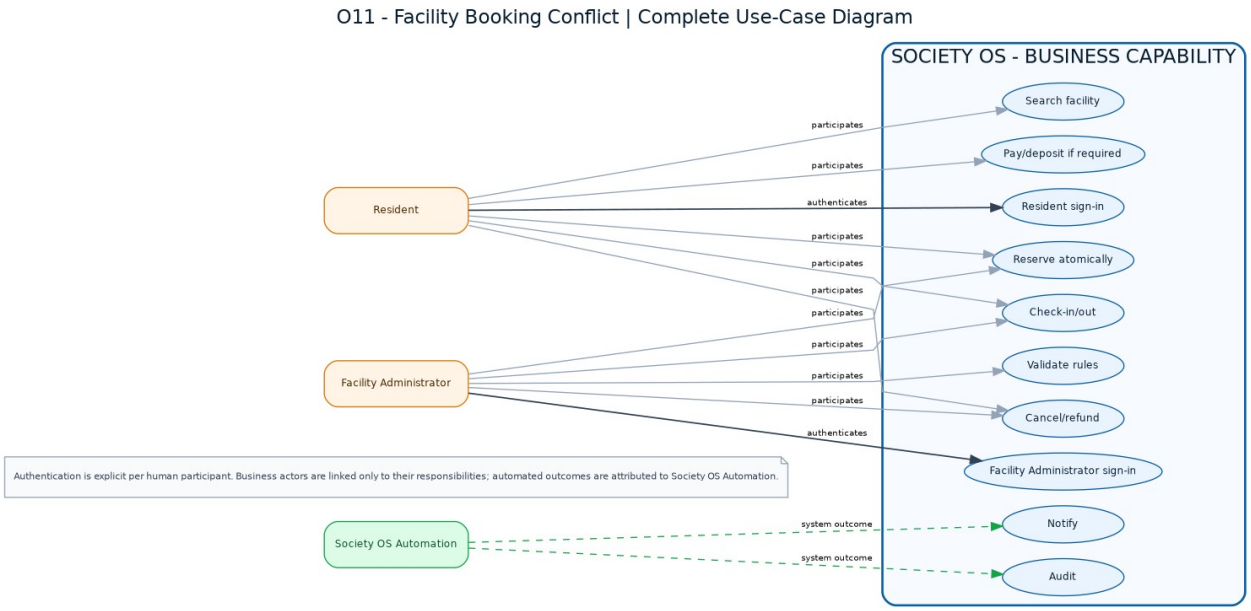
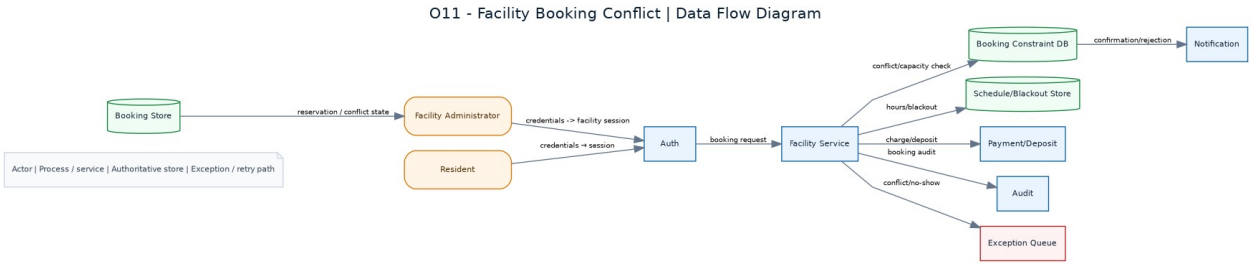


Figure O11-B. Complete data flow diagram - actors, services, authoritative stores, external boundaries and exception path.


O11 - Facility Booking Conflict | Data Flow Diagram
Booking Store
reservation / conflict state
Facility Administrator
Resident
credentials -> facility session
credentials → session
Auth
booking request
Facility Service
conflict/capacity check
hours/blackout
charge/deposit
booking audit
conflict/no-show
Booking Constraint DB
Schedule/Blackout Store
Payment/Deposit
Audit
Exception Queue
confirmation/rejection
Notification
Actor | Process / service | Authoritative store | Exception / retry path

O12 - Notice and Emergency Broadcast

Domain: Communication

Primary participants: Society Administrator, Resident, Notification Provider

Business objective: Publish targeted notices with priority, schedule, expiry and emergency override semantics.

Real-world scenario: A scheduled water shutdown notice is published at 10:55 AM for a repair beginning at 11:00 AM. A critical fire notice can bypass resident quiet hours under policy.

Preconditions: Notice template/audience and channel configuration exist.

Normal flow: Admin signs in → drafts notice → selects audience/priority/effective window → validates attachments/localization → publishes/schedules → dispatches channels → tracks delivery/acknowledgement → expires/withdraws while preserving revision history.

Exception / alternate flow: Provider throttling, invalid audience, emergency override misuse or withdrawn notice must be auditable.

State progression: DRAFT → SCHEDULED → PUBLISHED → DELIVERING → DELIVERED/PARTIAL → EXPIRED/WITHDRAWN

Key invariant: Emergency override requires an explicit policy reason and audit record.

QA acceptance intent

- Verify the protected human actor signs in before performing privileged operations such as approval, verification, financial adjustment or NOC issuance.
- Verify the happy path creates exactly one authoritative business outcome and downstream notifications are triggered from confirmed state.
- Verify retries, duplicates and provider callbacks cannot create duplicate business effects.
- Verify exception states are visible, recoverable and auditable, without corrupting historical records.

Figure O12-A. Complete use-case diagram - participants, authenticated entry points and business interactions.

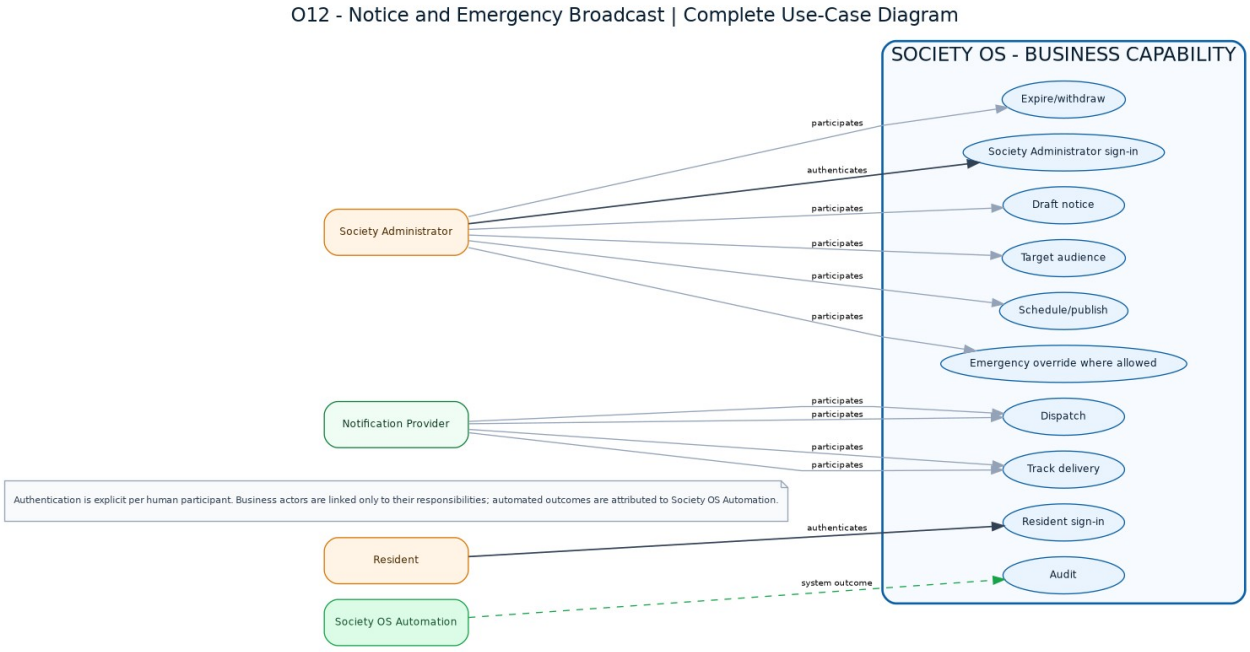
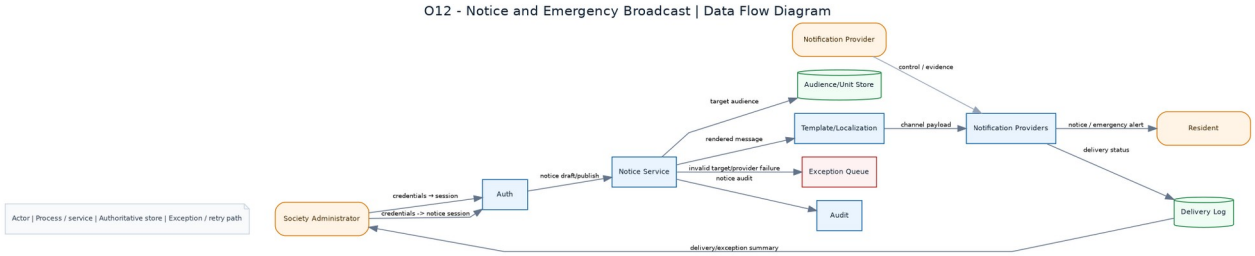


Figure O12-B. Complete data flow diagram - actors, services, authoritative stores, external boundaries and exception path.


O12 - Notice and Emergency Broadcast | Data Flow Diagram
Actor | Process / service | Authoritative store | Exception / retry path
Society Administrator
Auth
Notice Service
Notification Provider
Audience/Unit Store
Template/Localization
Notification Providers
Resident
Exception Queue
Audit
Delivery Log
credentials → session
credentials -> notice session
notice draft/publish
target audience
rendered message
invalid target/provider failure
notice audit
control / evidence
channel payload
notice / emergency alert
delivery status
delivery/exception summary

O13 - Resident Communication Approval

Domain: Communication

Primary participants: Resident, Authorized Contact, Moderator / Administrator

Business objective: Enable controlled resident-to-resident communication without exposing private contact information by default.

Real-world scenario: A committee member wants to contact a resident about a parking issue. The resident accepts the first-contact request before a persistent chat channel opens.

Preconditions: Privacy/contact policy is configured.

Normal flow: Sender signs in → requests contact → system checks privacy policy → recipient receives consent request → recipient accepts/rejects → accepted channel created → messages are append-oriented → report/moderation controls remain active.

Exception / alternate flow: Abuse report, block, revoked consent or message retention policy must close/limit the channel appropriately.

State progression: REQUESTED → PENDING_CONSENT → ACCEPTED/REJECTED → ACTIVE → BLOCKED/CLOSED

Key invariant: Phone numbers remain hidden unless explicitly permitted by policy.

QA acceptance intent

- Verify the protected human actor signs in before performing privileged operations such as approval, verification, financial adjustment or NOC issuance.
- Verify the happy path creates exactly one authoritative business outcome and downstream notifications are triggered from confirmed state.
- Verify retries, duplicates and provider callbacks cannot create duplicate business effects.
- Verify exception states are visible, recoverable and auditable, without corrupting historical records.

Figure O13-A. Complete use-case diagram - participants, authenticated entry points and business interactions.

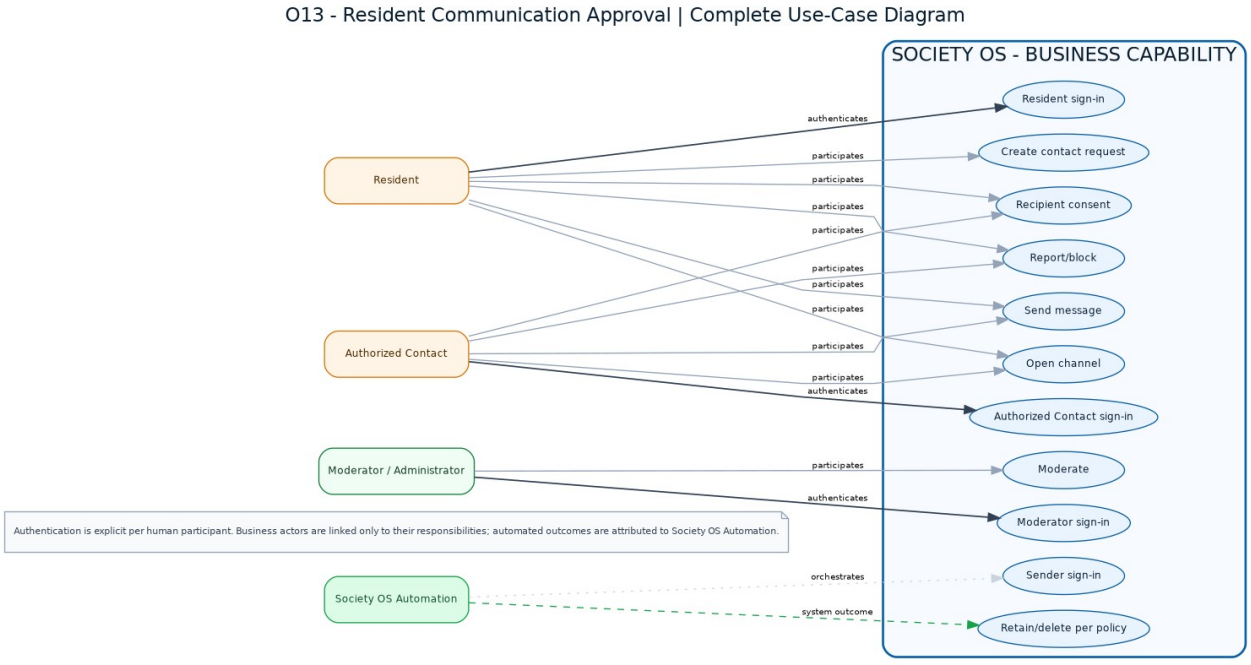
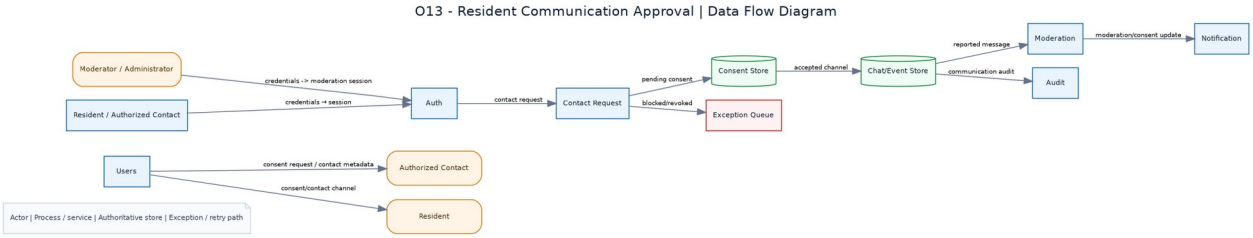


Figure O13-B. Complete data flow diagram - actors, services, authoritative stores, external boundaries and exception path.


O13 - Resident Communication Approval | Data Flow Diagram
Moderator / Administrator
Resident / Authorized Contact
credentials -> moderation session
credentials → session
Auth
contact request
Contact Request
pending consent
blocked/revoked
Consent Store
Exception Queue
accepted channel
Chat/Event Store
reported message
communication audit
Moderation
Audit
moderation/consent update
Notification
Users
consent request / contact metadata
consent/contact channel
Authorized Contact
Resident
Actor | Process / service | Authoritative store | Exception / retry path

O14 - Move-In Workflow

Domain: Move-In

Primary participants: Resident, Society Administrator, Guard / Facility Team

Business objective: Coordinate a move-in date, access requirements and operational conflicts.

Real-world scenario: A tenant schedules Saturday move-in. Loading access is approved while a lift maintenance block is detected, so the system requests an alternate slot.

Preconditions: Unit/occupancy approval and move-in policy exist.

Normal flow: Resident signs in → submits move-in → admin signs in → verifies occupancy/docs → selects date/time → scheduling engine checks lift/loading conflicts → approvals complete → gate/facility instructions generated → move-in executed → event closed.

Exception / alternate flow: Maintenance conflict, overdue documents, missing deposit or restricted timing must block or route to exception approval.

State progression: REQUESTED → VERIFIED → SCHEDULED → APPROVED → IN_PROGRESS → COMPLETED / CANCELLED

Key invariant: Move-in cannot be marked complete without validated occupancy and required operational clearances.

QA acceptance intent

- Verify the protected human actor signs in before performing privileged operations such as approval, verification, financial adjustment or NOC issuance.
- Verify the happy path creates exactly one authoritative business outcome and downstream notifications are triggered from confirmed state.
- Verify retries, duplicates and provider callbacks cannot create duplicate business effects.
- Verify exception states are visible, recoverable and auditable, without corrupting historical records.

Figure O14-A. Complete use-case diagram - participants, authenticated entry points and business interactions.

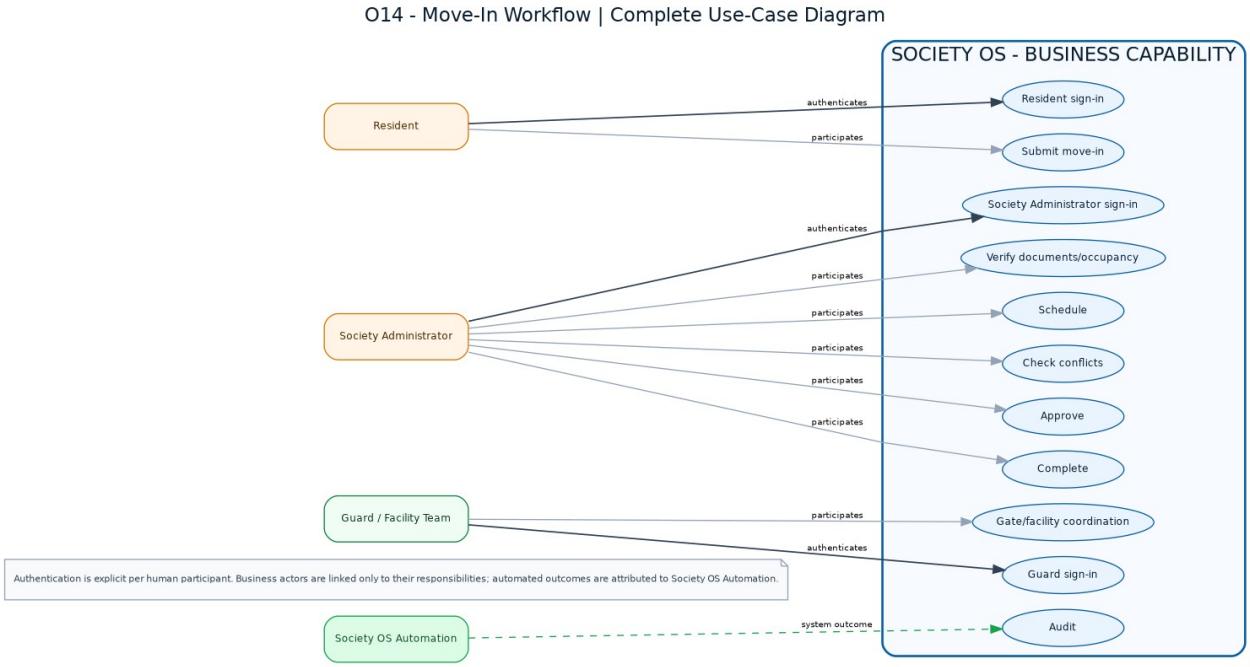
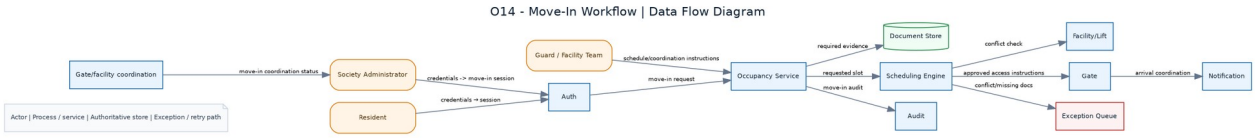


Figure O14-B. Complete data flow diagram - actors, services, authoritative stores, external boundaries and exception path.


O14 - Move-In Workflow | Data Flow Diagram
Gate/facility coordination
move-in coordination status
Society Administrator
Resident
credentials -> move-in session
credentials -> session
Auth
Guard / Facility Team
schedule/coordination instructions
move-in request
Occupancy Service
required evidence
requested slot
move-in audit
Document Store
Scheduling Engine
Audit
conflict check
approved access instructions
conflict/missing docs
Facility/Lift
Gate
Exception Queue
arrival coordination
Notification
Actor | Process / service | Authoritative store | Exception / retry path

O15 - Move-Out with NOC

Domain: Move-Out / NOC

Primary participants: Resident, Society Administrator, Finance Administrator, Security Administrator

Business objective: Clear financial, document, parking and access obligations before issuing a verifiable NOC.

Real-world scenario: A resident requests NOC with ₹18,000 outstanding. The system blocks issuance until finance clearance is completed, then admin approves and digitally signs the NOC; access is revoked and history retained.

Preconditions: Move-out request exists; NOC policy/checklist is configured.

Normal flow: Resident signs in → submits move-out/NOC → admin signs in → system evaluates dues, documents, parking, assets and disputes → exceptions resolved → finance clearance → admin approval/signature → NOC PDF/digest generated → gate/credentials revoked → occupancy archived.

Exception / alternate flow: Disputed damage, outstanding dues, active access asset, missing document or failed signature must keep NOC pending.

State progression: REQUESTED → CLEARANCE_CHECK → EXCEPTION/READY → APPROVED → SIGNED → ISSUED → ACCESS_REVOKED → ARCHIVED

Key invariant: NOC issuance must reference the clearance snapshot; revocation must not delete historical records.

QA acceptance intent

- Verify the protected human actor signs in before performing privileged operations such as approval, verification, financial adjustment or NOC issuance.
- Verify the happy path creates exactly one authoritative business outcome and downstream notifications are triggered from confirmed state.
- Verify retries, duplicates and provider callbacks cannot create duplicate business effects.
- Verify exception states are visible, recoverable and auditable, without corrupting historical records.

Figure O15-A. Complete use-case diagram - participants, authenticated entry points and business interactions.

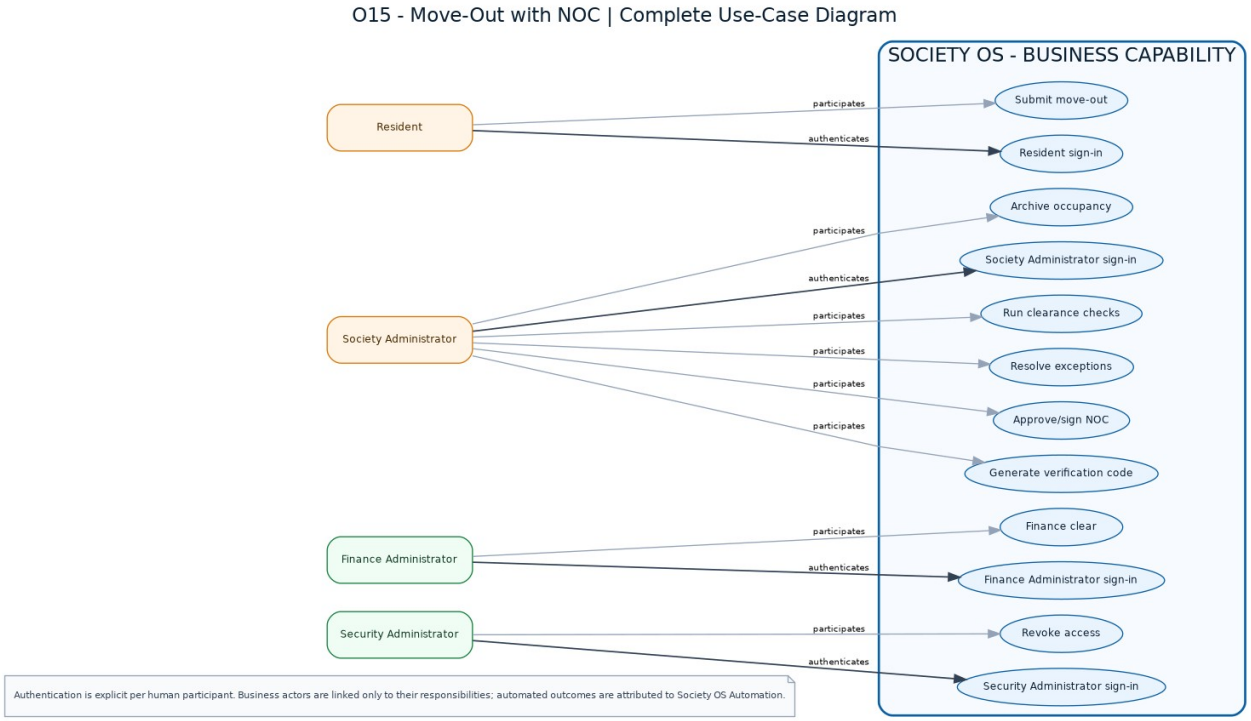
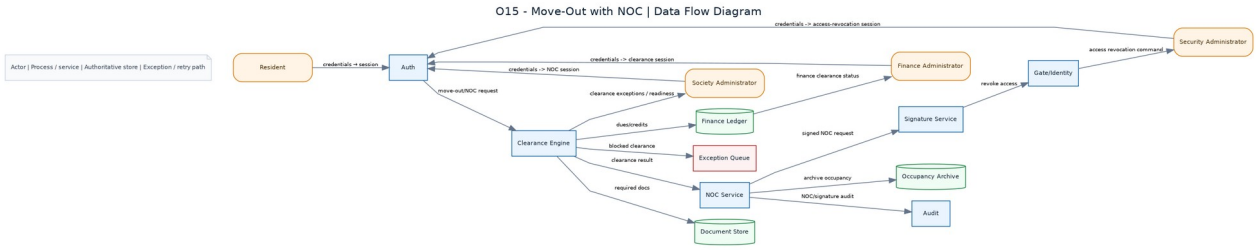


Figure O15-B. Complete data flow diagram - actors, services, authoritative stores, external boundaries and exception path.



O16 - Parking Allocation / Transfer

Domain: Parking

Primary participants: Resident, Parking Administrator, Guard

Business objective: Manage parking entitlements, slot allocation, transfers and temporary assignments without conflicting active allocations.

Real-world scenario: A resident sells one vehicle and temporarily receives another slot for two weeks; the old vehicle association ends and the new one becomes active.

Preconditions: Parking inventory and allocation policy exist.

Normal flow: Resident/admin signs in → selects slot → eligibility checked → existing allocation conflict checked → allocation/transfer approved → vehicle association updated → RFID/ANPR mapping updated where configured → guard can validate active assignment.

Exception / alternate flow: Duplicate vehicle, overlapping allocation, reserved/visitor slot misuse or expired temporary allocation must be rejected/expired.

State progression: AVAILABLE → REQUESTED → ALLOCATED → TRANSFERRED/TEMPORARY → ENDED → AVAILABLE

Key invariant: An active physical slot may not have conflicting active resident allocations.

QA acceptance intent

- Verify the protected human actor signs in before performing privileged operations such as approval, verification, financial adjustment or NOC issuance.
- Verify the happy path creates exactly one authoritative business outcome and downstream notifications are triggered from confirmed state.
- Verify retries, duplicates and provider callbacks cannot create duplicate business effects.
- Verify exception states are visible, recoverable and auditable, without corrupting historical records.

Figure O16-A. Complete use-case diagram - participants, authenticated entry points and business interactions.

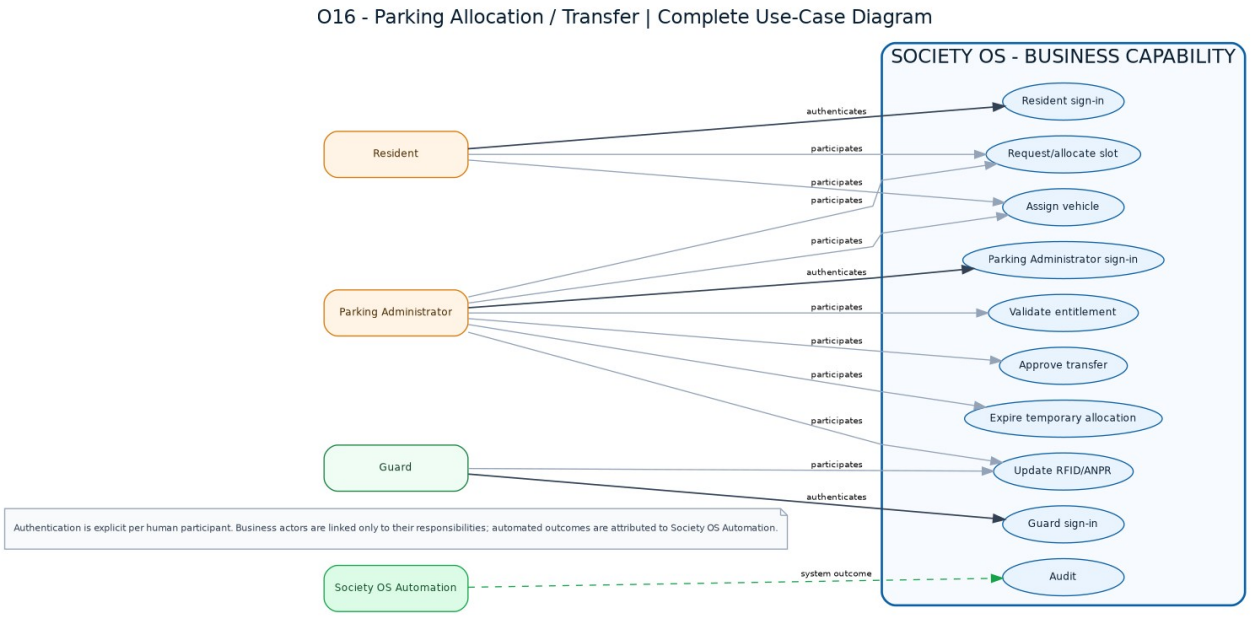
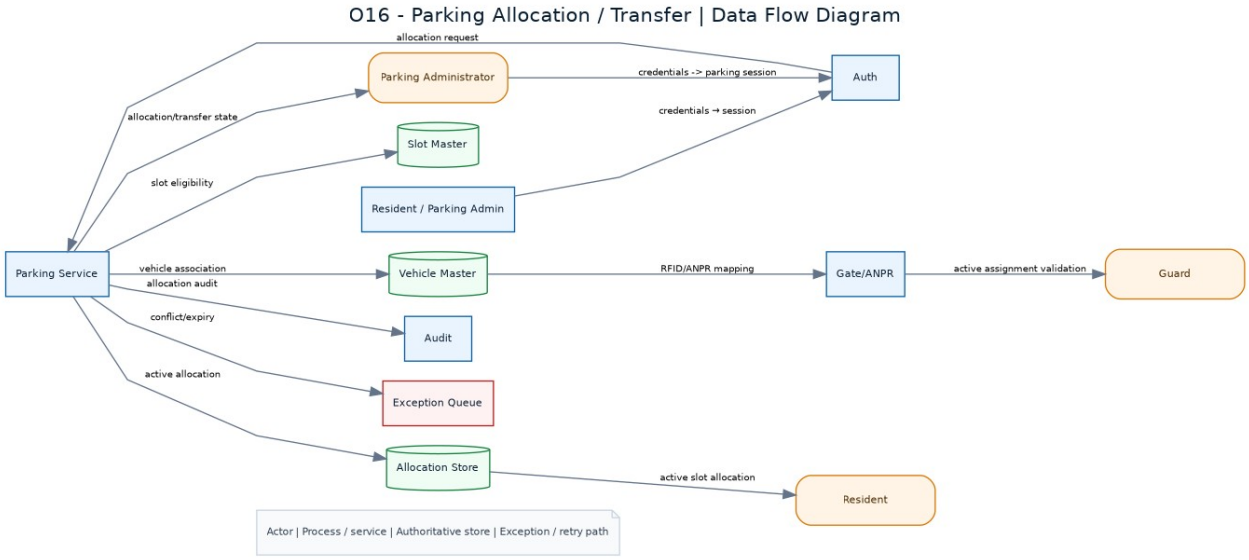


Figure O16-B. Complete data flow diagram - actors, services, authoritative stores, external boundaries and exception path.


O16 - Parking Allocation / Transfer | Data Flow Diagram
allocation request
Parking Administrator
credentials -> parking session
Auth
allocation/transfer state
credentials → session
Slot Master
slot eligibility
Resident / Parking Admin
Parking Service
vehicle association
allocation audit
Vehicle Master
RFID/ANPR mapping
Gate/ANPR
active assignment validation
Guard
conflict/expiry
Audit
active allocation
Exception Queue
Allocation Store
active slot allocation
Resident
Actor | Process / service | Authoritative store | Exception / retry path

O17 - Asset Lifecycle

Domain: Asset Management

Primary participants: Asset Manager, Technician, Vendor

Business objective: Track an asset from acquisition through allocation, maintenance, impairment and retirement.

Real-world scenario: A generator is received, assigned to the society utility area, serviced monthly, repaired after a failure and eventually retired with its full service history intact.

Preconditions: Asset category and lifecycle statuses exist.

Normal flow: Manager signs in → create/receive asset → capture serial/warranty → assign location → maintenance events update history → incidents link to asset → transfer/return/retire → final history preserved.

Exception / alternate flow: Duplicate serial, transfer to invalid location, warranty mismatch or retirement with open work order requires exception.

State progression: PLANNED → RECEIVED → ACTIVE → UNDER_MAINTENANCE → TRANSFERRED → RETIRED/DISPOSED

Key invariant: Asset identity and historical events remain immutable; current location/owner is derived from the latest valid allocation.

QA acceptance intent

- Verify the protected human actor signs in before performing privileged operations such as approval, verification, financial adjustment or NOC issuance.
- Verify the happy path creates exactly one authoritative business outcome and downstream notifications are triggered from confirmed state.
- Verify retries, duplicates and provider callbacks cannot create duplicate business effects.
- Verify exception states are visible, recoverable and auditable, without corrupting historical records.

Figure O17-A. Complete use-case diagram - participants, authenticated entry points and business interactions.

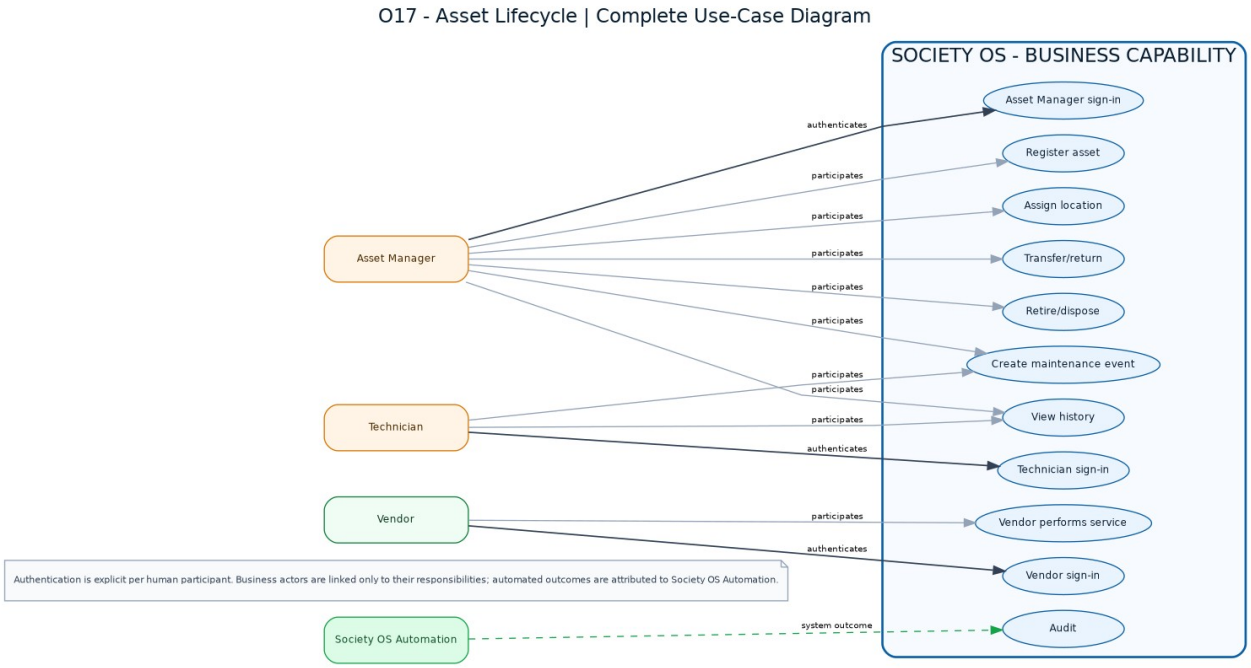
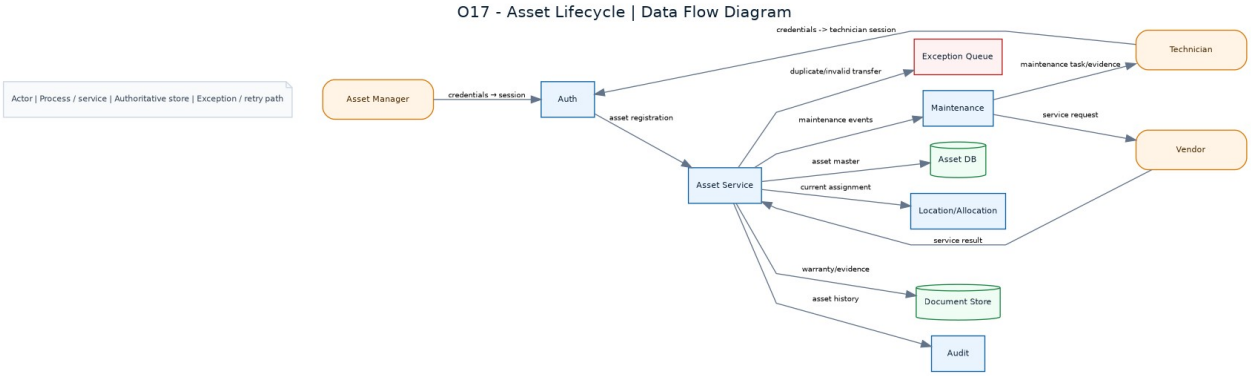


Figure O17-B. Complete data flow diagram - actors, services, authoritative stores, external boundaries and exception path.


O17 - Asset Lifecycle | Data Flow Diagram
Actor | Process / service | Authoritative store | Exception / retry path
Asset Manager
credentials → session
Auth
credentials -> technician session
asset registration
Exception Queue
duplicate/invalid transfer
Technician
maintenance task/evidence
Maintenance
maintenance events
service request
Vendor
Asset DB
asset master
Asset Service
current assignment
Location/Allocation
service result
warranty/evidence
asset history
Document Store
Audit

O18 - Document Vault Upload / Version / Secure Access

Domain: Documents

Primary participants: Resident, Society Administrator, Document Service

Business objective: Store sensitive documents privately, version them, verify integrity and expose them only after fresh authorization.

Real-world scenario: A tenant uploads a rental agreement version 2. The previous agreement remains immutable version 1; admin access requires sign-in and authorization. A configured approval can include an admin digital signature.

Preconditions: Document category, owning entity, ACL, retention and size/type rules are configured.

Normal flow: Resident/admin signs in → upload initiated → malware/size/type validation → private object stored → checksum recorded → metadata/version saved → admin signs in for verification/approval where required → secure access issues short-lived retrieval → access is audited.

Exception / alternate flow: Failed upload, duplicate version, unauthorized access, expired document or checksum mismatch must be blocked and logged.

State progression: DRAFT_UPLOAD → VALIDATING → STORED → VERIFIED/REJECTED → ACTIVE_VERSION → SUPERSEDED/EXPIRED

Key invariant: Object storage remains private; every retrieval requires authorization and version integrity remains verifiable.

QA acceptance intent

- Verify the protected human actor signs in before performing privileged operations such as approval, verification, financial adjustment or NOC issuance.
- Verify the happy path creates exactly one authoritative business outcome and downstream notifications are triggered from confirmed state.
- Verify retries, duplicates and provider callbacks cannot create duplicate business effects.
- Verify exception states are visible, recoverable and auditable, without corrupting historical records.

Figure O18-A. Complete use-case diagram - participants, authenticated entry points and business interactions.

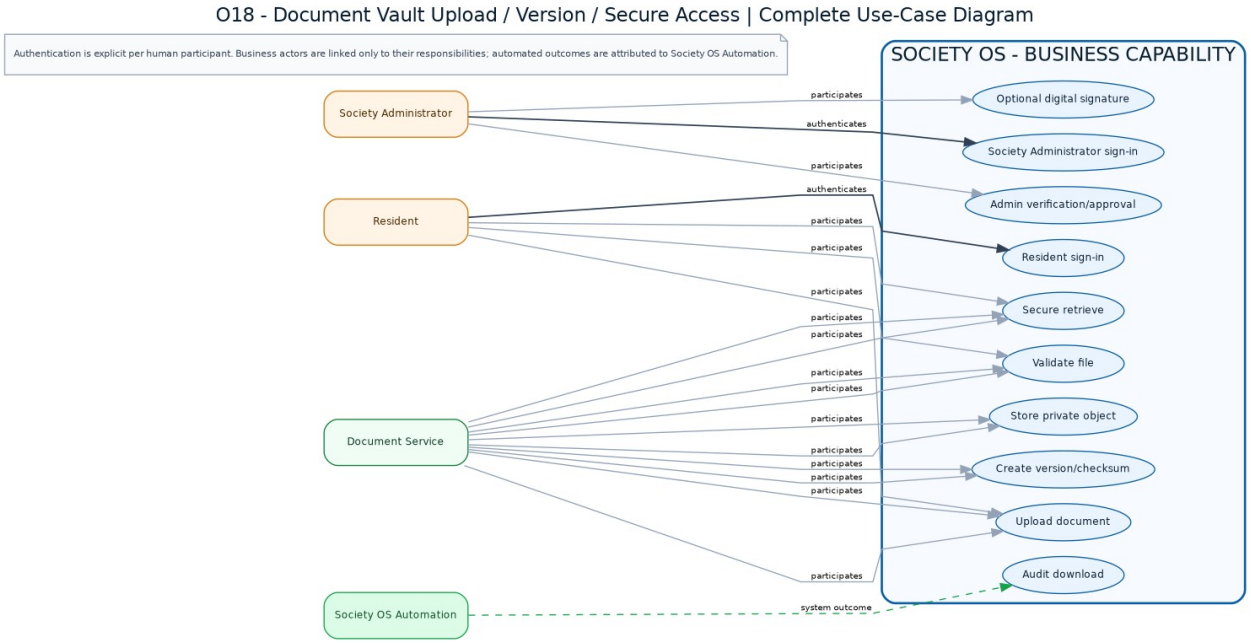
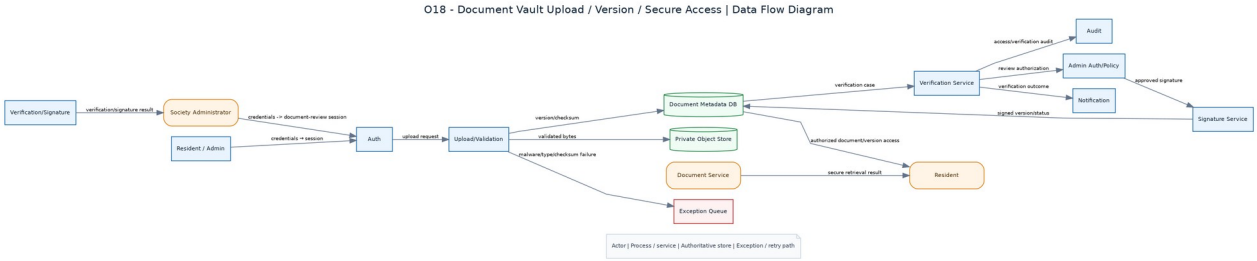


Figure O18-B. Complete data flow diagram - actors, services, authoritative stores, external boundaries and exception path.


O18 - Document Vault Upload / Version / Secure Access | Data Flow Diagram
Verification/Signature
verification/signature result
Society Administrator
credentials -> document-review session
Resident / Admin
credentials -> session
Auth
upload request
Upload/Validation
version/checksum
validated bytes
malware/type/checksum failure
Document Metadata DB
Private Object Store
Exception Queue
Document Service
verification case
Verification Service
access/verification audit
Audit
review authorization
Admin Auth/Policy
verification outcome
Notification
approved signature
signed version/status
Signature Service
authorized document/version access
secure retrieval result
Resident
Actor | Process / service | Authoritative store | Exception / retry path

O19 - Biometric Attendance Sync / Correction

Domain: Biometric / Attendance

Primary participants: HR / Administrator, Biometric Device, Employee

Business objective: Ingest attendance events idempotently and support authorized corrections without rewriting source punches.

Real-world scenario: A biometric terminal is offline for 20 minutes, stores punches locally, then uploads them. HR later corrects a missed punch with reason and audit.

Preconditions: Device is registered; integration credentials and attendance policy exist.

Normal flow: Device captures punch → local buffer if offline → reconnect → signed/batched sync → server validates device/employee/time → deduplicates → attendance record derived → admin signs in for correction if needed → corrected record versioned/audited.

Exception / alternate flow: Unknown device, duplicate event, clock drift, offline replay or unauthorized correction must be isolated.

State progression: CAPTURED → BUFFERED → SYNCED → VALIDATED → ATTENDANCE_POSTED → CORRECTION_REQUESTED → CORRECTED

Key invariant: Raw biometric templates remain outside Society OS; attendance events are idempotent and correction does not delete source history.

QA acceptance intent

- Verify the protected human actor signs in before performing privileged operations such as approval, verification, financial adjustment or NOC issuance.
- Verify the happy path creates exactly one authoritative business outcome and downstream notifications are triggered from confirmed state.
- Verify retries, duplicates and provider callbacks cannot create duplicate business effects.
- Verify exception states are visible, recoverable and auditable, without corrupting historical records.

Figure O19-A. Complete use-case diagram - participants, authenticated entry points and business interactions.

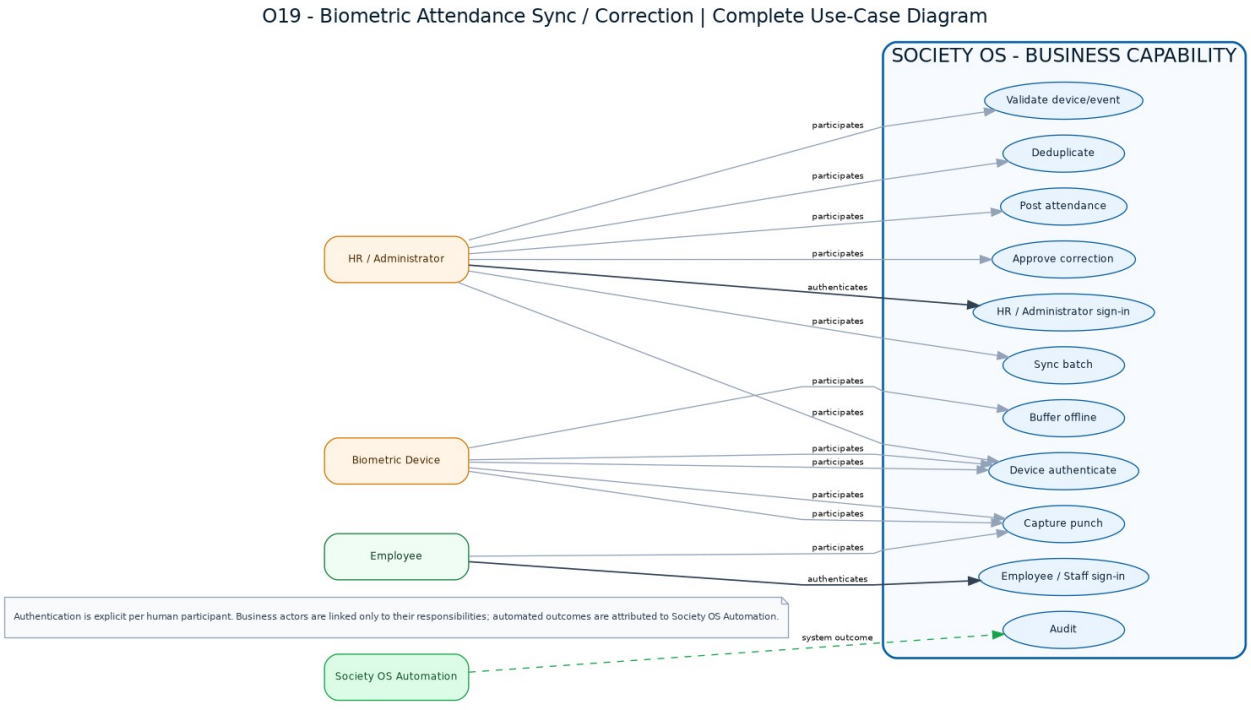
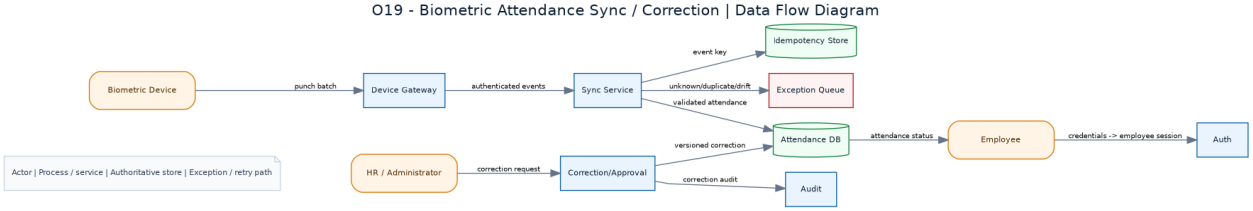


Figure O19-B. Complete data flow diagram - actors, services, authoritative stores, external boundaries and exception path.

O19 - Biometric Attendance Sync / Correction | Data Flow Diagram



O20 - Import / Export / Migration

Domain: Platform Data

Primary participants: Administrator, Migration Administrator

Business objective: Safely import and export operational masters and transactions with preview, error reporting and rollback/compensation semantics.

Real-world scenario: A society imports residents, vehicles, parking and inventory. 4,912 rows are accepted and 88 fail with downloadable row-level explanations.

Preconditions: Template and mapping rules exist.

Normal flow: Admin signs in → upload → detect schema → validate columns/business rules → duplicate check → preview → confirm → asynchronous commit → success/failure report → history retained; exports apply authorization filters.

Exception / alternate flow: Partial batch failure, duplicate import, invalid foreign key or revoked authorization must not produce silent corruption.

State progression: UPLOADED → VALIDATING → PREVIEWED → COMMITTING → COMPLETED / COMPLETED_WITH_ERRORS / FAILED

Key invariant: Every import has a batch identity and row-level lineage; exports respect privacy and authorization.

QA acceptance intent

- Verify the protected human actor signs in before performing privileged operations such as approval, verification, financial adjustment or NOC issuance.
- Verify the happy path creates exactly one authoritative business outcome and downstream notifications are triggered from confirmed state.
- Verify retries, duplicates and provider callbacks cannot create duplicate business effects.
- Verify exception states are visible, recoverable and auditable, without corrupting historical records.

Figure O20-A. Complete use-case diagram - participants, authenticated entry points and business interactions.

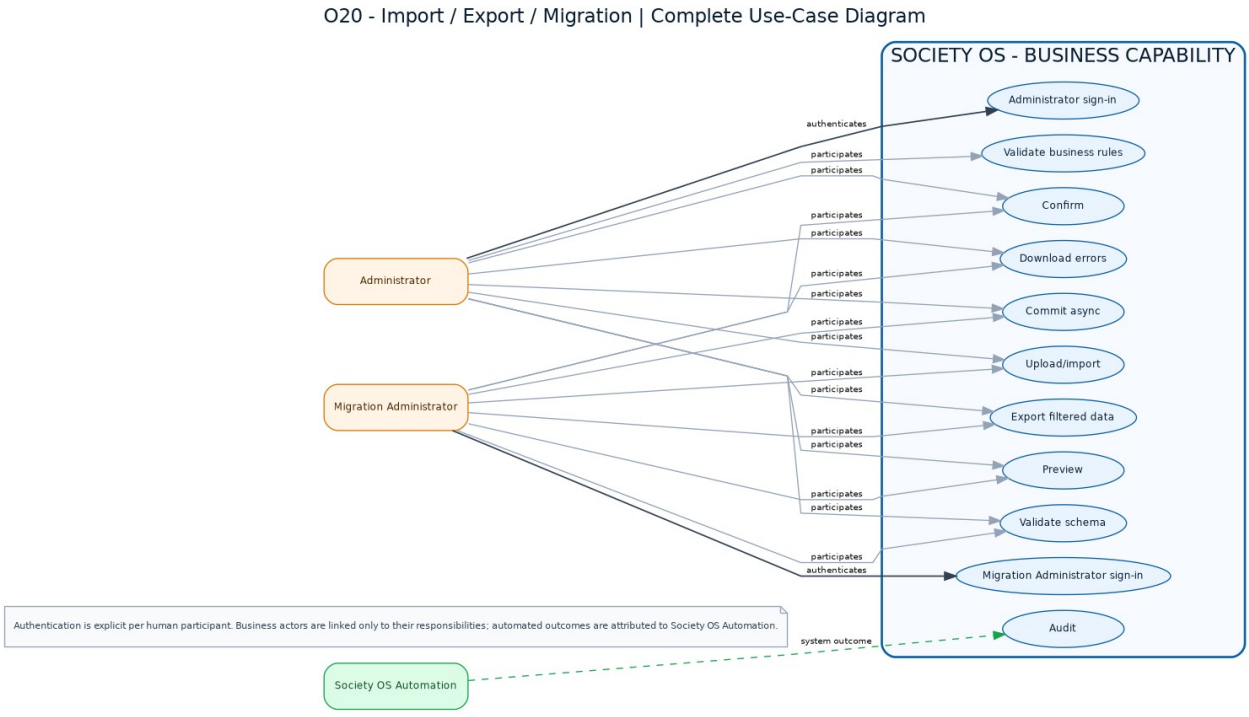
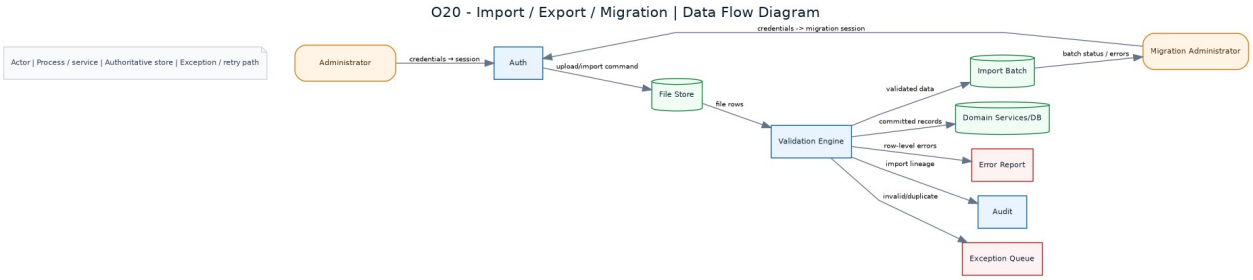


Figure O20-B. Complete data flow diagram - actors, services, authoritative stores, external boundaries and exception path.


O20 - Import / Export / Migration | Data Flow Diagram
Actor | Process / service | Authoritative store | Exception / retry path
Administrator
credentials → session
Auth
upload/import command
credentials -> migration session
File Store
file rows
Validation Engine
validated data
committed records
row-level errors
import lineage
invalid/duplicate
Import Batch
Domain Services/DB
Error Report
Audit
Exception Queue
batch status / errors
Migration Administrator

O21 - Offline Gate Sync / Conflict

Domain: Offline Operations

Primary participants: Guard Device, Resident, Server / Gate Service

Business objective: Continue critical gate operations offline and reconcile safely when connectivity returns.

Real-world scenario: A gate tablet loses connectivity for 20 minutes during a busy period. Guards capture admissible events locally and synchronize when the network returns.

Preconditions: Offline cache and device encryption enabled; device registered.

Normal flow: Guard signs in locally → records event → event queued with client UUID/time/device metadata → connectivity returns → batch sync → server authenticates/device-validates → idempotency/conflict strategy applied → authoritative state returned → local queue marked synced.

Exception / alternate flow: Visitor revocation that predates offline admission, duplicate sync, clock skew or incompatible schema must route to a domain-specific conflict outcome.

State progression: LOCAL_CAPTURED → QUEUED → SYNCING → ACCEPTED / CONFLICT / REJECTED → ACKNOWLEDGED

Key invariant: Conflict strategy is domain-specific; financial ledgers never use generic last-write-wins.

QA acceptance intent

- Verify the protected human actor signs in before performing privileged operations such as approval, verification, financial adjustment or NOC issuance.
- Verify the happy path creates exactly one authoritative business outcome and downstream notifications are triggered from confirmed state.
- Verify retries, duplicates and provider callbacks cannot create duplicate business effects.
- Verify exception states are visible, recoverable and auditable, without corrupting historical records.

Figure O21-A. Complete use-case diagram - participants, authenticated entry points and business interactions.

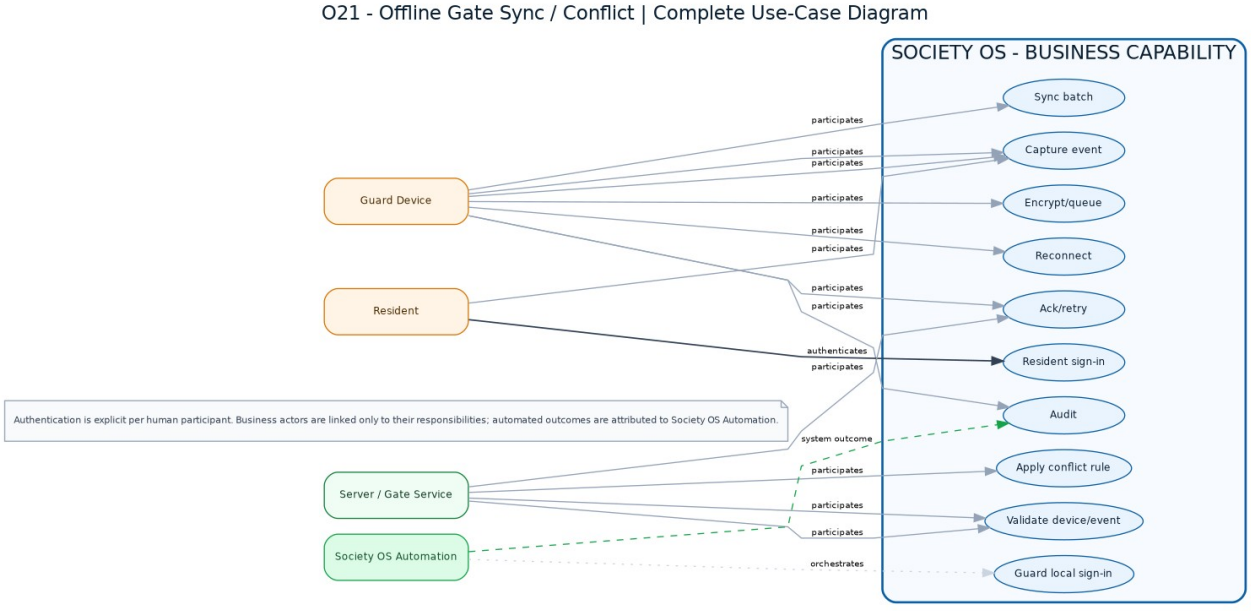
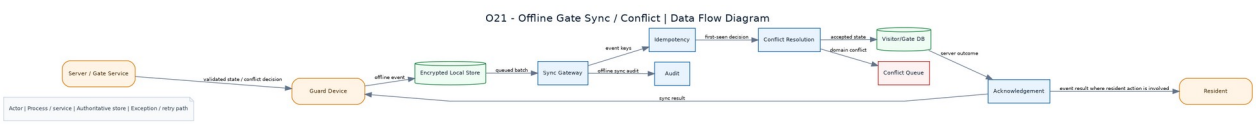


Figure O21-B. Complete data flow diagram - actors, services, authoritative stores, external boundaries and exception path.



O22 - Notification Retry / Fallback

Domain: Platform Messaging

Primary participants: Notification Engine, Notification Provider, Administrator

Business objective: Deliver important notifications reliably with retries, fallback channels, templates and observable failure states.

Real-world scenario: Payment reminders are delayed because the push provider throttles requests. The engine retries, then falls back to SMS for residents whose preferences permit it.

Preconditions: Template, user preferences, channel credentials and retry policy exist.

Normal flow: Domain event emitted → notification rule evaluated → template/localization rendered → preferred channel attempted → delivery result recorded → retry/backoff → fallback if configured → final success/DLQ → admin operations can inspect.

Exception / alternate flow: Provider outage, invalid token, quiet hours, critical override or dead-letter condition must be visible and auditable.

State progression: CREATED → QUEUED → SENT → DELIVERED or RETRYING → FALLBACK → DLQ/EXHAUSTED

Key invariant: Notification delivery must not change authoritative business state; it is a downstream effect.

QA acceptance intent

- Verify the protected human actor signs in before performing privileged operations such as approval, verification, financial adjustment or NOC issuance.
- Verify the happy path creates exactly one authoritative business outcome and downstream notifications are triggered from confirmed state.
- Verify retries, duplicates and provider callbacks cannot create duplicate business effects.
- Verify exception states are visible, recoverable and auditable, without corrupting historical records.

Figure O22-A. Complete use-case diagram - participants, authenticated entry points and business interactions.

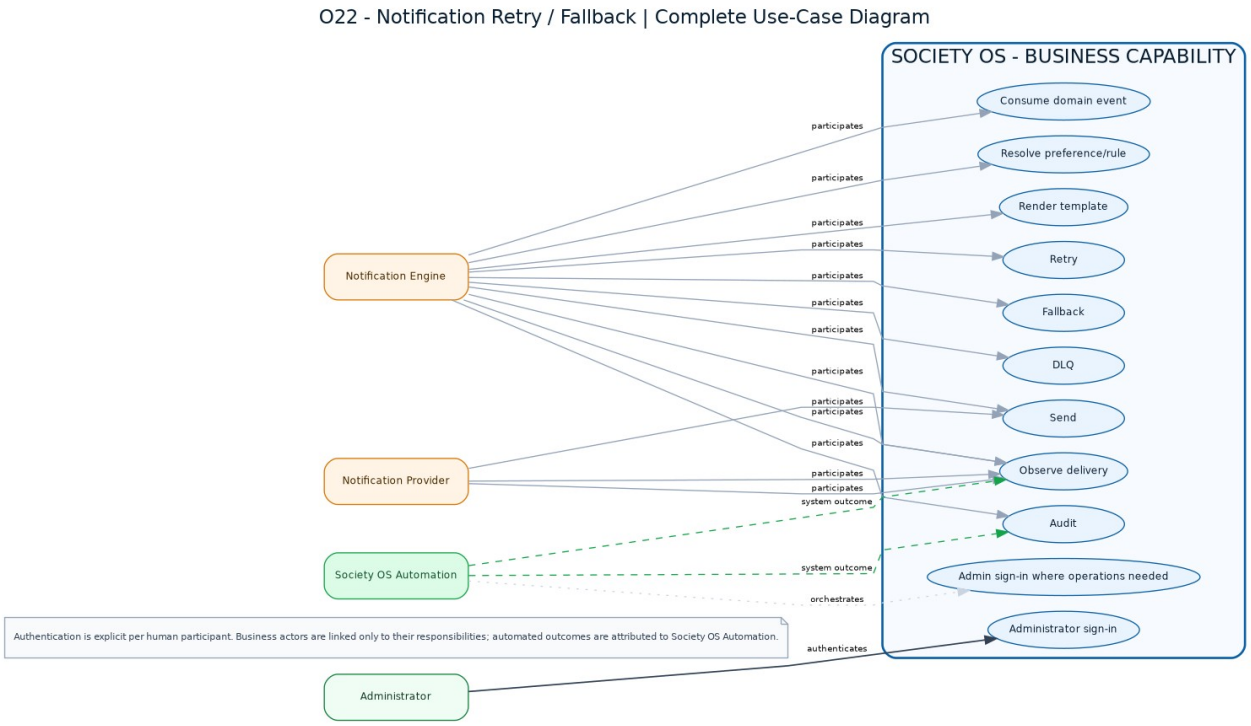
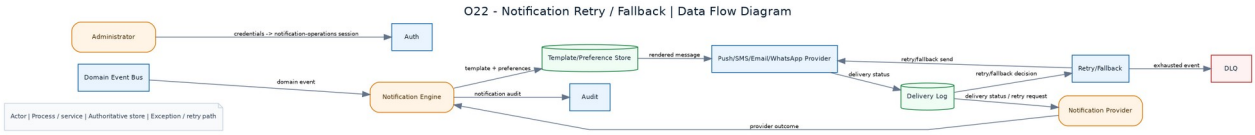


Figure O22-B. Complete data flow diagram - actors, services, authoritative stores, external boundaries and exception path.


O22 - Notification Retry / Fallback | Data Flow Diagram
Administrator
credentials -> notification-operations session
Auth
Domain Event Bus
domain event
Notification Engine
template + preferences
Template/Preference Store
notification audit
Audit
rendered message
Push/SMS/Email/WhatsApp Provider
retry/fallback send
delivery status
Delivery Log
retry/fallback decision
Retry/fallback
exhausted event
DLQ
delivery status / retry request
Notification Provider
provider outcome
Actor | Process / service | Authoritative store | Exception / retry path

O23 - Global Search and Privacy Filtering

Domain: Search

Primary participants: Authorized User, Search Service

Business objective: Provide fast cross-domain search without bypassing underlying privacy and tenant boundaries.

Real-world scenario: An administrator searches for “A-1203” and sees the unit, active occupant, vehicle and open complaint permitted by policy, but not a former tenant’s restricted KYC document.

Preconditions: Search indexes and authorization policy are available.

Normal flow: User signs in → enters search → server resolves society context and permissions → searches allowed indexes → filters/sorts/paginates → field-level privacy filters applied → results returned → access may be audited.

Exception / alternate flow: Cross-society query, restricted document, stale index or export attempt must not bypass authorization.

State progression: REQUESTED → AUTHORIZED → SEARCHED → FILTERED → RETURNED / PARTIAL / DENIED

Key invariant: Search is a projection; it must never become a side-channel around source authorization.

QA acceptance intent

- Verify the protected human actor signs in before performing privileged operations such as approval, verification, financial adjustment or NOC issuance.
- Verify the happy path creates exactly one authoritative business outcome and downstream notifications are triggered from confirmed state.
- Verify retries, duplicates and provider callbacks cannot create duplicate business effects.
- Verify exception states are visible, recoverable and auditable, without corrupting historical records.

Figure O23-A. Complete use-case diagram - participants, authenticated entry points and business interactions.

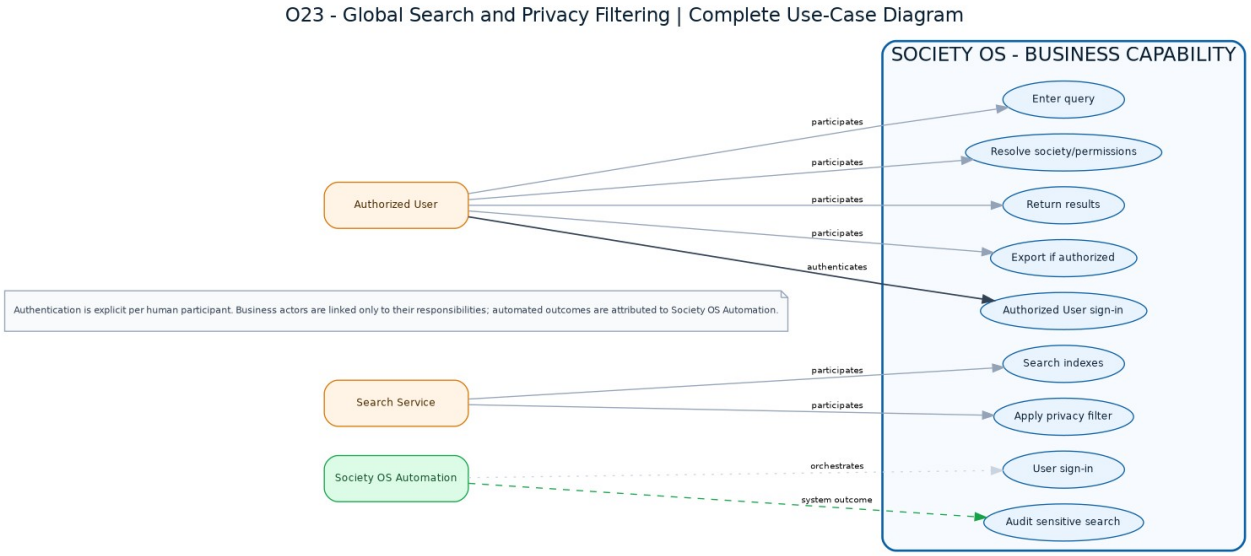
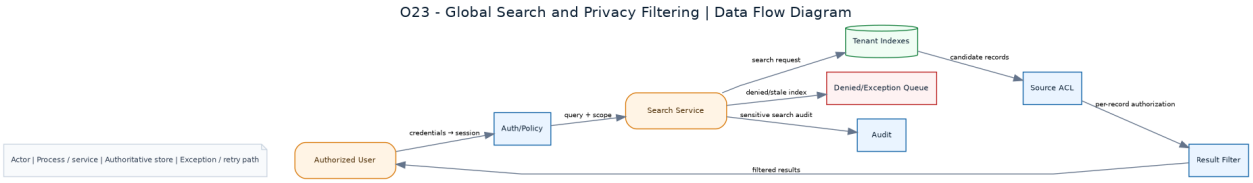


Figure O23-B. Complete data flow diagram - actors, services, authoritative stores, external boundaries and exception path.



O24 - Backup Restore / Disaster Recovery

Domain: Platform Operations

Primary participants: Operations Engineer, Storage/Database Services

Business objective: Recover the platform to a validated point while preserving tenant isolation and audit integrity.

Real-world scenario: A deployment corrupts configuration data. Operations restores the affected state to a known point, validates tenant isolation and replays safe operational jobs.

Preconditions: Backups, PITR/logs, restore runbooks and RPO/RTO targets exist.

Normal flow: Engineer signs in → declares incident → selects recovery target → restores database/object data → validates schema/checksum/RLS → replays eligible jobs → reconciles queues/audit → validates application health → closes incident.

Exception / alternate flow: Corrupt backup, partial object restore, missing audit events or replayed financial callbacks require stop/rollback/escalation.

State progression: INCIDENT → RECOVERY_PLANNED → RESTORING → VALIDATING → RECONCILING → VERIFIED → CLOSED

Key invariant: Recovery must preserve tenant boundaries and must not replay financial effects without idempotency checks.

QA acceptance intent

- Verify the protected human actor signs in before performing privileged operations such as approval, verification, financial adjustment or NOC issuance.
- Verify the happy path creates exactly one authoritative business outcome and downstream notifications are triggered from confirmed state.
- Verify retries, duplicates and provider callbacks cannot create duplicate business effects.
- Verify exception states are visible, recoverable and auditable, without corrupting historical records.

Figure O24-A. Complete use-case diagram - participants, authenticated entry points and business interactions.

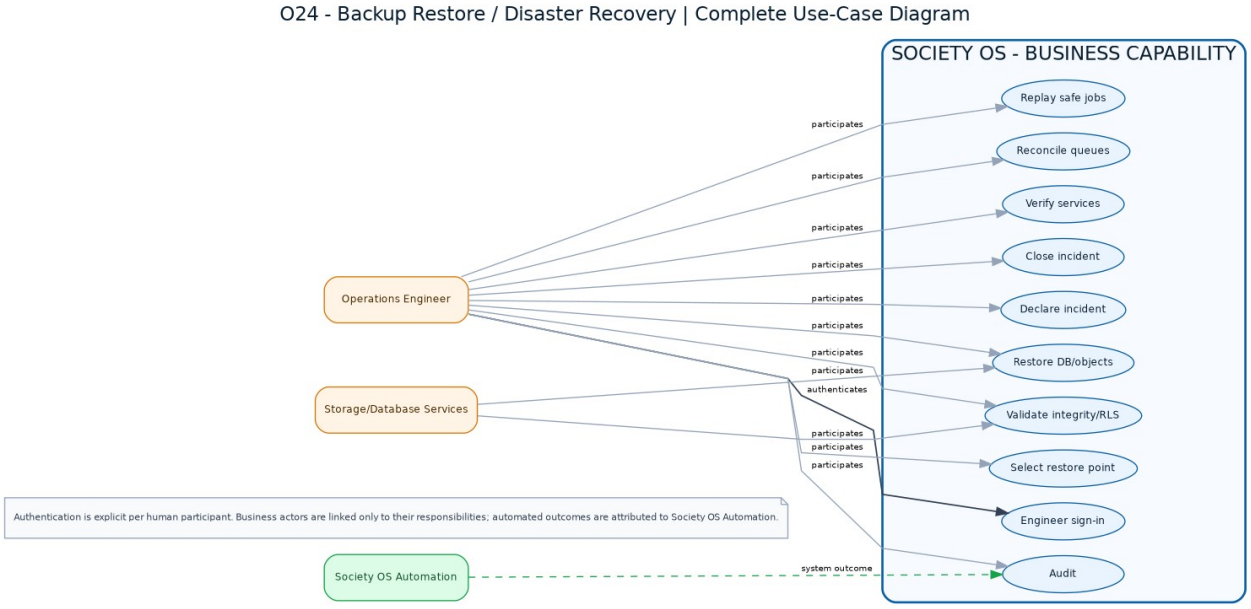
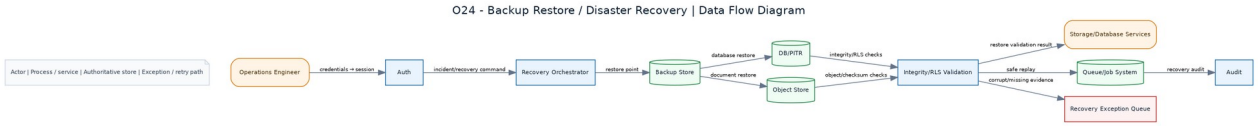


Figure O24-B. Complete data flow diagram - actors, services, authoritative stores, external boundaries and exception path.



O25 - AI-Assisted Complaint Classification

Domain: Intelligence

Primary participants: Operator, AI Service, Helpdesk Administrator

Business objective: Use AI to assist classification while keeping final business state deterministic and human-correctable.

Real-world scenario: A complaint text and photo strongly suggest water leakage. AI predicts the category and confidence, but the operator remains able to correct it before assignment.

Preconditions: AI model/version and classification taxonomy exist.

Normal flow: Operator signs in → submits complaint → AI service analyzes text/media → classification and confidence returned → operator reviews/corrects → deterministic category/SLA assigned → audit stores model/version and final human decision.

Exception / alternate flow: Low confidence, model timeout, hallucinated detail or unsafe recommendation must fall back to manual classification.

State progression: CREATED → AI_ANALYZED → REVIEWED → CLASSIFIED → ASSIGNED

Key invariant: AI output is advisory metadata; it never directly authorizes payments, access, NOC or other consequential state.

QA acceptance intent

- Verify the protected human actor signs in before performing privileged operations such as approval, verification, financial adjustment or NOC issuance.
- Verify the happy path creates exactly one authoritative business outcome and downstream notifications are triggered from confirmed state.
- Verify retries, duplicates and provider callbacks cannot create duplicate business effects.
- Verify exception states are visible, recoverable and auditable, without corrupting historical records.

Figure O25-A. Complete use-case diagram - participants, authenticated entry points and business interactions.

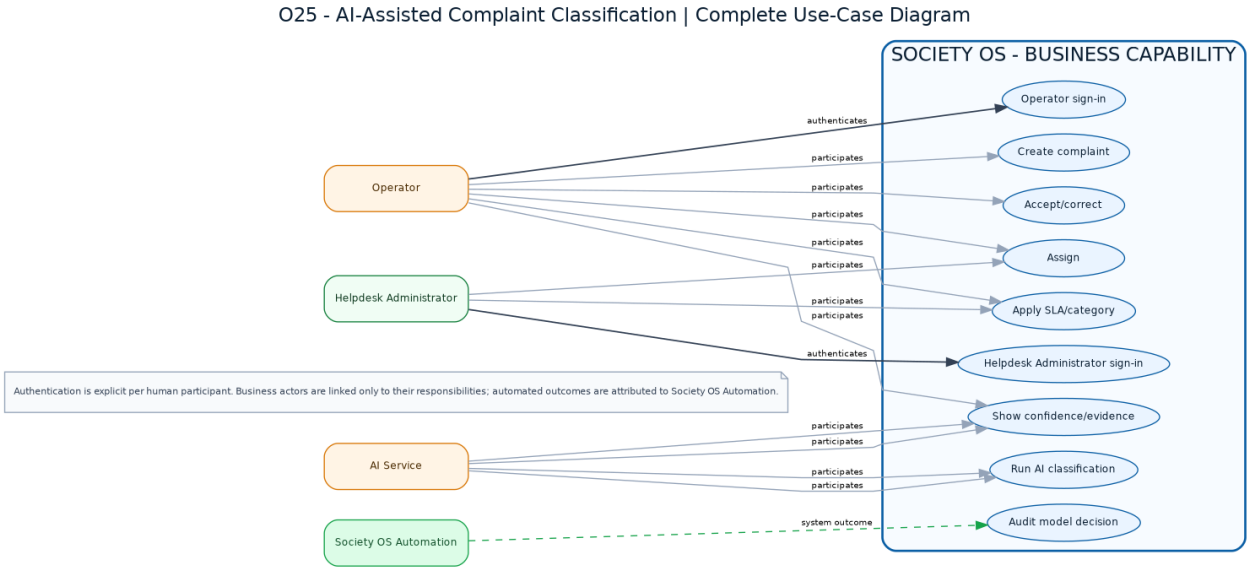
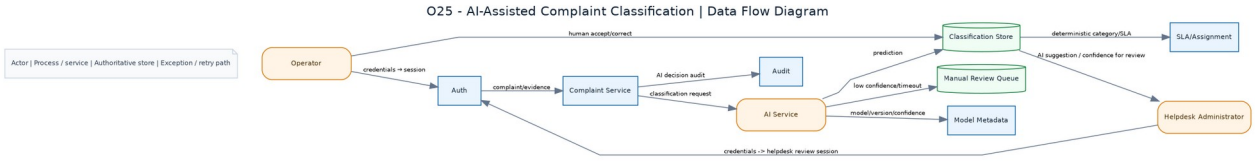


Figure O25-B. Complete data flow diagram - actors, services, authoritative stores, external boundaries and exception path.



O26 - Document Verification / Admin Approval & Digital Signing

Domain: Documents / Verification

Primary participants: Resident, Society Administrator, Verification Service, Signature Service

Business objective: Complete a formal document verification workflow in which the administrator is explicitly authenticated, reviews evidence and signs/approves the outcome when configured.

Real-world scenario: A tenant uploads a rental agreement and identity document. The admin signs in, checks the files and metadata, rejects the agreement for a missing page, the tenant resubmits, and the admin approves and digitally signs the verification outcome.

Preconditions: Document category, verification checklist, approval threshold and signature method are configured.

Normal flow: Resident signs in → uploads document → file validation/checksum → verification case created → admin signs in → opens verification case → reviews document/metadata → accept/reject/resubmit → if approved, admin signs approval/certificate where configured → verification status recorded → resident notified → audit trail sealed.

Exception / alternate flow: Expired credential, tampered file/checksum mismatch, unauthorized verifier, signature failure or resubmission timeout keeps the case pending/rejected and never marks the document verified.

State progression: UPLOADED → VALIDATING → PENDING_ADMIN_REVIEW → RESUBMISSION_REQUIRED or VERIFIED → SIGNATURE_PENDING → SIGNED → ACTIVE

Key invariant: Only an authenticated authorized admin/verifier may approve; verification and signature are separate auditable events.

QA acceptance intent

- Verify the protected human actor signs in before performing privileged operations such as approval, verification, financial adjustment or NOC issuance.
- Verify the happy path creates exactly one authoritative business outcome and downstream notifications are triggered from confirmed state.
- Verify retries, duplicates and provider callbacks cannot create duplicate business effects.
- Verify exception states are visible, recoverable and auditable, without corrupting historical records.

Figure O26-A. Complete use-case diagram - participants, authenticated entry points and business interactions.

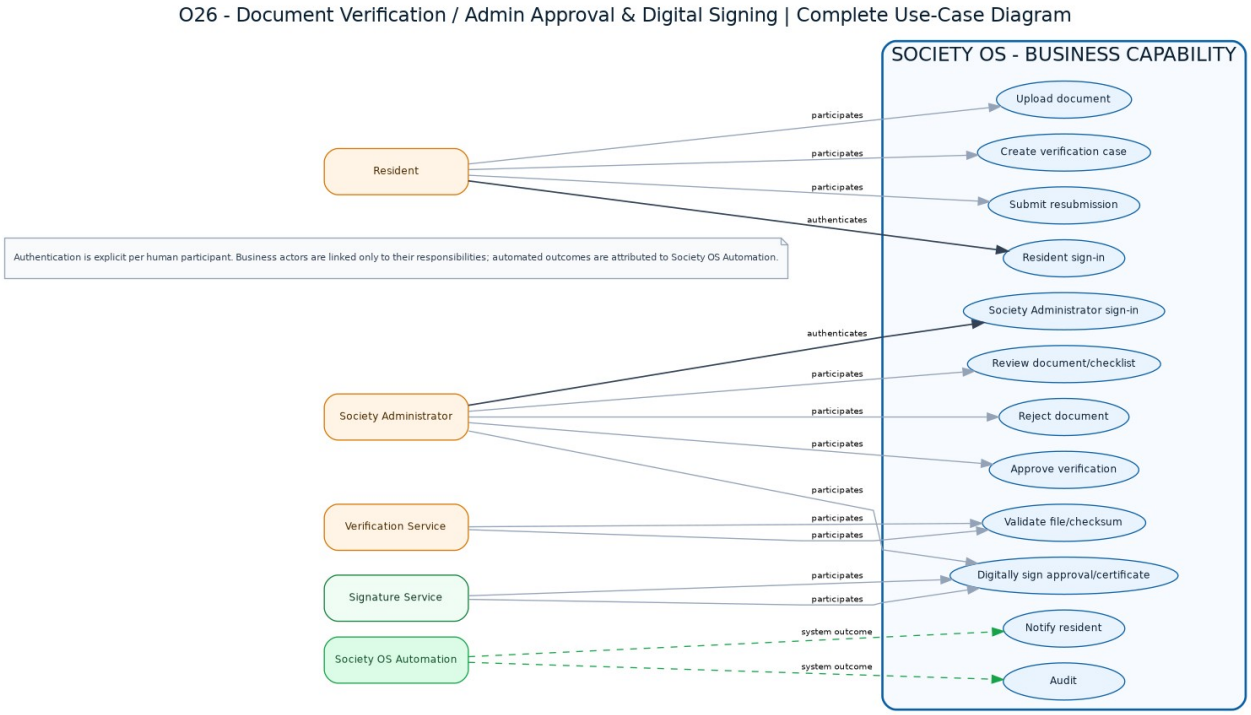
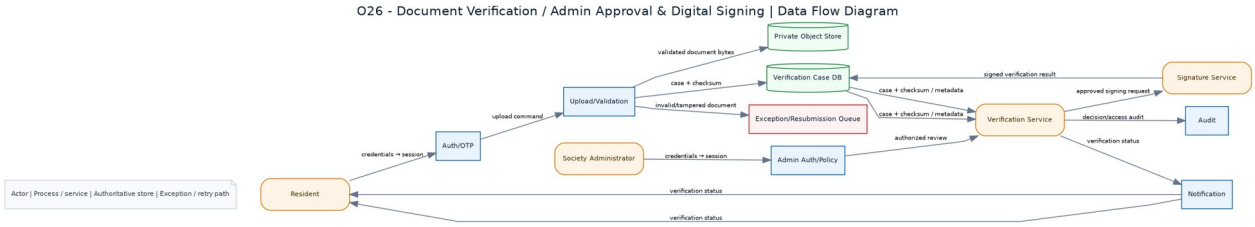


Figure O26-B. Complete data flow diagram - actors, services, authoritative stores, external boundaries and exception path.



19.11 Financial Scenario Completion Standard

- Finance is the hero product domain and must be treated as a controlled accounting system, not simply a collection of payment screens or gateway integrations.
- Every financial scenario explicitly identifies source transaction, calculation/matching rule, approval boundary, ledger effect, reconciliation state, correction/reversal path, notification effect and audit evidence.
- Every posted journal is balanced; posted history is append-only; corrections are linked reversal/adjustment transactions; refunds reference originating payments; duplicate callbacks are idempotent; reconciliation never edits authoritative ledger history.
- Every financial reportable number is traceable to source transactions and effective-dated rules so that historical periods remain reproducible.
- Financial exception states must end in a deterministic operator action: retry, wait for provider, reconcile, approve, reverse, refund, dispute, escalate or close.

19.12 Explicit Document Verification Control

Mandatory rule: Document verification is not complete merely because a document was uploaded. The administrator/verifier must authenticate into the system before opening the verification case, perform the configured checklist, record approve/reject/resubmit, and - where digital approval/signature is configured - execute the signature action as a separate auditable event.

Why this matters: This prevents the common diagram gap in which a document jumps directly from resident upload to VERIFIED without showing the authenticated administrative decision, the verification service, the approval/signature boundary and the resulting audit evidence.

SOCIETY OS

Document Trust, Verification & Automation Engine

Production-grade engineering addendum to the Society OS consolidated enterprise blueprint

Revision v4.4 | 04 November 2026 | Proposed implementation specification

Document control	Value
Baseline assessed	Society OS_Documentation.pdf: 133-page consolidated blueprint; cover labelled v4.0, body includes v4.3 expansion notes.
Older reference	Society_OS_Enterprise_Product_Blueprint_v3(1).docx: superseded conceptual product baseline; preserved separately.
Change scope	New detailed requirements for secure generated documents, uploaded-evidence checks, KYC intake, NOC issuance, procurement/RFQ/tender packages, expiry/compliance, APIs, security, QA and low-cost self-hosted processing.
Effective relationship	This addendum supplements existing blueprint sections; no unrelated requirement is removed. A conflict must be resolved through product change control before implementation.
Owner / audiences	Product, Engineering, Society Operations, Security, QA, Compliance, Finance and Implementation teams.
Implementation status	SPECIFIED ONLY. Nothing in this document asserts that the application code is implemented, audited or legally certified.
Non-negotiable constraint	Core operations must run with ZERO paid OpenAI or other third-party LLM API calls. Self-hosted OCR is optional; deterministic checks and authorized people remain the decision makers.

Executive decision

Society OS requires a reusable Document Trust & Automation Engine rather than scattered PDF-generation endpoints. Each issued document is a governed business outcome backed by authorized records, policy/eligibility checks, multi-party decisions where required, a frozen evidence snapshot, controlled PDF generation, issuance authentication, and revocation/verification support. Uploaded documents can be checked for format, structural integrity and cryptographic signatures when a trusted standard exists; the system MUST NOT claim authenticity of arbitrary IDs, ownership deeds, police checks or GST registrations without authoritative verification.

Crosswalk to retained master blueprint

Existing master section / location	v4.4 augmentation
7.1 onboarding and 8 occupancy	Purpose-limited identity evidence; ownership/tenancy and household document policies.
7.11 procurement, p.31	RFQ/tender release, vendor bidding, bid locking, comparative statements, selection and three-way matching.
10 Document Vault and NOC, pp.37-38	Whole new issuance, verification and evidence lifecycle; signatures, status and revocation.

Existing master section / location	v4.4 augmentation
11 reusable workflows / 12 APIs and DDL	Policy engine, state machines, API contracts, entity models, event/outbox and concurrency.
14 privacy / 16 automation / 18 acceptance	Identity-proof controls, legally meaningful verification levels, zero-cost stack, failure behavior and test suites.

Authority: the baseline blueprint remains the primary product scope. This v4.4 supplement specifies previously underspecified document/verification behavior. Screen designs and final role-permission matrices remain in their dedicated specifications, consistent with the master blueprint.

Contents / engineering reading order

20. Capability scope and document catalogue
21. Canonical document lifecycle and workflow states
22. Document-template and PDF issuance subsystem
23. Verification taxonomy and no-paid-API operating model
24. Identity, tenancy, ownership, staff and vendor evidence
25. NOC and administrative certificate scenarios
26. Tender, RFQ, procurement and contract records
27. Compliance, expiries and recurring document obligations
28. Reference architecture, execution and integration events
29. Data model, persistence and integrity rules
30. API and event contracts
31. Privacy, legal, signatures and threat model
32. Self-hosted implementation options and operating costs
33. Product experience and localization outcomes
34. Test, performance, reliability and release acceptance
35. Prioritized delivery backlog and rollout
36. Decision log, ownership and reference sources

Normative language

MUST is a security, correctness or required business contract; SHOULD is the recommended implementation unless a documented exception is approved; MAY is optional. “Verified” is a narrowly defined, evidence-backed assertion. “Approved” is a business decision; “Generated” is only a file-production event; “Issued” is an authorized immutable record. None of these is interchangeable.

Non-goals and truthful scope

- No automatic government database validation through an unavailable or unofficial endpoint. No invisible background scraping of government portals.
- No claim that OCR, image recognition, name similarity, checksum, PAN/GSTIN syntax or manual stamping proves identity or title.
- No unsolicited collection of full Aadhaar number, raw biometrics, facial templates or indefinite ID photocopies.
- No creation of a government, statutory, legal or CA-issued certificate by a society when the society lacks authority to issue it.
- No issuance of legally required qualified/CA-backed electronic signatures via a drawn signature image, simple QR or SHA-256 hash.

20. Capability scope and document catalogue

Every document type MUST be backed by a configurable DocumentTypePolicy: purpose, legal/issuer boundary, eligible entity, requester relationships, required inputs/evidence, checks, approval path, content template, serial

pattern, delivery audience, expiry, retention, privacy and verification mode. The catalogue below describes proposed product capabilities, not a universal legal requirement for every society.

Document family	Examples / generated output	Minimum check	Truth / legal boundary
Residence / ownership	Residence/occupancy certificate; membership confirmation; address letter	Active relationship + documented evidence	Society-authored factual letter; do not certify legal title
Owner / property	Share certificate copy request; share transfer acknowledgment; ownership-change record	Registered member, transfer workflow, administrator review	Do not replace statutory registered deeds/certificates
Rent / tenants	Tenant registration confirmation; tenant entry NOC; tenancy expiry reminder	Owner consent, agreement dates, configured KYC checks	Police verification is external evidence, not a society-issued result
Move-in	Move-in authorization letter; lift/truck booking pass; checklist	Clear move-in prerequisites, booking and deposits	Access starts only from authorized effective timestamp
Move-out	Move-out NOC; no-dues certificate; handover/clearance summary	Fresh financial, facility, parking, access and disputes checks	Block where mandatory clearance fails
Maintenance / finance	Maintenance demand notice; invoice; receipt; payment statement; dues letter	Published ledger, settled payment and rule version	Receipts only after authoritative posting
Parking / vehicles	Parking allocation letter; temporary permit; parking clearance	Active allocation, vehicle/unit relationship, dues/violations	Expiry and revocation required
Renovation	Renovation commencement NOC; worker pass; completion/return of deposit	Society rules, timing, deposits, contractor and damage inspection	Do not imply municipal construction permission
Utilities	Water connection request/approval; meter reading statement; EV charger permission	Eligibility, capacity, technical inspection, charges	External utility authority approval remains independent
Facilities	Facility booking confirmation; deposit receipt; usage/damage report	Booking state, settled payment, venue restrictions	Do not treat pending booking as confirmed
Governance	AGM/SGM notice; agenda; minutes; resolution extract; voting result	Approved meeting record, configured quorum/decision rules	Signing and member eligibility controlled
Complaints	Complaint acknowledgment; resolution report; mediation settlement	Case timeline and evidence; party permissions	Sensitive case content not widely published
Security	Visitor pass; material-gate pass; incident report; loss/damage record	Valid guest request and security approval	QR pass is not a government identity proof
Staff / workforce	Employment/engagement letter; staff ID slip; attendance statement	Approved staff assignment and verification status	Platform ID does not certify police verification
Vendor	Vendor onboarding checklist; engagement letter; AMC reminder; work completion	Vendor profile, insurance/licence/contract status, work order	GST/PAN format checks only unless authority checked
Procurement	Purchase requisition; RFQ; RFP/tender pack; bid summary; comparison sheet	Approved scope, budget and procurement rule pack	Tender is society-issued, not public-government tender
Procurement / contracts	Purchase order; work order; goods receipt; service acceptance; invoice approval	Approved sourcing, vendor, three-way verification	Separate requester, approver, vendor submitter
Compliance	Fire/lift/insurance licence tracker; certificate request; renewal report	Uploaded external certificate and expiry audit	Society stores/tracks, does not forge regulator issuance

Document family	Examples / generated output	Minimum check	Truth / legal boundary
Emergency	Incident summary; drill record; emergency response report	Incident source, approved report and redactable PII	No automatic emergency-cause claim
Legal / financial requests	Loan/charge/no-objection response; society confirmation to bank	Authority, financial check and committee approval	Bank/legal adviser review as required
Society administration	Circular, policy amendment, disciplinary notice, official correspondence	Authorized publisher, versioned policy and audience	No impersonation of public authority
Internal platform	Verification result receipt; document audit pack; export manifest	Evidence provenance, actor, timestamp, access scope	No embedded raw secrets in exports

20.1 Document kinds: separate generation from validation

Class	Canonical input and output	Allowed claim
A - Society-issued	Template + authoritative records + approval => controlled PDF	Society issued this version on this date; current status known
B - Submitted evidence	Private upload + malware/schema validation + review	Document received; integrity and review status known
C - Externally issued / locally cryptographically signed	Trusted signing certificate / approved offline verification format	Signature validates over these exact bytes, with declared signer/trust context
D - Externally authoritative, not electronically verified	Scans of PAN, voter ID, police report, agreement, sale deed, GST record	Automated structure check only; independent authority/manual verification may be required
E - Operational export	Deterministic snapshot of permitted records + export authorization	Records exported at cutoff, not certified by a third party

20.2 Default minimum workflow for all document types

1. DocumentTypePolicy configured by qualified administrator with version, legal category and template.
2. Request scoped to authenticated society/unit/object context; purpose and allowed requester checked.
3. Required files and master-data references collected; uploads quarantined until validated.
4. Deterministic checks and reviewer tasks run on immutable evidence versions.
5. Approvers resolve from policy; maker-checker separation applied to high-impact types.
6. A final issuance transaction re-checks time-sensitive records; snapshot and serial number reserved atomically.
7. Immutable signed/controlled PDF bytes created; SHA-256, template/rule version and approvers stored.
8. Access policy and verification status activated; event/outbox publishes notification.
9. Revocation, expiry, supersession, corrections and retention continue after issuance.

21. Canonical document lifecycle and workflow states

21.1 Request, uploaded evidence, issuance and verification are different state machines

Aggregate	State path (illustrative)	Forbidden shortcuts
DocumentRequest	DRAFT -> SUBMITTED -> PRECHECK -> REVIEW -> APPROVAL_PENDING -> APPROVED -> ISSUING -> ISSUED; side paths NEEDS_INFO, REJECTED, CANCELLED, EXPIRED	ISSUED without approval and fresh rules; REJECTED directly to ISSUED

Aggregate	State path (illustrative)	Forbidden shortcuts
EvidenceFile	UPLOADING -> QUARANTINED -> SCAN_PASSED -> VALIDATED -> REVIEWED; side paths UPLOAD_FAILED, MALWARE_BLOCKED, INVALID, SUPERSEDED	PUBLIC/REVIEWABLE before scanner passed
IssuedDocument	ISSUED -> ACTIVE -> EXPIRED / REVOKED / SUPERSEDED; optional correction creates a NEW version	Mutation of already issued bytes or serial
VerificationAssessment	NOT_STARTED -> RUNNING -> PASSED / PARTIAL / NEEDS_REVIEW / FAILED / NOT_SUPPORTED / UNAVAILABLE	UNAVAILABLE translated to PASSED
Tender	DRAFT -> APPROVED -> PUBLISHED -> BIDS_OPEN -> SUBMISSION_CLOSED -> OPENED -> EVALUATED -> AWARDED / CANCELLED	Submission after deadline; bid visibility before authorized opening

21.2 Trigger / transition discipline

- Every mutation MUST provide aggregate version (If-Match / optimistic lock), tenant context, actor, scope and idempotency key when retriable. Invalid transitions return 409 or 422 with machine-readable details.
- Deadline-sensitive, eligibility-sensitive and financial values MUST be read from authoritative services at the approval/issuance boundary, not trusted from a stale approval UI.
- Events emitted only by a durable outbox in the same transaction as the authoritative transition; workers retry with deterministic idempotency and dead-letter inspection.
- Cancellation can withdraw a request before issuance; afterwards only revoke/supersede a recorded issued document. No record-level hard deletion of issued versions.
- Blocking policy failure should attach evidence ID, rule ID, reason, next action and as-of timestamp. Sensitive causes remain redacted in resident-facing notifications.

21.3 Duplicate NOC submission

Situation: Resident double-clicks Request NOC while mobile connection stalls.

Normal path: Both calls contain same tenant-scoped Idempotency-Key; backend returns same requestId and current state.

Exceptions: If distinct keys target the same active lifecycle, unique business constraint returns existing request or explicit conflict.

Invariant: Never generate two active NOCs from one cleared move-out without a new approval cycle.

21.4 Financial race during issuance

Situation: NOC approved at 10:00; new penalty is posted at 10:01 before issuance.

Normal path: Issue command obtains rule-relevant snapshot/lock, re-evaluates blocking balance, records snapshot cutoff and issues only when clear.

Exceptions: If balance changes, return NEEDS_RECHECK and require decision again.

Invariant: Approval alone must not bypass late financial change or issue stale no-dues certificate.

21.5 Revoked document verification

Situation: A previously issued renovation permission is revoked after a serious violation.

Normal path: Issuer marks document REVOKED with reason and authority; QR status endpoint immediately returns revoked/verification failed.

Exceptions: Old downloaded PDF remains printable and byte-valid but is no longer currently valid.

Invariant: Content hash proves byte integrity only; current lifecycle status must be checked separately.

22. Document-template and PDF issuance subsystem

22.1 Versioned templates with deterministic merges

- Template identity: templateCode, documentType, jurisdiction, locale, society branding, version, effective dates, active flag, content checksum and policy binding.
- Use allowlisted merge tokens mapped to backend DTOs only (society.displayName; unit.number; approvedSignatory.name; issuedAt; balanceSnapshot), never executable expressions, file paths or arbitrary SQL.
- Author templates as sanitized structured HTML or XSL-FO with controlled fonts, a fixed approved base layout and accessible print/PDF render. Escape all user-supplied values.
- Required clauses, annexure list, conditions, privacy notice and date formats are governed by jurisdiction/society policy; translations are approved human-authored variants.
- Preview is watermarked PREVIEW - NOT VALID; no live issuer signature or active verification code, and download permissions are scoped.
- PDF/A or accessible tagged PDF may be configured when supported/tested; do not claim either without actual validation. PDF must embed permitted font subsets and support readable multilingual text.
- Stable document IDs separate from display serial numbers. Sequential visible numbering is transactional per society + document type + financial year where configured. Deleted/revoked docs never free a used serial.
- Signature modes: (1) approved officer name + organizational digital seal/QR trust in platform; (2) CA-backed digital signature when legally required and a licensed integration exists. Modes must be labeled honestly.

22.2 Required fields of an issued document

Field	Purpose
Issuer + legal role	Society registered name, address, registration reference, issuer role (not government unless applicable).
Document type + unique serial	Human reference number + immutable version UUID.
Subject/scope	Authorized person/unit/vehicle/work order and effective dates.
Basis and scope clause	What facts the society checked and exact limits of the assertion.
Approver and issue timestamp	Identity/authority at issuance, timezone Asia/Kolkata or declared locale.
Validity	Valid from/to and conditions; expiry is explicit or N/A as policy permits.
Verification instructions	QR/short code to official first-party Society OS verification endpoint.
Evidence anchor	Link only within secure portal to the frozen evidence snapshot; no private file URLs.
Current lifecycle disclaimer	Public page shows status; printed PDF can be revoked later.
Footer classification	Confidentiality, contact process and document-version terminology.

22.3 File creation and anti-tamper controls

1. Prepare a canonical data snapshot from authorized database reads (policyVersion, templateVersion, evidenceIds/checksums, approvals, ledger position, cutoff).
2. Obtain an issuance reservation with tenant-scoped unique request/version constraint; reserve visible serial in transaction.
3. Render PDF from a sanitized pre-approved template in isolated worker; apply verification QR that contains only opaque document token/URL, no personal IDs.

4. For CA digital signing, sign the finalized PDF using approved private-key custody and verify CMS/PAdES outcome; otherwise store server-trust HMAC/QR and clearly label “Platform-verified issuance,” NOT legal eSign.
5. Calculate SHA-256 over final immutable PDF bytes; store object with immutability/versioning and checksum; publish issuance as atomic metadata state only after file is durably stored.
6. Perform asynchronous reconciliation for orphan objects/reservations; avoid duplicates; on failure leave request ISSUING_FAILED with safe retry.
7. Log actor, input snapshot digest, template/rule versions, issued bytes digest, signature validity data, trace id, timestamps and notification outcome.

22.4 PDF sample copy and assertions

“Issued by [Registered Society Name] on [Date] for [Unit/Subject]. This document confirms only the records and approvals specifically listed herein as at [Verification Cutoff]. Scan the QR code to check the current status and authenticity of the Society OS issue record. This document is not a government-issued identity or title certificate.”

23. Verification taxonomy and zero-paid-API operating model

23.1 Five independently reported assurance levels

Level	What is actually checked	Appropriate status
L0 Receipt	File exists, object scan complete, checksum and permitted MIME	RECEIVED / CLEAN / INVALID
L1 Structural	Schema, readable dates, mandatory fields, OCR-extracted candidate values, visual completeness	STRUCTURE_PASS or NEEDS_REVIEW
L2 Local cross-record	Name, unit, dates, issuer description, ledger/owner/lease policy, duplicate hashes	MATCH / MISMATCH / INCONCLUSIVE
L3 Offline cryptographic	Trusted issuer signature, certificate chain and signing format with current trust material	CRYPTO_VALID / INVALID / UNKNOWN
L4 Authority / manual	Explicit external authority check, verifiable portal response or documented human review	SOURCE_VERIFIED / HUMAN_REVIEWED / NOT_VERIFIED

L0-L3 do not automatically imply L4. VerificationAssessment MUST retain checkType, rulesVersion, engineVersion, source, artifactHash, pass/fail/unknown, assessedAt, reviewer (if any), and check limitations. A human mark must never be displayed as “government verified”.

23.2 Checks feasible wholly within Society OS

Document / claim	Local no-paid-service check	Not proven locally
Society-issued NOC	First-party serial lookup, server-signed QR token, immutable PDF checksum, status	Legality or third-party recognition of contents
Aadhaar secure QR or offline XML	UIDAI public-key signature verification subject to latest standards and authorized consent	Live UIDAI authentication, current address or physical presenter identity
Rental agreement	Page/text availability, party names and dates extracted, unit match, expiry rules	Valid execution, stamp duty, police verification, forged signatures
Sale deed / share certificate	Required pages, dates and seller/unit names; metadata and hashes	Current title or encumbrance status
PAN	Character pattern, checksum only if officially applicable, name comparison	Tax department database ownership/authenticity

Document / claim	Local no-paid-service check	Not proven locally
GSTIN	Checksum/format consistency and PAN-shaped embedded segment	GST portal active registration status
Driving licence / voter / passport	Candidate structural/date checks if user provides data	Issuing authority authenticity or government verification
Police verification receipt	Evidence uploaded, reference/date capture, manual officer review	Police issued/approved, without trusted authority response
Vendor contract / insurance	Signed-PDF crypto validation if applicable; expiry and named-party comparison	Insurer-issued current policy unless verified with issuer
Fire or lift certificate	Certificate type, dates, society address, issuer text consistency	Government department registry validity without authority access
Tender bids	Timely submission, sealed file checksum, policy eligibility and arithmetic recomputation	Vendor truth of references / real-world capability
Maintenance receipt	Authoritative payment and ledger posting, receipt-to-allocation link	Bank settlement prior to authoritative callback/reconciling

23.3 Tooling and execution policy

- Baseline validation requires only deterministic software and first-party database queries; it works without any LLM model.
- ClamAV or equivalent local antivirus scans uploads; Apache Tika/PDFBox parse files with sandbox and time/size budgets. Malware scan failure is fail-closed: quarantined, not reviewed/issued.
- OCR is OPTIONAL for extracting candidate information: self-hosted Tesseract with language packs and trained models, no outside call. OCR scores are heuristics, not identity authenticity.
- QR/barcode ZXing or similar reads first-party document tokens and recognized safe data; untrusted QR URLs must never be fetched server-side without SSRF restrictions.
- UIDAI public-key verification must use authoritative trust anchors, test vectors and evolving formats; do not implement ad-hoc signature parsing.
- Document classification and semantic text comparisons may use local rules/dictionaries first; optional open models can be introduced only after documented CPU/RAM, safety and privacy tests.
- Avoid AI-generated legally binding wording for NOCs, tender clauses and committee decisions. These must come from approved templates and current policies.
- Assurance UI shows explicit NOT_CHECKED / NOT_SUPPORTED / MANUAL_REQUIRED when source verification is unavailable.

23.4 Aadhaar offline signed evidence

Situation: A tenant voluntarily supplies UIDAI offline signed XML or secure QR for an allowed verification purpose.

Normal path: Obtain purpose-specific consent; use official signed payload schema and trusted UIDAI public certificate; validate signature; optionally compare disclosed name/photo to presenter through authorized human workflow; store only minimized result.

Exceptions: Invalid format, unavailable trusted cert, or signature failure => NEEDS_REVIEW/FAILED; non-Aadhaar alternative must remain available.

Invariant: Never call this “online Aadhaar authentication”; never store full Aadhaar or raw biometrics.

23.5 Forged-looking PAN photo

Situation: Vendor uploads a PAN photograph and entered PAN-like number.

Normal path: Validate syntax, compare entered name, store only reviewed evidence status, ask for authorized follow-up when needed.

Exceptions: OCR may misread P/O/0; flag mismatch. No system claim of official authenticity without permissible verification source.

Invariant: Successful PAN pattern check is only structural, not tax authority confirmation.

24. Identity, tenancy, ownership, staff and vendor evidence

24.1 Evidence policy per subject and purpose

Subject / event	Minimum candidate evidence (configurable)	Authorized verification
Owner linking	Registered ownership/share proof, current society membership, relationship claim	Existing society record + manual/committee review if needed
Co-owner / heir	Relationship proof, society-approved change or lawful transfer evidence	Society governance and legal review; never infer title from OCR
Tenant move-in	Owner approval, lease/agreement metadata, ID alternative, police confirmation where applicable	Signed evidence integrity + owner consent + policy check; police only if substantiated
Family / caregiver	Owner/tenant sponsorship, effective dates, minimal ID evidence when needed	Relationship authorization, no ownership rights by default
Domestic worker	Employer association, permitted ID proof, emergency contact, access dates	Manual verification process; do not store raw biometrics
Vendor / contractor	Company identity, GSTIN/PAN where needed, licences, insurance, bank details, work authorization	Separate approvals; bank account changes heightened checker
Short stay	Host declaration, dates, ID only if allowed/required and proportionate	Time-bound access with auto-expiry and clear retention

24.2 Personal data minimization and consent contracts

- Identity proof MUST NOT be universal requirement for simple notices or routine resident actions; collect only when a documented purpose and society policy supports it.
- Aadhaar offline verification is voluntary; provide acceptable alternate methods, record consent and prescribed notices, and follow applicable UIDAI OVSE obligations. Do not collect/store full 12-digit Aadhaar numbers or biometrics in this module.
- If a proof image contains unnecessary identifiers, allow client-side masked/redacted submission, server-side protected previews, and authorized redaction of derivatives while retaining strictly justified originals only when required.
- Separate verifier access from applicant, gate guard and platform support; later unit occupants cannot access previous tenants' KYC.
- Use manual reviewer notes with restricted purpose; do not place original evidence in ordinary push messages or email attachments by default.
- Prevent fraudulent verification via self-approval: document uploader cannot be sole reviewer for high-risk identity changes; administrative overrides include reason and second authorization where policy demands.
- Track document expiry and tenancy end independently: expired agreement must trigger renewal task and a controlled reassessment, not erase tenancy history.

24.3 Ownership transfer while an active tenant remains

Situation: Unit has active tenant when previous owner transfers the property.

Normal path: Preserve tenant OccupancyPeriod and document archive, end old ownership relationship, create pending new owner relationship, validate transfer proof and owner permissions before activating.

Exceptions: Tenant access changes only by applicable relationship/policy, not merely by sale deed upload. Disputed ownership creates a HOLD and legal review.

Invariant: Stable unitId; never expose old owner ID proof to new tenant; never silently reassign financial history.

24.4 Tenant agreement renews after expiry

Situation: Tenant uploads renewed agreement two days after prior agreement expires.

Normal path: Existing version remains archived. New version is scanned/validated, dates checked, owner consent confirmed if required, and effective-dated occupancy updated.

Exceptions: Overlap conflicts, rent agreement unsigned, owner not reachable => review hold; no automatic acceptance from readable OCR.

Invariant: Every document/version, authorization decision and access adjustment remains traceable.

25. NOC and administrative certificate scenarios

25.1 NOC policy contract

For each NOC: require identifier, legal issuer caveat, event trigger, eligible requester, unit/asset/context, rules, evidence checklist, clearance services and cutoff, multi-step approval, effective period, fee if any, template, QA, final issuance/revocation, privacy, retention and escalation owner. The same approval engine services all NOCs; no dedicated hard-coded admin bypass.

NOC / letter	Required authoritative inputs	Exception and approval
No-dues	As-of unit ledger, billed/unbilled approved charges per policy, pending cheque, refunds, deposits	Unsettled or provisional payment -> blocked/pending; finance checker required
Move-out	Finance + facility damage + parking + cards/tags + pending liabilities + lease closure	Recalculate at issue; permitted override records policy rule and justification
Tenant registration	Owner consent, agreement period, required ID evidence/police reference as configured	Missing police result -> explicitly Pending External Verification
Renovation commencement	Society time/noise limits, contractor entries, deposit, structural plan where required	Society NOC not municipal building consent; expiry at work-window
Renovation completion	Facility inspection, common area damage, material exit and deposit settlement	Dispute remains open or a supported conditional release is approved
Parking no objection	Allocation and vehicle history, no outstanding violation, asset return	Temporary pass expired / slot dispute -> block or reviewed exception
Residence confirmation	Effective-dated active occupancy and approved relationship	Historical occupant receives a historical period letter, never current address assertion
Sale / transfer society NOC	Dues and membership records, society bylaws, legal authorization and property-change approval	Legal/title limitations; bank/registrar requirements separately reviewed
Bank loan consent / lien-related letter	Unit/share/member facts, existing society encumbrance records if maintained, authorized legal template	Never assert nationwide encumbrance-free title based only on internal data
Utility / EV / material movement	Technical capacity and inspection, payments, vendor permissions, configured rules	External regulator or utility approval remains distinct

25.2 Deep scenario: ₹18,000 maintenance outstanding

1. Applicant requests move-out NOC for Unit B-804; policy requires a fresh no-dues confirmation.
2. Finance service returns ₹18,000 outstanding with authoritative ledger positions and cutoff.

3. Request status becomes BLOCKED_CLEARANCE, showing amount and payable source but not editing/waiving the ledger.
4. Resident settles payment; only verified gateway/manual clearing entry updates ledger.
5. Rule engine re-evaluates finance and other clearances. Approvers review frozen evidence and current state.
6. Issuance transaction rechecks all required invariants; commits numbered NOC and verification artifact.
7. Move-out access revocation follows configured effective date; historical occupancy and original ledger remain preserved.

25.3 Deep scenario: conditional NOC with dispute

A resident has no financial dues but an unresolved ₹12,000 renovation-damage claim. A society policy MAY allow “conditional clearance” with explicitly worded exclusions after committee approval. If not enabled, request remains blocked. If enabled, generated PDF must display conditional status, unresolved items, approving persons, clause version and associated review; it cannot be labelled “unconditional no dues”.

25.4 Attestation, signature and legal effect

Society officer name/approval and PDF/QR trust establish the society system-of-record, not a government digital-signature service. For legal situations requiring a recognized electronic signature, the document must be signed with an applicable CA-issued DSC / licensed eSign mechanism and independently verified under current CCA/IT Act rules. If no such integration exists, support wet-signature print/upload with provenance and mark method MANUAL_SIGNED - NOT_CRYPTOGRAPHICALLY_VERIFIED.

26. Tender, RFQ, procurement and contract records

26.1 Society procurement capability, not an e-procurement government gateway

Society OS SHOULD support a society-created procurement process for lift AMC, painting, housekeeping, security vendor, firefighting systems, plumbing, solar and capital repairs. A “tender” in Society OS is an internal/private RFP/RFQ issued by the society. Statutory public procurement rules, GST requirements, audit and municipal approval depend on society type and jurisdiction; the platform must offer configurable rule packs rather than assume universal government tender regulations.

26.2 End-to-end sourcing flow

1. Create requisition with need, budget estimate, budget owner, asset/work-order linkage and conflict-of-interest disclosure.
2. Approve solicitation type (direct quotation / ≥N comparative quotes / restricted tender / open RFP) and threshold rule.
3. Compile controlled RFQ/tender from approved clauses: scope, quantities/BoQ, specifications, technical criteria, site visit, submission cutoff, validity, commercial terms, performance guarantees where applicable.
4. Publish with version and immutable timestamp; amendments create corrigendum and possibly extend close time with clear vendor notifications.
5. Authorized vendors submit sealed bid documents and structured line items through a secure upload path. Files remain inaccessible to evaluation team until formal opening event as policy requires.
6. Close submission by trusted server time; do not accept late edits or backdated bids.
7. Open bids with authorized committee/maker-checker record; generate bid-opening report with submission hash and exceptions.
8. Evaluate eligibility independently of commercial totals; compute tax/line arithmetic and comparative statements using exact decimal amounts and effective quote rules.

9. Record selection, conflicts of interest, disqualifications, reasons, approvals and any committee resolution.
10. Issue purchase order/work order as a new controlled document linked to selected bid and accepted contract version.
11. Receive goods/services, inspect and accept, reconcile quantity/price/tax with PO and invoice (three-way matching), then authorize finance workflow.
12. Retain unsuccessful bids, corrigenda, scoring and award history per approved retention schedule; produce audit export.

26.3 Tender controls and evidence boundaries

- Unsealed bids must be encrypted and role-limited. Hidden bids MUST not leak through search, admin exports, preview URLs, thumbnails or logging.
- Supplier cannot view competing bids, scores, personal contact details or sensitive tender estimates unless published policy explicitly permits.
- A vendor amendment after cutoff must be rejected; earlier version remains immutable; separate authorized clarification channel can operate during evaluation.
- Pricing and GST calculations use Decimal/NUMERIC and configuration; never float math, never rank a quotation based only on OCR.
- Technical evaluation and commercial evaluation may be separated, with scoring rubric/version frozen before opening, audit of score changes and justification.
- Template and contract clauses require society/legal approval; generate documents deterministically without generative AI fabricated terms.
- Purchase order approval MUST prevent duplicate obligations and budget overspend in concurrency. If full encumbrance ledger is unavailable, track explicit committed amounts and mark budget checks as informational.
- Three-way matching supports partial deliveries, returns, quality rejects, rate variance tolerances and partial invoice payment; exceptions use approved adjustments.

26.4 Lift AMC tender: three competing bids

Situation: Society requests annual lift AMC bids from three vendors.

Normal path: RFQ issued with deadline and technical criteria. Three sealed bid hashes stored. On opening, authorized panel reviews technical criteria; comparative statement calculates yearly contract cost, taxes and exclusions; selected bid is approved, PO generated and linked to AMC contract.

Exceptions: One bid has expired safety licence -> DISQUALIFIED_PENDING_REVIEW. One bid contains PDF malware -> quarantined, cannot be opened. Bid received after cutoff -> rejection with audit.

Invariant: No bid visible before opening; winner and rationale auditable; no “AI approved vendor” label.

26.5 Goods receipt quantity mismatch

Situation: 100 valves were ordered; 90 arrived; invoice requests 100.

Normal path: Goods receipt records 90, inspection signs off accepted count; three-way match identifies 10 short and withholds unmatched amount.

Exceptions: Authorized tolerance or manual discrepancy approval recorded; no false full-delivery receipt.

Invariant: No duplicate stock increment, no invisible invoice settlement.

27. Compliance, expiries and recurring document obligations

27.1 Compliance obligation registry

The system tracks obligations from a configurable RulePack for registration renewal, fire/lift certificates, insurance, structural audit, water/pollution/environmental evidence where applicable, AMC, pest control and

workforce/vendor licences. Society OS cannot assert that any specific certificate is universally mandatory or legally valid; each obligation stores jurisdiction, source, applicability rule, evidence, frequency, issuing authority, accountable owner, expiry and legal-review status.

Event	Required behavior	Outcome
Expiring within reminder window	Time-zone aware scheduler creates task and notices	OPEN_RENEWAL_TASK
Uploaded renewal evidence	Quarantine, parse, compare issuer/period, review	PENDING_VERIFICATION
Certificate already expired	Compliance badge changes, escalation logged	EXPIRED - DO NOT MISSTATE
External regulator portal unavailable	Do not invent renewal confirmation	SOURCE_UNAVAILABLE / MANUAL_REQUIRED
New certificate approved	Old version archived; new effective dates applied	ACTIVE with review provenance
Inspection fails	Create corrective action/work order and exception	NON_COMPLIANT / REVIEW_REQUIRED

27.2 Risk-based renewal scoring

Risk signal = deterministic policy (criticality × days-overdue × operational impact) with transparent inputs; do not use opaque AI risk scores. A fire safety certificate due in 7 days may be “critical”, but score must expose policy version, due date, reminder success and evidence status. For stale/source-missing data report UNKNOWN, not GREEN.

28. Reference architecture, execution and integration events

28.1 Service boundaries

```
Resident / Admin / Vendor Clients
  -> Auth Gateway -> Tenant Context + ABAC/RBAC -> Document API
  -> DocumentRequest Service -> Policy & Rules Engine
    |-- Finance read model (dues / allocation / refund status)
    |-- Residence/Occupancy service (dates, owner consent)
    |-- Facilities/Parking/Work Orders (clearance)
    |-- Procurement/Compliance/Calendar (documents & obligation)
  -> Evidence Service -> quarantine -> AV/parse/OCR -> assessments
  -> Approval Service -> maker/checker + delegated authority
  -> Issue Coordinator -> template renderer -> signer (optional)
    -> private object storage + metadata + immutable audit
  -> Outbox -> notification worker / reminders / export / verification
Public verification gateway (rate limited, opaque token, minimal disclosure)
```

Transaction integrity uses PostgreSQL as source of truth; object storage is private; Redis is optional for transient cache and rate limits only. Document bytes, rendering jobs and untrusted parsing run in isolated worker queues, separated from online resident and gate critical paths. New modules may be implemented inside a modular monolith initially while preserving interfaces.

28.2 Issuance sequence / failure isolation

1. REQUEST_SUBMITTED persists metadata and outbox validation event.
2. Worker validates files, structure and rules, records assessment with provenance; then creates tasks according to approval graph.
3. Approver approves exact requestVersion and checks snapshot; changes to evidence invalidate dependent approval if policy requires.

4. IssueCoordinator obtains fenced/optimistic issuance lock, re-runs volatile checks and commits issuance intent + reserved number.
5. Rendering worker receives signed internal job and immutable input snapshot; writes content-addressed object and verifies checksum.
6. Finalize transaction confirms file durability, unique active issuance business key, approval validity and activates document; emits ISSUED event.
7. Queue retries are idempotent. Crashes during rendering reconcile orphan files; no public ACTIVE status until finalize transaction succeeds.
8. Public verifier resolves opaque token, checks current issued metadata and status, optionally compares uploaded-file SHA-256 if supported.

28.3 Canonical event catalogue

Event	Business key and payload minimum	Consumer
document.requested	societyId, requestId, requestVersion, documentType, traceId	validation worker
evidence.scan_completed	evidenceId, checksum, scanEngineVersion, result	assessment service
verification.assessed	assessmentId, sourceCategory, checkVersion, result	workflow engine
approval.changed	requestId, approvalTaskId, decision, authorizedActor	issue coordinator
document.issued	issuedDocumentId, serial, issuedAt, status	notification / audit / verifier cache
document.revoked	issuedDocumentId, effectiveAt, reasonCode	verifier cache / notification
document.expiring	documentId, deadline, obligationId	reminder worker
tender.bid_closed	tenderId, closeAt, bidCount, policyVersion	bid-opening workflow

29. Data model, persistence and integrity rules

29.1 Proposed new entities (compatible with baseline, migrate deliberately)

Entity	Required fields / key relationships	Consistency notes
document_type_policies	id, society_id (nullable for platform default), type_code, version, evidence_policy, approvals, template_code, retention, status	Effective-dated; immutable once activated for a published request
document_requests	id, society_id, requester_user_id, unit_id nullable, type_code, context_id, state, version, idempotency_key, created_at	Scope derived from membership; unique idempotency within society/requester
document_evidence	id, society_id, request_id, object_id, doc_kind, checksum_sha256, scan_status, classification, sensitivity, version	Uploaded bytes immutable; encrypted storage
verification_assessments	id, society_id, evidence_id, method, level, result, rule_version, details_redacted, assessed_at	Separate automated finding from manual reviewer
approval_tasks	id, society_id, request_id, step, assignee_scope, decision, decision_at, request_version, reason	No actor can approve their own sensitive request when disallowed
issued_documents	id, society_id, request_id, document_type, serial, object_id, sha256, issued_at, state, valid_to, version, template_version	Unique active business key; immutable bytes; revoke separately

Entity	Required fields / key relationships	Consistency notes
document_status_events	id, society_id, issued_id, old_state, new_state, actor_id, reason, at	Append-only state history
document_verification_tokens	id, society_id, issued_id, token_digest, key_version, expires_at, revoked_at	Random token hashed at rest; no person ID in QR
document_access_events	id, society_id, issued_id, action, actor, purpose, at, trace_id	Minimize content; tamper-resistant audit
compliance_obligations	id, society_id, obligation_code, jurisdiction, subject_id, due_at, evidence_id, owner_id, status	Policy and evidence provenance
tenders	id, society_id, procurement_id, policy_version, state, deadline_at, version	Deadline by trusted server time
tender_bids	id, society_id, tender_id, vendor_id, bid_object_id, digest, submitted_at, sealed, state	Unique vendor+round/version; immutable until opening
document_render_jobs	id, society_id, request_id, input_digest, render_key, status, retries, object_id	Idempotent worker result
document_templates	id, society_id, code, version, approved_at, checksum, locale, object_id	Approved template registry
document_outbox	id, society_id, aggregate_id, event_id, payload_version, delivered_at	At-least-once delivery; idempotent consumers

29.2 Representative PostgreSQL schema and isolation constraints

```
-- Illustrative migration. Apply matching composite tenant FKs in full schema.
CREATE TABLE document_requests (
 id uuid PRIMARY KEY,
 society_id uuid NOT NULL REFERENCES societies(id),
 requester_user_id uuid NOT NULL,
 unit_id uuid NULL,
 document_type varchar(80) NOT NULL,
 context_id uuid NULL,
 state varchar(32) NOT NULL,
 request_version bigint NOT NULL DEFAULT 1,
 idempotency_key uuid NOT NULL,
 policy_version integer NOT NULL,
 created_at timestamptz NOT NULL DEFAULT now(),
 UNIQUE (society_id, requester_user_id, idempotency_key),
 UNIQUE (society_id, id)
);
CREATE TABLE issued_documents (
 id uuid PRIMARY KEY,
 society_id uuid NOT NULL REFERENCES societies(id),
 request_id uuid NOT NULL,
 serial_no varchar(120) NOT NULL,
 document_type varchar(80) NOT NULL,
 object_storage_key text NOT NULL,
 file_sha256 char(64) NOT NULL,
 state varchar(32) NOT NULL,
 template_version integer NOT NULL,
 valid_from timestamptz NOT NULL,
 valid_to timestamptz NULL,
 issued_at timestamptz NOT NULL DEFAULT now(),
 UNIQUE (society_id, serial_no),
 UNIQUE (society_id, id),
 FOREIGN KEY (society_id, request_id)
   REFERENCES document_requests(society_id, id)
);
ALTER TABLE document_requests ENABLE ROW LEVEL SECURITY;
```

```
ALTER TABLE issued_documents ENABLE ROW LEVEL SECURITY;
ALTER TABLE document_requests FORCE ROW LEVEL SECURITY;
ALTER TABLE issued_documents FORCE ROW LEVEL SECURITY;
CREATE POLICY docreq_tenant ON document_requests
 USING (society_id = nullif(current_setting('app.current_society_id',true),'')::uuid)
 WITH CHECK (society_id = nullif(current_setting('app.current_society_id',true),'')::uuid);
CREATE POLICY issued_tenant ON issued_documents
 USING (society_id = nullif(current_setting('app.current_society_id',true),'')::uuid)
 WITH CHECK (society_id = nullif(current_setting('app.current_society_id',true),'')::uuid);
```

Important: sample DDL is a starting fragment, not a runnable full migration. The production schema must implement matching tenant composite FKs for evidence, approvals, jobs, verification tokens and outbox; least-privilege DB roles; session-scoped SET LOCAL app.current_society_id inside database transactions; row-level security and FORCE RLS; immutable triggers or privileges for issued bytes and audit; a business-key active issuance uniqueness constraint; approved transition enforcement; and reviewed migration tests. Table constraints are not a substitute for service-level permission checks.

29.3 Invariants and concurrency

- No issuance without active approved request with all required clearances passed at cutoff.
- Each request/renewal or approved re-issue action has at most one active issued document of same business key.
- An immutable evidence object cannot be replaced in-place; new upload creates a version.
- No issuing file can be published when object checksum differs from database digest.
- An expired/revoked/superseded document can be downloaded by authorized parties for history but never returned as ACTIVE.
- No platform super admin automatically inherits tenant private-document viewing permission.
- No committed document event has missing tenantId, actor, policyVersion or traceId.

30. API and event contracts

30.1 Endpoints (illustrative REST v1)

HTTP endpoint	Purpose	Required controls
POST /api/v1/documents/requests	Create NOC/certificate/letter request	Auth, context, idempotency, policy, rate limit
GET /api/v1/documents/requests/{id}	Get request status, redacted checks and tasks	Object relationship + sensitivity
POST /api/v1/documents/requests/{id}/evidence-upload	Create scoped upload intent	Size/MIME quotas + encrypted storage + quarantine
POST /api/v1/documents/requests/{id}/submit	Validate and start workflow	Version/transition checks
GET /api/v1/documents/requests/{id}/assessments	View allowed verification results	Purpose/permission filtering
POST /api/v1/documents/approvals/{id}/decision	Approve/reject/need-info	Step-up auth, actor separation, If-Match
POST /api/v1/documents/requests/{id}/issue	Issue approved controlled PDF	Current clearance + serial/worker idempotency
POST /api/v1/documents/issued/{id}/revoke	Revoke validity, not bytes	Elevated permission, mandatory reason
GET /api/v1/documents/issued/{id}/download	Fresh authorized retrieval	Short-lived URL/controlled stream, audit

HTTP endpoint	Purpose	Required controls
GET /v/{opaqueToken}	Public authenticity/status check	Rate limit, no identity/PII disclosure, anti-enumeration
POST /api/v1/tenders	Create draft RFQ/tender	Procurement policy/budget
POST /api/v1/tenders/{id}/bids	Upload sealed bid by authorized vendor	Deadline, upload scan, anti-leak ACL
POST /api/v1/tenders/{id}/open	Authorized bid-opening event	Dual authorization, trusted time
GET /api/v1/documents/audit-export	Controlled evidence/audit manifest	Explicit export permission, legal hold check

30.2 Canonical create-request example

```
POST /api/v1/documents/requests
Authorization: Bearer <access-token>
X-Society-Id: <uuid>  (input only; backend MUST validate)
Idempotency-Key: <uuid>
{
  "documentType": "MOVE_OUT_NOC",
  "unitId": "<uuid>",
  "businessContextId": "<moveOutRequestId>",
  "purposeCode": "MOVE_OUT_CLEARANCE",
  "requestedEffectiveDate": "2026-11-15"
}
201 {"requestId":"<uuid>","state":"DRAFT",
    "policyVersion":7,"requiredEvidence":["LEASE_CLOSURE"],
    "traceId":"<uuid>"}
```

30.3 Deterministic rule failure contract

```
422 {
  "code": "DOCUMENT_CLEARANCE_BLOCKED",
  "requestId": "<uuid>",
  "checks": [{"ruleCode":"FINANCE_OUTSTANDING",
          "status":"BLOCKED", "amountMinor":1800000,
          "currency":"INR", "asOf":"2026-11-04T12:00:00+05:30"}],
  "retryable": true,
  "nextAction": "SETTLE_DUES_AND_REEVALUATE",
  "traceId": "<uuid>"
}
```

Use 400 invalid data, 401 unauthenticated, 403 unauthorized, 404 hidden-not-found when needed to prevent enumeration, 409 version/idempotency conflict, 422 valid request blocked by business rules, 429 throttling, 503 dependent system unavailable. Never return raw evidence path, signed storage URLs in error logs or internal OCR text to unauthorized clients.

30.4 Public verifier minimal JSON

```
200 {"documentType":"MOVE_OUT_NOC",
    "issuerDisplayName":"Example Registered Society",
    "documentReference":"NOC/2026/000123",
    "issuedOn":"2026-11-04", "currentStatus":"ACTIVE",
    "expiresOn":null,
    "fileHashMatched":null,
    "verificationMeaning":"Society OS issue-record match only"}
```

Do not return the subject name, flat number, full ID, ledger, or attachments publicly by default; public disclosure depends on documented purpose. Upload-to-compare hash, if supported, is separately

authorized/consented. Random tokens must be unguessable and stored only as digests. A QR is a pointer/proof of system record, not a government-issued seal.

31. Privacy, legal, signatures and threat model

31.1 Key legal boundary

- DPDP Act 2023 and notified 2025 Rules have phased commencements. Implement purpose notices, data minimization, access/retention workflows now; maintain counsel-approved compliance configuration aligned to actual effective dates rather than claiming every provision is immediately in force.
- Under UIDAI Aadhaar Act section 8A and OVSE guidance, offline Aadhaar verification requires informed consent, prohibited collection of full Aadhaar/biometrics by offline-verification seeking entities, purpose limitation and alternative identity methods.
- CCA guidance distinguishes an Aadhaar/e-KYC identity check from an independent legally recognized electronic signature. CA-issued DSC / licensed eSign requires appropriate mechanism; static scanned signatures, checkboxes, hashes and plain QR alone do not equal it.
- Society-issued NOCs/letters must use approved scope language; no blanket assertion of property title, government license validity, police clearance or statutory compliance without competent authority.
- State-specific CHS/RWA rules, police requirements, stamp/registration, records retention and disclosure must be confirmed with counsel and society policy before enforcing a universal rule.

31.2 Security checklist (OWASP-style)

- Private object storage; encrypt at rest and transit; use server-side keys and audited rotation, no secrets in app binaries.
- Presigned uploads bound to user, tenant, size, content hash and expiry; final server scans type, content and malware before availability.
- Defend against polyglot documents, embedded JavaScript, macros, zip/decompression bombs, XML external entities, PDF parser CVEs, EXIF PII, malicious SVG and path traversal; isolate parser with CPU/memory/timeout limits.
- Never use user-provided URLs as backend fetch targets without SSRF protections and strict host allowlists; default no remote fetching.
- Scoped ABAC object checks apply to versions, previews, search indexes, thumbnails, exports and audit endpoints; cache keys always include tenant and permission scope.
- Worker queues use minimal signed job payloads and per-service accounts; parsing workers have no direct authority to issue NOCs or post money.
- Harden public verification with opaque tokens, enumeration resistance, request throttling, PII-minimal responses and observability.
- Approval accounts require MFA/step-up for high-risk issuance and revocation where feasible; no shared admin accounts.
- Audit includes actor, subject, document/request refs, decision/rule versions, before/after state and reason, without dumping raw KYC.
- Retention policy includes purpose, legal hold, correction and right-request handling; backups must not silently recreate erased eligible data.

31.3 Threat-focused acceptance

Attack / error	Prevention and test result
Cross-society UUID probing	403/404; no object metadata or signed URL exposed; RLS regression
Forged QR / guessed serial	Signature/opaque token validation and current issuer-record lookup fail closed

Attack / error	Prevention and test result
Deleted old QR still screenshot-shared	Verifier returns REVOKED/EXPIRED even if original bytes unchanged
Malware or zip bomb upload	Quarantine and parser resource ceilings; worker survives
Two approvers same credentials	Distinct authorized actors and verified login/session required
Fake offline government ID	At most STRUCTURAL_PASS; no SOURCE_VERIFIED without trusted proof
Log leak	Tests scan logs/export for secrets, full Aadhaar and private evidence content
Untrusted template expression injection	Allowlist binding and HTML escaping; no executable DSL in document templates
Payment reversal after prior NOC	Current no-dues verifiability and re-assessment/revocation policy, never silently alter historical issuance

32. Self-hosted implementation options and operating costs

32.1 Practical server-side components (no OpenAI API required)

Capability	Recommended local / open-source option	Important limitation
Backend orchestration	Java 21 + Spring Boot or existing backend framework; state machines and transaction services	Integrate with current architecture, do not create duplicate platform auth
PDF generation	Apache PDFBox and/or Apache FOP with approved XSL-FO templates	Complex typography/large reports need rendering QA
Document metadata/text	Apache Tika + PDFBox parsers in sandboxed workers	Text extraction is not authenticity
Optional OCR	Tesseract OCR deployed in private infrastructure	CPU cost, imperfect Indic language/poor scans
Barcode and QR	ZXing	QR parsing is not the same as trusted signature validation
Crypto checks	JCA/JCE + Bouncy Castle/PDFBox signature tools + official trust anchors	CA trust/revocation policies and qualified signatures need expertise
Malware scanning	ClamAV with managed signatures / sandbox	Updates, isolation and false positives require operations
Rules	Versioned deterministic Java rules / JSON configuration	Rules must be tested and cannot evaluate arbitrary code
Templates	Checked-in or admin-approved versioned catalog (HTML/XSL-FO) + token allowlist	Complex template edits require review
Storage	PostgreSQL + encrypted private S3-compatible storage (e.g. MinIO) + outbox	Infra and backups are not literally free
Jobs & metrics	Durable queue + retry/dead-letter, OpenTelemetry, Prometheus/Grafana	Observability has operational cost

32.2 Minimal deployment profile

Starting production architecture: one API service with tenant/context enforcement; one document worker deployment with upload antivirus, local parsing/OCR and PDF rendering; PostgreSQL for metadata, policy, approvals and outbox; private S3-compatible object store; scheduled tasks; a public minimal verification endpoint; centralized logs/metrics with PII scrubbing. Add separate OCR workers only when workloads justify it. Set per-file/tenant quotas, burst limits, upload throughput and job retention; do not run large OCR jobs synchronously inside the resident UI request.

32.3 Cost reality

Zero paid AI endpoint fees does not equal zero operating cost. Storage, object egress, CPU for OCR, database, server bandwidth, malware-signature updates, OTP/SMS, legally recognized signatures and compliance/legal reviews still cost money. Default no-LLM mode avoids model inference expense entirely. Establish monthly budgets for scans, render jobs, storage per society and optional OCR; alert on abnormal processing.

32.4 Non-LLM automation capabilities

Task	Deterministic mechanism
No-dues letter	Read immutable ledger + snapshot + configured threshold, generate standard template
Agreement reminders	Expiry date rule + scheduled job + idempotent notification
Document rejection	Missing page/expired document/incorrect format + review reasons
Vendor renewal	Calendar + insurance/licence expiry + AMC obligations
Tender comparison	Structured bid lines + Decimal arithmetic + rubric and human panel
Complaint paperwork	Verified ticket fields + approved report template
Minutes draft	Structured agenda, attendance and resolutions entered by authorized secretary
Identity assessment	Signature/QR where supported + owner consent + manual review; no free-form AI verdict

33. Product experience, transparency and localization outcomes

- Separate “Request a document”, “Upload evidence”, “Track approval”, “Issued documents”, “Verify authenticity” and “Compliance/renewals” experiences logically; avoid overloading resident home dashboard.
- Use an explicit status timeline with reviewer/authority type, evidence missing, policy reason, available next action, original request date and expected due date only where SLA policy defines one.
- Show verification chips that say exactly what was checked: “File scanned”, “Details match society record”, “UIDAI offline signature valid”, “Pending manual review”, never generic “100% verified”.
- Support English/Marathi/Hindi and configurable regional templates using approved translated clauses, UTF-8, locale-formatted dates/currency and accessible high contrast.
- Offer downloadable clear PDF and accessible HTML metadata; show first-party verification status and history of replaced/revoked versions.
- Admin approvals use side-by-side source record, extracted candidate fields clearly marked, checks/limitations, risk-free manual corrections and explicit approve/reject/need-info reasons.
- For tender bids, vendor sees own submissions and deadline; evaluation team sees bids only after authorization to open.
- Support low-connectivity evidence upload drafts with resume and confirmation after server acceptance; never show “Verified” from local-only file pick.

34. Test, performance, reliability and release acceptance

34.1 Traceable acceptance suite

Case	Condition	Expected proof
DOC-001	Resident requests no-dues and has zero settled balance	Successful after finance checker/rules; one immutable serial and proof

Case	Condition	Expected proof
DOC-002	₹18,000 due when no-dues requested	422 CLEARANCE_BLOCKED; no issued PDF
DOC-003	Duplicate submit/retry with same idempotency key	Same request; exactly one issued business outcome
DOC-004	Ledger changes after admin approval before issuance	Forced recheck; stale approval cannot issue certificate
DOC-005	Concurrent issuance requests for same move-out	One active certificate; other receives deterministic conflict
DOC-006	Manual override where forbidden	403/422; audit rejection and no issue
DOC-007	Valid approved manual exception	Different checker, reason, policy clause and conditional footer captured
DOC-008	Malformed/PDF JS/malware/zip bomb file	Quarantined or rejected; worker bounded; never downloaded normally
DOC-009	Upload interrupted at 80%	No valid document record; safe resume and exactly-once finalize
DOC-010	Old tenant accesses next tenant KYC	Denied irrespective of URL/UUID or caching
DOC-011	Society A member requests Society B doc	Tenant scope denied across all APIs and object-store links
DOC-012	Aadhaar offline XML valid signed with trusted official key	CRYPTO_VALID with exact assurance scope; no full Aadhaar retained
DOC-013	Aadhaar signature invalid / unknown key	FAIL / UNKNOWN; never “verified”
DOC-014	Non-Aadhaar ID voluntarily supplied instead	Alternate review allowed; no forced Aadhaar
DOC-015	OCR name looks correct on forged ID	No authoritative verification; manual/external check remains
DOC-016	Expired police report uploaded	Expired evidence flagged, no “police verified” claim
DOC-017	Revoked certificate QR scanned	Current status REVOKED with minimal fields
DOC-018	Printed PDF tampered after issuance	Hash/file comparison fails when enabled; original bytes unmodified
DOC-019	Private QR endpoint enumerated at scale	Rate limited; no PII or valid-ID leakage
DOC-020	Two templates issue during version rollout	Pinned template version reproducible per request
DOC-021	Signed PDF check	Validates against configured trust policy; revoked/unknown handled
DOC-022	Signer image pasted but no CA signature	Not labelled legally CA-signed
DOC-023	Tender bid submitted after server deadline	Rejected with audit; no backdated acceptance
DOC-024	Two vendors submit bids; sealed before opening	No cross-vendor read or pre-opening committee access
DOC-025	Bid edited after tender close	Rejected; prior version checksum retained
DOC-026	PO 100 items, receive 90, invoice 100	Variance flagged; 90 inventory counted; invoice not fully matched
DOC-027	AML/insurance certificate expires	Reminder and EXPIRED state without false renewal
DOC-028	Antivirus unavailable during upload	Quarantine / retry; no valid artifact
DOC-029	Object storage fails after serial reservation	ISSUING_FAILED / retry reconcile; no ACTIVE phantom file
DOC-030	Worker retries outbox events	No duplicate serial/notification effects after dedupe
DOC-031	Document key rotated	Old tokens verified under supported key version or safely marked unavailable

Case	Condition	Expected proof
DOC-032	User changes active home during download	Fresh authorization uses requested document tenant; no leakage
DOC-033	Legal hold and erasure request	Purpose/hold documented; eligible data erased/minimized safely
DOC-034	Report/export with sensitive IDs	Redacted by policy, access logged and scope checked
DOC-035	Forced failure on restore / backup	Restore drill verifies file checksums, DB FK consistency, tenant RLS and audit
DOC-036	Local OCR unavailable in no-LLM mode	Manual entry/review works; core NOC generation still operates

34.2 Non-functional targets (initial SLO proposals, to benchmark)

Measure	Target and test note
Request creation	p95 <= 800 ms under an agreed load with file upload excluded; budget subject to sizing
Rule precheck	p95 <= 2 s for typical no-dues when dependent services healthy
PDF generation	p95 <= 10 s for a standard <= 5-page non-OCR certificate, queued when complex
Reliability	Zero duplicate active issued certificates under concurrent retry and crash tests
Quarantine	No unscanned new uploaded document downloadable via ordinary endpoint
Audit	100% high-risk transitions have actor, tenant, rule version, timestamp, trace and outcome
Retention	Deletion/hold job produces verifiable per-record outcome and exception queue
Recovery	Define and prove RPO/RTO by criticality; scheduled restore test checks document bytes and metadata
Observability	Dashboard shows scan backlog, job retry/dead-letter, issue errors, signature failures, expiring docs and abnormal download events

Targets are acceptance proposals, not measured capabilities. QA must test low-end Android upload, Marathi fonts, 10MB+ scanned PDFs, slow networks, interrupted storage, antivirus downtime, clock skew, concurrency and policy variants. Release readiness requires signed security/legal reviews for high-risk document types.

35. Prioritized delivery backlog and rollout

Slice / priority	What to build	Production exit gate
P0 / MVP foundation	Private vault, scan/quarantine, versioning, entity-scope ACL, document policy registry, checksum, audit, upload/download APIs	Cross-tenant denial, malware, retention, restore and access tests
P0 / MVP document output	Deterministic invoices/receipts/statements/acknowledgments from existing authoritative records	Exactly-once issue, ledger tie-out, verified snapshot
P1 / Phase 1A	Agreement/tenant/vendor evidence checklists, local structural parsing, manual verification dashboard, expiry reminders	Honest assurance labels, privacy boundary and document version tests

Slice / priority	What to build	Production exit gate
P1 / Phase 2 NOC	Shared approval engine, no-dues/move-out/tenant/parking/renovation/residence NOC templates, QR verification, revoke/supersede	Finance race, maker/checker, public verifier, issuance E2E
P2 / Phase 2 procurement	RFQ/tender pack, sealed bids, comparison sheet, PO, goods receipt / three-way match	Bid confidentiality, decimal math and auditable award
P2 / Phase 2/3 compliance	Rulepack obligations, renewal tasks, certificate evidence, escalation, exports	Expiry/timezone and human override tests
P3 / optional advanced	Self-hosted OCR, signed UIDAI offline evidence support, CA-backed eSign adapters if commercial/legal need	Consent/legal review, trust-anchor test vectors, CPU cost / failure tests
Never default paid	Do not enable third-party LLM inference; local optional models only with explicit feature flag	Core works with model provider disabled/unavailable

35.1 Definition of done for each implementation slice

- Business trigger + normal/exception/rollback states documented and tested.
- API contract + schema migration + row-level security + object-permission checks complete.
- All outgoing events idempotent and from durable outbox; worker failure/dead-letter observable.
- Approved template/rule versions reproducible; checksum and verification status testable.
- Manual review and external dependency unavailable paths do not claim false successful verification.
- Data retention, DPDP/UIDAI/privacy and legally binding signature use reviewed by responsible specialists.
- Accessibility/localization and operational dashboards reviewed; backup and recovery drill passed.
- Implementation status updated separately from the blueprint as NOT_STARTED, IN_PROGRESS, API_READY, INTEGRATED or PRODUCTION_READY; no hypothetical status inflated.

36. Decision log, ownership and authoritative references

36.1 Decisions to resolve before coding

Decision	Recommended default	Decision owner
Which document types can issue automatically?	Informational letters only; financial/legal NOCs require policy-validated authorized approval	Product + committee/legal
Is Aadhaar required?	No. Voluntary offline method plus alternatives	Privacy/legal
Which documents need CA-backed signature?	Per jurisdiction/purpose legal assessment; do not infer	Legal + security
Does society issue tenders?	Support private RFQ/RFP with sealed bidding configured by threshold	Finance/procurement
What is retention by document type?	Purpose- and legal-hold-based configurable schedule, not one global number	Privacy/records
Which roles review police/owner docs?	Dedicated authorization matrix with least privilege and separation of duties	Security/product
What is accepted authenticity claim?	L0-L4 discrete evidence outcomes; explicit NOT_VERIFIED otherwise	Security/legal
How are keys protected?	KMS/HSM-backed or protected managed secret, rotation tested	Security/operations
What is the expiry/revocation policy?	Mandatory current-status verification for QR-dependent documents	Product/legal

Decision	Recommended default	Decision owner
What runtime stack is already used?	Implement adapter interfaces and reuse existing framework; Java/Spring option is reference only	Architecture

36.2 Source hierarchy and legal verification notes

Product source: Society OS_Documentation.pdf (133 pages): Sections 7.11, 10, 11, 12, 14, 16, 17, 18; superseding conceptual v3 DOCX is retained as a reference. External public sources below were reviewed on 04 November 2026 for specific claims; live compliance must be revalidated before release.

Primary source	URL	Relevance
UIDAI - Aadhaar Paperless Offline e-KYC	https://uidai.gov.in/en/aadhaar-paperless-offline-e-kyc	User-supplied signed offline XML and data minimization.
UIDAI - Offline Verification Seeking Entity FAQ	https://uidai.gov.in/images/FAQ_OVSE.pdf	Offline cryptographic verification, license and identity-purpose limits.
UIDAI - Aadhaar Act section 8A	https://www.uidai.gov.in/hi/aadhaar-act	Consent, notice and restricted collection in offline verification.
UIDAI - Aadhaar QR use	https://uidai.gov.in/kn/aadhaar-online-services	Secure QR signature and offline scanners.
CCA - eSign	https://cca.gov.in/eSign.html	Distinction between eSign, DSC and ordinary electronic approval.
CCA - Digital Signature FAQ	https://cca.gov.in/digital_sign.html	Cryptographic authentication of electronic documents.
MeitY - DPDP Rules, 2025	https://www.meity.gov.in/documents/act-and-policies/digital-personal-data-protection-rules-2025-gDOxUjMtQWa	Phased rule applicability; regulator-issued text.
Government of India Gazette DPDP Rules PDF	https://www.meity.gov.in/static/uploads/2025/11/53450e6e5dc0bfa85ebd78686cadad39.pdf	Commencement clauses and privacy rules.

36.3 Closing product commitment

Society OS will be trustworthy not because it produces many PDFs, but because every document can be traced to a real request, eligible subject, approved policy, authorized decision, exact evidence, immutable issued bytes and current verifiable status. The engine must remain useful without commercial AI APIs and must acknowledge uncertainty wherever the software cannot actually validate external facts.

SOCIETY OS

Document Operations, Records & Evidence Trust

Enterprise Product & Engineering Specification | VERSION 4.5

Production architecture • business scenarios • rule packs • document catalogue • trust boundaries • API/data contracts • QA acceptance

Document control	Definition
Revision	v4.5	04 November 2026
Baseline	v4.4 consolidated master (159 pages) plus v4.4 document trust addendum (25 pages).
Authority	New sections 37-64 supplement v4.4 sections 20-36 and the inherited v4.0/v4.3 blueprint. More specific approved rules prevail on conflict.
Maturity	SPECIFICATION / DESIGN INTENT ONLY. No software deployment or legal certification is asserted.
Intended audience	Product owner, engineering, QA, administration, committee, finance, procurement, records/compliance, privacy and security.
Core constraint	Deterministic backend-first workflow; paid OpenAI/other generative AI APIs NOT required.
Regional/legal status	India-first configurable templates. Statutory duties require counsel/authority confirmation for each state, society category and effective date.

EDITORIAL RULE Do not mistake an uploaded file, OCR extraction, format check, society-issued letter, cryptographic signature or official authority verification for one another. Each conveys a different assurance.

Revision brief and scope

This revision deliberately expands beyond NOC generation and document vault storage. Society OS must operate an evidence-backed record system that understands why a record exists, who is entitled to act, which authoritative facts support it, what legal or society-policy requirements apply, and when the result expires, becomes disputed or must be retained.

Where v4.4 was strong, and what v4.5 closes

Existing coverage retained	Additional v4.5 capability / closure
Versioned documents, QR lookup, restricted downloads	First-class registers, signed record extracts, case bundles, corrections, event-time state and certified-copy semantics.
NOC, rental documents, procurement RFQs	Broader society legal/business cases: resale, inheritance, lender letters, nominations, complaints, elections, disaster claims and management transitions.
Local validation + optional OCR	Evidence-level provenance and page-level findings, trust provider controls, challenge workflow, measurable human review and robust no-AI routing.
Workflow, maker/checker, audit	Delegation, recusal/conflict of interest, quorum and resolution evidence, response clocks, reapproval expiry and authority matrices.
Expiry tracking, archive and secure links	Jurisdictional obligations, records schedules, data-subject rights, legal hold, breach/revocation propagation, printable physical records.
Reference APIs, limited proposed schema	Expanded multi-tenant data model, packet/registry/event tables, API safety, event ordering, migration and rollout contract.

Non-negotiable product outcome

- Every document has an explicit origin and trust statement: SOCIETY_GENERATED, USER_SUBMITTED, EXTERNAL_EVIDENCE, ATTESTED_COPY, REPORT_EXPORT or LEGALLY_SIGNED.
- All business decisions use server-authoritative society/unit relationships, ledger, policy and approval data rather than OCR or generated narratives.
- Every issued artifact is traceable to an immutable decision snapshot and versioned template; a changed ledger or new evidence invalidates a stale clearance.
- No society-issued letter asserts legal title, government identity validation, police clearance, statutory compliance or licensed digital signing without a genuinely supported basis.
- All modules function without any paid LLM provider; optional self-hosted OCR/search only assist staff and do not issue automatic legal findings.
- Each scenario has normal, alternate, blocked and dispute paths; each is testable before public rollout.

v4.5 contents

37. Scope boundaries, document classification and canonical claims

38. Lifecycle document catalogue and recommended default templates

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46. Approval authority, delegation, maker/checker and committee resolutions
47. Templates, PDF generation, language and accessibility
48. Electronic signing, seals, external attestations and authority boundaries
49. KYC, identity/ownership verification and fraud escalation
50. Controlled sharing, public verification and external recipient evidence
51. End-to-end tender, contract and vendor documentation
52. Finance-linked attestations, receipts and recoveries
53. Property, resale, loan, succession, tenants and renovation cases
54. Governance, elections, audits, disputes and legal holds
55. Compliance and safety document operations
56. Privacy, retention, deletion and disclosure governance
57. Zero-paid-AI search, OCR, rules and local intelligence
58. Reference architecture, queues and integration contracts
59. Extended PostgreSQL data model and migration controls
60. Versioned REST / verification API contracts
61. Security, observability, capacity and disaster recovery
62. Scenario-driven acceptance, boundary tests and negative paths
63. Prioritized build backlog and launch gates
64. Decision registry, source hierarchy and traceability

37. Scope boundaries, document classification and canonical claims

A document is not always a PDF. The platform must distinguish record-of-truth, evidence attachment, human request, governed decision, published certificate and recipient-facing disclosure. A record can exist with no downloadable file; a file can exist without establishing an authoritative fact.

Object type	Source of truth / examples	What the system may claim
Source register	Membership, share register, unit ledger, incident log, compliance obligation, vendor agreement index	An operational record existed in this society at the specified point-in-time.
Submitted evidence	Resident-supplied ID, lease, sale deed scan, police receipt, vendor insurance copy	File received, malware scanned; machine/manual checks as separately documented.
Case/packet	Move-out file, resale request, renovation application, claims packet, AGM record bundle	Collects linked evidence, checklist, decisions and communication.
Decision	Approval, rejection, no-objection, waiver, committee resolution, legal hold	An authorized decision was recorded with reason, vote/authority and timestamps.
Issued artifact	NOC, receipt, letter, certificate, acknowledgement, work order, signed extract	The stated issuer issued precisely these bytes/fields in this version.
External document	Government certificate, contract, registered deed, licence, inspection result	Third-party statement; trust requires independent validation, not society generation.
Public verification assertion	Opaque token/QR and revocation-status lookup	Only current issue-record status and explicitly allowed public summary.
Certified copy	Approved reproduction of society-held record with attestation if permissible	Verified copy from this society record; not necessarily a statutory certified copy.

37.1 Separation of responsibilities

- Document Vault owns secure bytes, upload state, checksums, versioned metadata and access operations.
- Record Registry owns durable registers and business identifiers; document rendering is a projection of records.
- Case/Packet Engine owns request-specific evidence, checklist completeness, correspondence and exceptions.
- Policy/Obligation Engine owns criteria, effective dates, escalation, legal applicability flag and dependency graph.
- Approval Engine owns signer authority, maker/checker, delegation, recusal, resolution references and challenge.
- Issuance/Verification Engine owns serial, template version, signed status assertions, public verifier and lifecycle.
- Domain systems remain authoritative for money, ownership, occupancy, access, vendors, incidents and property.

NO AI COST Local checks must remain viable with no LLM whatsoever. OCR and classification are optional candidate extraction; a result can be manually keyed without blocking the workflow.

38. Lifecycle document catalogue and default templates

Every row below is a proposed product capability, not a statement that a specific document is universally mandated. Document types are registered with issuer category, trigger, sensitive fields, policy version, signature

option, expiry mode and retention schedule. G=generate; R=record/register; U=upload/review; P=packet; X=external source; V=verify.

38.1 Society formation and institutional records

Capability / artifact	Type	Trigger	Required boundary
Society registration certificate	U/X	Onboarding; authority issue	External-origin proof; never generate as government authority
Registered bylaws, amendments and resolutions	U/R	Initial setup; amendment approved	Version and effective date; link filing proof
Society PAN/GST registration evidence	U/V	Finance/tax setup	Pattern check only unless issuer checked
Committee roster, appointment and term record	R/G	Election, co-option, resignation	Validity dates, signed minutes, pending disputes
Authorized signatory specimen / mandate	R/G	Committee changes, bank operations	Maker/checker and bank acceptance separate
Bank mandate/official correspondence	G/P	Account opening, signatory update	Requires lawful authorization; not bank approval
Society policy handbook and change notice	G/R	Policy published/updated	Policy diff, acknowledgement, historical versions
Asset handover / builder-to-society deed index	P/U	Builder handover, transition	Received vs legally accepted separately

38.2 Member, share and property history

Capability / artifact	Type	Trigger	Required boundary
Membership application and acknowledgement	G/P	Member joins or transfers	Identity & relationship proof; no title assertion
Membership acceptance/rejection decision	G/R	Committee decision	Reason, authority and rulepack revision
Share certificate issuance register / duplicate log	R/G	Share process or lost certificate	Share number, serial, cancellation, indemnity when applicable
Nomination filing/acknowledgement/changes	R/G	Member nominates or updates	Not equivalent to transfer of legal title
Resale/transfer checklist and clearance packet	P	Sale/transfer request	Society-specific legal checks only
Transfer/membership acknowledgement	G	Valid society action completed	State-specific applicability; do not invent NOC obligation
Unit relationship/occupancy extract	G	Authorized owner/resident request	Effective-dated data; explicit source and limitation
Historical record/certified extract request	P/G	Lawful inspection/copy request	Permission, scope and fee if permitted

38.3 Resident, tenant and household

Capability / artifact	Type	Trigger	Required boundary
Address/residence confirmation letter	G	Bank/address administrative request	Society-confirmed occupancy, not government proof

Capability / artifact	Type	Trigger	Required boundary
Tenant onboarding and police-referral checklist	P	Tenancy begins	Police outcome remains EXTERNAL_NOT_VERIFIED
Owner consent to tenant onboarding	G/R	Owner signs or approves	Version and identity of consenting owner
Lease/rental agreement evidence	U	Tenant adds agreement	Review dates/parties, not legal execution
Tenancy renewal/expiry notice	G	Expiry window	Time-bound, extension checked against owner consent
Short-stay authorization and checkout	G/P	Short stay allowed by policy	Time-limited and gated by society rules
Family/caregiver association acknowledgement	G	Person relationship starts/ends	No ownership rights by default
Move-in access and logistics pass	G	Pre-move checklist complete	Valid dates and security instructions
Move-out clearance and no-dues certificate	G/P	Exit checklist complete	Fresh finance/access/asset/dispute snapshot

38.4 Finance and recoveries

Capability / artifact	Type	Trigger	Required boundary
Maintenance invoice/charge note	G	Bill run published	Ledger-derived charges, tax treatment per rulepack
Receipt / credit acknowledgement	G	Settled allocation posted	Never produce on client redirect/provisional cheque
Unit statement of account / ledger extract	G	Resident/treasurer request	As-of statement, credits and reversals visible
Demand notice / late dues notice	G	Dues aging threshold	Disputed vs undisputed shown as policy requires
No-dues/no-objection finance letter	G	Configured policy request	As-of balances and pending obligations
Refund/deposit settlement statement	G	Refund decision/settlement	Refund status not equated to actual payout
Bank reconciliation/exception report	R/G	Period-end finance close	State clear: unmatched vs verified
Annual income-expense/balance-sheet draft	G	Year-end reporting	Draft vs audited externally; sign-off tracking

38.5 Vehicles, parking, gate and facilities

Capability / artifact	Type	Trigger	Required boundary
Parking allocation/renewal/termination letter	G/R	Slot allocated/revoked	Effective dates; no absolute property title claim
Vehicle sticker or access card issuance record	R/G	Security credential allocated	Lost credential revocation and replacement
Temporary material gate-pass	G	Equipment in/out	Gate validity, inward/outward reconciliation
Parking violation and dispute evidence packet	P	Violation recorded	Photos/witnesses, response and appeal
Amenity booking confirmation and conditions	G	Facility booking confirmed	Payment/approval if required; not just reservation hold

Capability / artifact	Type	Trigger	Required boundary
Facility damage or deposit claim packet	P/G	Post-use inspection	Evidence, reviewer, appeal and deduction posting
Visitor/contractor activity extract	G	Authorized inquiry	Privacy filtered, not public export

38.6 Renovation, construction and technical safety

Capability / artifact	Type	Trigger	Required boundary
Renovation request and contractor dossier	P	Resident plans renovation	Scope, time, safety, permitted approvals
Society renovation NOC/conditional permission	G	Society approves work	Municipal/structural permission separate
Work permit / safety checklist	G/R	Hot work, height work, electrical work	Designated qualified sign-off; stop-work path
Lift protection booking and deposit acknowledgement	G	Large move/construction materials	Slot and damage evidence controls
Structural engineer report	U/X	High-risk renovation or damage	External licensed professional record
Work-completion inspection report	G/R	Resident/vendor marks completion	Independent inspection vs self-attestation
Common-area restoration and deposit release	G	Inspection passes	Financial settlement still ledger-governed

38.7 Vendors, staff and workforce

Capability / artifact	Type	Trigger	Required boundary
Vendor registration and due-diligence packet	P	New vendor onboarding	Registration/insurance/licence verification trail
Vendor KYC, GST, licence and insurance copies	U/V	Vendor starts/renews	External issuer-check where applicable
Domestic worker association/access permission	G/R	Resident hires or removes help	Per-unit consent and revocation
Employment/contractor letter and attendance export	G	Authorized HR/workforce request	Employment fact limited to available records
Shift incident and disciplinary evidence packet	P	Operational incident	Role-restricted, appeal and data minimization
Vendor insurance expiry / compliance notice	G	Expiry scheduler	No claim of continued coverage without proof

38.8 Procurement and contracts

Capability / artifact	Type	Trigger	Required boundary
Purchase requisition + budget approval	P/G	Department requests purchase	Budget reservation and policy threshold
RFQ/RFP/tender invitation and addenda	G	Sourcing approved	Deadline/version and sealed terms

Capability / artifact	Type	Trigger	Required boundary
Bid submission acknowledgement	G	Vendor submits before close	Hash/time receipt, no visibility to competitors
Bid opening/evaluation comparative statement	G/R	Procurement closing	Scoring rubric, exceptions, conflict declarations
Award/LOI/work order/PO	G	Authorized award	Approvals and budget; not external statutory tender
Vendor contract, amendment and renewal notice	P/G	Contract executed/extended	Signature and legal effect separately proven
Delivery challan, GRN and inspection record	R/G	Goods/services delivered	Three-way match; variance and dispute
Retention/security deposit release authorization	G	Milestone and defect checks done	Finance posting and bank payment separate

38.9 Governance, correspondence and elections

Capability / artifact	Type	Trigger	Required boundary
AGM/SGM notice, agenda and annexure pack	G/P	Meeting scheduled	Notice lead time configured/legal reviewed
Attendance/quorum/proxy register	R/G	Meeting held	Eligibility effective as of meeting cutoff
Minutes, resolutions, voting/tally extract	G/R	Meeting decision recorded	Draft/proposed vs confirmed distinct
Committee circular, policy bulletin, public notice	G	Communication approved	Audience snapshot, effective dates
Committee decision/action register	R	Resolution passed	Task ownership and closure evidence
Election notice and nomination record	P/G	Election policy enabled	Election readiness and legal requirements reviewed
Official incoming/outgoing correspondence log	R/G	Letter/notice arrives or sent	Proof of dispatch and acknowledgement

38.10 Compliance, assets and insurance

Capability / artifact	Type	Trigger	Required boundary
Fire NOC/licence and inspection evidence	U/X	Mandatory if law/rulepack applies	Authority-issued record; validation cannot be assumed
Lift licence/inspection/AMC certificate	U/X	Inspection/renewal	Vendor/authority source and expiry
Building insurance policy and claim packet	U/P	Policy renewal/claim	Coverage not assumed from receipt only
Water/electric/solar/EV clearance records	P/U	Utility/asset work	External utility/engineer approval separate
Waste disposal, housekeeping, pest control logs	R/G	Scheduled service completed	Evidence and quality review
Asset register/AMC checklist & defects	R/G	Asset service/inspection	SLA, work order and repeat-defect trace
Regulatory compliance calendar and evidence pack	R/P	Obligation trigger	Jurisdiction and validity date visible

38.11 Incidents, disputes, insurance, legal

Capability / artifact	Type	Trigger	Required boundary
Incident/accident witness statement pack	P	Safety/security incident	Chain of custody, consent, restricted access
Water seepage/damage claim dossier	P	Inter-flat dispute	Evidence and mediation separate from final liability
Insurance claim proof and handover pack	P/G	Loss event and insurer request	Society facts vs insurer determination distinct
Legal notice acknowledgment / response register	R/P	External legal notice received	Legal review/hold and service proof
Dispute mediation minutes and settlement terms	G/P	Mediation completed	Mutual assent, confidentiality and version history
Handover between committees/auditors	P/G	Office-bearer change or audit	Inventory, books, credentials, outstanding exceptions

38.12 Privacy and information rights

Capability / artifact	Type	Trigger	Required boundary
Consent notice / purpose declaration record	R/G	Personal evidence requested	Separate purposes and alternatives
Personal-data access/correction request packet	P	Resident requests data rights	Legal basis, identity, deadline and result
Data retention/legal hold decision record	R	Archive or deletion triggered	Justified hold and constrained access
Secure record disclosure / certified-copy log	R/G	Authorized share with external recipient	Scope, expiry, recipient, proof, no hidden PII
Breach/unauthorized access evidence packet	P	Security incident	Incident response, impact tracking, mandated notices if applicable

Total proposed catalogue entries: 86. These are implementable product candidates, not obligations automatically imposed on every society.

38.13 Recommended launch bundles

Bundle	Initial artifacts	Default priority
Every society	Registered documents index; member/occupancy record; invoices/receipts; unit statement; notices; secure upload; issue/correction log	P0
Resident lifecycle	Ownership verification packet; tenant agreement register; onboarding; move-in; no-dues/move-out; residence letter	P1
Committee and compliance	Minutes and resolutions; statutory compliance register; renewal tasks; correspondence; evidence and retention	P1
Facility/procurement	Renovation permit; vendor packet; RFQ/PO/GRN; AMC/inspection reports; insurance claims	P2
Advanced external trust	Licensed eSign/DSC options; external government lookups; complex document automation	P3 / provider/legal dependent

39. Real-world triggers and required paperwork packs

The resident or administrator should not need to guess which PDF to request. A business-event trigger creates a configured CaseFile and tells the user which records are needed, who must verify them and which outputs can eventually be issued. A packet can complete with no generated document if the outcome is only an official register update.

Trigger ID / actual event	Minimum evidence bundle	Produced outcome
RESIDENT_LINK_OWNER - Existing owner links a flat	Relationship proof; share/membership record; verification decision	Approved unit relationship; not title adjudication
TRANSFER_SALE - Resale or sale transfer	Transfer checklist; dues statement; member decision; succession/legal evidence if necessary	Membership transfer record; legally permissible letter only
TRANSFER_DEATH - Owner dies	Death/succession evidence; nomination; claimant authorizations; committee/legal review	Historical relationship closed; lawful successor process, not automatic inheritance
COOWNER_ADD - Add co-owner	Requested share/membership modification, parties and consent	Authorized change register; no title assertion
BANK_LOAN - Mortgage/bank asks for society letter	Bank request, member consent, society dues/records	Limited factual letter; lender requirements external
LOST_SHARE_CERT - Member reports lost share certificate	Affidavit/indemnity if rulepack requires, certificate serial and review	Original cancelled; duplicate recorded and issued if permitted
TENANT_NEW - New rental begins	Owner consent, agreement metadata, ID alternatives, required police evidence	Tenant association and move-in checklist; police status explicit
TENANT_RENEW - Lease extension	Extension/renewal evidence, owner consent and effective dates	Amended occupancy period and notifications
TENANT_END - Tenant moves out	Dues, pending charges, keys/cards, facility, disputes, assets	Move-out certificate if clearance passes; old access revoked
SHORT_STAY - Owner authorizes short stay	Stay details, consent, occupants if required, access periods	Time-limited stay; no permanent tenancy
RENOVATION_START - Resident plans interior work	Scope, drawing, contractor proof, permits, deposits, safety checklist	Conditional society permission, not civic approval
RENOVATION_STOP - Safety violation during renovation	Inspection, violation evidence and safety reviewer	Stop-work decision and recheck, no hidden renewal
EV_INSTALL - Resident requests EV charging	Slot rights, load calculations, licensed technical approvals as needed	Society clearance only subject to external norms
VISITOR_DAMAGE - Vehicle scratches gate	Incident photos, custody, parties, gate log	Incident report; liability adjudication stays separate
PARKING_CHANGE - Parking allotment change	Allocation history, slot eligibility, outstanding disputes	New time-bounded allocation letter
PET_REG - Pet registration or complaint	Permitted policy details, owner declaration and incident evidence if any	Pet record/mediation; no unsupported restrictions
FIRE_ALARM - Fire event or mock drill	Incident chronology, attendance/evacuation, responders, equipment	After-action report and compliance tasks
LIFT_FAILURE - Repeated lift breakdown	Fault log, AMC contract, technician report, safety evidence	Corrective action and vendor notice
VENDOR_ONBOARD - New facility vendor	Registration, GST/licence as relevant, insurance, conflicts, contact, payment controls	Approved vendor record with evidence/provenance

Trigger ID / actual event	Minimum evidence bundle	Produced outcome
RFQ_START - Society requests repair quotes	Approved requisition, budget, scope/BoQ, bidder policy	RFQ or tender pack
RFQ_AWARD - Tender evaluation ends	Bid hashes, opening register, scoring, approval and minutes	Award/PO/contract after authorized signoff
PURCHASE_RECEIVED - Vendor supplies goods	PO, GRN, inspection, invoice	Match/exception and finance authorization
AGM_SCHEDULE - AGM announced	Meeting basis, notice rules, agenda, voter eligibility source	Notice pack with immutable audience snapshot
AGM_RESOLUTION - Resolution approved	Quorum, attendance, votes/minutes, dissent where applicable	Approved resolution extract; draft not official
COMMITTEE_CHANGE - Office bearers change	Election/resignation proof, keys/credentials, open obligations	Handover packet and signatory update tasks
ANNUAL_AUDIT - Audit begins	Financial books, vouchers, bank reconciliation, evidence exceptions	Scoped auditor packet with access log
INSURANCE_CLAIM - Flood/fire/damage claim	Policy, photos, incident, estimates, invoices	Claims packet; insurer decides coverage
COMPLIANCE_EXP - Lift/fire certificate near expiry	Applicable rulepack, current certificate and renewal owner	Reminder + evidence task, escalate missing renewal
LEGAL_NOTICE - Legal notice received	Original notice, receipt time, related evidence, counsel instructions	Legal case and preservation hold
DATA_REQUEST - Resident requests their data	Verified requester, requested scope, exemptions and rights pathway	Controlled export/correction/denial with reason
BUILDER_HANDOVER - Builder transfers operations	Asset list, warranties, approvals, defects, meter maps, signed minutes	Handover acceptance exceptions and inventory history
REDEVELOPMENT - Society considering redevelopment	Meeting minutes, consent register, consultant docs, counsel-advised records	Case and approval records, no automatic authority
EMERGENCY_REPAIR - Water pipe bursts	Emergency authority, work evidence, quotation exception, damage facts	Urgent PO and audit with ex-post review

39.1 Scenario contracts that are deliberately difficult

S39-01 Resale with disputed maintenance dues

Trigger / initiating event: A seller requests a resale-related clearance while ₹26,400 is posted but under formal dispute.

Expected path: Open a property-transfer packet; snapshot the account, dispute status and local legal/society policy. Show the difference between balance and disputed liability. Route to authorized finance and committee decision.

Exceptions: Do not issue a false no-dues statement. A case-specific conditional factual statement may be possible only under a reviewed rulepack.

Authorized outcome: ISSUE_WITH_DISCLOSURE or BLOCKED_BY_POLICY with reason; never convert outstanding dues into zero.

Acceptance proof: As-of ledger extract and decision note are retained; later adjustments generate new versions.

S39-02 Death, nomination and succession

Trigger / initiating event: A nominated family member submits a death certificate and asks to become the owner.

Expected path: Receive sensitive evidence, link existing membership/share register and nominee record, create legal-review task and apply society/state transfer rules.

Exceptions: Nomination must not be treated as conclusive title. Multiple claimants or court orders freeze disputed changes.

Authorized outcome: Create lawful society-record changes only after approved legal/member workflow.

Acceptance proof: Existing records remain unchanged until authorization; audit includes the succession basis and objections.

S39-03 Retrospective certificate request

Trigger / initiating event: Owner asks today for a residence certificate stating they lived in the flat two years ago.

Expected path: Use effective-dated historical occupancy records and the policy for retrospective assertions.

Exceptions: Current authorized unit access does not imply access to former occupants; missing history yields insufficient evidence.

Authorized outcome: Produce a clearly dated historical fact extract if policy permits, or refuse with documented reason.

Acceptance proof: Output distinguishes requested past date, actual source period and issue date.

S39-04 Official-looking document uploaded from WhatsApp

Trigger / initiating event: A contractor submits a photographed lift licence with seal-like graphics.

Expected path: Malware-scan, preserve source/provenance, extract issuer/date candidates, check format and optionally validate through an authority-approved mechanism.

Exceptions: OCR, image quality and a stamp image cannot establish licensing authority.

Authorized outcome: RECEIVED / REVIEW_REQUIRED until actual external verification; no VERIFIED badge.

Acceptance proof: Review decision cites page evidence and authority-check method or its absence.

S39-05 Committee member attempts self-approval

Trigger / initiating event: A committee member drafts their own reimbursement approval and tries to sign it.

Expected path: Policy rejects self-approval and routes to independently eligible approvers with conflict disclosure.

Exceptions: Absent quorum or recusals can block the process; admin impersonation must not bypass it.

Authorized outcome: No official artifact or payment release before policy-compliant approval.

Acceptance proof: Signer identity, role, recusal and decision version are stored.

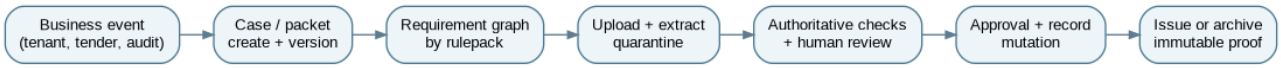


Figure A. A case file is a durable workflow anchored to a real business event. Generating a PDF is a possible terminal action, not the starting point.

40. Registry of record, certified copies and information requests

Digital generation cannot replace formal society registers. Build first-class, append-audited registers that support effective-dated corrections, opening balances, certified/attested extracts when permitted, read-only historical snapshots and reconciliation to source transactions.

Register / system of record	Canonical keys and events	Control / data privacy
Membership and share register	society, member number, share class/serial, admission/termination, transfer event	Committee authorization; no deletion on transfer; nominee separate

Register / system of record	Canonical keys and events	Control / data privacy
Property and occupancy register	unit stable UUID, period, legal/occupancy relationship, evidence reference	Privacy boundary between occupant generations
Nomination and co-owner register	person, filing date, effective state, acknowledgement, conflict	Access minimized; nomination not a title determination
Cash/bank, voucher and financial registers	journal ID, amount minor, posting date, payer/beneficiary, reversal link	Must reconcile to immutable ledger, not typed PDF totals
Minutes and resolutions register	meeting ID, notice version, quorum, resolution number, votes, confirmation	Draft vs adopted; amendments through subsequent resolution
Contracts and assets register	vendor, asset/contract, warranty/AMC, obligations and expiry	Changed vendor ownership and renewals versioned
Compliance obligations register	regulator/rulepack, due date, frequency, owner, evidence and accepted outcome	No universal mandate without applicability decision
Correspondence register	incoming/outgoing serial, sender, recipient, subject, dispatch proof	Legal notices/time-sensitive documents flagged
Inspection/accident/ dispute register	incident source, timeline, evidence, custody and final outcome	Access restrictions, legal hold and redaction
Document issuance/revocation register	serial, issuer, reference, version, status and effective time	Public status lookup decoupled from private source records

40.1 Authenticated copy request workflow

1. Create a request specifying register, period, subject, purpose and requester basis; verify relationship or legal authority.
2. Check document-level and field-level rights. Protected personal information of unrelated members must be redacted or excluded.
3. Use an immutable as-of snapshot or transactionally consistent export; mark DRAFT, EXTRACT or ATTESTED_COPY as applicable.
4. Require competent attestation before labelling a copy society-certified. Never advertise a “government-certified” copy unless an actual governmental authority issued it.
5. Log disclosure scope, record count, authorized recipient, format, reason, fees when lawfully configured, and download expiry.
6. Support correction or refusal with reason and recorded appeal/escalation process; protect original source state.

STATUTORY NUANCE A member record inspection or copy right may depend on the society category, local statute, amended bylaws, legitimate requester status and applicable fees/response period. The product uses configurable legal rulepacks and publishes only verified obligations.

41. Intake, scanning, physical archive and chain of custody

Societies may have 20 years of paper files, low-quality scans, loose originals and WhatsApp photos. Intake must distinguish source medium, chain of custody, document completeness, evidence authenticity and the separately identified custodian of any physical original.

Step	Machine / operator action	Failure and proof
1. Capture	Mobile photo, PDF, scanner, structured form, bulk import or authorized API; record capture device, submitter and time.	Reject unsupported MIME, over-limit files and suspicious extensions; no public preview.
2. Quarantine	Store new bytes in private quarantine, record SHA-256, client and server timestamps, source type, claimed document type.	File cannot become reviewable before malware and parser checks.
3. Normalize	Extract permitted text/page count/rotation and optional local OCR; build privacy-constrained preview.	OCR fails? Manually enter fields, mark confidence NOT_APPLICABLE/LOW.
4. Inspect	Capture per-page missing/illegible/mismatch/expired/signed? findings and reviewer note.	Do not silently merge pages from different people or documents.
5. Cross-link	Link evidence to case, unit, person, vendor, contract, obligation and revision.	Prevent cross-tenant links, ownership confusion and hidden inherited exposure.
6. Original tracking	For hardcopies, track envelope/file number, custodian, intake/return signature and physical location.	Missing original escalates; scan is not represented as original.
7. Retention and disposition	Apply classification, purpose, schedule, hold and deletion workflow.	Disposal proof includes eligible scope; legal holds block deletion.

41.1 Backfile migration (500-50,000 documents)

Import manifests MUST specify society ID, class, declared owner, file source, logical record, checksum and date range. Stage 1 inventories without assigning legal authority. Stage 2 quarantines and deduplicates byte-identical files without accidental cross-household linking. Stage 3 maps fields and resolves duplicate people/units. Stage 4 samples quality and approvals. Stage 5 activates valid records and produces rejected-row files. Do not overwrite original byte streams.

S41-01 Missing tenancy annexure

Trigger / initiating event: A 22-page agreement is uploaded as 21 pages and a clause references annexure B.

Expected path: Extractor flags the page/annexure candidate gap and requests supplementary evidence.

Exceptions: No unsupported automatic “forgery” label; manual override must explain why processing continued.

Authorized outcome: NEEDS_MORE_EVIDENCE until resolved, then new evidence set version.

Acceptance proof: Original file retained; second upload does not silently replace it.

S41-02 Physical original checked out

Trigger / initiating event: A committee member removes a paper share register for audit and it is not returned.

Expected path: Check-out requires authorized custodian and due date; overdue warning escalates.

Exceptions: Digital scan remains available only by permission, but must not masquerade as original.

Authorized outcome: Open custody exception until signed return.

Acceptance proof: Physical location, actor, receipt, case link and timestamps audit.

42. Evidence packets, case files, checklists and missing-evidence resolution

A Packet is the reusable container for one business decision and must not be confused with a ZIP attachment. It captures references to existing evidence, avoids duplicating sensitive bytes, and maintains the exact fact basis used for every approval. The same evidence may link to multiple cases only after purpose and scope checks.

Packet concept	Required invariant
Case identity	societyId + caseId + caseType + stable businessContextId; one active case per unique scope where policy says so.
Checklist item	Requirement ID, effective policy version, mandatory/conditional flag, evidence cardinality, owner, status and reason.
Evidence link	Independent access decision and provenance; link can be removed from a case without destroying records under hold.
Rule evaluation	Authoritative state snapshot with source versions, engine version and precisely evaluatedAt timestamp.
Issue resolution	Missing/mismatch/expired/unclear is a typed exception with responsible actor, due date and closure evidence.
Decision challenge	Applicant can submit corrections or appeal; approval decision cannot be edited in place.
Packet closure	Closure reason, final state, issued outputs, retention classification and accessible audit timeline.

42.1 Canonical packet state and edge behavior

```
DRAFT -> SUBMITTED -> INTAKE_VALIDATION -> EVIDENCE_PENDING / UNDER_REVIEW
UNDER_REVIEW -> APPROVAL_PENDING -> APPROVED -> ISSUANCE_PENDING -> COMPLETED
Any nonterminal state -> WITHDRAWN / REJECTED / ON_HOLD (policy-governed)
ISSUED output -> ACTIVE / EXPIRED / REVOKED / SUPERSEDED (separate state aggregate)
Rule/source version changed: ASSESSMENT_STALE -> RECHECK_REQUIRED
Disputed decision: CASE_CHALLENGED -> REOPEN_REVIEW (no mutation of old decision)
```

Offline upload and partial synchronization are allowed only for non-authoritative drafts and evidence capture. Final verification, decision, issuance and public verification need server authority; no offline “NOC issued” state.

43. Versioned policies, jurisdiction rule packs and forms

A society can operate as CHS, RWA, apartment owners association, condominium, villa community or mixed-use entity. Requirements vary by state, constitutive documents, building type, regulator and effective date. Each document policy must be selected from immutable versions with explicit precedence: applicable law/verified directions -> society governing instrument -> approved committee policy -> case-level exceptions allowed by that policy.

Policy field	Meaning / guardrail
jurisdictionCode + authorityRef	India/state/city and source URL/file; owner, lastLegalReviewedOn, effectiveFrom/to.
societyLegalForm	CHS, RWA, AOA, condominium, developer-managed or custom type. Not derived only from name.
documentType + purpose	Separate policy for no-dues, renovation NOC, tenant acknowledgement, statement, share record, etc.

Policy field	Meaning / guardrail
requirements[]	Typed expression over existing domain facts; required evidence kind, date and issuer; no arbitrary code eval.
approvalGraph	Steps, quorum, signatory authority, delegation and maker/checker policy.
externalCheckStrategy	NONE, MANUAL, OFFLINE_SIGNATURE, LICENSED_PORTAL; availability/permission flag and fallback.
templateVersion + locale	Published template and language; source variables limited to approved field dictionary.
retention / sharing / privacy	Classification, allowed viewers, purpose, expiry and legal-hold behavior.
effective interval + change record	No retroactive mutation of completed decisions; migration rule explicit.

43.1 Safe deterministic policy example

```
policy: MOVE_OUT_NOC
scope: SOCIETY_UNIT_OCCUPANCY
appliesWhen: societyProfile.moveOutClearanceEnabled == true
requirements:
 - id: FINANCE_CLEARANCE
   source: ledger.outstandingAsOf
   comparator: EQUALS
   valueMinor: 0
   onUnknown: BLOCK
 - id: GATE_ASSETS_RETURNED
   source: accessAssets.unreturnedCount
   comparator: EQUALS
   value: 0
   onUnknown: REVIEW
 - id: OPEN_DISPUTES
   source: disputes.mandatoryOpenCount
   comparator: EQUALS
   value: 0
   onUnknown: REVIEW
approval: [FINANCE_CHECKER, SOCIETY_AUTHORIZED_SIGNATORY]
recheckBeforeIssue: true
snapshotConsistency: REQUIRED
signature: AUTHORIZED_SOCIETY_ISSUER
legalApplicability: LOCAL_REVIEW_REQUIRED
```

The YAML above is a contract example, not an executable policy DSL. Compile policies from an allowlisted set of typed operators and known fact providers; configuration editing must never permit arbitrary JavaScript, SQL, network calls or code execution.

43.2 Dynamic application forms

- Form metadata supports field type, validation, data classification, help text, localization, evidence capture and conditional display.
- Public and sensitive questions are separated; collect only information needed for a declared purpose and authorized workflow.
- No freeform HTML/script insertion into PDF templates. Reject external URLs, local file references and template injection.
- Migration maps prior field keys, retains original submission and tells staff about new mandatory fields.
- Publish draft/test/approved templates with preview and rollback; disable new issuance on a revoked unsafe template.

43.3 Legally sensitive rulepacks

Maharashtra government sources list revised model housing-society bye-law publications dated August 2026. Rules about NOCs, transfers, member access and meeting processes must therefore reference the applicable current enactment and registered society bylaws, not be carried forward from a generic 2014 template. Flag jurisdiction/legal review as a release blocker for any statutory certification or mandatory requirement.

44. Document obligations, deadlines, expiry and dependent tasks

Expiry reminder alone is insufficient. The obligation system must represent who owes a filing, which entity issued it, when it becomes due, dependencies, renewal lead time, permitted evidence, compliance reviewer and escalation chain. The system may automate chasing evidence; it must not manufacture compliance or quietly waive a missed filing.

Obligation state	Allowed transitions and automation
NOT_APPLICABLE	Recorded exemption basis and reviewer; can become APPLICABLE if law/society context changes.
NOT_DUE / DUE_SOON	Lead-time scheduler opens responsible task with idempotent key (obligation, period, society).
EVIDENCE_PENDING	Upload and quarantine; staff sees actual renewal submission and supporting record.
UNDER_REVIEW	Check issuer/date, requirement coverage, validity and authenticity method.
FULFILLED	Qualified reviewer accepts proof; condition and next deadline are calculated.
OVERDUE / BREACHED	Escalation and operational visibility; service limitations governed separately.
DISPUTED / WAIVED	Time-limited and authorized decision with legal basis; no generic admin override of law.

For recurring duties, calculate timezone-aware recurrence from legal trigger date, not from the last alert. If renewal evidence is rejected, retain the previous status and the rejection trace. Evaluate weekends, holidays, custom grace periods and official portal outages explicitly. Monitoring needs dashboards for high-severity missed obligations.

S44-01 Lift inspection expires during festival week

Trigger / initiating event: An authorized compliance record is due on Sunday. The reminder job has already been retried twice.

Expected path: Create exactly one obligation instance keyed by due period; notify the designated reviewer and management by configured severity.

Exceptions: No accepted new certificate means status remains OVERDUE; an inspection booking is not proof of renewal.

Authorized outcome: Escalate and apply an appropriate operational restriction only if a reviewed safety policy requires it.

Acceptance proof: One task, multiple delivery attempts, no false VERIFIED status; logs show issuer and previous expiry.

45. Issuance, supersession, correction and revocation invariants

The critical unit of issuance is a unique authorized business outcome. Documents must not become official merely because a background worker wrote a PDF. Use a transactional outbox, stable serial allocation, committed state, immutable bytes and a verifier projection. Any correction results in a new version/issue, not an overwrite.

7. Lock the relevant business record / read a transactionally consistent snapshot with explicit source version(s), effective date and financial cutoff.
8. Evaluate applicable policy and live domain facts. Approval is valid only for the specific evidence set, request version, rulepack version and source snapshot constraints.
9. Allocate serial idempotently and persist an ISSUANCE_RESERVED state; generate bytes in a controlled worker from template and stable snapshot.
10. Store private bytes with checksum and metadata; commit ISSUED when storage existence and checksums confirm, then publish verifier state/event.
11. Immediately before activating an NOC, re-evaluate freshness where stipulated. A changed ledger/access state causes STALE_RECHECK_REQUIRED, not silent issuance.
12. Correct material errors by SUPERSEDE -> new issue with parent pointer and reason, or REVOKE with effective time and notification; old QR must show historic state.
13. Never reuse an invalidated serial or delete the original bytes from the evidentiary chain except under legally applicable disposition.

Special case	Required handling
PDF worker crashes after reservation	Retry same issue ID and exact snapshot; no new serial/duplicate issuance.
Object uploaded, DB transaction rolled back	Orphan candidate quarantined; sweeper reconciles, never publicly serves.
Approver changes at signing	Revalidate assigned signing authority and delegation effective time.
Issued NOC later disputed	No silent content edit; add appeal, recheck, revocation or explicit amended issue.
Revocation cache stale	Public verification must fetch authoritative status or use bounded cache with urgent invalidation.
User deletes local copy	Record and status remain server-governed under retention rules.

46. Approval authority, delegation, maker/checker and resolutions

Authorization is contextual: one person can be a resident, vendor owner, committee member and treasurer. The system must identify conflicts and distinguish preparer, reviewer, approving signatory, finance checker, witness and legal adviser. Role name alone is not signing authority.

Decision control	Enforcement
Maker/checker	Self-approval prohibition where policy applies, distinct principals and separate authenticated sessions.
Delegation	Start/end date, delegator authority, authorized business types and maximum thresholds; revocable and audited.
Quorum/resolution dependency	Meeting ID, notice legality review, eligibility snapshot, quorum result, approved resolution version.
Conflict of interest	Declaration before vendor selection/reimbursement; recusal captured; no read of sealed competing bids if conflicted.
Financial threshold	Policy-specific aggregate value including taxes/deposits; anti-splitting detection suggested, never auto guilt.
Reapproval on change	Any material attachment, amount, selected vendor, ledger, subject or template change invalidates stale approvals.
Emergency action	Explicit break-glass reason and bounded approval, with mandatory post-action review where policy authorizes.

Decision control	Enforcement
Decision appeal	Separate challenge/decision history; old record remains immutable.
Signature evidence	Approved document hash, actual signer, authentication strength, effective delegation and signing method.

S46-01 Compromised approval link

Trigger / initiating event: A person forwards an approval email link to a different employee.

Expected path: Require fresh authenticated session and exact assigned identity, valid role, case/version and step-up for sensitive action.

Exceptions: Email ownership/link possession alone is not signatory authority.

Authorized outcome: 403 and security event; never approve.

Acceptance proof: Audit contains actor, expected approver, trace and denied attempt without leaking private data.

47. Templates, PDF generation, language and accessibility

Templates are reusable controlled assets, not ad-hoc string concatenation. Input fields must be typed and sourced from approved view models. Test rendering against fonts/language, pages, repeated rows and injection; preserve preview versus official output semantics.

Output characteristic	Required behavior / acceptance
Letterhead and identity	Society legal name, registered address, registration number where relevant, issuing office, contact and approved logo; tenant identity must not be inferred.
Numbering and references	Human-readable serial within society/type/year + global immutable ID; no collision on concurrent jobs or year rollover.
Facts and assertions	Explicit purpose, as-of time, source basis and limitations; no blanket claim of ownership or compliance.
Languages	English, Marathi, Hindi and expandable rulepack localization; fonts embedded/licensed; translated templates individually reviewed.
Formatting	A4/letter as configured; accessible headings, selectable text, proper page counts, tables and continuation labels.
Multi-page packets	Cover index, document IDs, annexure references, page count, section bookmarks and per-attachment privacy review.
Template changes	Semantic version, reviewer, sample fixtures, preview-to-published diff; no in-place mutation of historical output.
Printable artifacts	Watermark DRAFT or COPY where applicable; printability without exposing internal or masked PII.
QR + serial	Safe verification URL carries an opaque reference/status only, not subject name/ID/data in cleartext.
Archive	Preserve source template, rendering engine version, locale, exact serialized input and PDF hash.

47.1 Example generated no-dues letter skeleton

```
[SOCIETY LEGAL NAME + REGISTRATION INFORMATION]
```

Document: NO-DUES STATEMENT Ref: [ISSUER]-[YEAR]-[SERIAL]
Subject: Unit [DISPLAY ID] As of: [2026-11-04 14:00 IST]
This society confirms the account position recorded in its unit ledger
at the stated cutoff, subject to [listed exclusions/conditions].
Outstanding posted dues: INR [amount]
Unapplied credit: INR [amount]
Pending/reconciling items: [explicit status]
The statement does not certify ownership, mortgage status or government title.
Approved by: [authorized signatory and role] on [timestamp]
Verification: [opaque QR] / [verification URL] / [status]
Document ID: [immutable UUID] Template v[integer] Page 1 of N

If the statement title is NO_DUES, policy MUST block a non-zero eligible outstanding balance unless a specifically reviewed and legally permitted disclaimer/case type is used. The example skeleton describes conditional display, not an invitation to label a disputed debt as settled.

47.2 Print and digital signature are independent

An image of a seal or handwritten signature may improve a paper workflow, but is not a cryptographic guarantee. PDF generation, officer authorization, organizational server seal, certificate-based signing and third-party electronic signing are independently configured and evidenced.

48. Electronic signing, seals and external attestations

Signature / validation level	Implementation	Truth label and limitation
A. Approval click + audit	Authenticated decision with session, actor, context, signed-off digest and timestamp.	SOCIETY_APPROVED; no claim of statutory eSign.
B. Server-authenticated artifact seal	Back-end HMAC / asymmetric signature on verification token or JWS; secure key storage/rotation.	SOCIETY_OS_VERIFIABLE; protects internal token, does not validate document legal content.
C. PDF cryptographic integrity signature	Standards-based signed bytes with certificate chain, time and revocation checking where supported.	CRYPTOGRAPHICALLY_SIGNED; signer trust must be independently established.
D. Licensed eSign / DSC	Integrate with CCA-recognized CA/service under applicable contractual/regulatory framework.	QUALIFIED/LEGALLY_RECOGNIZED only if specific method/legal purpose satisfies relevant law.
E. Stamp duty, registrar or authority filing	External regulated process, document receipt and official verification status.	EXTERNAL_AUTHORITY_CONFIRMED only on actual evidence of completion.

The Controller of Certifying Authorities distinguishes ordinary digital-signature machinery from the regulated eSign service; vendor/provider contracts, subscriber identity, key custody, cost and legal suitability remain separate engineering decisions. For a no-paid-API product, deploy A and optionally B first; expose C/D as disabled capabilities until properly implemented. No uploaded stamp image is a substitute for C or D.

48.1 Signature continuity

- Track signer authority and certificate validity at the signing time; later expiry of the signer role does not rewrite historic acts.
- Store certificate chain/verification report and revocation/time-stamp evidence where applicable for long-term validation.
- Key rotation does not invalidate prior QR verification; publish versioned public verification keys as required.
- Never let administrators upload arbitrary signing keys to a multi-tenant shared runtime.

- Any fallback from licensed eSign to unsigned internal approval must be clearly visible and potentially block issuance based on document policy.

49. KYC, identity, ownership verification and fraud escalation

“Verified” must always identify WHAT was verified, HOW, WHEN, BY WHOM and the limits of assurance. Do not build a single green checkmark that conflates readable text with governmental validation. Face recognition, biometric matching, scraping government portals and opaque risk-scoring are not default features.

Evidence / claim	Safe local verification without paid AI	Unverified until separate proof
Aadhaar secure QR / offline XML	Use currently permitted UIDAI-signed format, genuine cryptographic verification and consent/alternatives; follow OVSE applicability and latest specifications.	Presenter match, live identity state and unauthorized Aadhaar authentication.
PAN photograph/card	Readable content, allowed format, name comparison and duplicate intake warning.	Income Tax issuer status or ownership without official check.
GSTIN	Format/check digit, internal vendor naming comparison; timestamp checks if authoritative registry accessed by lawful mechanism.	Live GST registration, turnover, returns or tax compliance.
Driving licence / voter card / passport scan	Image quality, expiry candidate, field structure and manual review.	Authenticity or issuer database validity.
Registered sale deed	Page/party/unit/registration reference candidates and manual comparison to lawful record.	Current title, encumbrance, transfer legality or mutation approval.
Share certificate	Society register record and serial revocation; source chain.	Statutory property ownership where external law controls.
Rental agreement	Date ranges, unit, parties, annexures, signature appearance and duplicate hashes.	Proper stamping, executed signature, police report or lawful registration.
Police acknowledgment/report	Receipt reference, declared issuing station and upload provenance.	Actual police verification completed/accepted unless corroborated.
Vendor licence/insurance	Expiry, business name and policy reference in submitted evidence.	Issuer confirmation/current policy coverage.

49.1 Verification result object

```
{
  "evidenceId": "uuid",
  "claim": "REGISTERED_OWNER_OF_UNIT",
  "method": "LOCAL_RECORD_MATCH",
  "result": "INCONCLUSIVE",
  "assurance": "CROSS_RECORD_MATCH_ONLY",
  "matchedSourceRecordId": "uuid-or-null",
  "issuerValidation": "NOT_PERFORMED",
  "ruleVersion": 12,
  "observedAt": "2026-11-04T12:00:00+05:30",
  "reviewRequired": true,
  "redactedFindings": ["UNIT_LABEL_DIFFERENT"],
  "reviewerDecisionId": null
}
```

49.2 Human-review and fraud escalation

- Escalate unusual cases as POSSIBLE_MISMATCH or POTENTIAL_TAMPERING, never as proven fraud based only on OCR anomaly.

- Present the original page region, extracted candidate value, authoritative comparator and reviewer instructions.
- Restrict allegations and sensitive evidence to a dedicated security/compliance role; provide correction/challenge path.
- De-identify observations in analytics and prevent bias via unrelated personal traits, religion, caste, family status or location.
- Capture false-positive and overturned decisions to refine rules; no autonomous adverse action without appropriate human authority.
- Delete/redact Aadhaar content according to applicable UIDAI rules and consent withdrawal; prefer alternative IDs and never require full Aadhaar number by default.

2026 UIDAI CHANGE UIDAI publishes an Offline Verification Seeking Entity registration framework (including housing/real estate use cases). Whether a particular Society OS deployment may perform an Aadhaar verification method, or needs registration and specific consent handling, must be checked before shipping. This specification does not authorize unregistered use.

S49-01 False positive PAN tampering

Trigger / initiating event: A low-quality scanned PAN card has imperfect OCR and inconsistent name spacing.

Expected path: Classify as LOW_CONFIDENCE / REVIEW_REQUIRED with page crop and compare typed name.

Exceptions: Do not deny resident access solely on OCR, and do not call it a fake PAN.

Authorized outcome: Manual review or alternative evidence path.

Acceptance proof: Review record shows source, OCR confidence, final reason and right to correct.

S49-02 Aadhaar number accidentally uploaded unmasked

Trigger / initiating event: Resident uploads full-number card after consent notice requested redaction.

Expected path: Place in secure quarantine, avoid search indexing and routine previews, perform approved minimization/redaction or reject with clear instructions.

Exceptions: Never send to external AI; never store full Aadhaar unnecessarily; comply with UIDAI method and applicable rights.

Authorized outcome: Only permissible derived/minimized evidence becomes available to reviewers.

Acceptance proof: Access log, deletion/redaction proof and reviewer role recorded.

50. Controlled sharing, public verification and recipient evidence

Sharing must be an authorized issuance event, not a long-lived publicly accessible object URL. A resident, admin or external recipient may have different legal bases to view a document. Public verification should answer authenticity/status without exposing identity attributes, financial balance or the underlying file.

Sharing mode	Controls
Resident private download	Fresh access check on active context, purpose, document sensitivity; short-lived signed URL or controlled streaming.
External secure share	Explicit recipient name/contact, specific artifact/version, minimum fields, expiry, view count and withdrawal.
Society-to-vendor packet	Redacted need-to-know evidence only; do not reveal unrelated residents or historic tenant documents.
Committee/auditor dossier	Case-scoped permission, time-bound review workspace, export log and legal hold behavior.

Sharing mode	Controls
Public QR status	Opaque lookup ID; ACTIVE/REVOKED/EXPIRED/SUPERSEDED + issuer/type/date and limited public fields.
File-byte verification	Optional SHA-256 comparison on client-submitted bytes within size limits; expose result only, never original.
Proof of delivery	Dispatch attempt, provider accepted vs delivered vs acknowledged; signed receipt where policy applies.

50.1 Public verification response

```
GET /verify/v2/documents/{opaqueToken}
200 {
  "documentType": "RESIDENCE_LETTER",
  "issuer": "Example CHS",
  "reference": "CHS-2026-000087",
  "issueDate": "2026-11-04",
  "currentStatus": "SUPERSEDED",
  "supersededByPublicReference": "CHS-2026-000091",
  "verificationScope": "SOCIETY_ISSUANCE_AND_STATUS_ONLY",
  "personalDetailsReturned": false
}
```

Token guessing must be rate-limited; error timing and visible metadata should not facilitate membership enumeration. If a document is revoked, previous verification links MUST show the revocation status. A copied screenshot is not current proof.

51. Tender, contract and vendor documentation - complete lifecycle

v4.4 introduced RFQ and sealed bid controls; v4.5 completes the record chain from procurement need to vendor termination and liability documents. Treat procurement as a regulated business process where conflicts and confidentiality matter even for non-government societies.

Stage	Documents and authoritative decision	Non-negotiable controls
Demand and budgeting	Requisition, scope/BoQ, estimate, budget availability, approval threshold	Policy against splitting purchases to bypass signoff; cost estimates versioned.
Sourcing design	RFQ/RFP/tender template, vendor qualification, selection method, conflict declarations	Procurement mode and evaluation rubric fixed before bids open.
Publication/ addendum	Invitations, mandatory clarifications and addenda; vendor acknowledgement	Every bidder gets controlled current version; deadline changes audited.
Bid receipt	Sealed upload, bid receipt/hash, bidder time, late flag	No pre-opening view of commercial details; deadline server authoritative.
Opening	Opening committee register, members present, document integrity summary	No hidden replacement after close; exceptions independently approved.
Technical evaluation	Mandatory compliance, deviations, evidence and reviewer notes	Failure of mandatory requirement excludes or routes lawful waiver.
Commercial comparison	Line-item price, taxes, alternatives, total cost, scoring method, summary	Decimal arithmetic and published weighting; no black-box winner.
Award	Decision minutes, conflict and approval proof, LOI/PO, contract signing	Winning bid alone is not award; signatory and budget requirements.

Stage	Documents and authoritative decision	Non-negotiable controls
Execution	Mobilization, licences, insurance, milestones, work orders, safety docs	Expiry hold if vendor lacks needed approvals.
Acceptance and invoice	GRN, completion certificate, quality report, invoice/PO/GRN match	Variances route to dispute; funds disbursed only by finance subsystem.
Renewal/closeout	Performance review, defects, retention, final settlement, keys/access revocation	Historical record and obligations retained under policy.

51.1 Anti-collusion and evidence safety

- Do not email raw bid submissions to all committee members; use controlled sealed-access and audited opening with key custody or encrypted private objects.
- Support vendor declarations of related-party status; restrict conflicted decision makers and record reasons for overrides where allowed.
- Allow correction/addendum to a bid only before deadline and according to the tender policy; preserve all versions.
- Record evaluation score components and source evidence. A deterministic formula may recommend a rank but cannot independently finalize vendor selection.
- Make exceptional single-source/emergency procurement a separate audited workflow rather than faking three quotations.
- For contracts, distinguish society document preparation from legal execution, stamping, regulatory approval and countersigned receipt.

S51-01 Late bid arriving during upload

Trigger / initiating event: Vendor starts a 50 MB bid upload before 5 PM but finishes at 5:00:03 PM.

Expected path: Policy defines whether server-received completion time or an explicit sealed submission commit time governs; preserve both and show receipt.

Exceptions: Client clock is not authoritative; no implicit grace period.

Authorized outcome: LATE or ACCEPTED according to published policy, with audit and appeal path.

Acceptance proof: A replayed idempotency key cannot change the verdict.

S51-02 Emergency plumbing purchase

Trigger / initiating event: A main water pipe bursts at 2 AM, requiring immediate replacement without a regular tender.

Expected path: Apply declared emergency procurement exception with bounded authority, ceiling, incident ID and retroactive review deadline.

Exceptions: No unchecked override for unrelated procurement; preserve purchase orders and vendor invoice mismatch.

Authorized outcome: Emergency record + required ex-post approval; no fictitious sealed bidding.

Acceptance proof: Case cross-links outage, quote, approval, PO, inspection and bank payment status.

52. Finance-linked attestations, receipts and recoveries

Official financial letters are only as correct as their ledger read model. Finance must reconcile charges, payments, credits, suspense, refunds, advance balances, reversals and period locks before claiming dues cleared. A PDF is not an independent accounting source.

Financial artifact	Issue-time check and exception
Payment receipt	Only after settled posting, unique allocation and valid source reference. A cheque awaiting clearance shows provisional acknowledgement, never paid receipt.
No-dues certificate	Balance as-of cutoff and eligible unpaid/pending components exactly evaluated; policy may require unbilled charges/asset deposit clearance.
Final settlement	Separate accrued charges, posted payments, refundable deposit and bank payout; unmatched refund remains PENDING.
Demand notice	Calculate interest/late fees with versioned effective policy; reveal amount breakdown, dispute route, and how to pay.
Statement for lender or buyer	Authorization and disclosure consent where required, as-of date, historical adjustments, period scope, legal limitations.
Audited annual statement	Internal draft vs independently audited record, auditor signed evidence and adoption minute separated.
Budget/utilization certificate	Budget source, commitment vs actual, head-level breakdown and approving authority.

S52-01 Last-minute payment and duplicate issue

Trigger / initiating event: Resident pays at 11:59:59 and clicks issue-NOC twice before the payment webhook confirms settlement.

Expected path: Payment stays PENDING; NOC recheck reads ledger in consistent transaction and blocks until posted.

Exceptions: Do not mark as paid using frontend redirect or gateway attempt.

Authorized outcome: Exactly one resulting issue after settlement; retries return same request reference.

Acceptance proof: Snapshot records settled transaction and no duplicate generated serial.

S52-02 Month-end late fee correction

Trigger / initiating event: A delayed adjustment retroactively changes dues after a statement was issued.

Expected path: Keep original issued document and ledger; mark material statement superseded if policy requires and issue corrected version.

Exceptions: Do not mutate old PDF or ledger journal.

Authorized outcome: Historical verifier points to supersession and current document.

Acceptance proof: Both calculations and adjustment reference are audit-visible.

53. Property, resale, loans, succession, tenancy and renovation

The underlying model must keep property title, society membership, share rights, possession, tenant occupancy, nominee status, parking allocation, bank security interests and delegated authority as different legal/operational dimensions. Documents may be required by specific banks, registrars or society policy, but Society OS cannot substitute for their decisions.

Case	Packet and decision model	Caveat
Resale / transfer	Seller application, permitted NOC or factual dues letter, buyer application, member register decision, share/certificate lifecycle.	Never assume an NOC is legally mandatory for every transfer; check latest jurisdiction rules and registered bylaws.
Inheritance / nomination	Death/succession evidence, nominee filings, other claims, legal review and share register event.	Nominated person not automatically legal owner.

Case	Packet and decision model	Caveat
Home loan / bank NOC	Authenticated member consent, bank request, charges and property/membership statement.	No representation about mortgage priority or encumbrance absent independent records.
Rental and police process	Owner consent, signed agreement evidence, identity alternatives, police reference status, period and access.	Police verification cannot be generated internally.
Move-in / move-out	Lift access, vehicles, deposits, dues, handover assets, dispute exception, gate revocation.	NOC only after relevant clearance and approval.
Renovation	Drawings, contractor permit, insurance, deposit, schedule, sound rules and inspections.	No replacement for municipal or structural permission.
Redevelopment	Project decision, meeting eligibility/consent, consultant proposals, legal documents and time-line.	High-risk legal undertaking; expert review and no AI-generated certification.

53.1 Unit timeline is the key to document truth

Every issued letter referencing residence or occupancy must use effective-dated owner/tenant/authorized-occupant records. A user associated with a unit today cannot automatically view a former resident's agreement, police evidence, contact details or sensitive complaints. A committee can access historical records only under the granted business purpose and audit control.

54. Governance, elections, audits, disputes and legal holds

Meetings and resolutions are a document-intensive chain, not a minutes template. Preserve notice -> circulation -> eligibility -> quorum -> attendance/proxy -> agenda version -> deliberation -> proposed resolution -> votes/approval -> signed minutes -> action items -> amendment/objection.

Governance document	Workflow and truth boundary
AGM/SGM notice	Law/society policy notice period; attendees/eligibility are a snapshot, with delivery/acknowledgement tracked.
Meeting agenda/annexures	Stable agenda version and permission-filtered distribution; late additions explicitly approved.
Proxy and attendance	Capture eligibility cutoff, identity, revocation and duplicates; do not count one vote twice.
Minutes and resolution register	Draft/approved states; preserve material dissents, chairman/secretary approval and correction/ratification.
Election paperwork	Independent legal/procedural readiness gate; ballot secrecy and e-voting controls separate from simple polls.
Audit requests	Time-limited read-only access, exception requests, evidence trace and export/report signatures as configured.
Disputes/legal notices	Incoming notice proof, case number, response owner, deadline, privileged content restriction and hold.
Management transfer	Handover checklist for assets, bank access, vendor contracts, society seal, records, keys, unclosed incidents.

54.1 Evidence preservation and litigation hold

- Legal holds apply to source record, object version, case event, audit and relevant backups where feasible; log affected scope and hold owner.
- Hold does not imply broad user visibility. A matter-specific legal workspace accesses only approved items and logs each access.
- Export bundles include manifest with hash, generated time, date range, files, provenance and chain of custody; never label as court-admissible automatically.
- When a hold ends, reevaluate ordinary retention schedule and deletion rights; do not immediately destroy records while a legal review is incomplete.
- Resolve redaction conflict by producing separately versioned redacted disclosure objects, never overwriting the confidential original.

S54-01 AGM minutes disputed six months later

Trigger / initiating event: A member alleges a resolution was never voted on, while a new committee has already edited the display text.

Expected path: Retrieve the immutable agenda, attendance/quorum evidence, votes, original minutes and each amendment as separate changes.

Exceptions: No backdating or overwriting adopted minutes.

Authorized outcome: Dispute case links signed historic evidence; amended record identifies correction date and authority.

Acceptance proof: Hashes, issue dates, participants, authority and actions independently traceable.

55. Compliance and safety document operations

Build a compliance register with legal applicability review rather than a list of fixed Indian statutory requirements. Housing society registration form, common-area safety, lift inspection, fire safety, structural audit, water quality, workplace/vendor labour, insurance and environmental duties vary materially by locality/building and ownership arrangement.

Compliance family	Evidence to model	Automated action / limit
Fire and emergency	Fire equipment checklist, drill logs, authority-issued inspection/licence, incident report.	Notify renewal and corrective tasks; issuer authenticity remains separate.
Lift and mechanical	Licence, inspection, AMC contract, maintenance events, safety defect and downtime.	Escalate overdue critical inspection; do not certify lift safe automatically.
Building structure	Structural/technical audit report, defects, repairs and engineer signoff.	Track professional source and remediation; software cannot be engineering certification.
Water / electrical / solar / EV	Meter reports, contractor qualification, utility permissions, maintenance and safety logs.	Rulepack-defined intervals; no universal utility licence assumption.
Housekeeping / waste / pest control	Work checklist, vendor proof, sampling, service SLA and complaints.	Calculate completion vs validated outcome; attendance alone not service quality.
Insurance / claims	Policy period, premium status, coverage summary, claim evidence and settlement.	Highlight potential gap; insurer acceptance and actual coverage separate.
Resident/contractor safety	Permit-to-work, safety induction, PPE/precautions, risk checklist.	Stop-work/escalate through authorized safety officer, not autonomous AI.

55.1 Evidence completeness is not compliance

COMPLIANCE_DOCUMENT_RECEIVED, COMPLIANCE_REVIEWED, COMPLIANCE_AUTHORITY_VERIFIED and COMPLIANCE_FULFILLED are distinct statuses. An unverified upload cannot make the compliance dashboard green. If source unavailable, show UNKNOWN or UNCONFIRMED and create an owned exception instead of silently lowering the numerator.

56. Privacy, retention, deletion and disclosure governance

This module will contain identity documents, legal evidence, minor-related records, financial data, private household documents and incident photos. Every file must have a purpose code, controller/processor context, data category, subject/relationship, permitted viewers, retention class and audit requirements. A single seven-year or eight-year policy for all files is not appropriate.

Data category	Indicative policy design (NOT a fixed statutory retention rule)
Identity proof / KYC	Purpose-specific minimum data; alternatives to Aadhaar; explicit method eligibility; short retention where legally allowed.
Owner/transfer / legal	Long-term historic chain where lawful; separate secure access from current tenant/unit view.
Financial journals / receipts	Record class tied to statutory/tax/accounting obligations, accounting lock and audit requirements.
Visitor and transient photos	Short configurable operational retention; longer only for incident/hold or specific lawful duty.
Tender/contract	Procurement threshold and legal defense/contract retention policy; protect rejected bids.
Incident / complaints	Sensitivity-based duration; holds for claims, police matters or disputes where appropriate.
Consent and rights requests	Purpose, consent/withdrawal, requests, decision evidence and limited audit provenance.
Backups / search indexes	Retention and deletion propagation windows documented; no unbounded unindexed shadow copies.

56.1 Data subject requests and lifecycle

Capture an identity-verified data access/correction/erasure/withdrawal/nomination/grievance request as an independent case, with a legally versioned rights policy and applicability timeline. The DPDP Act and the DPDP Rules have phased commencement; engineering must track effective legal dates and current official notifications rather than assuming all obligations began simultaneously.

- Identify requester and scope, including multi-society context, without exporting other household members or unrelated society records.
- Map applicable rights, exceptions, legal hold and record of lawful basis; communicate expected status according to valid policy.
- Produce minimized export; correct inaccurate records by versioned change; delete/anonymize only eligible data.
- Withdraw consent for a particular voluntary verification method without falsely assuming all accounting/operational records must be deleted.
- Propagate permitted removal to generated previews, indexes, temporary files and backups according to controlled retention/recovery design.
- Keep a minimal rights-request decision log for audit; do not persist unnecessary copies of sensitive source bytes.

56.2 Privacy and consent notice fields

Field	Requirement
purposeCode	Specific business purpose; avoid generic “all future services” default.
evidenceCategory and fields	Disclose what information is requested and alternatives when relevant.
reviewers and recipients	Named category of persons/organizations with purpose-limited sharing.
retentionPolicyRef	Effective schedule, review obligation and lawful exceptions.
withdrawalRoute	Contact and in-product action; warn of necessary effect on unsupported voluntary service.
consentEvent	Granular opt-in where required, timestamp, locale, notice version, subject or guardian and proof.

57. Zero-paid-AI search, OCR, rules and local intelligence

Core requirement: full operational value with no external OpenAI, Gemini, Claude or comparable per-token API. A conventional Java backend, relational queries, policy engine and open-source parsers are sufficient for the critical tasks. Optional local inference must degrade to manual handling and never change financial/legal truth without a governed approval.

Capability	No-paid-AI implementation	Quality and reliability boundary
Template generation	PDFBox / OpenHTMLToPDF / Apache FOP + versioned fixtures.	No model required. Layout tests and input escaping mandatory.
Metadata extraction	Apache Tika/PDFBox, JSON/PDF parsers, regular expressions and typed validators.	Extraction proves readability, not authenticity.
Image OCR	Optional Tesseract in resource-limited private worker and image preprocessing.	Quality varies; cap CPU/memory, sample accuracy; never raw auto-KYC acceptance.
Document classification	Configured taxonomy; filename/metadata hint; deterministic rule matching; human selection.	Unknown remains UNCLASSIFIED; no destructive auto-filing.
Search	PostgreSQL full-text search / trigram, indexed approved metadata; per-record ACL at query time.	Do not index raw unredacted Aadhaar, secrets or legal privilege by default.
Duplicate review	SHA-256 byte duplicate, optional perceptual similarity for manual review.	Matching hash means same bytes, not same validity or identity.
Anomaly detection	Date/amount/party mismatch, missing annexure, expiration, impossible state transitions, checksum.	Explainable flags only; no automatic fraud finding.
Rules / reminders	Typed policy evaluator, cron/Quartz, transactional outbox, idempotent jobs.	No model required; monitor missed scheduled runs.
Signature & QR	JCA/Bouncy Castle where warranted, ZXing, internal signed token.	Use trusted issuer keys and legal signability checks; QR text alone not enough.
Optional local LLM	Self-host only for opt-in internal summaries/semantic search, budgeted hardware and eval.	Deny mutations; retrieve authorized data only; summary is never evidence.

57.1 Trust-aware search architecture

14. Index only ACL-approved fields and sanitized OCR; store per-document visibility, societyId, subject and classification alongside index.
15. Apply society membership, role, per-unit relationship, purpose and sensitivity filters before retrieval, not after ranking or result assembly.
16. Log sensitive search queries and resulting disclosure when policy requires, without repeating full secret keywords.
17. Search result snippet must be redacted and its link must still require a new authorization decision.
18. Never share raw document text with any hosted general-purpose AI endpoint absent explicit approved contracts and policy.
19. If embeddings are enabled locally later, enforce the same tenant-aware prefilters and delete vectors with source records.

57.2 Verification accuracy measurement

Maintain labeled test sets by source type and language: clean native PDFs, Marathi/English mixed documents, phone photos, low-light gate scans, multi-page legal documents and handwritten stamps. Track recall on detecting required expiry dates, false-positive mismatch rate, unreadable rejection and human correction rate. Targets must be measured on representative deployment devices; no assumed 99% OCR accuracy.

58. Reference architecture, queues and integration contracts

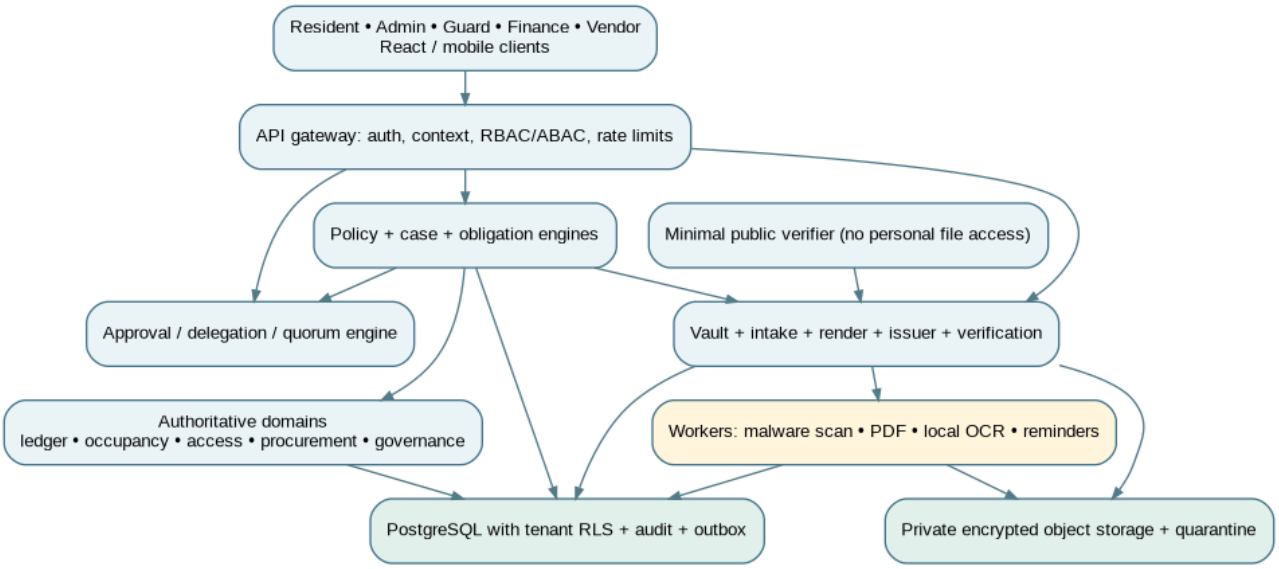


Figure B. Reference boundaries; logical design rather than a claim about currently deployed Society OS services.

58.1 Service contracts

Service	Owned state	Requests / events
CaseService	Case, evidence requirement graph, challenge and state	case.created; requirements.changed; case.closed
EvidenceService	Private object, scan state, page findings, provenance and custody	evidence.received; scan.completed; review.decided
PolicyService	Immutable jurisdiction/society policy version and rule execution	policy.published; rule.evaluated; rule.stale
ApprovalService	Step/task, delegation, recusal, decision and committee basis	approval.decided; approval.invalidated

Service	Owned state	Requests / events
IssueService	Unique serial, immutable bytes, snapshot and status	document.issued; document.superseded; document.revoked
RegisterService	Membership, nomination, correspondence and document register records	register.entryCreated; register.correctionRecorded
ObligationService	Applicability, due period, task owner and compliance finding	obligation.due; obligation.breached; obligation.fulfilled
DisclosureService	Share grant, recipient, proof, verification/exports	disclosure.granted; disclosure.revoked; download.logged

58.2 Failure boundaries and event ordering

- Use outbox-in-one-transaction for business state and event intent; queue delivery is at-least-once, so consumers must be idempotent.
- Event payload carries tenant, aggregate ID, aggregate version, correlation ID, causation ID and schema version; source of truth remains PostgreSQL.
- File antivirus unavailable: upload remains QUARANTINED, never auto-clean; viewing and processing are blocked per policy.
- PDF worker unavailable: issuance remains PROCESSING/RETRY_PENDING, not ISSUED; no user-facing success claim.
- Local OCR unavailable: evidence review proceeds manually; no NOC or legal-critical workflow depends on OCR accuracy.
- Public verification unavailable: show service temporarily unavailable, never VALID by cached assumption beyond configured staleness window.
- Policy service unavailable: high-risk issuance fails closed; ordinary published documents remain viewable by authorized users.

58.3 Event ordering contract

```
{
  "eventId": "uuid",
  "eventType": "document.issue_committed",
  "schemaVersion": 1,
  "societyId": "uuid",
  "aggregateId": "uuid",
  "aggregateVersion": 8,
  "businessContext": {"unitId": "uuid", "caseId": "uuid"},
  "occurredAt": "2026-11-04T12:00:00Z",
  "correlationId": "uuid",
  "payloadRef": "private-event-payload-reference"
}
```

59. Extended PostgreSQL data model and migration controls

The existing blueprint already defines document_records and other core tables. These proposed additions complement rather than silently replace them. Every tenant-scoped object must carry society_id, direct RLS, composite foreign keys or equivalent invariant checks, and an intentional retention policy. Global user identities may remain outside society scope but access relationships are always checked.

New entity	Critical data / invariants
document_definitions	typeCode, tenant/default scope, product family, issuer category, version, permissions, effective interval, activeFlag.
document_templates	definitionId, locale, templateVersion, renderingEngine, contentObjectId, hash, publication, safe variables, approver.
record_registers / register_entries	registerType, sequence, societyId, subject references, effective time, correction pointer, immutable event sequence.
document_cases	caseType, businessContextId, requestor, state, idempotency, policyVersion, optimisticLock, retention class.
case_requirements / evidence_links	requirements, mandatory conditional logic, evidenceId, subject, review state, purpose justification.
object_records / object_versions	private key, checksum, scanState, encryptionKeyVersion, origin, retention/hold, version chain.
page_assessments / verification_results	page index, extracted candidates, method/trust result, issuerSource, ruleVersion, assessor and note.
approvals / delegation_grants	caseId, decision sequence, actor, quorum reference, recusal, signed approvedDigest and revocation.
issued_artifacts	issueId, serial, case, snapshotRef, templateHash, objectVersion, status, issuer identity and validity.
public_verifier_tokens	opaque ID hash, issueId, keyVersion, scope, expiry, revokedAt and verification policy.
obligations / obligation_instances	rulepack, applicability, due, recurrence key, owner, evidence, compliance status and source.
sharing_grants / disclosure_events	recipient, purpose, artifact version, fields disclosed, expiry, download, revoke and trace.
legal_holds / disposal_jobs	matter, subject record graph, authority, hold interval, eligibility, deletion proof.
physical_custody_events	hardcopy identifier, drawer/location, custodian, check-out/return, signature and exception.
import_manifests / migration_items	source hash, mapping, validation, external system, row state, quarantine and retry idempotency.

59.1 Core relational example (illustrative migration)

```
-- Target: PostgreSQL. Adapt references to the deployed schema.
CREATE TABLE document_cases (
  id UUID NOT NULL,
  society_id UUID NOT NULL REFERENCES societies(id),
  case_type VARCHAR(70) NOT NULL,
  context_id UUID,
  requester_user_id UUID NOT NULL,
  state VARCHAR(32) NOT NULL,
  policy_version INTEGER NOT NULL,
  version BIGINT NOT NULL DEFAULT 1,
  client_idempotency UUID NOT NULL,
  created_at TIMESTAMPTZ NOT NULL DEFAULT now(),
  PRIMARY KEY (id),
  UNIQUE (society_id, id),
  UNIQUE (society_id, requester_user_id, client_idempotency)
);
CREATE TABLE case_evidence_links (
  id UUID PRIMARY KEY,
  society_id UUID NOT NULL,
```

```
  case_id UUID NOT NULL,
  document_record_id UUID NOT NULL,
  requirement_code VARCHAR(80) NOT NULL,
  review_state VARCHAR(32) NOT NULL,
  created_at TIMESTAMPTZ NOT NULL DEFAULT now(),
  FOREIGN KEY (society_id, case_id)
    REFERENCES document_cases(society_id, id)
);
-- Deployed schema MUST also constrain document_record_id
-- to the same society, using composite FK or equivalent strict invariant.
ALTER TABLE document_cases ENABLE ROW LEVEL SECURITY;
ALTER TABLE document_cases FORCE ROW LEVEL SECURITY;
ALTER TABLE case_evidence_links ENABLE ROW LEVEL SECURITY;
ALTER TABLE case_evidence_links FORCE ROW LEVEL SECURITY;
CREATE POLICY doc_cases_tenant ON document_cases
  USING (society_id = NULLIF(current_setting('app.current_society_id',true),'')::uuid)
  WITH CHECK (society_id = NULLIF(current_setting('app.current_society_id',true),'')::uuid);
CREATE POLICY case_evidence_tenant ON case_evidence_links
  USING (society_id = NULLIF(current_setting('app.current_society_id',true),'')::uuid)
  WITH CHECK (society_id = NULLIF(current_setting('app.current_society_id',true),'')::uuid);
```

The snippet is illustrative and intentionally not a complete deployable migration. Production DDL must include all required composite references/indexes and validated role privileges, plus immutability mechanisms. Set tenant context transaction-locally from an authenticated server claim; never accept arbitrary client-provided societyId as authority. Database administrators, migration roles and superuser connections require separately controlled access.

59.2 Migration and backfill strategy

- Use expand/migrate/contract deployment. Add versioned enums/metadata before enabling new workflow routes.
- Map existing document_records to object versions; do not regenerate hashes or rename object keys without recorded migration provenance.
- Backfill documentType and issuerKind with UNKNOWN where uncertain; never infer verified status from file names.
- Recompute index and retention with dry-run reports; quarantine errors and keep a signed migration report.
- Apply RLS on all new tables and test cross-tenant joins, background worker connections and native SQL.
- Keep old APIs functional until consumer migration and mobile offline sync compatibility tests pass.
- Provide rollback only where state semantics support it; never delete legitimate post-migration issued artifacts.

60. Versioned REST and verification API contracts

All private endpoints require authenticated identity, server-resolved society/unit context and granular object authorization. Every write supports an idempotency key where retry is possible, request version/If-Match for conflict avoidance, rate limits and trace metadata. Public verification has its own narrowly scoped contract.

Method + endpoint	Meaning	Controls
POST /api/v2/document-cases	Create type-specific case	Auth, context, policy, idempotency, duplicate business key
GET /api/v2/document-cases/{id}	Read permitted timeline / requirements	Object authorization and redaction
POST /api/v2/document-cases/{id}/submit	Submit snapshot for processing	If-Match, required evidence, state transition

Method + endpoint	Meaning	Controls
POST /api/v2/document-cases/{id}/evidence-intents	Issue scoped signed upload intent	MIME/size, quota, quarantine and purpose
POST /api/v2/document-evidence/{id}/review	Record verification review findings	Reviewer scope, step-up auth, page-level proof
GET /api/v2/document-cases/{id}/assessments	Show permitted check results	Source certainty/limitations explicit
POST /api/v2/document-approvals/{id}/decide	Approve/reject/need-info	Separation of duties, version and delegation
POST /api/v2/document-cases/{id}/issue	Reserve idempotent issue	Fresh domain validation; checker and snapshot
POST /api/v2/issued-documents/{id}/revoke	Revoke prior artifact	Authority, reason, effectiveAt, urgent verifier invalidation
POST /api/v2/issued-documents/{id}/correct	New issue version from existing	No overwrite, parent pointer and reapproval policy
GET /api/v2/issued-documents/{id}/download	Controlled file streaming / signed URL	Fresh object relationship + audit
POST /api/v2/issued-documents/{id}/shares	Create limited recipient sharing	Purpose, minimal disclosure, expiry, recipient proof
DELETE /api/v2/document-shares/{id}	Revoke share immediately	Issuer authorization and audit
GET /api/v2/registers/{type}/entries	Scope-safe register query	Period filter, read-only, pagination, redaction
POST /api/v2/record-extracts	Request attested excerpt	Eligibility and human approval where required
POST /api/v2/obligation-instances/{id}/evidence	Link renewal/inspection proof	Applicability, source, typed requirement
POST /api/v2/obligation-instances/{id}/decide	Accept/reject compliance finding	Compliance authority; legal overrides forbidden
GET /verify/v2/documents/{token}	Minimal public status verification	Rate limit, opaque token and private detail exclusion
POST /api/v2/privacy-requests	Raise personal-data request	Identity, scope, rights policy, timeline, hold
GET /api/v2/exports/{id}	Status of controlled export	Scope, TTL, download permissions and audit

60.1 Issue request example

```
POST /api/v2/document-cases/{caseId}/issue
Authorization: Bearer <token>
X-Society-Id: <context-hint-only>
Idempotency-Key: <uuid>
If-Match: "case-version-19"
{
  "requestedDocumentType": "MOVE_OUT_NOC",
  "approvedDecisionId": "uuid",
  "templateVersion": 4,
  "clientExpectedEvaluationId": "uuid"
}
202 {
  "issueId": "uuid", "state": "ISSUANCE_RESERVED",
  "clientVisibleMeaning": "Request accepted; document not yet issued",
  "traceId": "uuid"
}
```

60.2 Invalidation / conflict errors

```
409 {
 "code": "DOCUMENT_DECISION_STALE",
 "currentCaseVersion": 21,
 "recheckRequired": true,
 "blockers": ["LEDGER_VERSION_CHANGED"],
 "traceId": "uuid"
}
422 {
 "code": "DOCUMENT_CLEARANCE_BLOCKED",
 "evaluatedAt": "2026-11-04T09:00:00Z",
 "checks": [
  {"code":"FINANCE_OUTSTANDING", "state":"BLOCKED",
   "outstandingMinor":2640000,"currency":"INR"}
 ],
 "nextAction": "OPEN_LEDGER_OR_START_DISPUTE_REVIEW"
}
```

60.3 Data minimization in errors

A 403/404 must not reveal whether a foreign society's private document exists. Use stable error codes rather than PII-rich descriptions. Error details may show a requester only their authorized own dues; public endpoints must never return personal balances, identity numbers, leases, case notes or private source PDFs.

61. Security, observability, capacity and disaster recovery

Threat / operational condition	Required mitigation and verification
Cross-tenant IDOR and join leakage	RLS and relationship authorization, including queues, signed URLs, search, export and reports.
Malware / parser exploitation	Quarantine, scanned content, isolated workers, hardened parsers, timeout/memory/file expansion caps.
Template injection / SSRF	Allowlisted placeholders; no external image/document fetching or arbitrary code/HTML from users.
Counterfeit QR / forged PDF	Signed opaque tokens, SHA-256 byte check and server-side revocation state.
Credential theft / forged admin approval	Step-up auth, device/session revocation, delegated authority checks and dual control.
Insider bulk export or harassment	Export budgets, privileged access alerts, four-eyes release for sensitive archives, audit reconciliation.
Tenant archive overexposure	Current occupant must not automatically read past tenancy/KYC or private complaints.
Billing race during no-dues	Snapshot/locking/optimistic concurrency, last-minute source freshness check.
Aadhaar/ID misuse	Purpose gating, consent/alternatives, redaction and no forbidden full-number storage.
Backup restores old verifier status	After restore, replay authoritative revocation outbox and validate public token status before accepting traffic.
Search indexing PII	Sensitive class filtering, encryption and purge on subject rights/retention events.
Audit tampering	Append-only event sink, signed/checksummed segments if configured, separate operator privileges.

61.1 SLO targets as starting hypotheses (benchmark before commitment)

Metric	Candidate target / verification
Document case creation	p95 < 800 ms at benchmark-defined concurrency; no upload content processing on request thread.
Simple policy evaluation	p95 < 2 seconds for no-dues on healthy domain services; report source freshness.
Normal <= 5-page PDF issuance	p95 < 10 seconds async excluding legal signing/provider calls; no duplicate issue on retries.
Public verifier	p95 < 400 ms in-region with authoritative revocation freshness policy; availability monitored.
Sensitive object exposure	0 unscanned/quarantined objects retrievable; 0 cross-tenant disclosures in negative test suite.
Important event reconciliation	100% of issued/revoked artifacts have verifier projection and audit/outbox event; retry repair job.
Scheduled obligation correctness	Exactly one obligation instance per period/key; missed reminder jobs visible in ops alerting.
Recovery objective	Set explicit RPO/RTO per tier after infra sizing; perform quarterly restore drills as policy permits.
Search/OCR	P95 indexed search and OCR job-time budgets based on representative size; bounded worker quotas.

61.2 Monitoring and cost constraints

- Metrics: case volume, completion p50/p95, NOC rejection reasons, stale approvals, expiry overdue, manual verification time, OCR corrections, malformed uploads, negative access attempts, issued/revoked status lag.
- Logs: trace IDs and rule IDs, not raw Aadhaar/PAN numbers or full sensitive document text. Separate security log access from normal support.
- Cost controls: storage lifecycle, maximum OCR pages per society/day, worker memory/CPU quotas, PDF generation queues, local cache with strict TTL, optional retrieval index.
- Incident process: triage, privileged disclosure response, immutable impact record, key rotation, user/recipient notices when legally required, forensic preserve and recovery tests.
- Backup process: database + object storage + encryption-key recovery + queue state; restore validation includes sample document hashes, RLS and financial references.
- Review service account keys and external signed URLs; revoke on committee transition, lost admin device and breach.

62. Scenario-driven acceptance and boundary tests

The following tests are additional to DOC-001...DOC-036 in v4.4. Each case must be automated where feasible; otherwise supply repeatable manual verification steps, fixture data, expected state transitions, audit events and download/verification proof. “Pass” means the actual backend implements the behavior in a production-like environment, not that a document has been written.

62.1 Event and case initiation

Test ID	Trigger / adversarial condition	Required proof
V45-001	Duplicate resale request under concurrent click	Exactly one active policy-defined case, stable idempotent response and one audit source.

Test ID	Trigger / adversarial condition	Required proof
V45-002	One user owns homes in different societies	Case context selected independently; unrelated home evidence never attached.
V45-003	Event trigger disabled by society feature flag	No backend case creation, predictable denial and no stale UI workaround.
V45-004	Retroactive residence letter with no history	INSUFFICIENT_EVIDENCE; no fabricated residency period.
V45-005	Case applicant withdraws after legal hold	Workflow stops but legally held evidence and audit remain preserved.
V45-006	Parent packet contains revoked evidence link	Re-evaluation invalidates dependent approvals; no issuance from stale evidence.
V45-007	Unrecognized society type / jurisdiction	Legal-dependent case blocked or manual review; never generic statutory assertion.

62.2 Physical intake and evidence

Test ID	Trigger / adversarial condition	Required proof
V45-008	Malicious executable renamed PDF	Quarantine and scan failure; zero normal downloads and safe worker termination.
V45-009	100 MB or zip-bomb PDF	Configured parser ceiling applies without worker OOM or server impact.
V45-010	21 of 22 contract pages with annexure reference	Completeness flag and human review path, not falsely complete.
V45-011	Repeated byte-identical upload	Provenance preserved, optional dedup, no cross-tenant evidence identity merge.
V45-012	Mixed Marathi-English document with low OCR confidence	Candidate extraction only; manual entry supported, no automatic ID acceptance.
V45-013	Photo with geolocation EXIF	Sensitive metadata stripped for ordinary preview while custody source tracked if lawful.
V45-014	Physical share register overdue checkout	Custody exception and escalation with original custodian/time recorded.
V45-015	5,000-record backfile import with 100 invalid associations	Valid rows previewed, errors downloadable, replay idempotent and auditable.

62.3 Verification and identity

Test ID	Trigger / adversarial condition	Required proof
V45-016	Aadhaar offline signature invalid	CRYPTO_INVALID and no verified identity claim.
V45-017	Full Aadhaar number accidentally shared	Minimization/quarantine path and no searchable full-number leakage.
V45-018	Resident rejects Aadhaar consent	Alternative allowed; no coerced Aadhaar-only onboarding.
V45-019	PAN OCR matches format but forged image	STRUCTURE_PASS only; no source verified badge.
V45-020	GST checksum valid but business dissolved	No active-registration assertion absent authoritative source proof.

Test ID	Trigger / adversarial condition	Required proof
V45-021	Police receipt uploaded without government confirmation	EXTERNAL_NOT_VERIFIED / reviewer state, never “police cleared”.
V45-022	Former resident requests previous occupant KYC	Denied with no metadata disclosure and security audit.
V45-023	Manual reviewer overturns mismatch after new evidence	Old assessment preserved, new decision/version and reason logged.
V45-024	Unlicensed offline QR verifier feature	Feature remains disabled until legal/vendor applicability cleared.

62.4 Approval, delegation and committee

Test ID	Trigger / adversarial condition	Required proof
V45-025	Maker approves own sensitive finance NOC	Denied with separation-of-duties error and no issuing side effect.
V45-026	Delegated signatory acts after grant expiry	403 even if cached UI still displays approval button.
V45-027	Bid reviewer declares conflict	Recusal blocks access to sealed bid and decision action.
V45-028	Committee resolution without quorum	Meeting-based approval invalid, no official resolution extract.
V45-029	Approved document amount changed	Existing approval stale and requires fresh signoff.
V45-030	Committee handover terminates old signer access	Immediate revocation of signing rights; historic signed records remain intact.
V45-031	Emergency break-glass approval	Reason, constrained scope, expiry and follow-up review are mandatory.

62.5 Finance and issuance

Test ID	Trigger / adversarial condition	Required proof
V45-032	NOC requested during UPI pending event	Blocked until authoritative ledger settlement.
V45-033	Financial ledger updates after approval	Issue-time recheck invalidates stale decision.
V45-034	PDF worker crash after serial reservation	Idempotent retry: one serial, one active artifact.
V45-035	Object saved but DB commit fails	Orphan quarantined; reconciliation avoids public exposure.
V45-036	Issued statement later corrected	Old bytes immutable, supersession record, verifier updated.
V45-037	Wrong payment reversed after receipt	Receipt state reflects reversal; ledger preserved with compensating entry.
V45-038	No-dues with pending cheque	Pending cannot count as fully cleared unless actual policy/evidence supports it.
V45-039	Thirty parallel issue requests across two societies	Tenant-isolated serial uniqueness and no duplicate active business outcomes.

62.6 Public verifier and controlled sharing

Test ID	Trigger / adversarial condition	Required proof
V45-040	QR token guessed via brute force	Rate limited/opaque; no private object or resident enumeration.
V45-041	Issued artifact revoked after PDF downloaded	Public verification shows REVOKED; old screenshot not proof of active status.
V45-042	Share recipient after expiry	Immediate deny; access log records rejected attempt.
V45-043	Share recipient from wrong domain/device if constrained	Access blocked according to explicit share policy; no bypass via signed URL.
V45-044	Admin shares whole household file for one vendor task	Field minimization prevents unnecessary PII disclosure.
V45-045	Public verification when backend outage	Explicit temporarily unavailable, never unauthenticated VALID.
V45-046	QR confirms issue but contents tampered	Checksum comparison fails or only serial status asserted; no false content validation.
V45-047	Revised artifact old QR	Old status SUPERSEDED with new public reference if policy allows.

62.7 Tender and contract

Test ID	Trigger / adversarial condition	Required proof
V45-048	Late RFQ bid completed after server deadline	LATE according to documented timestamp policy.
V45-049	Bid view attempted before sealed opening	Denied and audited, including privileged administrators unless authorized ceremony.
V45-050	Vendor changes signed bid after close	Original hash preserved; late replacement rejected.
V45-051	Award winner conflicts with configured scoring rubric	Requires documented review/justification and authority; no auto-award.
V45-052	Single source emergency purchase	Audited exceptional route; no fake competitive bid records.
V45-053	PO 10 items, GRN 8 items, invoice 10 items	Two-item variance visible, settlement cannot silently assume full receipt.
V45-054	Vendor insurance expired mid-contract	Compliance task and policy-based assignment restriction; no claim of active insurance.
V45-055	PO fragmented to bypass limit	Configurable exception flag for review; no automated allegation of misconduct.

62.8 Property and governance

Test ID	Trigger / adversarial condition	Required proof
V45-056	Nominee asks for automatic ownership based only on nomination	No title transfer; legal/committee review path.
V45-057	Bank asks for NOC not required by local law	Document may be optional factual letter; app does not assert legal mandate.
V45-058	Current tenant requests former tenant agreement	Denied; historical boundary applied to search, export and file endpoints.

Test ID	Trigger / adversarial condition	Required proof
V45-059	Renovation society NOC presented as civic approval	Document wording restricts issuer scope; verifier cannot claim municipal permit.
V45-060	AGM minutes edited after adoption	New correction record and signatures; old adopted text preserved.
V45-061	One member represented by conflicting proxies	Eligibility and quorum engine detects conflict; audit outcome.
V45-062	Builder handover includes missing warranties	Packet remains exception/open; no automatic acceptance.

62.9 Compliance, privacy and legal holds

Test ID	Trigger / adversarial condition	Required proof
V45-063	Expired lift certificate uploaded in newer file name	Actual expiration wins; remains overdue until valid evidence verified.
V45-064	Certificate issuer portal offline	UNAVAILABLE / REVIEW_PENDING; no green compliance claim.
V45-065	Legal hold and deletion request conflict	Hold decision logged; delete only eligible unrelated copies.
V45-066	Consent withdrawal with legitimate accounting record	Voluntary Aadhaar data handling changes, legitimate finance record not silently destroyed.
V45-067	Subject request spans two societies	Scope isolated; accurate redactions and per-society evidence.
V45-068	Sensitive document indexed then deleted	Document and derived snippets/vector index purged according to policy.
V45-069	Insurance claim file includes unrelated neighbour image	Export review redacts third-party data and preserves original privately.
V45-070	Updated law/rulepack with new effective date	Past decision remains explainable; new requests use correct applicable version.

62.10 Operations, failure recovery and performance

Test ID	Trigger / adversarial condition	Required proof
V45-071	OCR worker crashed for 10,000 queued files	Queue isolated, resource quotas respected; critical deterministic requests continue.
V45-072	Nightly obligation scheduler executed twice	One obligation instance and idempotent notification deliveries.
V45-073	Database PITR restores revoked document before revocation	Replay/reconciliation prevents verifier from returning ACTIVE incorrectly.
V45-074	Service account forgets tenant context	RLS denies records; no accidental unrestricted worker read.
V45-075	Old mobile client sends obsolete issue schema	Versioned recoverable error, no partial state mutation.
V45-076	Signed URL leaked via logs	No secret tokens in ordinary logs; expiry/revocation and incident path tested.
V45-077	100 parallel exports of high-sensitivity cases	Quota/rate limiting, audit and no resource starvation.
V45-078	Long-term audit evidence copied across storage tier	Hashes/manifest and metadata preserved and restorable.

v4.5 additional acceptance cases: 78; v4.4 retains its prior 36 tests. Combined inventory has at least 114 explicitly defined cases, with overlaps consolidated in the QA traceability tool before execution.

62.11 Acceptance evidence package

- Test environment details, service image hashes, seeded fixture versions, society and role setup, rulepack/template versions.
- Exact API requests/responses or reproducible UI steps; asserted status transitions; database and outbox receipts.
- Separate negative assertion: no improper document, payment, access, public response or data leak occurred.
- Automated report links to issue IDs/commit hash and remediation tickets; manual signoff retains reviewer/date/reason.
- Accessibility/localization smoke screenshots and PDF render comparison for Marathi, Hindi and English.
- Performance baselines and restore test log from a representative low-cost deployment profile.

63. Prioritized build backlog and production activation gates

This is sequencing, not a claim that a team can implement the entire product in a particular number of weeks. The engineering contract deliberately avoids requiring a paid model endpoint or an expensive government portal for baseline operation.

Delivery slice	Deliverable	Exit gate
P0-A: Evidence + private storage	Object metadata, quarantine, malware scanning, content policy, checksum, signed download, audit and RLS	V45-008/009/011/022/074; no unscanned downloads
P0-B: Records and core forms	Issue register, PDF templates, immutable input, serial, versioned output and corrections	V45-034/035/036; stable serial and bytes
P0-C: Case and packet framework	Business case keys, requirement graph, intake exception, manual verification and permissions	V45-001/002/006; authoritative case state
P0-D: Existing financial output migration	Ledger-derived statements/receipts with accurate allocation and as-of balances	V45-032/037/038; finance tie-out
P1-A: Society correspondence	Notice, meeting agenda/minutes, register extracts, announcements and controlled sharing	V45-040/042/060; disclosures logged
P1-B: Tenant & household evidence	Owner consent, agreement meta, move-in/out tasks, safe ID handling, conflict	V45-016/017/018/058; truth and privacy
P1-C: NOC clearance and approval	Rulepacks, dynamic evidence, maker/checker, issue-time fresh precheck, QR status	V45-025/033/039/041; exactly-once issuance
P1-D: Compliance and renewals	Obligation engine, expiry scheduler, evidence/inspection tasks and dashboard data	V45-063/064/070/072; no false green
P2-A: Procurement and tender	Requisition, sealed bids, comparison, contract, PO/GRN/invoice controls	V45-048-055; policy-bound award
P2-B: Complex member lifecycle	Transfer, succession, nomination, share certificate registry, statutory copy requests	V45-056/057; jurisdiction-reviewed rulepack
P2-C: Governance and legal archive	AGM/resolution, management transition, dispute/legal hold, audit packets	V45-060/061/065; immutable history
P2-D: Search and migration	Metadata/full-text indexing, bulk historical scanning and optional local OCR	V45-012/015/068/071; no privileged leakage
P3: Regulated integrations	Licensed eSign/DSC, lawful authority verification, optional hardware integrations	Separate commercial/legal/security gate; no silent substitution

63.1 Build-versus-buy and cost-aware engineering decisions

Component	Recommended default	Future integration and cost reality
Java backend	Reuse existing Java/Spring Boot conventions if deployed; otherwise integrate through existing service stack.	Do not rewrite auth or billing to build documents.
PDF generator	PDFBox/FOP/HTML rendering; design templates as versioned assets.	No per-document AI token fee; CPU/ops still cost money.
OCR	Manual first; self-hosted Tesseract only if input volume and accuracy justify it.	CPU, storage and accuracy-testing cost; cap demand.
Search	PostgreSQL metadata and FTS; start with exact filters.	Local search requires data privacy and indexing budget.
Malware scan	ClamAV and file processing isolation.	Signature update/operations and infra cost.
Storage	Private S3-compatible object store with encryption/versioning and lifecycle.	Cloud object/bandwidth/storage still have fees.
eSign/authority APIs	Disabled unless legally assessed and funded.	Often provider, licensing and per-transaction cost.
Verification/status	Signed opaque tokens + first-party DB status.	No external model required; cannot claim issuer authority.

63.2 Definition of done: no exceptions

- Documented state machine + business trigger + normal/alternate/blocked/appeal flows and clear assurance language.
- Tenant-scoped schema, migration, safe rollback/disposition plan, data classification, RLS and access tests.
- Immutable audit, issue/version semantics, idempotency/concurrency, source-of-truth checks and error codes.
- Privacy/security controls, file/QR hardening, key rotation, credential revocation and legal applicability signoff.
- Accessibility, localization, reference templates, PDF render tests, storage backup/restore and monitoring.
- Named product/security/finance/legal approvers as relevant; acceptance evidence and runbook in deployment repository.
- Clearly defined implementation status (NOT_BUILT, BACKEND_READY, INTEGRATED, QA_PASSED, PRODUCTION_ENABLED) and feature-flag rollout.

64. Decision registry, source hierarchy and traceability

The items below must be settled per deployment and reviewed when applicable law or local policy changes. A production-grade specification does not pretend to resolve unknown legal details or make an unsupported government-verification promise.

ID	Decision	Owner	Exit condition
D-01	Society legal type, registered state and operative bylaws	Product owner + local legal reviewer	Hard dependency on statutory forms, membership/transfer and governance
D-02	Who may certify copies, issue NOCs and sign financially sensitive letters?	Society committee + security	Approve delegation/authority matrix; signer role alone insufficient
D-03	Which document types genuinely require manual approval?	Product + finance + legal	Default high-risk approval; informational output can be deterministic

ID	Decision	Owner	Exit condition
D-04	Can Aadhaar offline verification be offered; by which legal method and entity?	Privacy/legal + UIDAI applicability reviewer	Alternative ID always available; no full-number storage by default
D-05	Which eSign/DSC provider and eligible document classes?	Legal + security + finance	No default assumption of provider-free legally recognized signing
D-06	Retention/hold schedule by society, category and purpose	Privacy/records/legal	Replace one-size-fits-all retention and document deletion
D-07	Does society operate procurement tender with committee opening?	Committee + procurement	Configure sealed-bid policy and financial limits
D-08	Which registers are statutory vs operational convenience?	Legal + compliance	Keep appropriate record structure and audit history
D-09	Which documents may be externally shared?	Privacy + committee	Recipient, purpose, expiry and minimum data
D-10	Is a physical original required and who is custodian?	Society administrator	Index/check-out/return instead of pretending scans are originals
D-11	Who reviews high-risk compliance expiry/overrides?	Compliance + safety	Human source-verification and escalation path
D-12	Is OCR enabled and on what infrastructure/languages?	Engineering + product	No paid AI endpoint, measured accuracy and bounded cost
D-13	How are user identity/global account and society membership represented?	Platform architecture	Reconcile v3 portable identities and consolidated master schema before migrations

64.1 Traceability to retained blueprint

v4.5 domain	Earlier section(s)	New required extension
Property, owner, tenant, household	Original master 7.1 and 8; v4.4 24-25	Registers, transfer/succession packets, source claims and historical privacy
Finance and NOC	Original 7.3, 8.7, 10; v4.4 25	As-of proof and corrections, jurisdiction-sensitive NOC, accurate refund/receipt semantics
Vendor, tender, procurement	Original operations/procurement; v4.4 26	Full sealed bid-to-contract-to-GRN lifecycle and conflicts
Documents / verification	Original 10; v4.4 20-24, 28-31	Case packets, evidence provenance, official extracts, signatures and field-level disclosure
Governance/meeting/legal	Original governance and compliance modules	Minutes, quorum, legal holds, certified copies and statutory records
Architecture/DB/API/security	Original 5/6/12/14; v4.4 28-32	Registers, cases, obligation engine, cross-society tenant security and versioned API
Automation and intelligence	Original 16; v4.4 32	Deterministic rules/metadata search, optional local OCR and explicit zero-paid-LLM boundary
Roadmap and QA	Original 17/18; v4.4 34/35	78 new negative/e2e acceptance cases, prioritized gates and production proof

64.2 Official reference sources (review date: 04 Nov 2026)

Source	Location / URL	Why it matters
Maharashtra Commissioner for Co-operation - model bylaws listing	https://sahakarayukta.maharashtra.gov.in/1105/Model-Bye-Laws?Doctype=1CC73BAD-36CA-45AA-BA03-2119E31B6337	Government listing includes revised model bye-law documents dated August 2026; registered society instruments and law control applicability.
UIDAI Offline Verification Seeking Entity registration	https://uidai.gov.in/pa/ovse	Official eligibility/registration and offline verification rules; product must not assume unrestricted lawful access.
UIDAI FAQ on Aadhaar offline verification	https://uidai.gov.in/images/FAQ_OVSE.pdf	Only last four digits where permitted, consent/alternatives, QR/XML signature validation and purpose limitations.
UIDAI Aadhaar Act section 8A	https://www.uidai.gov.in/hi/aadhaar-act	Consent and legal restrictions for offline verification.
CCA on eSign	https://cca.gov.in/eSign.html	Regulated eSign/CA distinction and legal signing process.
CCA on digital signatures	https://cca.gov.in/digital_sign.html	Cryptographic signature and certificate validation concepts.
MeitY official DPDP Rules 2025	https://www.meity.gov.in/documents/act-and-policies/digital-personal-data-protection-rules-2025-gDOxUjMtQWa?pageTitle=Digital-Personal-Data-Protection-Rules-2025	Data protection rules, source notifications and phased commencement; production counsel review required.
Society OS v4.4 existing master	Society_OS_Enterprise_Documentation_v4.4_Consolidated.pdf	Retained source blueprint; v4.5 adds without erasing prior content.
Society OS v4.4 document trust specification	Society_OS_v4.4_Document_Trust_Automation_Engineering_Addendum.docx	Retained 20-36 sections and 36 prior acceptance test cases.

64.3 Final product commitment

Document completeness means that any significant society event can be resolved with a traceable digital case, the right evidence, genuine authorized decisions, correct system-of-record updates and only the outputs that the society is permitted to issue. Society OS should never require OpenAI or any other paid generative AI API to perform the foundational workflow, and never replace legal, financial, professional or government authority with a convincing-looking PDF.

IMPLEMENTATION STATUS Specification updated; application code, provider integrations, security review, legal applicability and production test outcomes are not represented as completed. Features must retain a NOT_BUILT status until independently verified.